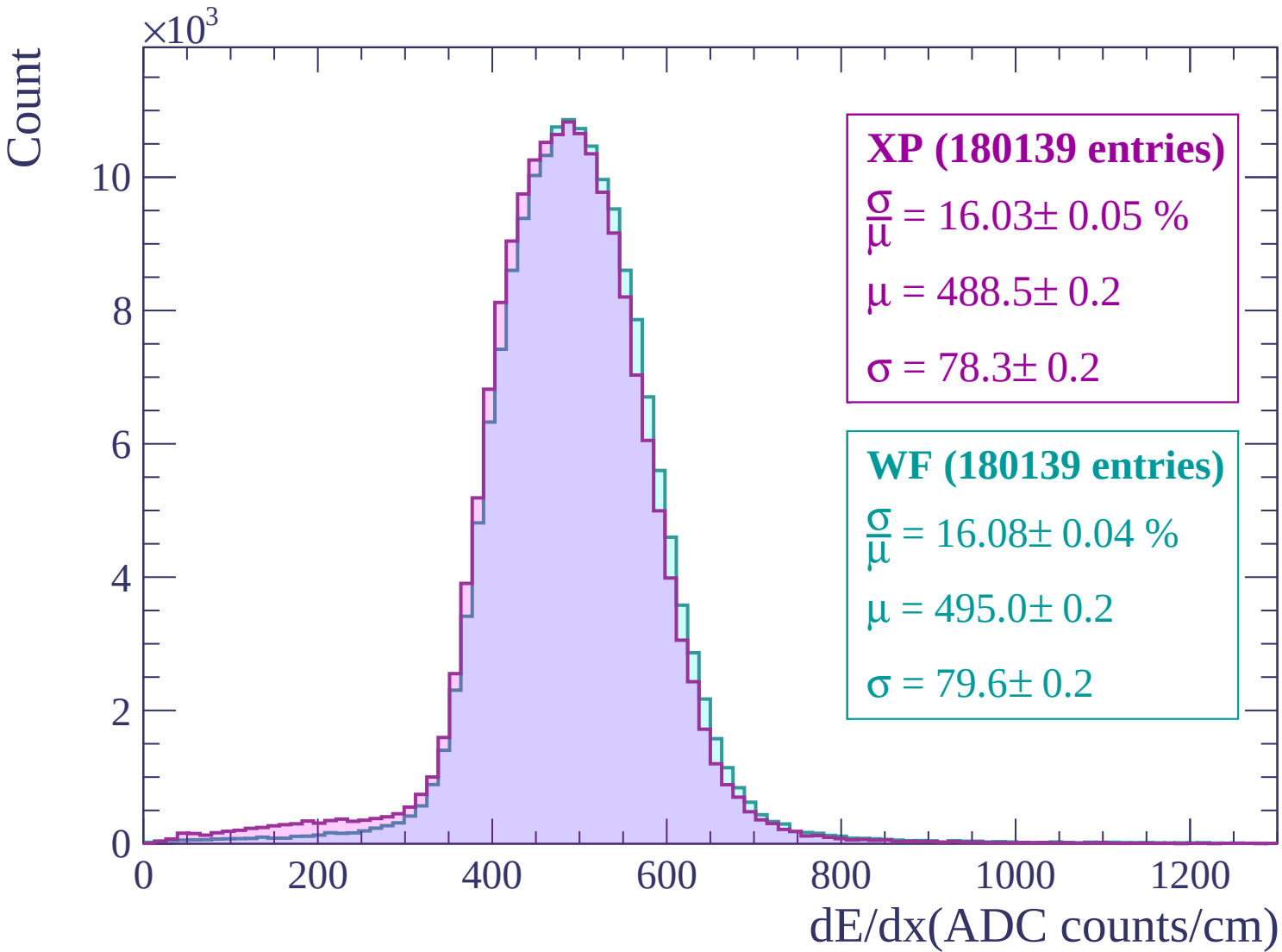
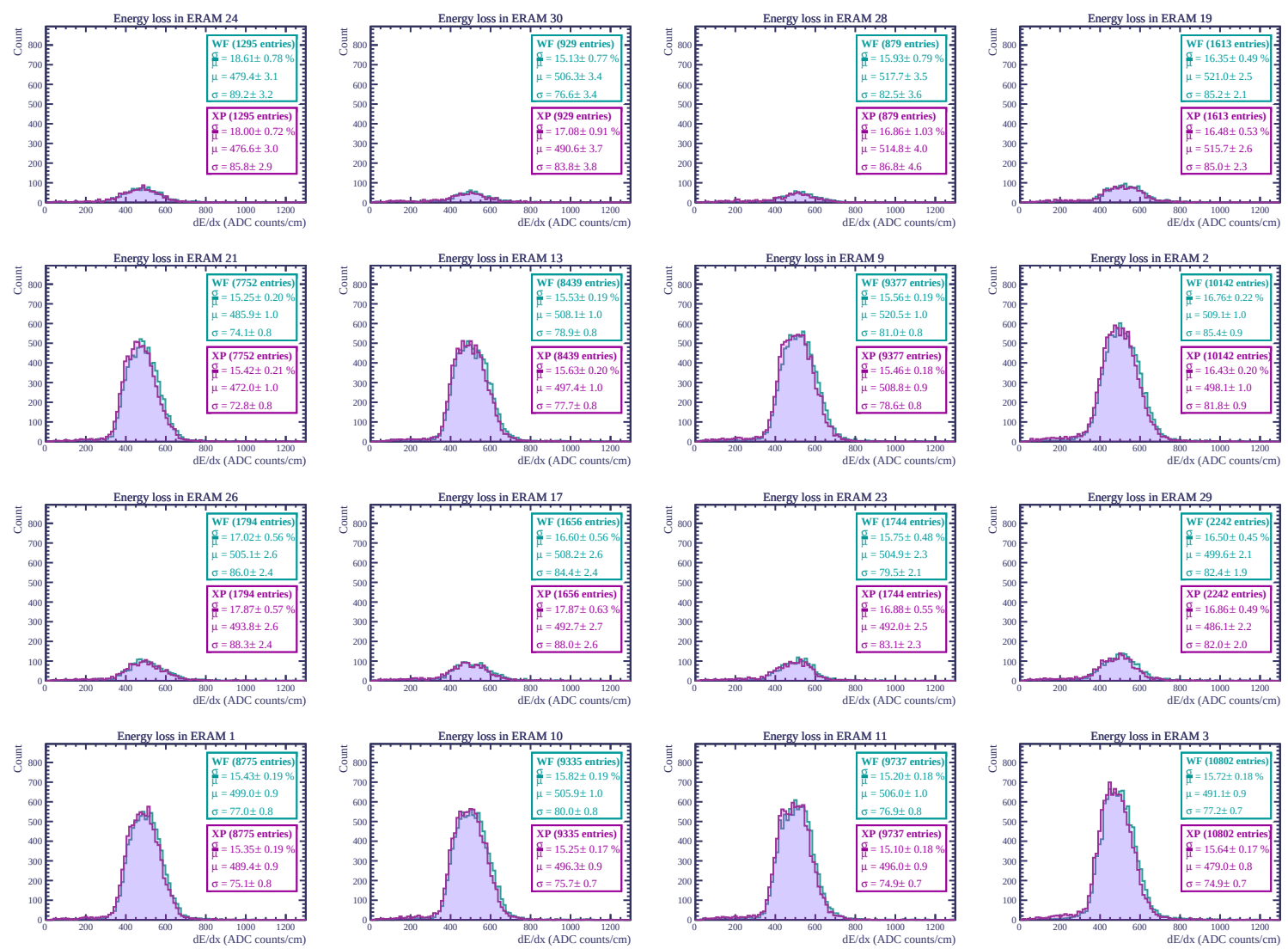
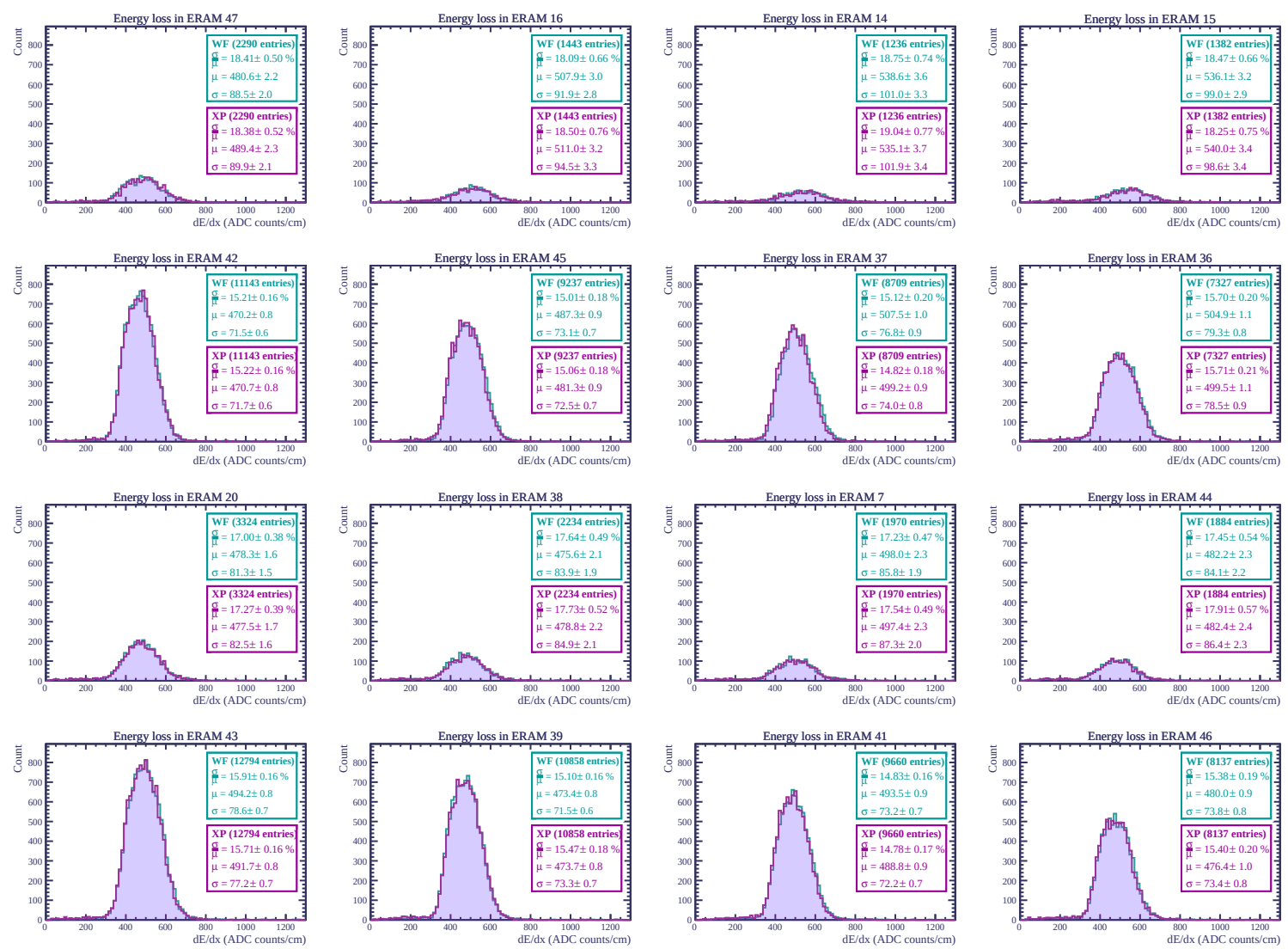
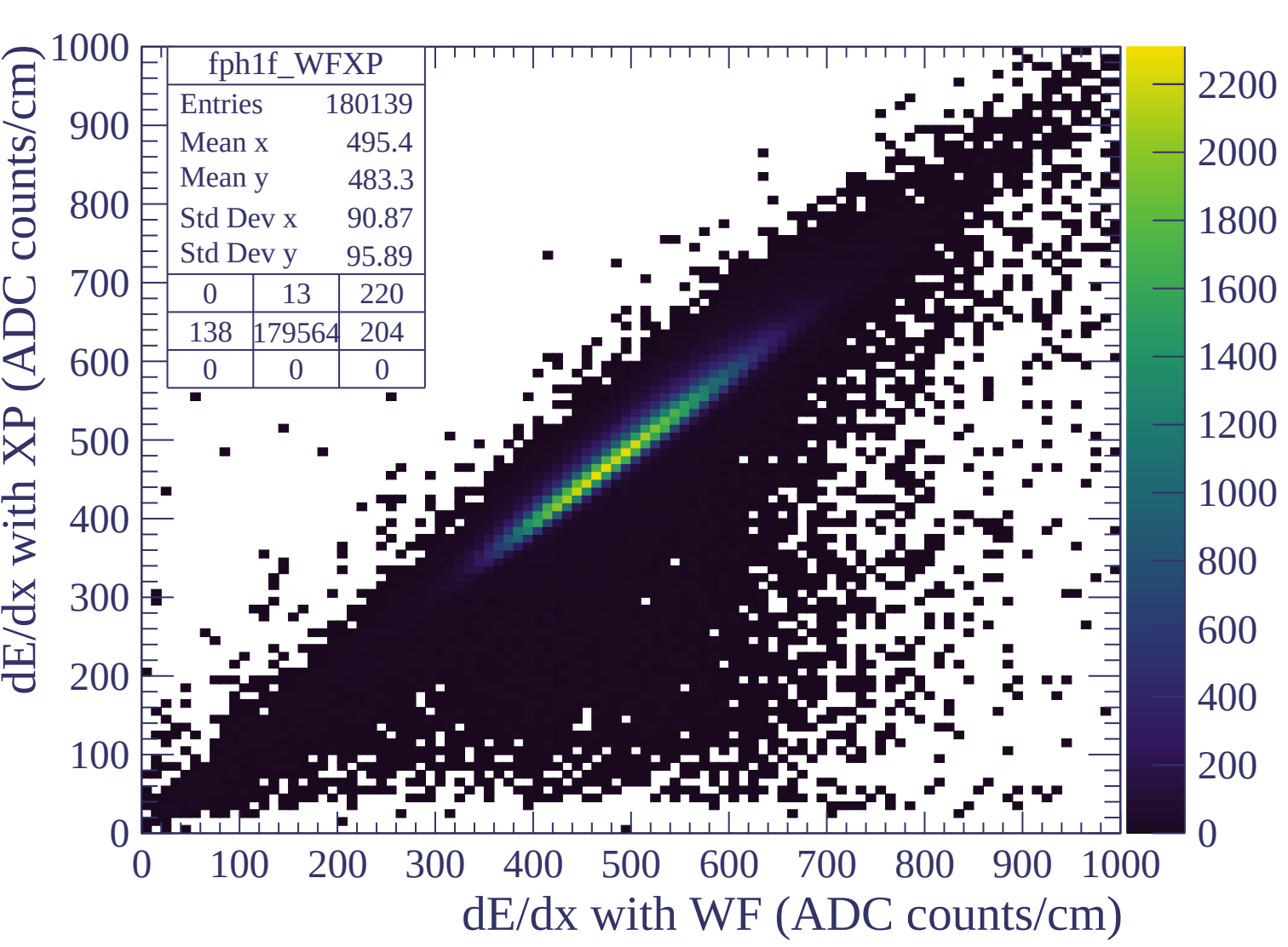
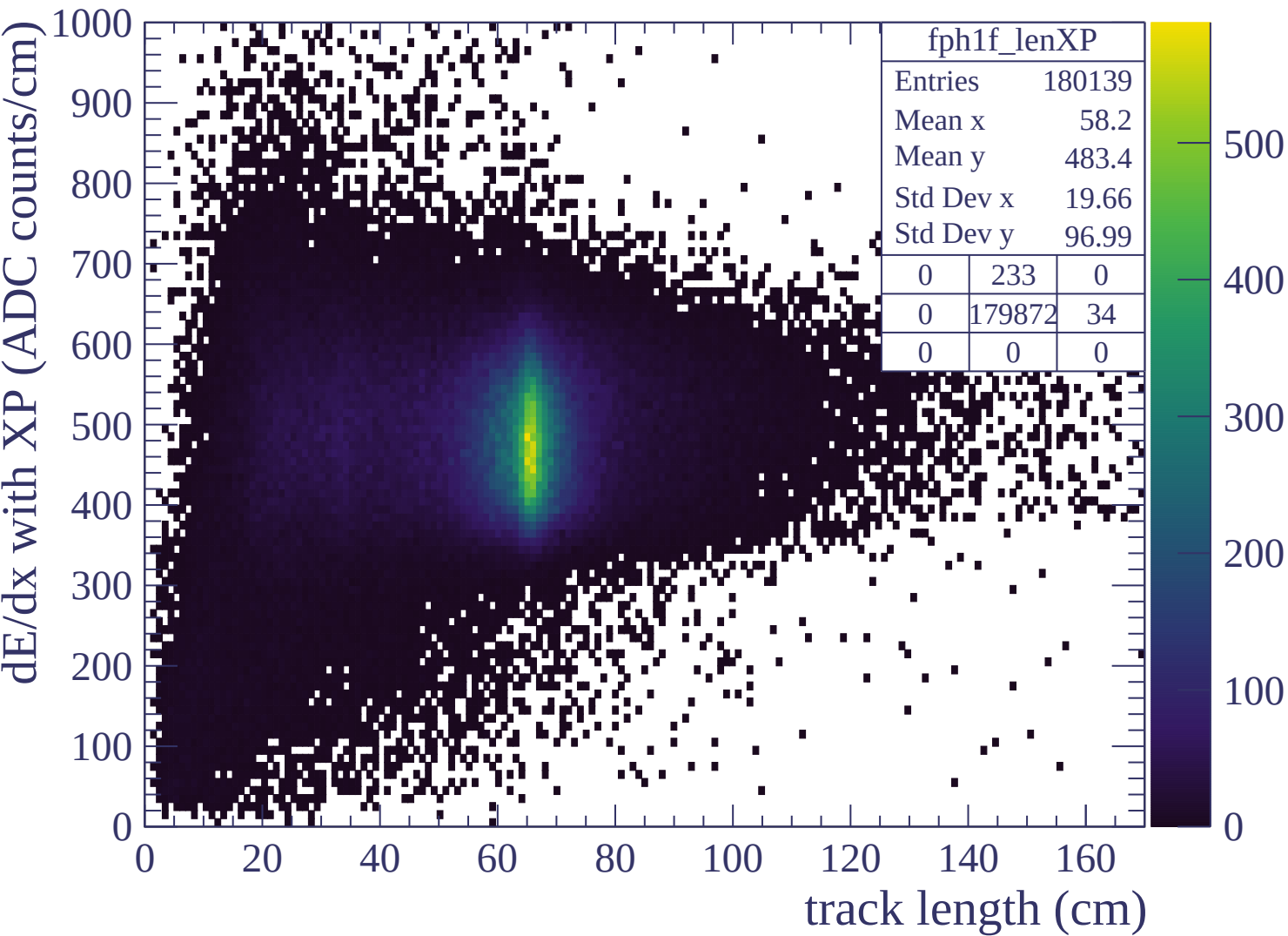


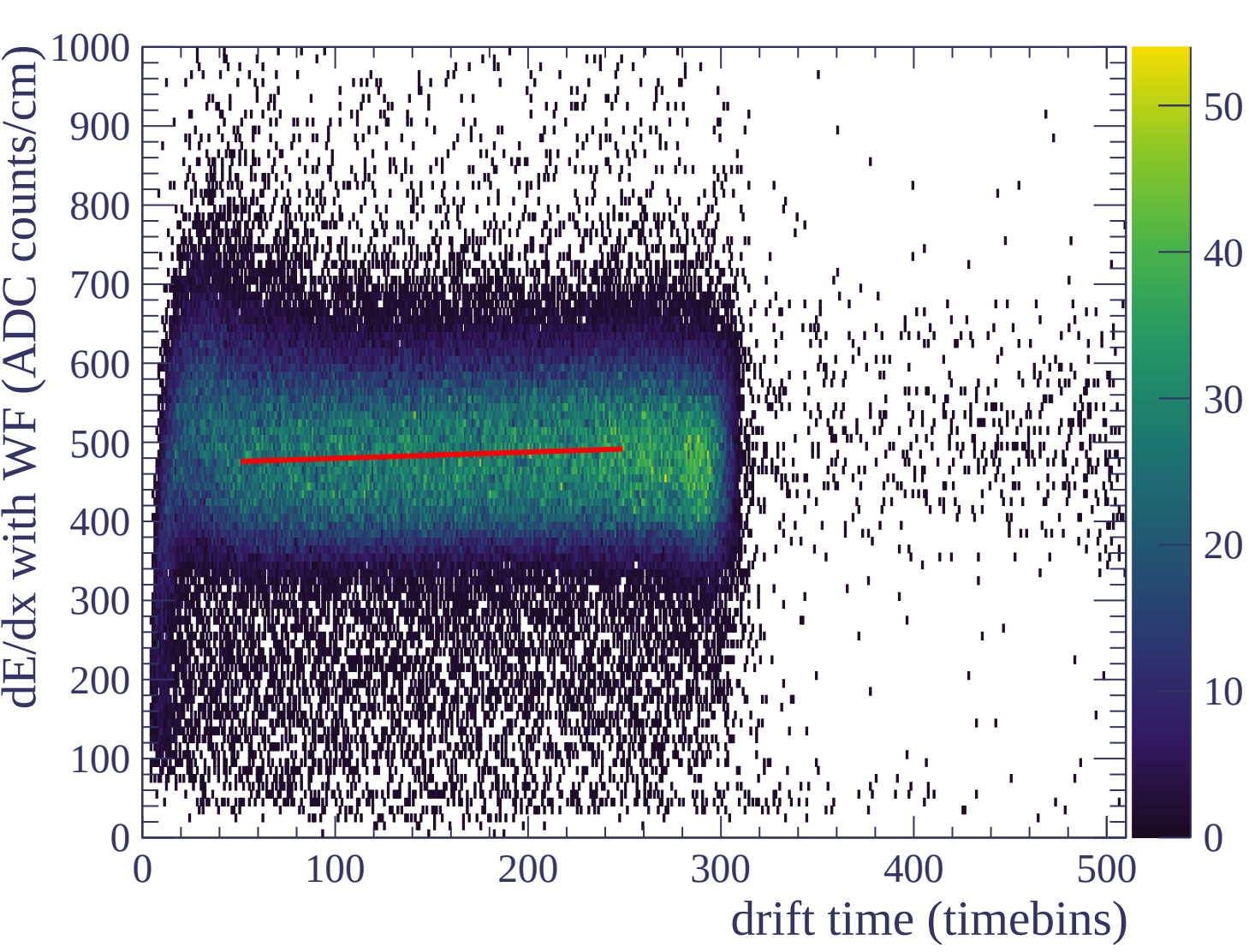


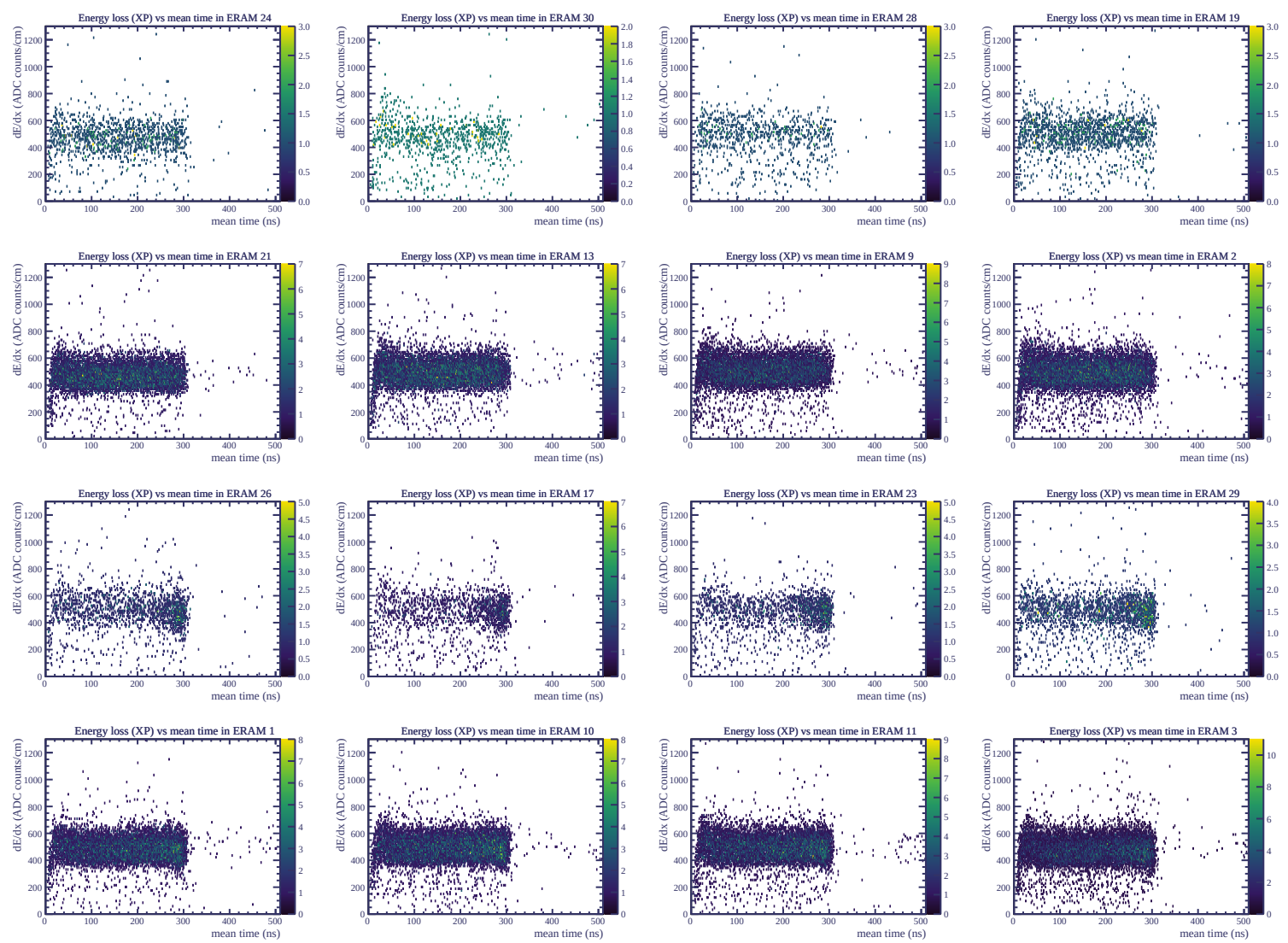


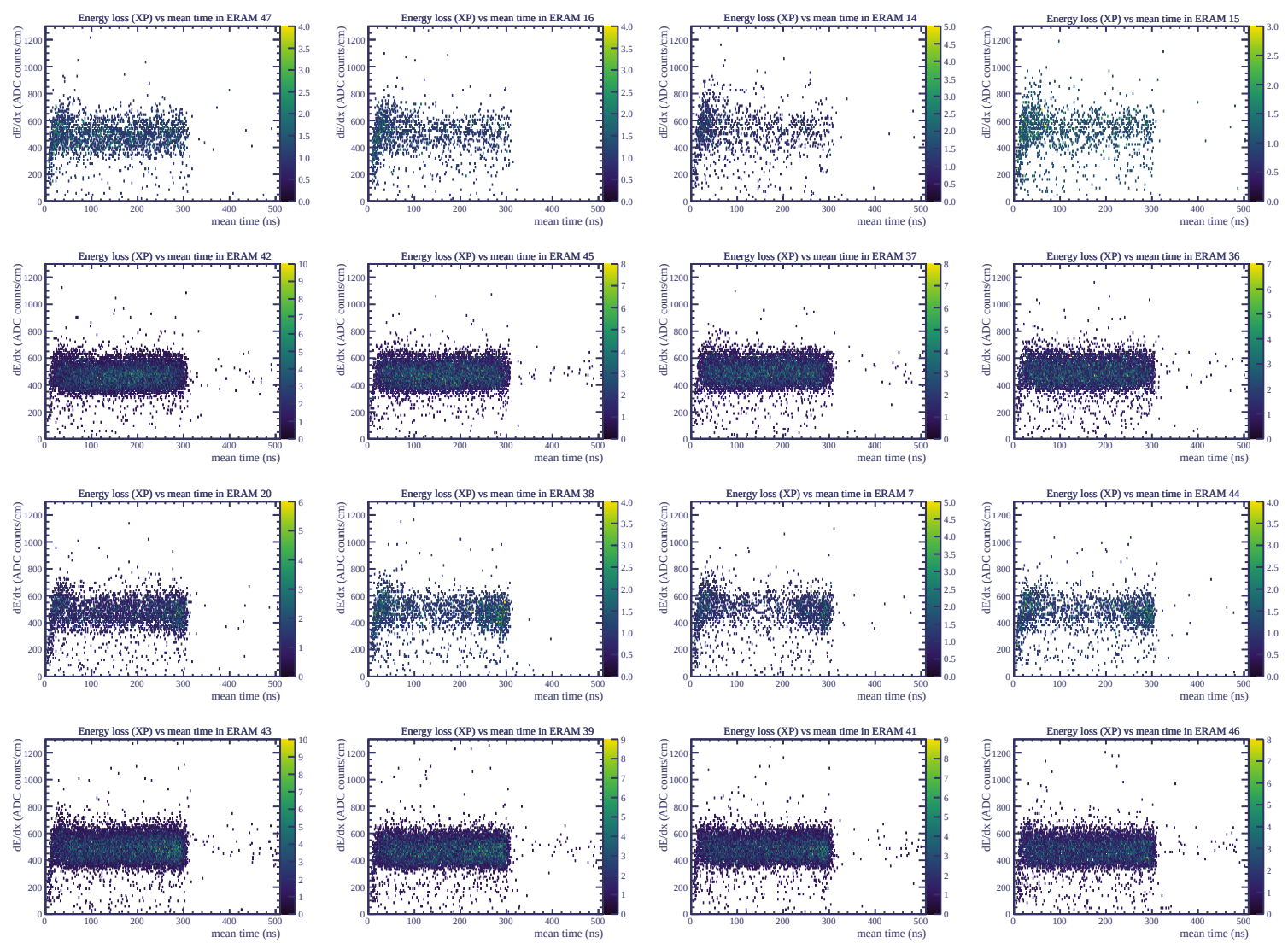




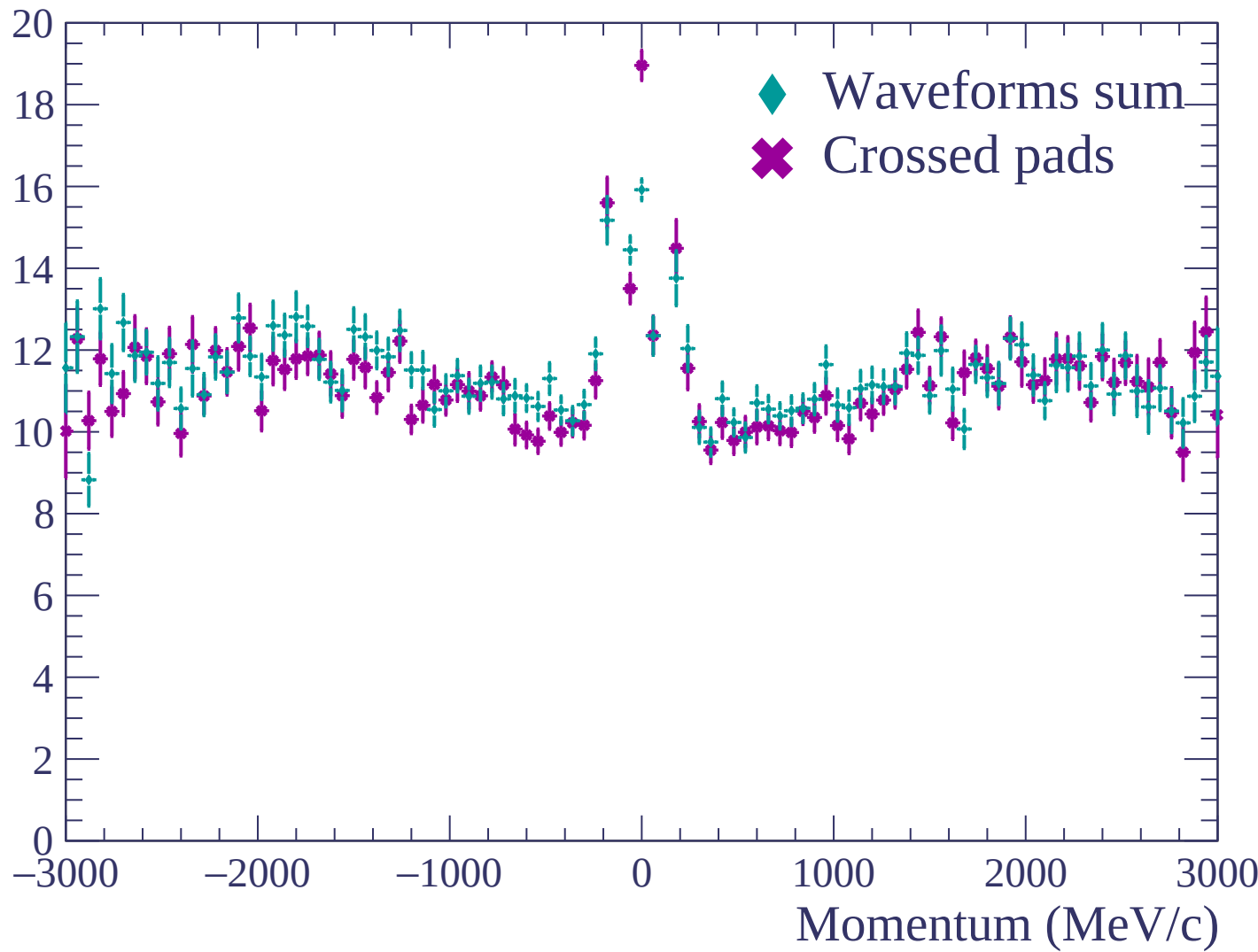


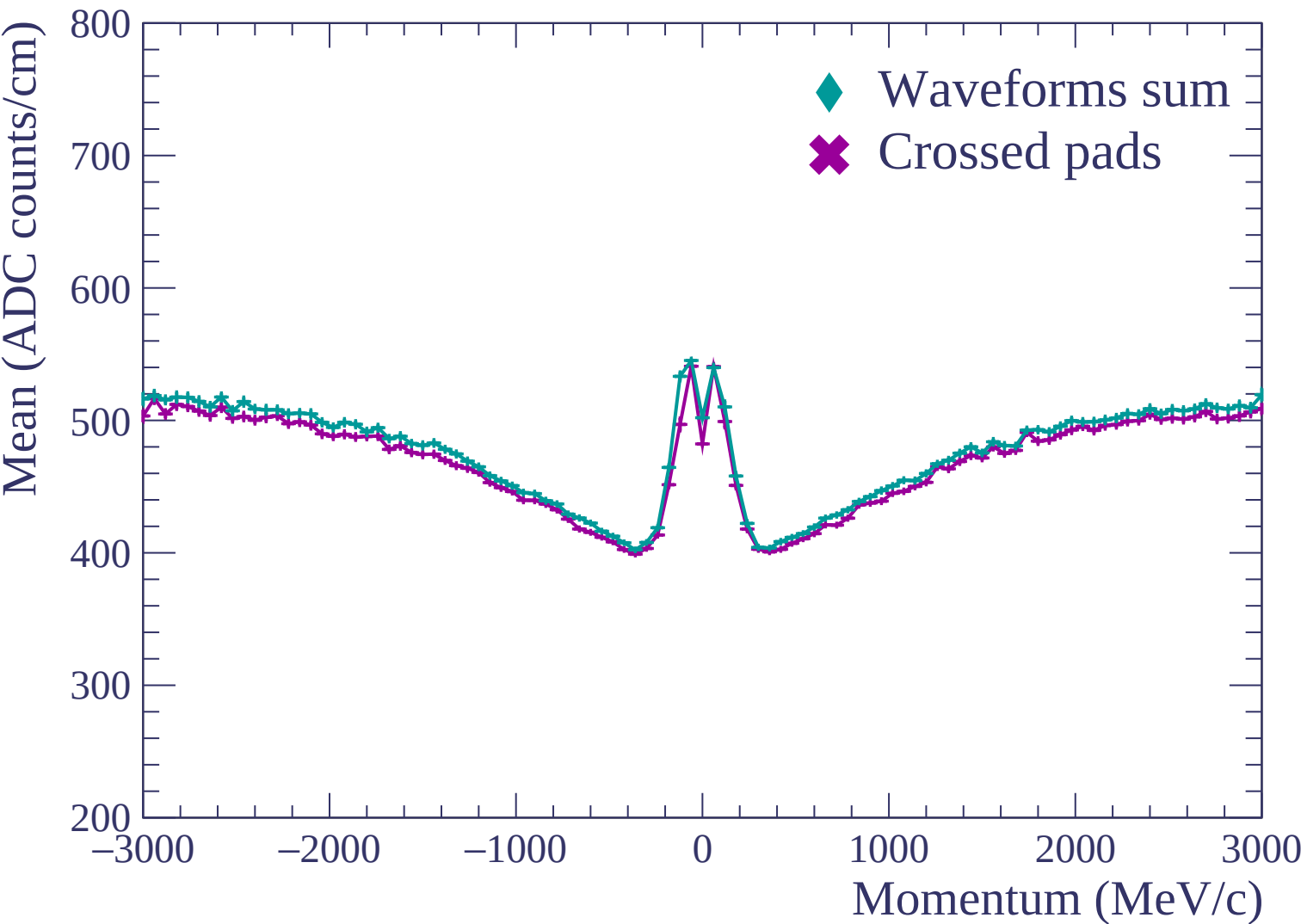


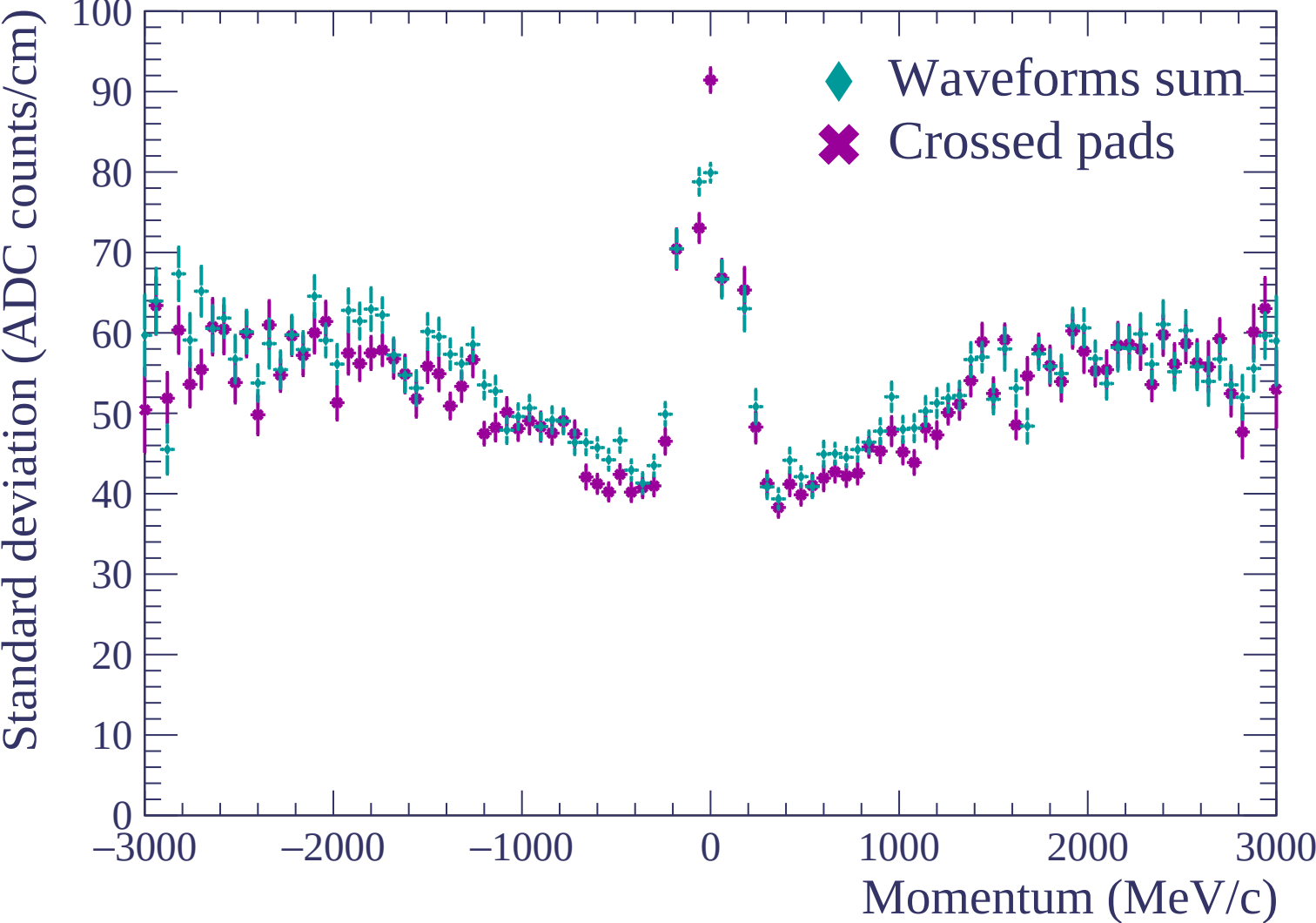


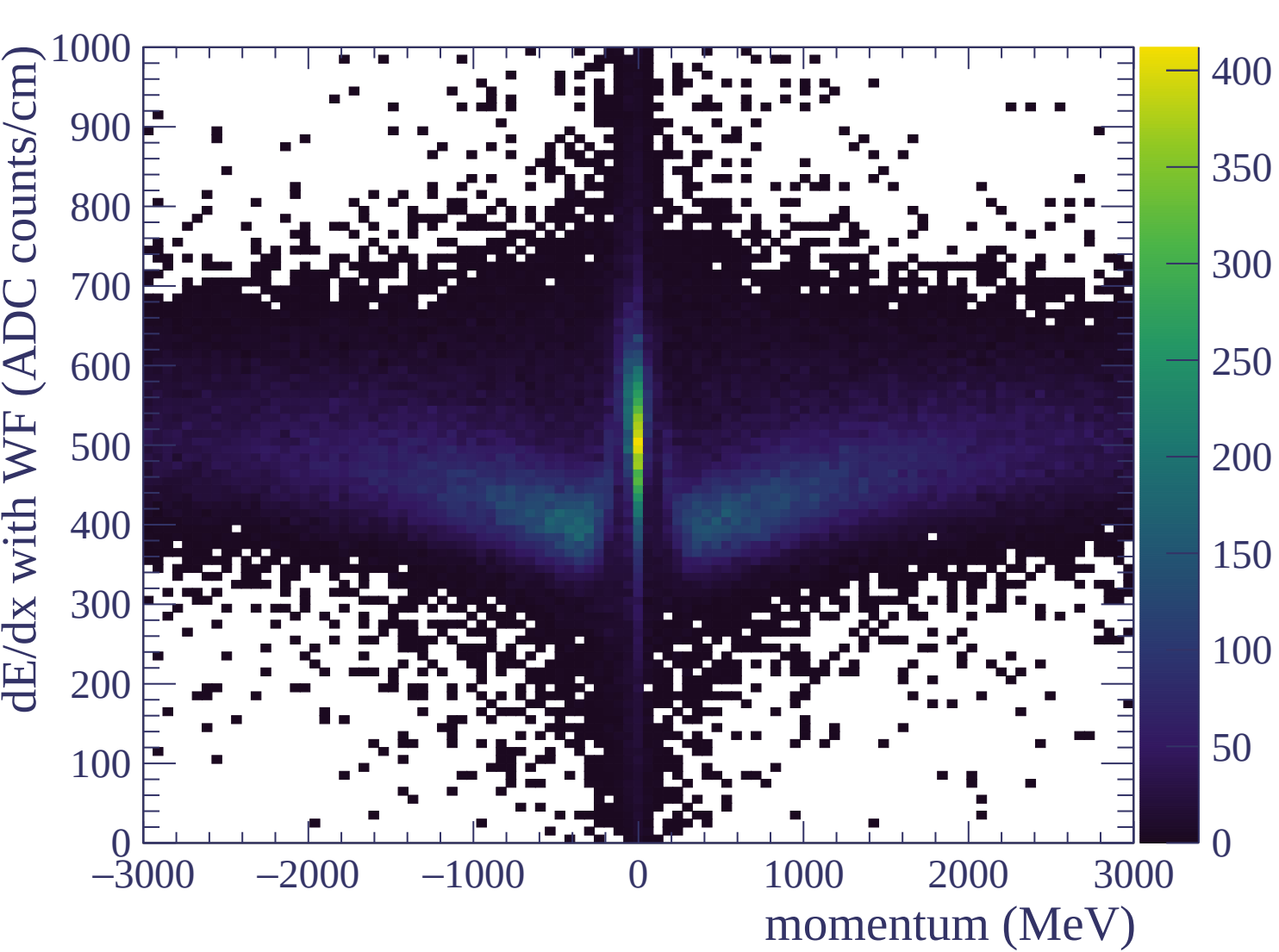


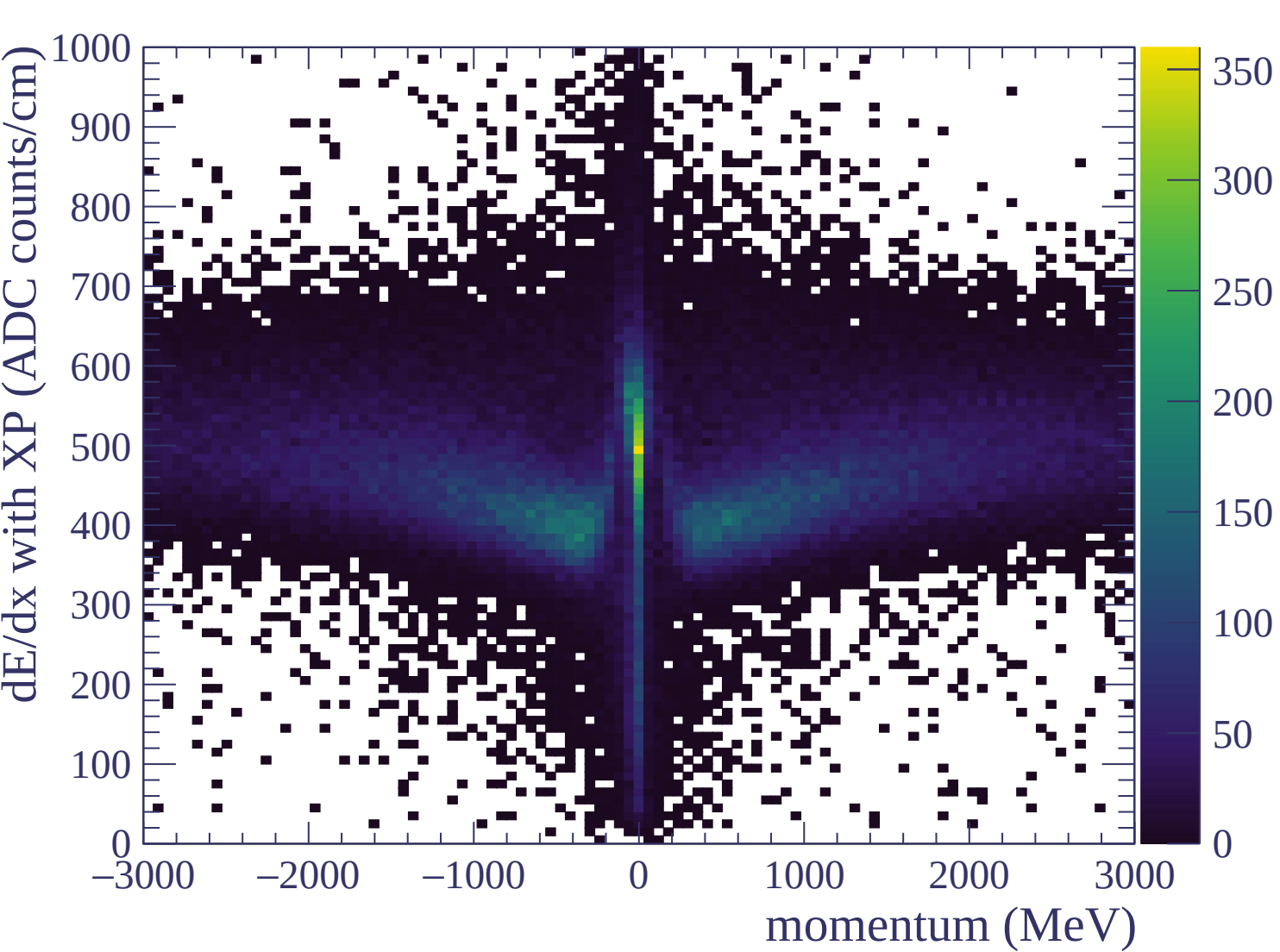
Resolution (%)

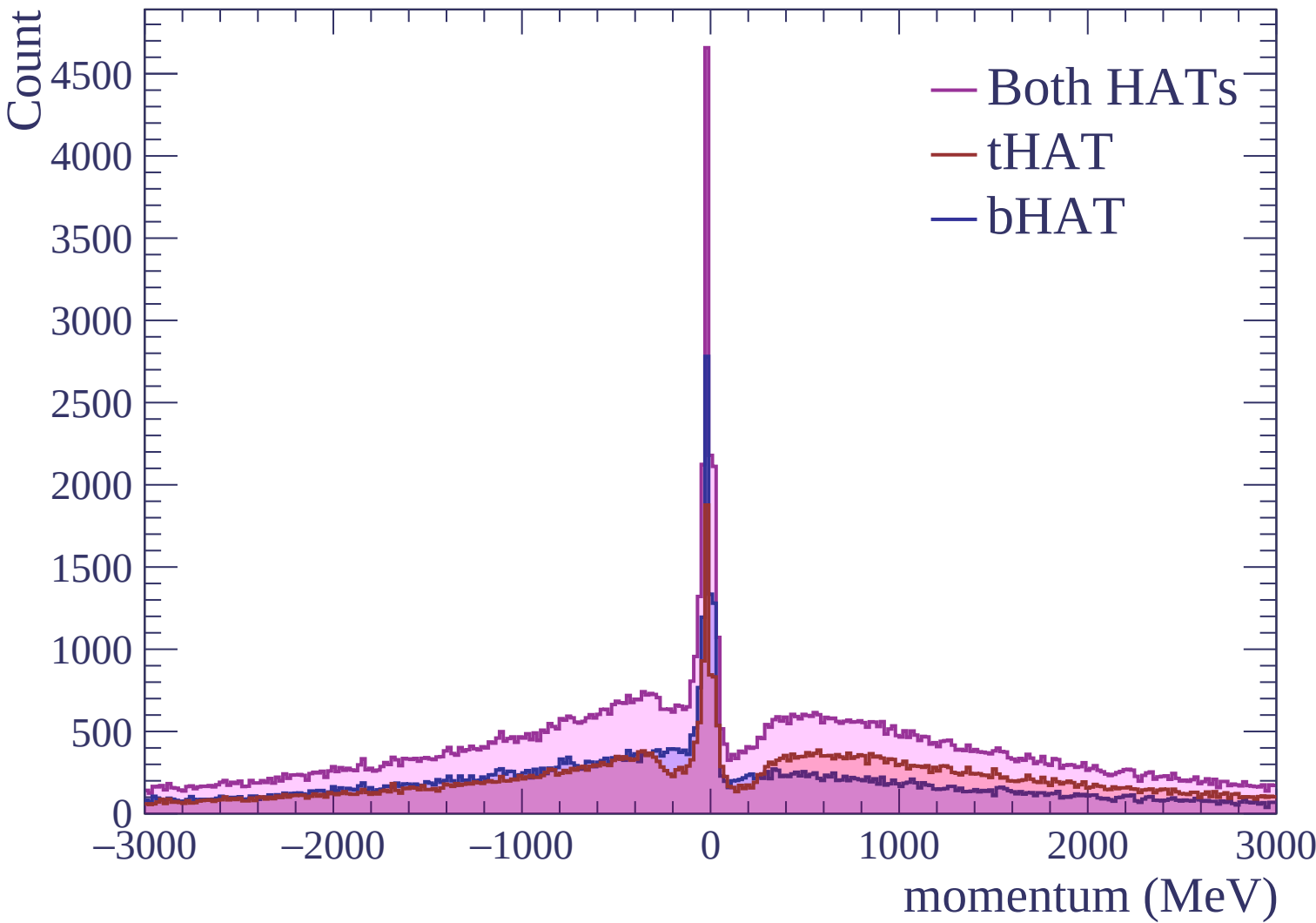


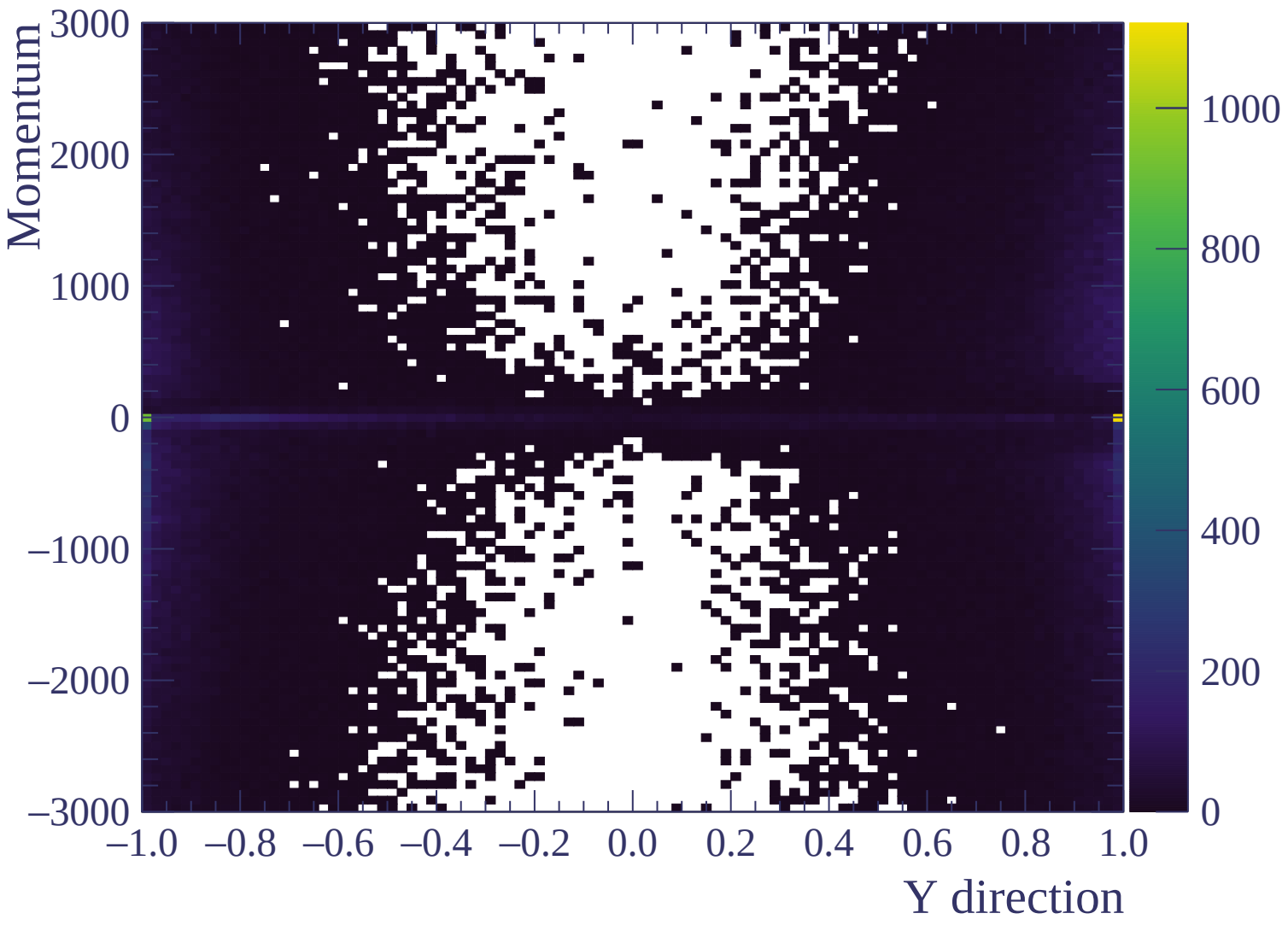


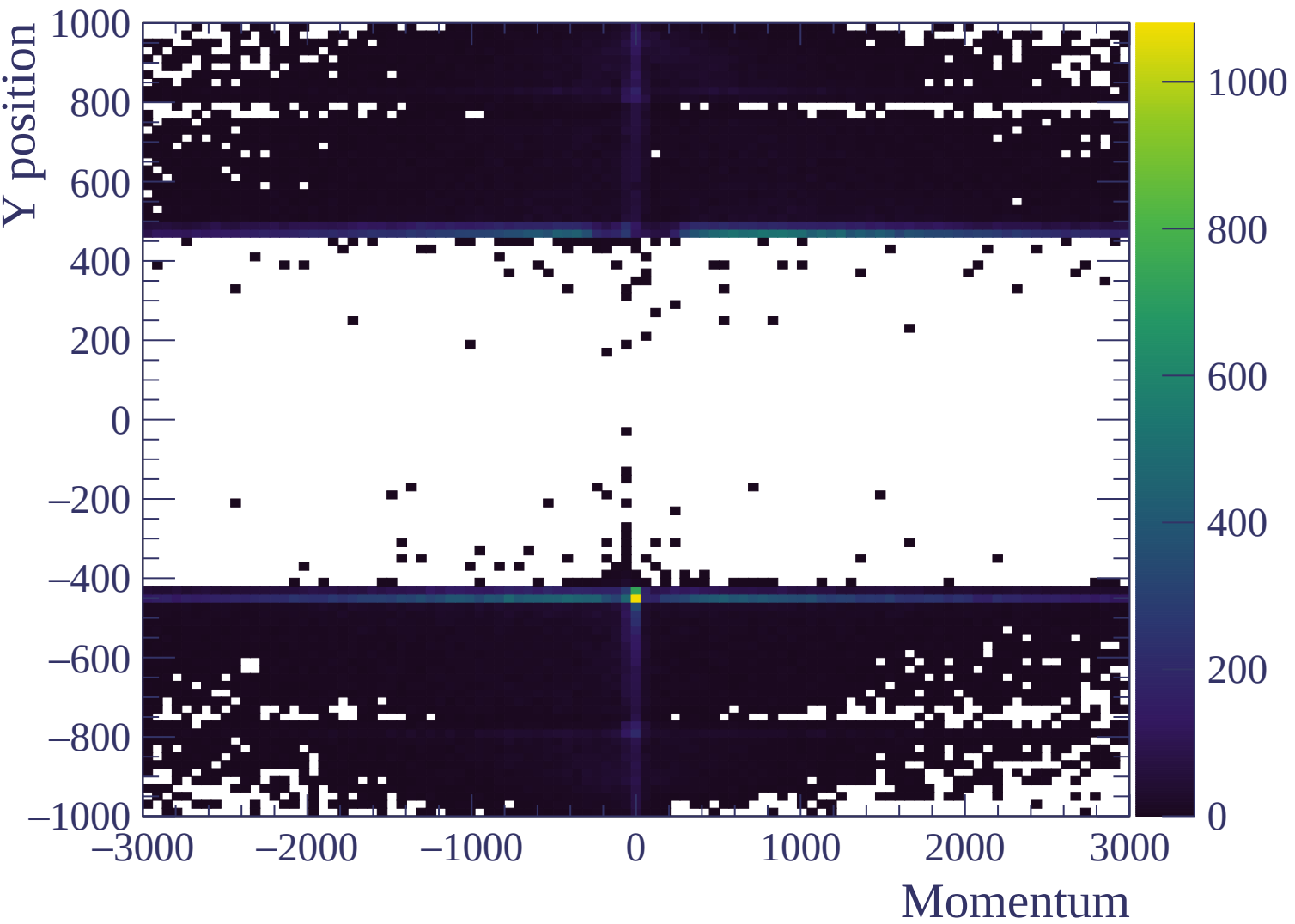


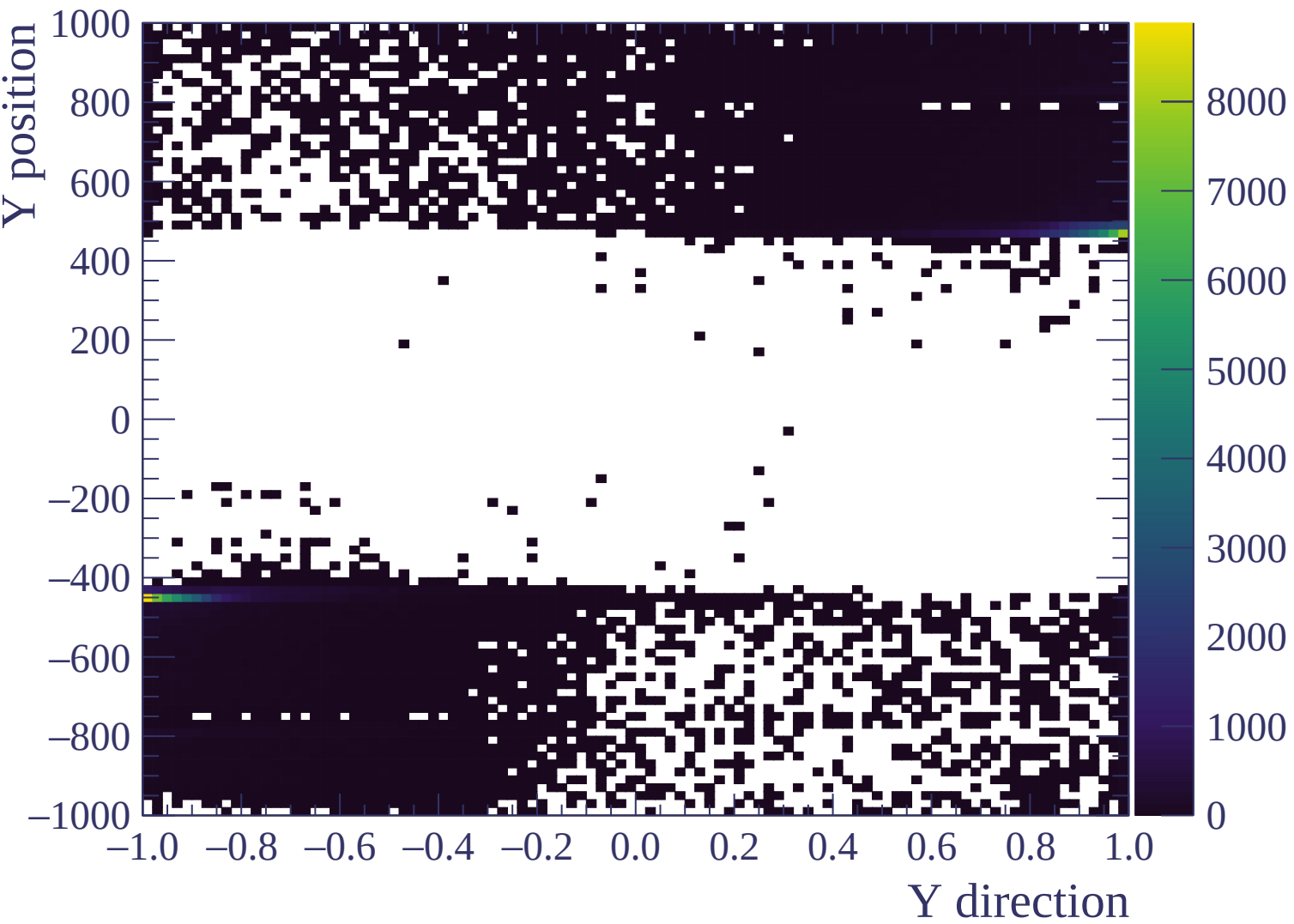






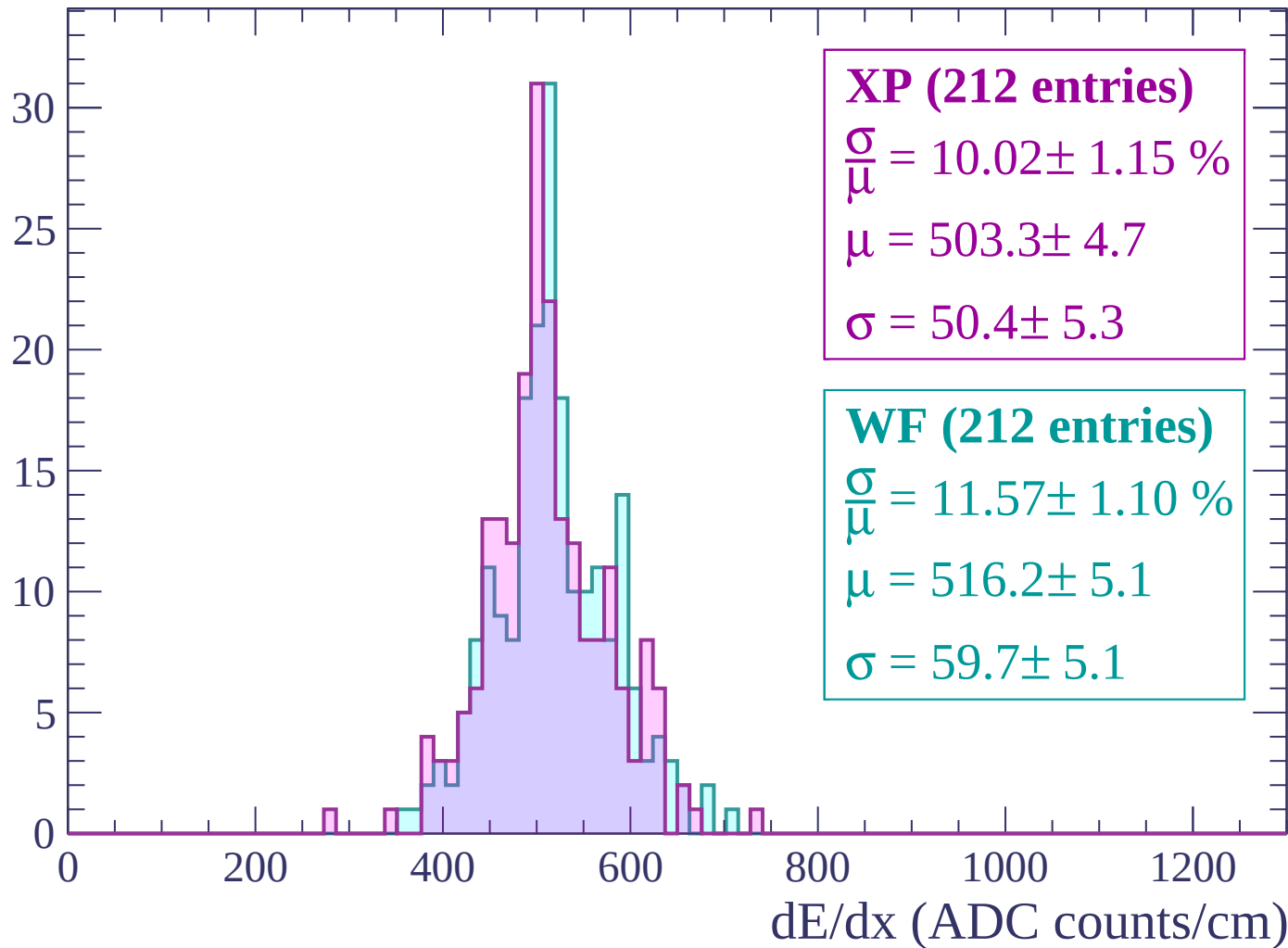






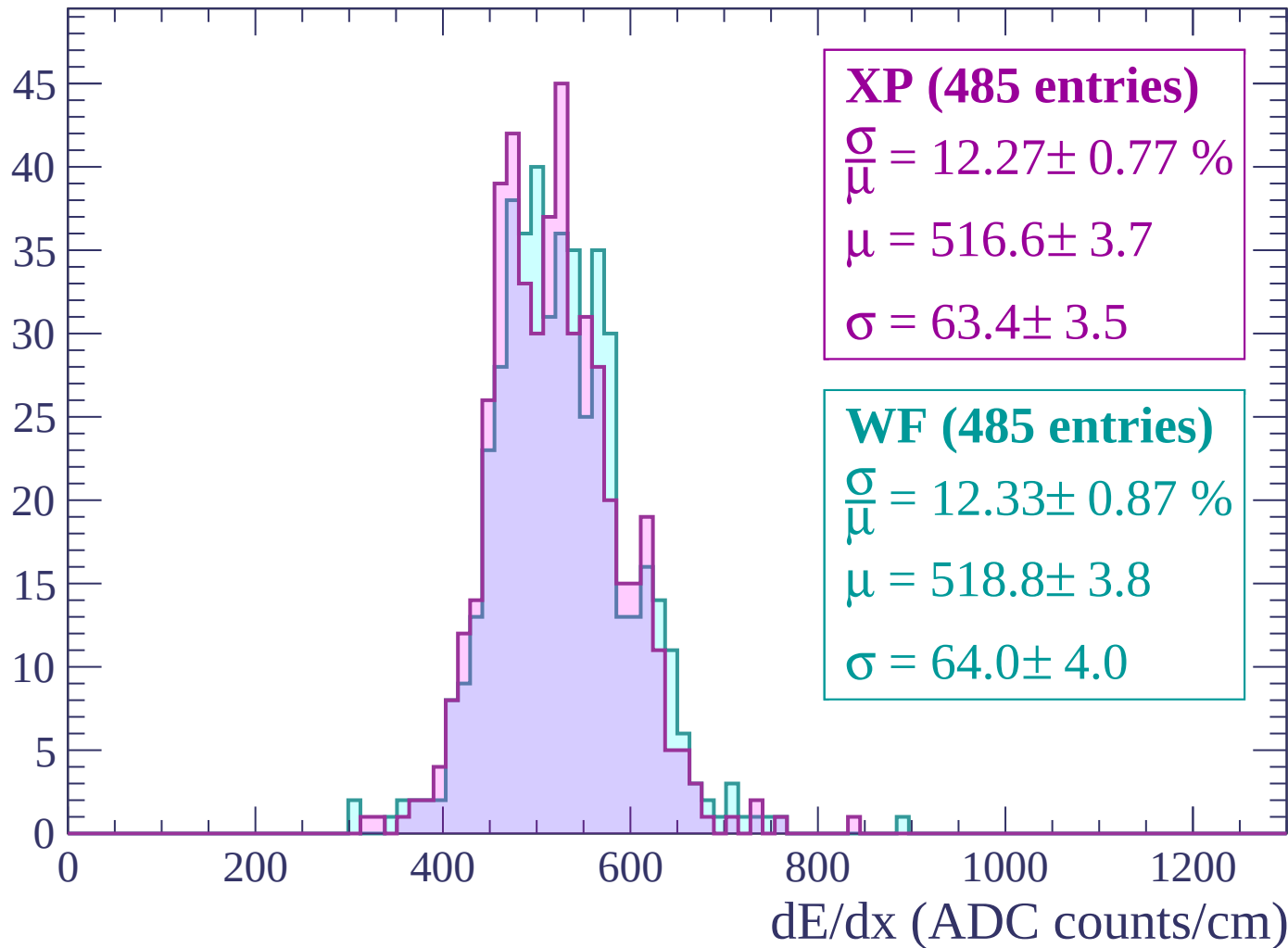
Energy loss | -3000 < p < -2940

Count



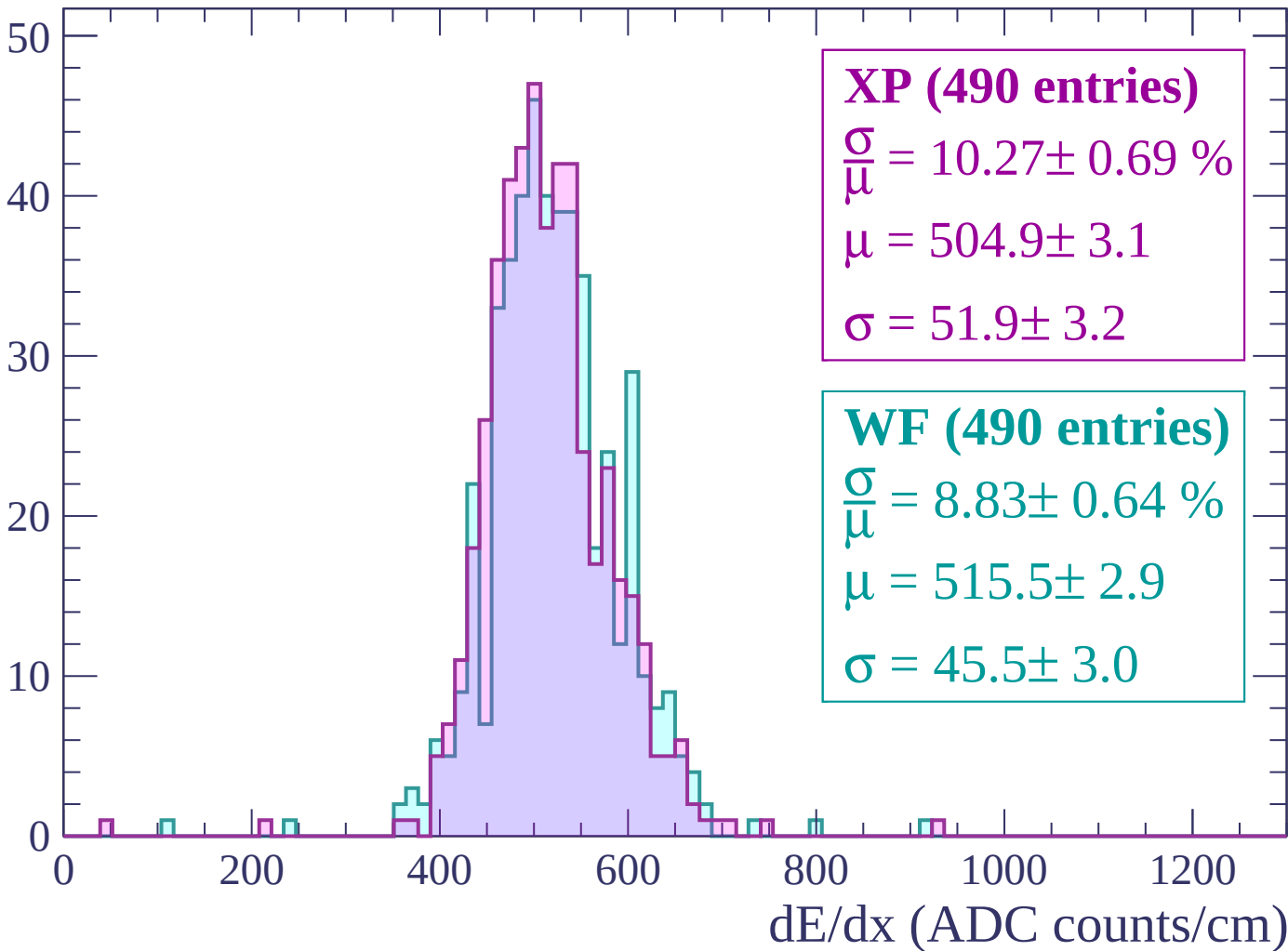
Energy loss | $-2940 < p < -2880$

Count



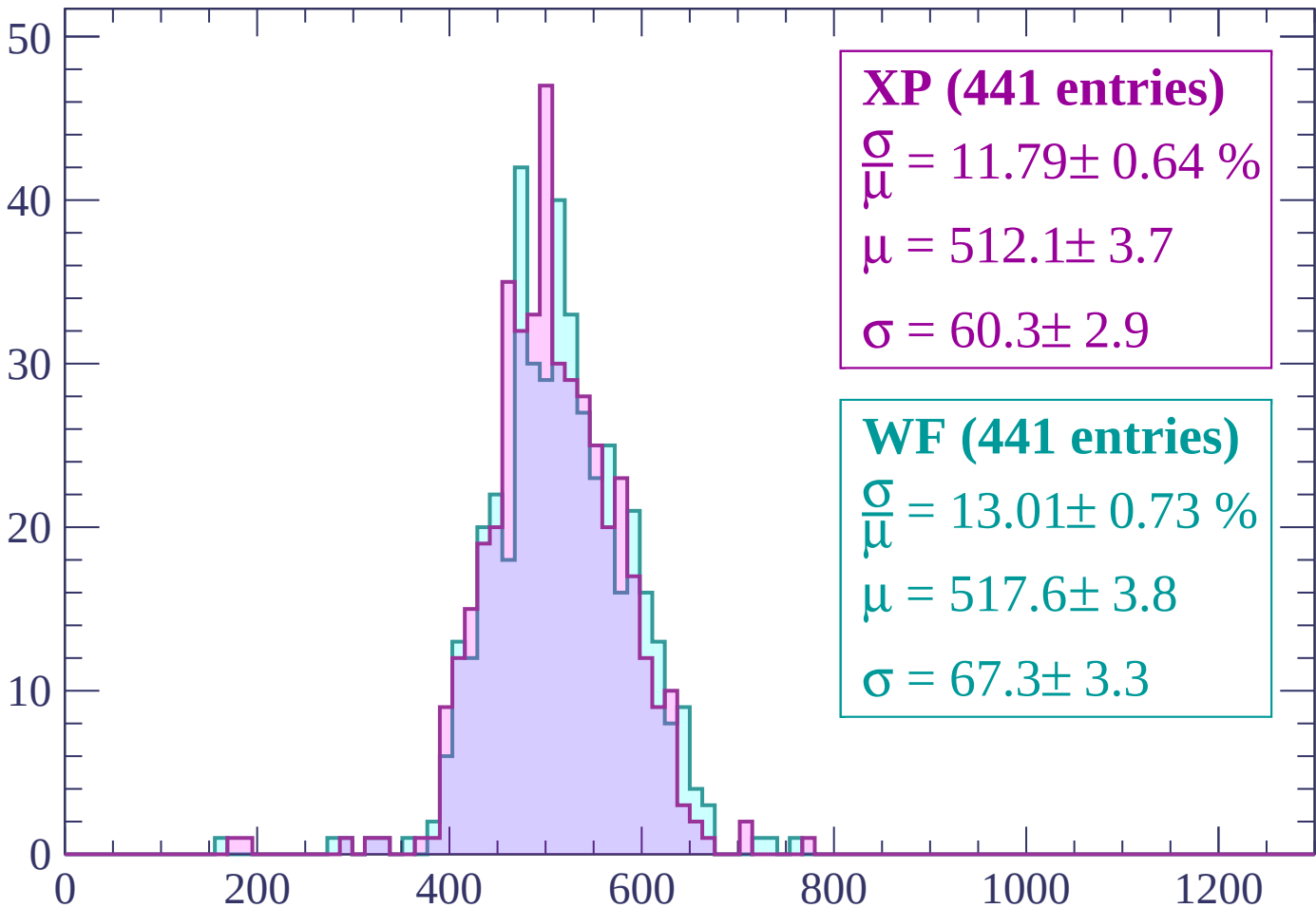
Energy loss | $-2880 < p < -2820$

Count



Energy loss | $-2820 < p < -2760$

Count



XP (441 entries)

$$\frac{\sigma}{\mu} = 11.79 \pm 0.64 \%$$

$$\mu = 512.1 \pm 3.7$$

$$\sigma = 60.3 \pm 2.9$$

WF (441 entries)

$$\frac{\sigma}{\mu} = 13.01 \pm 0.73 \%$$

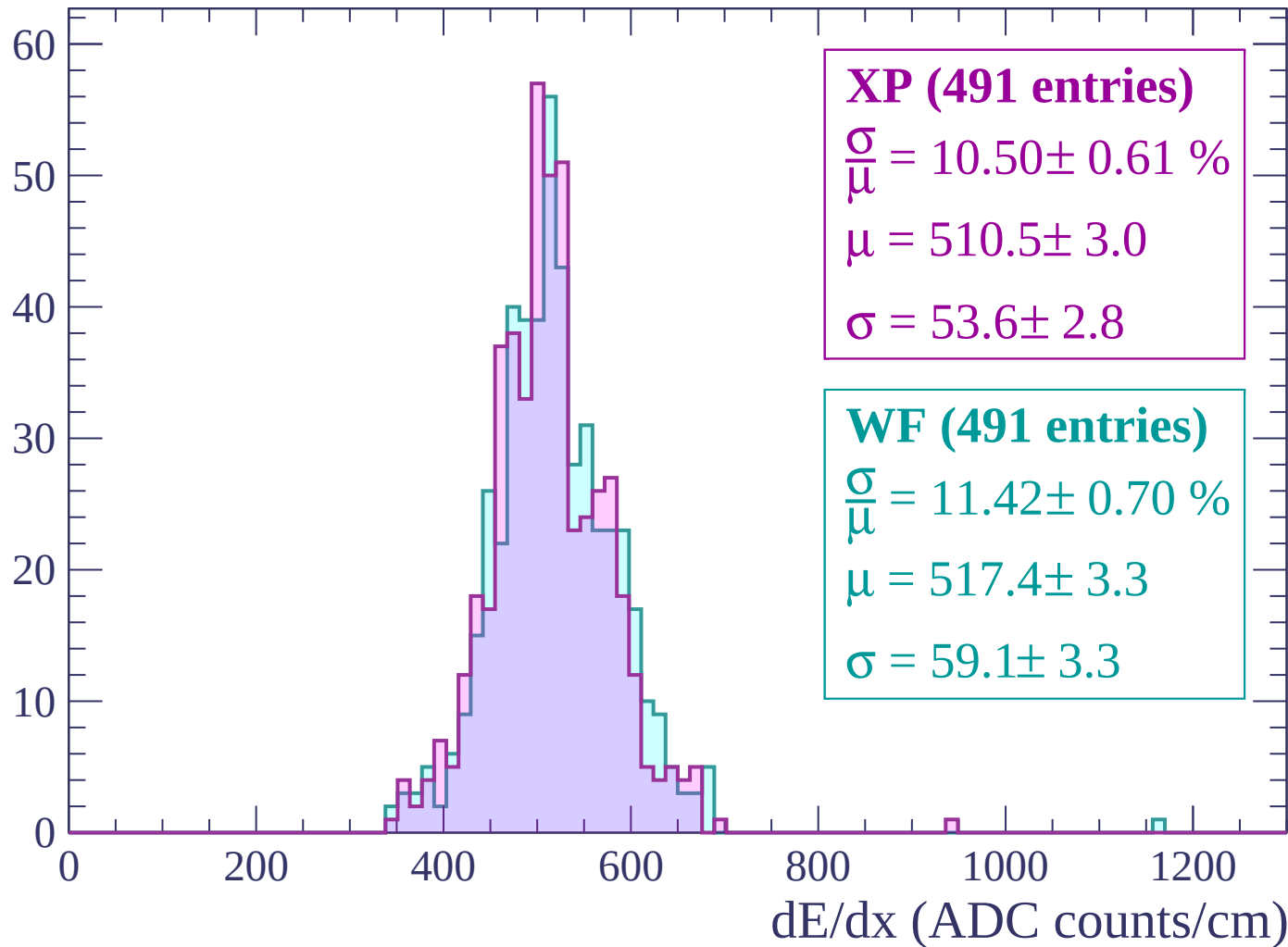
$$\mu = 517.6 \pm 3.8$$

$$\sigma = 67.3 \pm 3.3$$

dE/dx (ADC counts/cm)

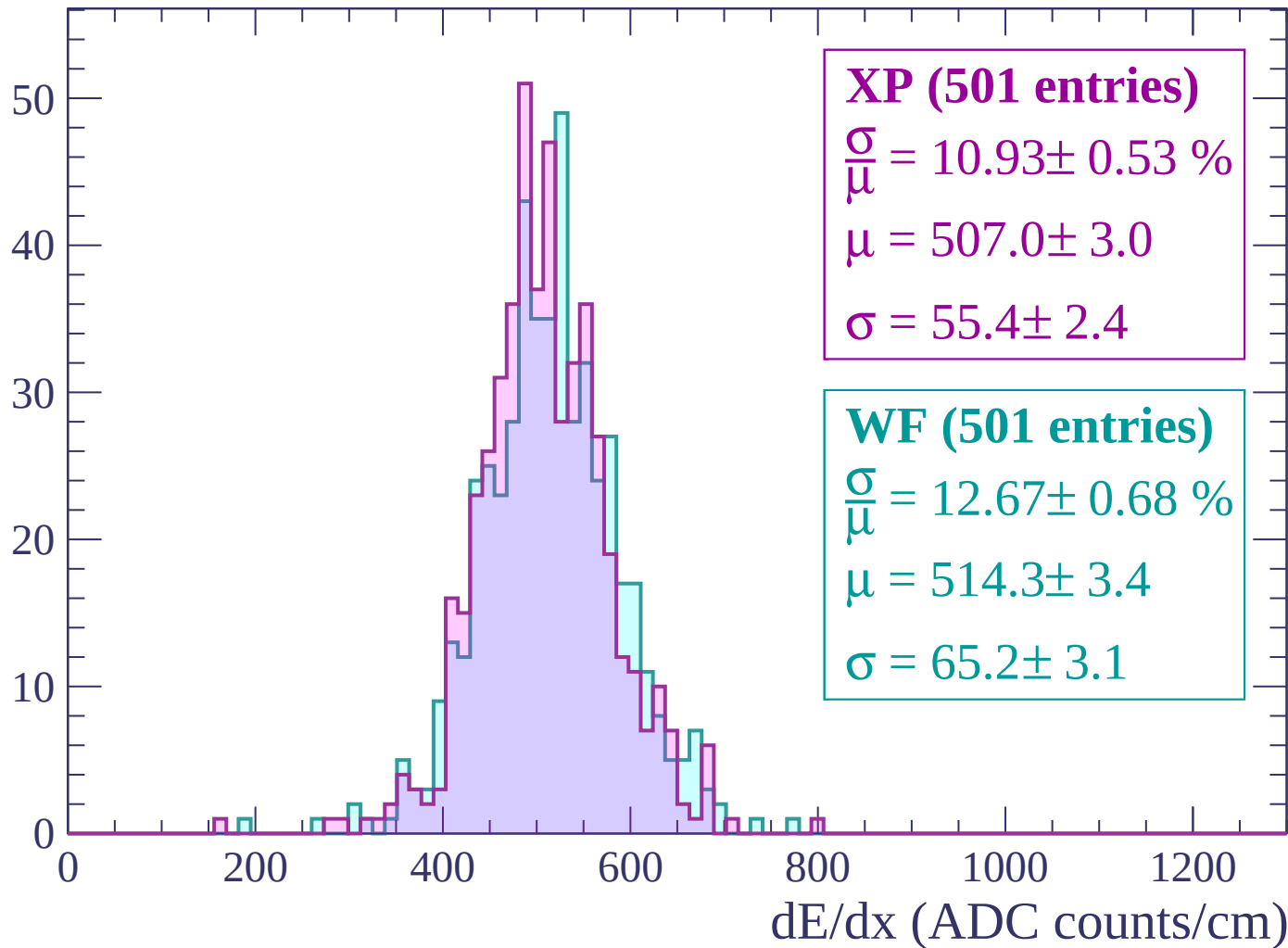
Energy loss | $-2760 < p < -2700$

Count



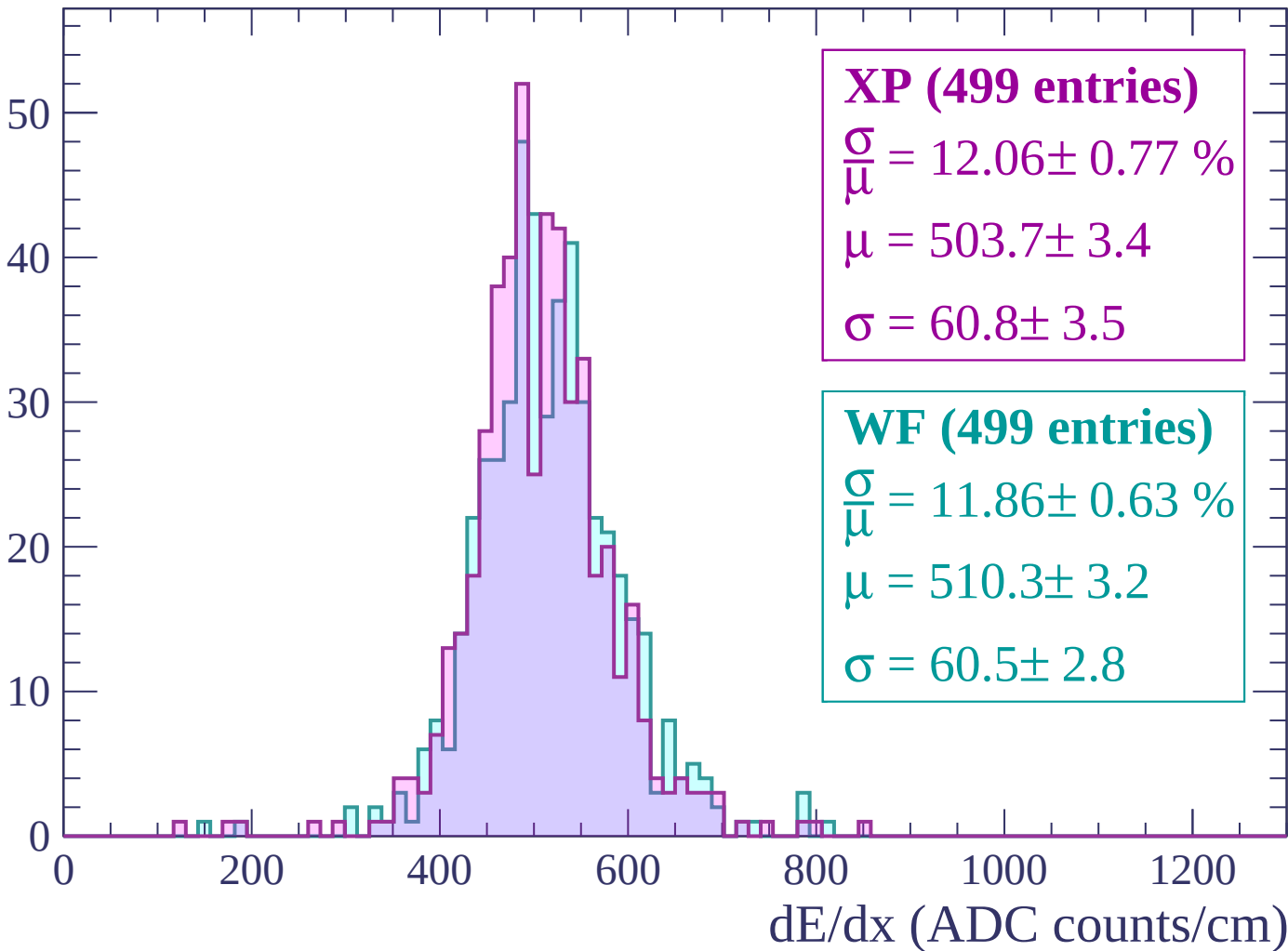
Energy loss | -2700 < p < -2640

Count



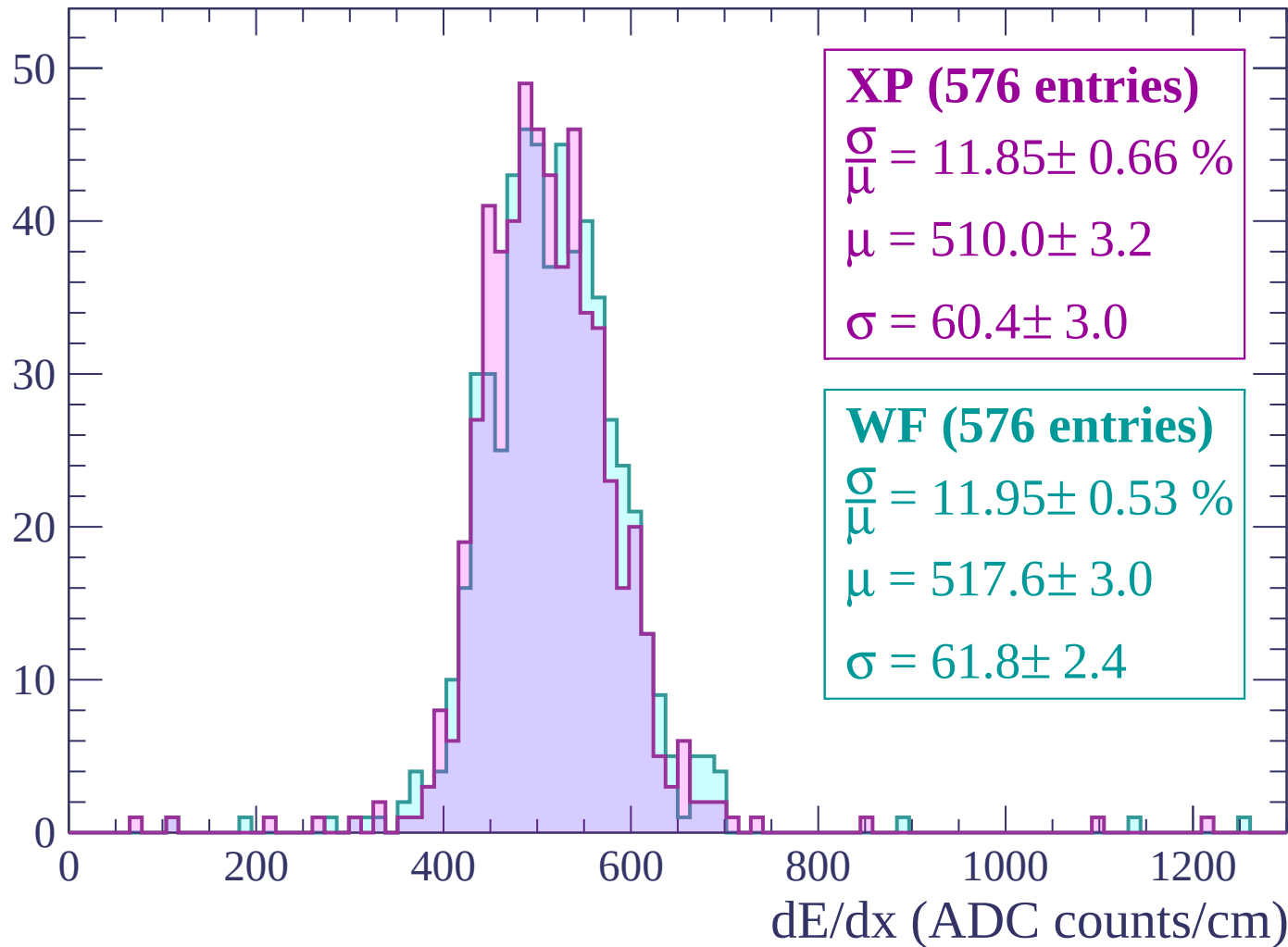
Energy loss | $-2640 < p < -2580$

Count



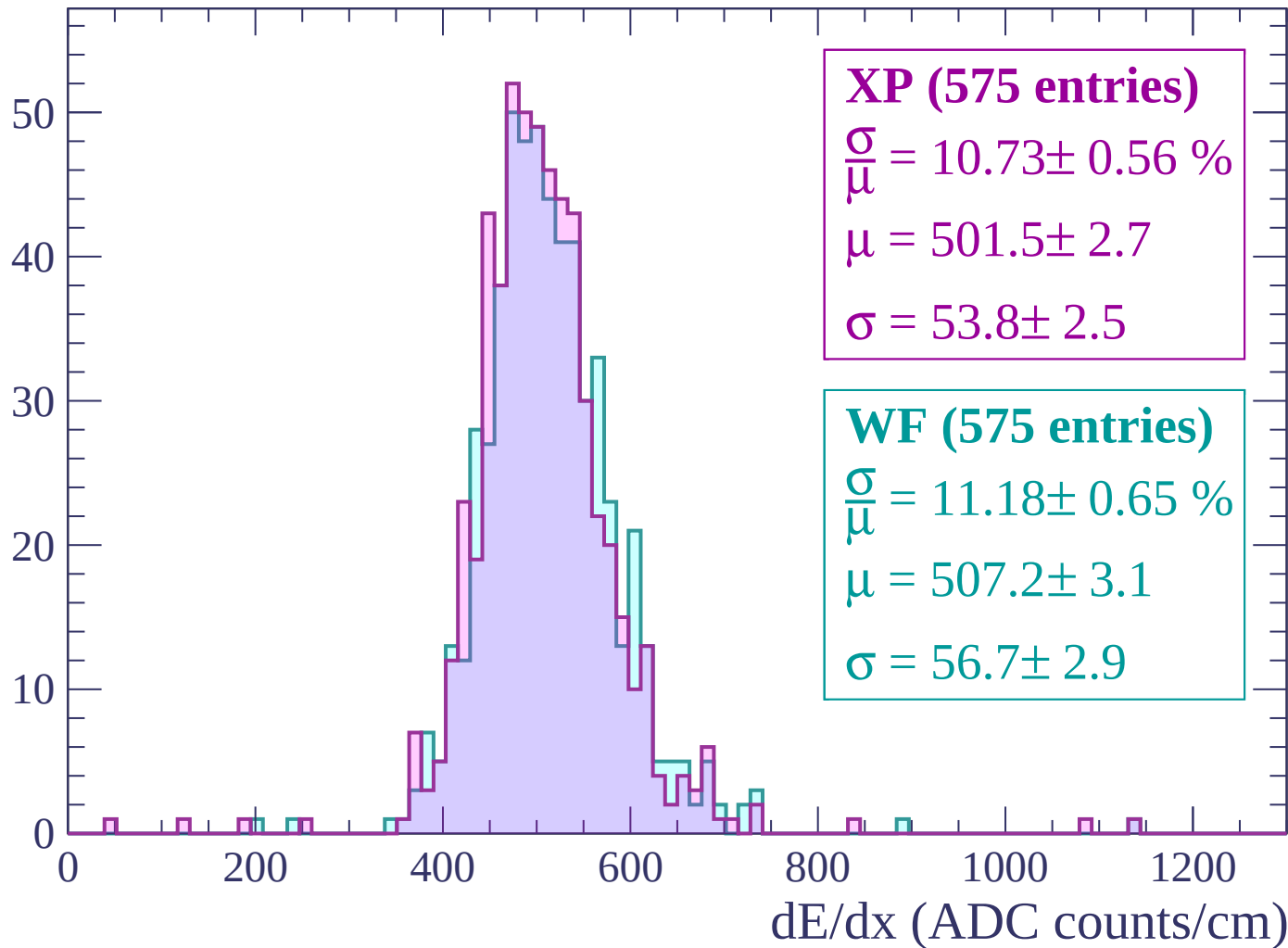
Energy loss | -2580 < p < -2520

Count



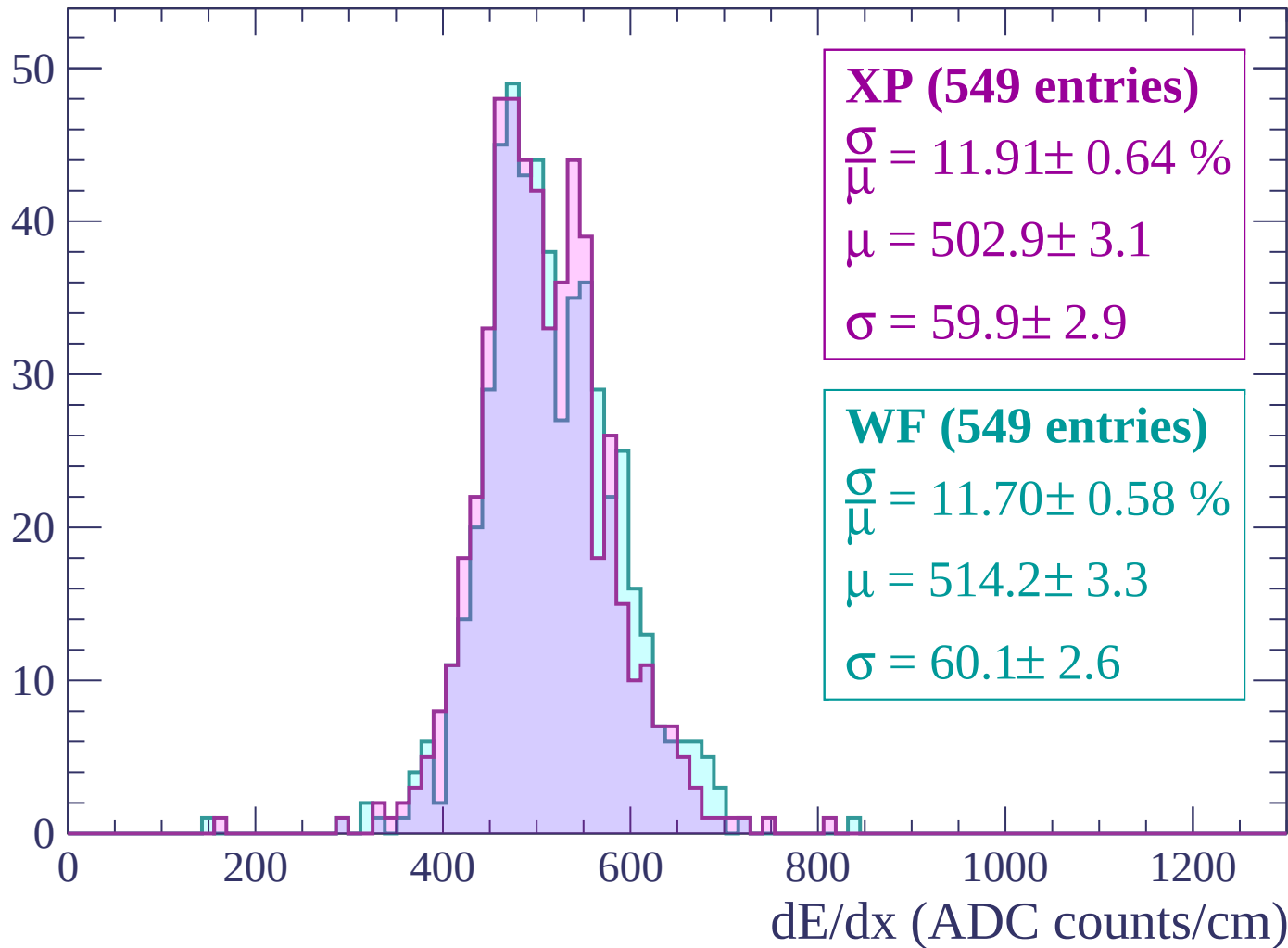
Energy loss | -2520 < p < -2460

Count



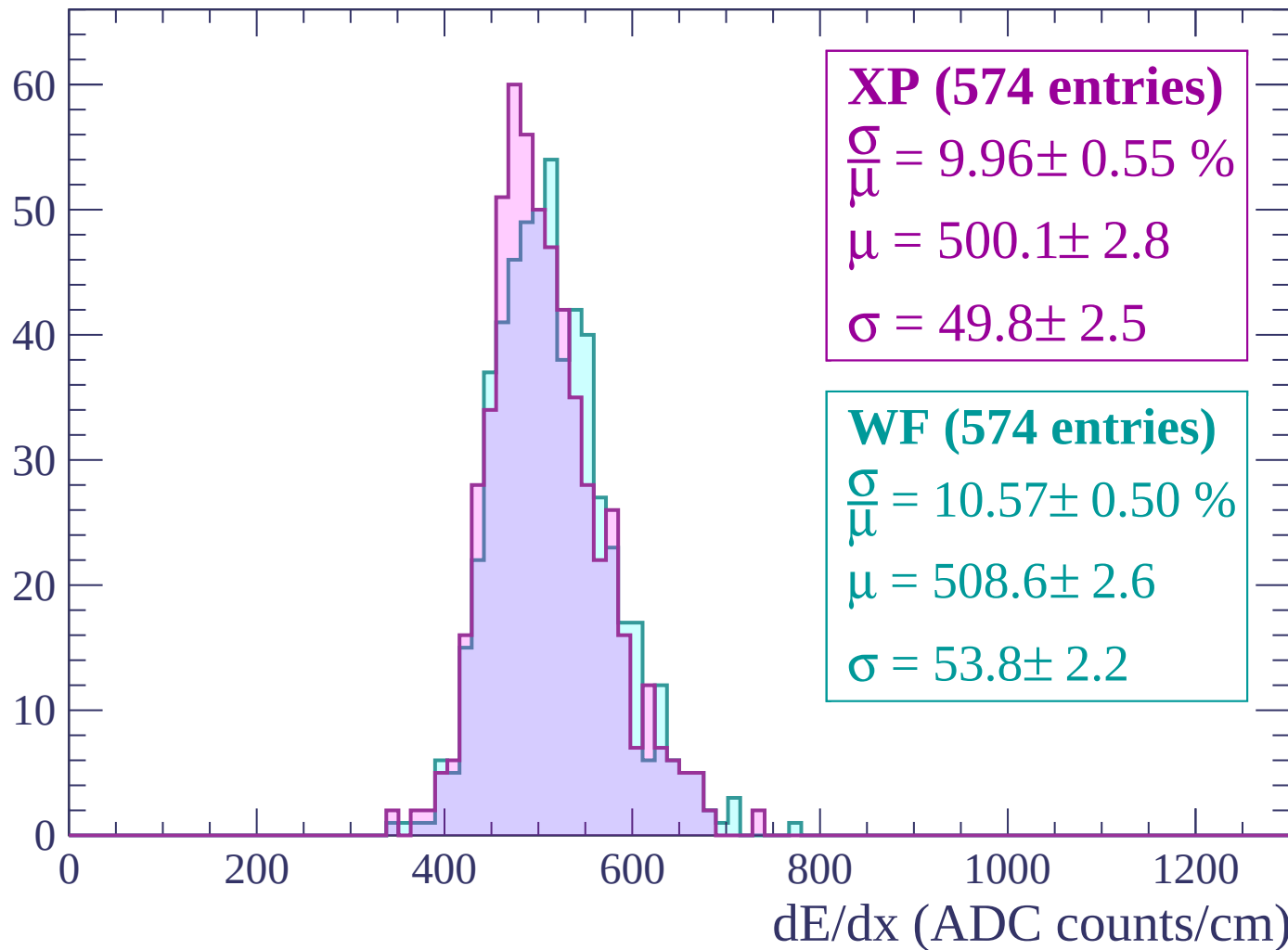
Energy loss | -2460 < p < -2400

Count



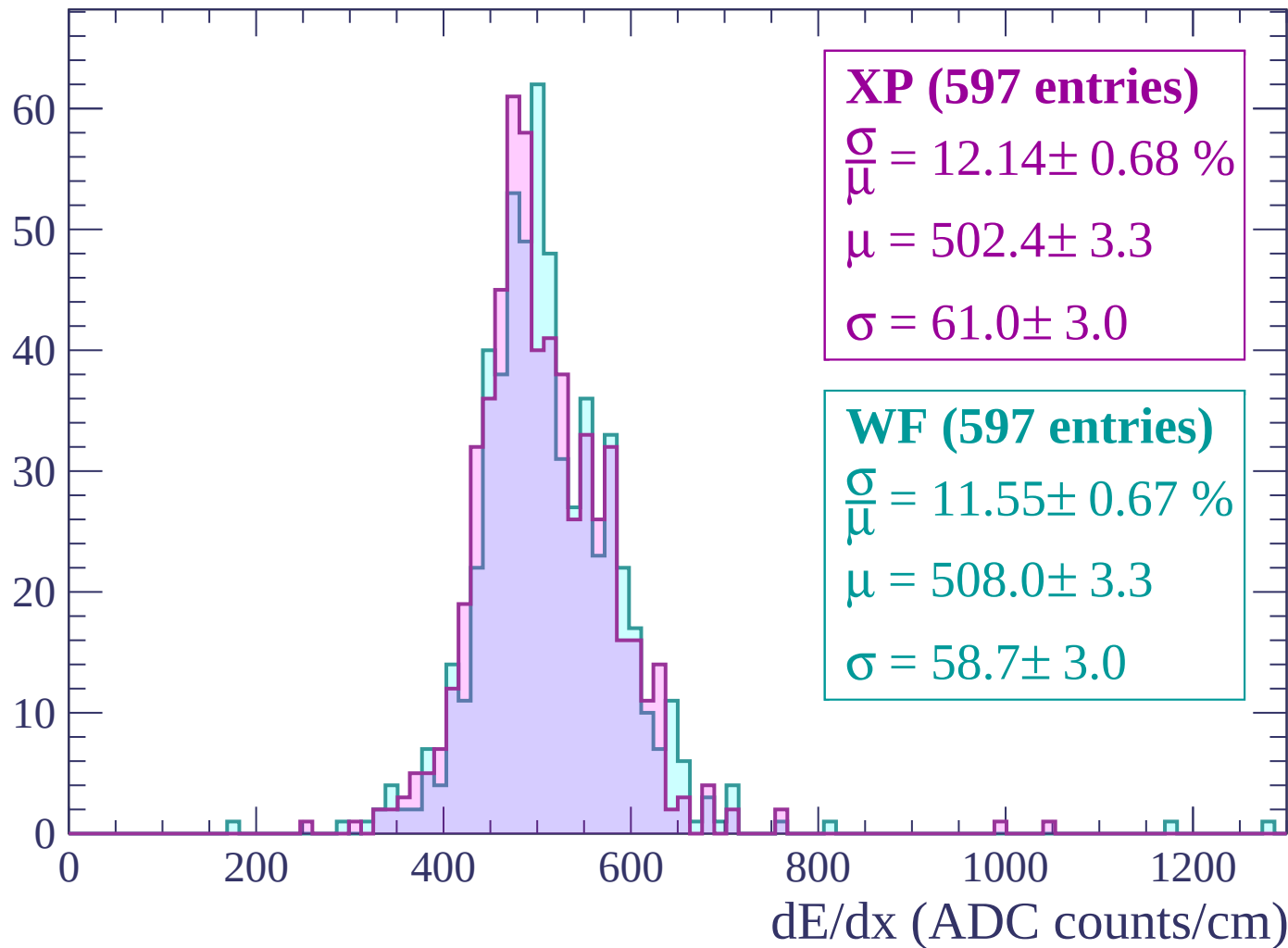
Energy loss | $-2400 < p < -2340$

Count



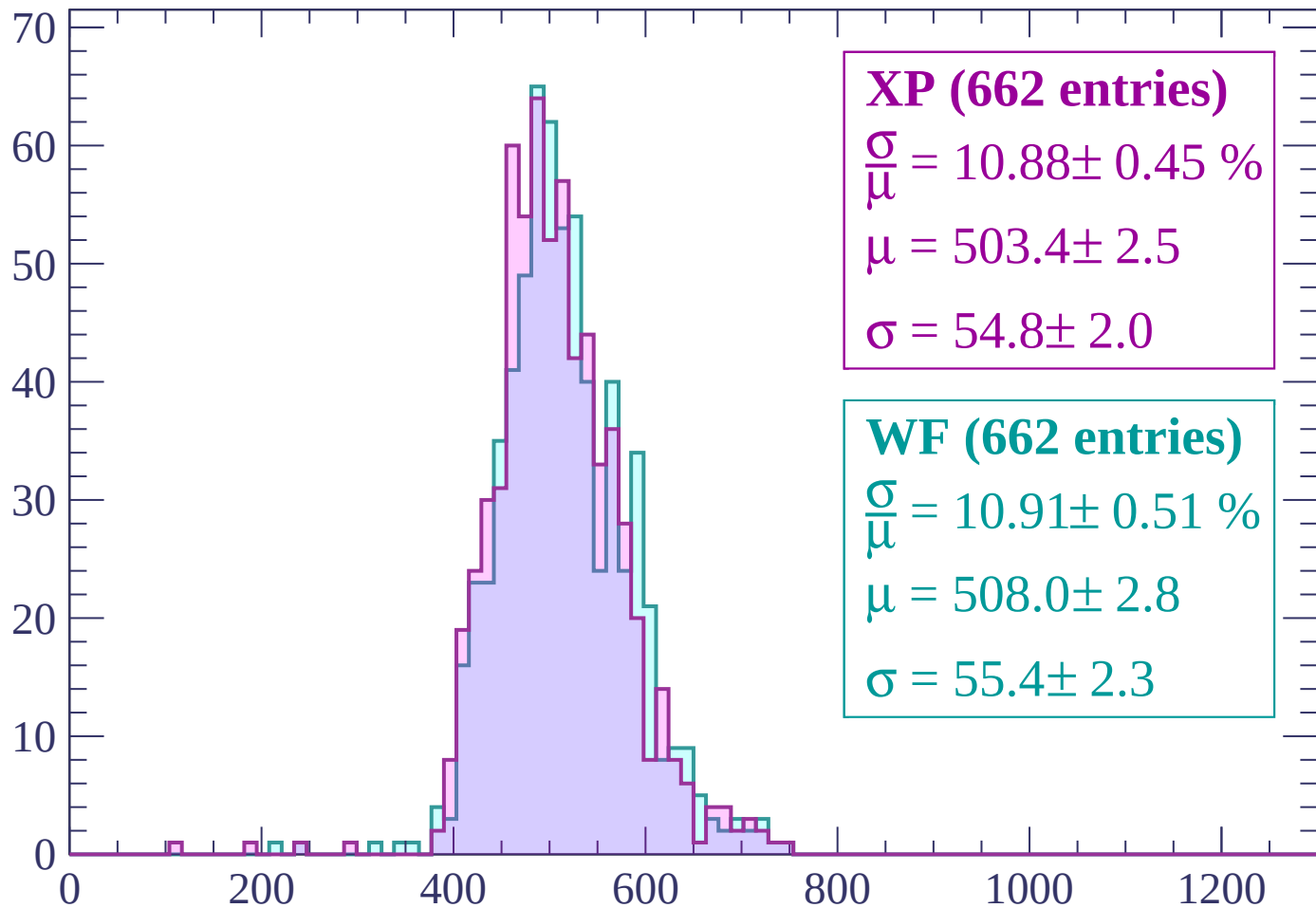
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (662 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.45 \%$$

$$\mu = 503.4 \pm 2.5$$

$$\sigma = 54.8 \pm 2.0$$

WF (662 entries)

$$\frac{\sigma}{\mu} = 10.91 \pm 0.51 \%$$

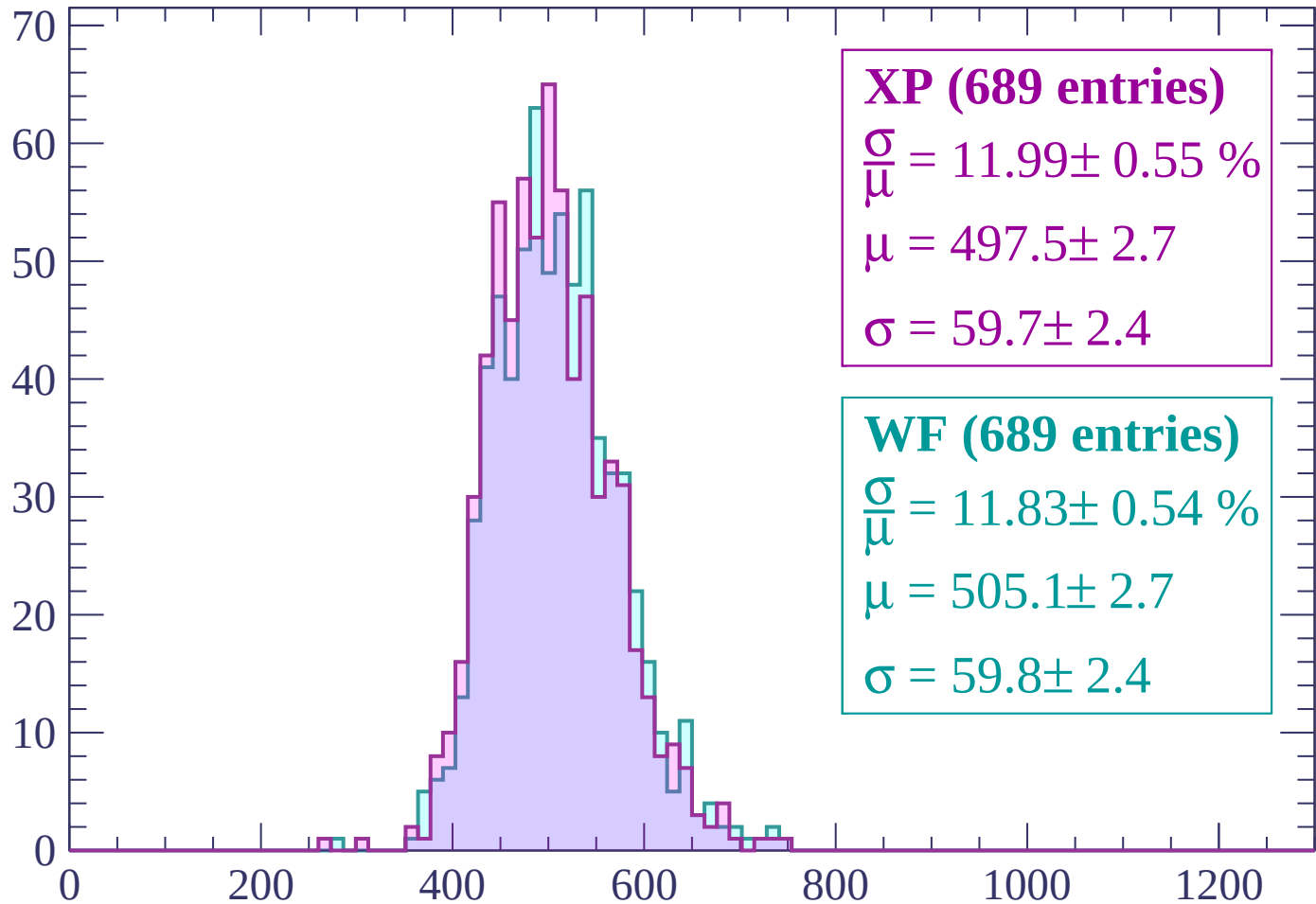
$$\mu = 508.0 \pm 2.8$$

$$\sigma = 55.4 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count



XP (689 entries)

$$\frac{\sigma}{\mu} = 11.99 \pm 0.55 \%$$

$$\mu = 497.5 \pm 2.7$$

$$\sigma = 59.7 \pm 2.4$$

WF (689 entries)

$$\frac{\sigma}{\mu} = 11.83 \pm 0.54 \%$$

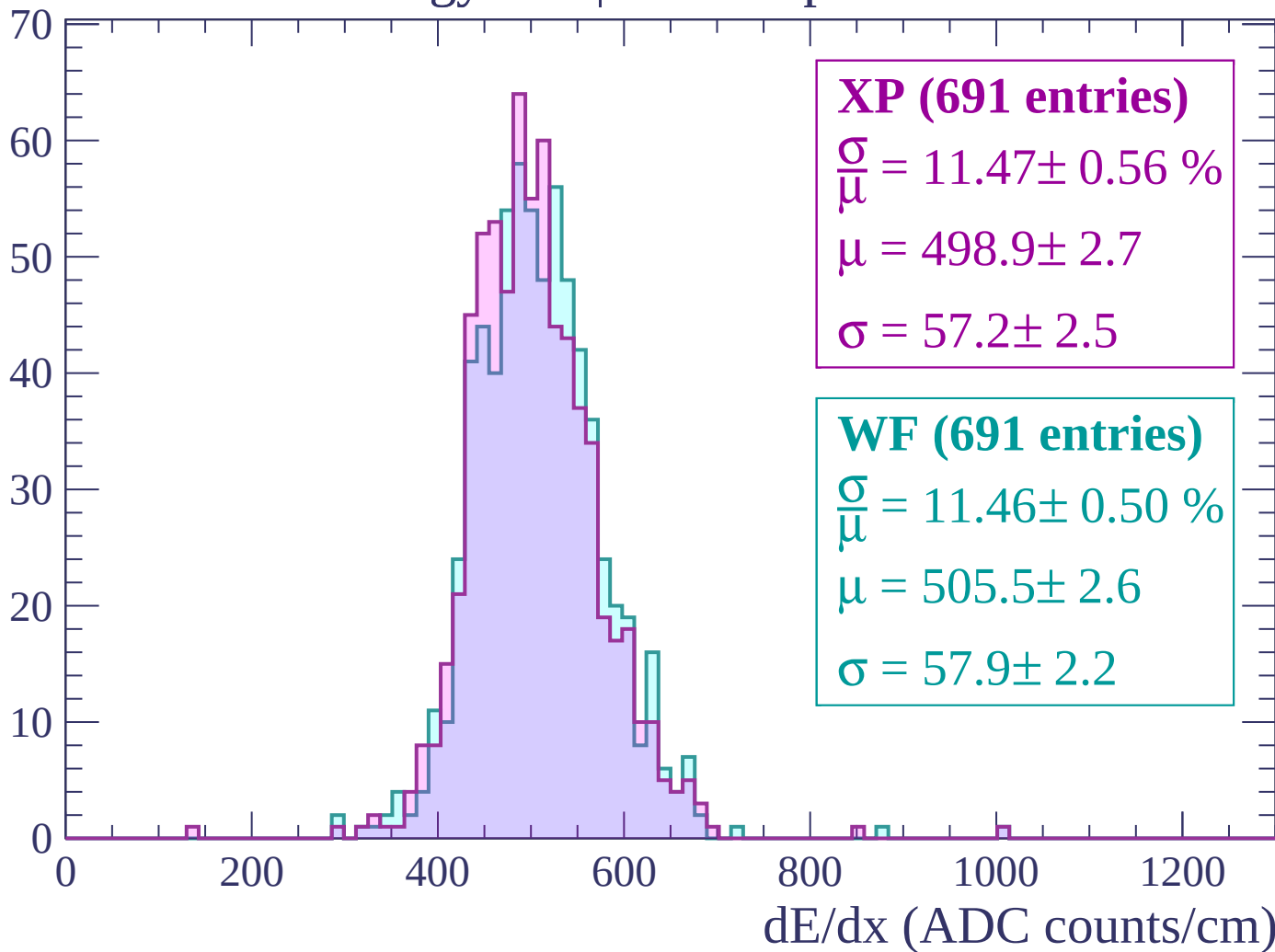
$$\mu = 505.1 \pm 2.7$$

$$\sigma = 59.8 \pm 2.4$$

dE/dx (ADC counts/cm)

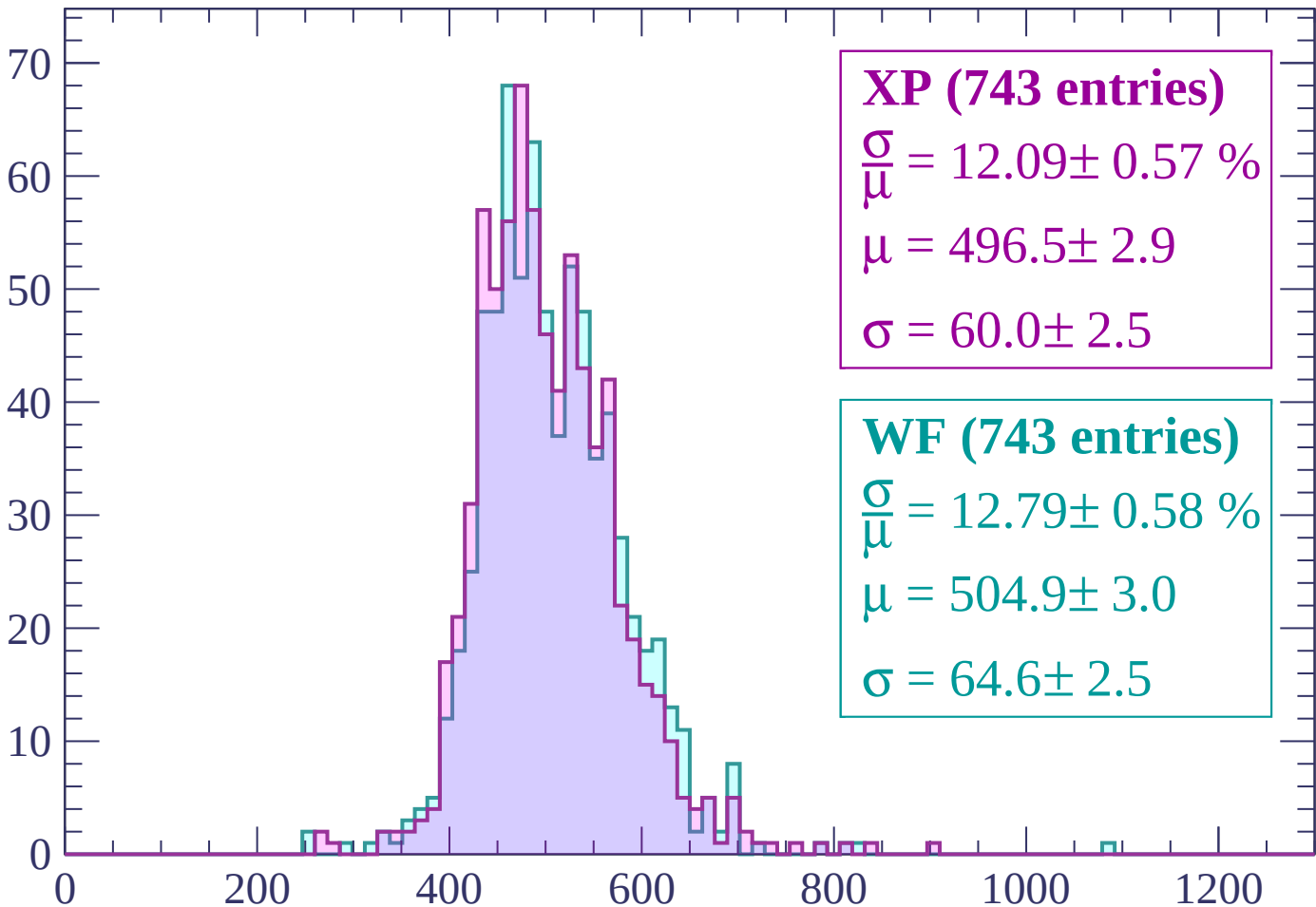
Energy loss | -2160 < p < -2100

Count



Energy loss | -2100 < p < -2040

Count



XP (743 entries)

$$\frac{\sigma}{\mu} = 12.09 \pm 0.57 \%$$

$$\mu = 496.5 \pm 2.9$$

$$\sigma = 60.0 \pm 2.5$$

WF (743 entries)

$$\frac{\sigma}{\mu} = 12.79 \pm 0.58 \%$$

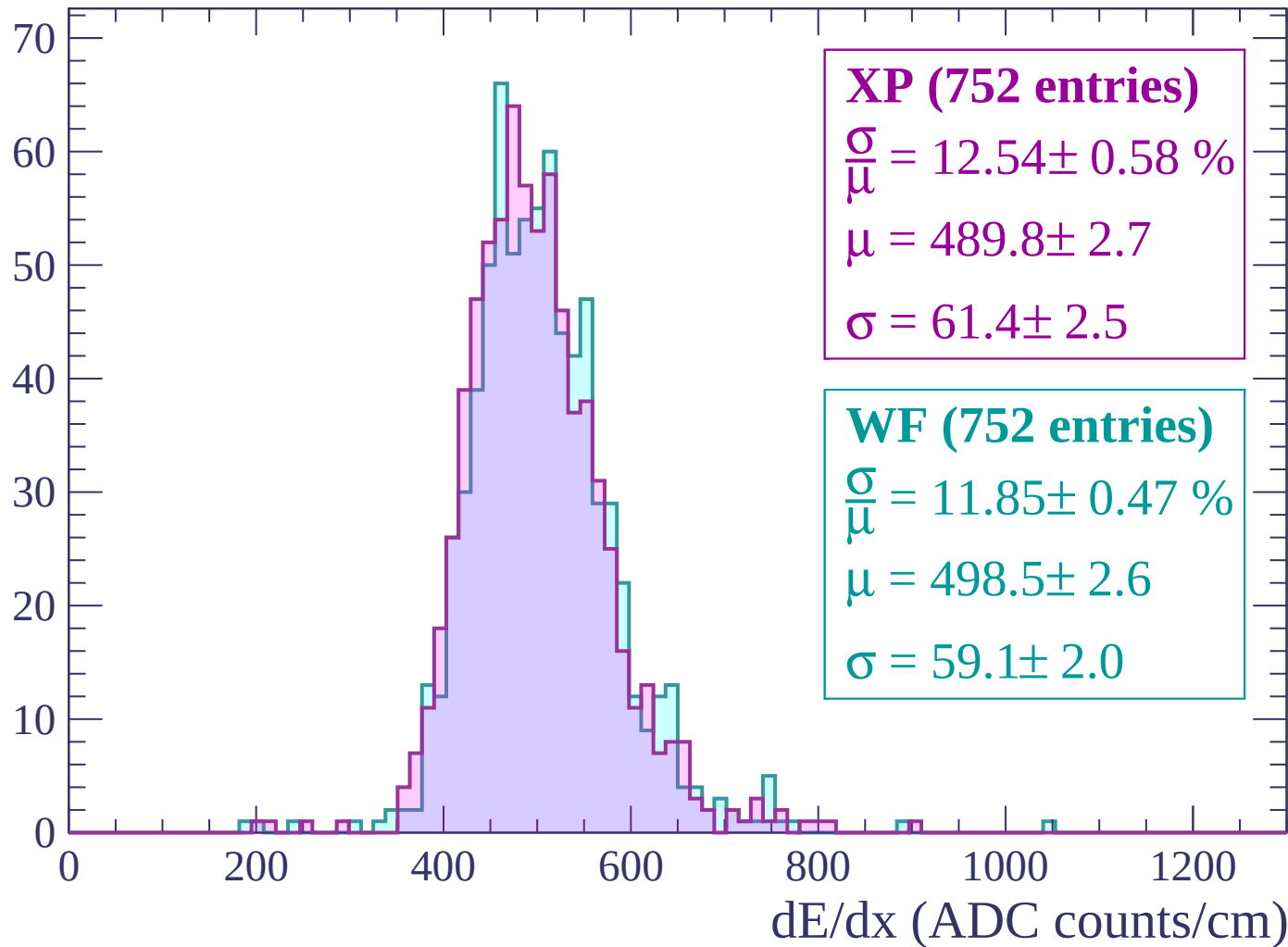
$$\mu = 504.9 \pm 3.0$$

$$\sigma = 64.6 \pm 2.5$$

dE/dx (ADC counts/cm)

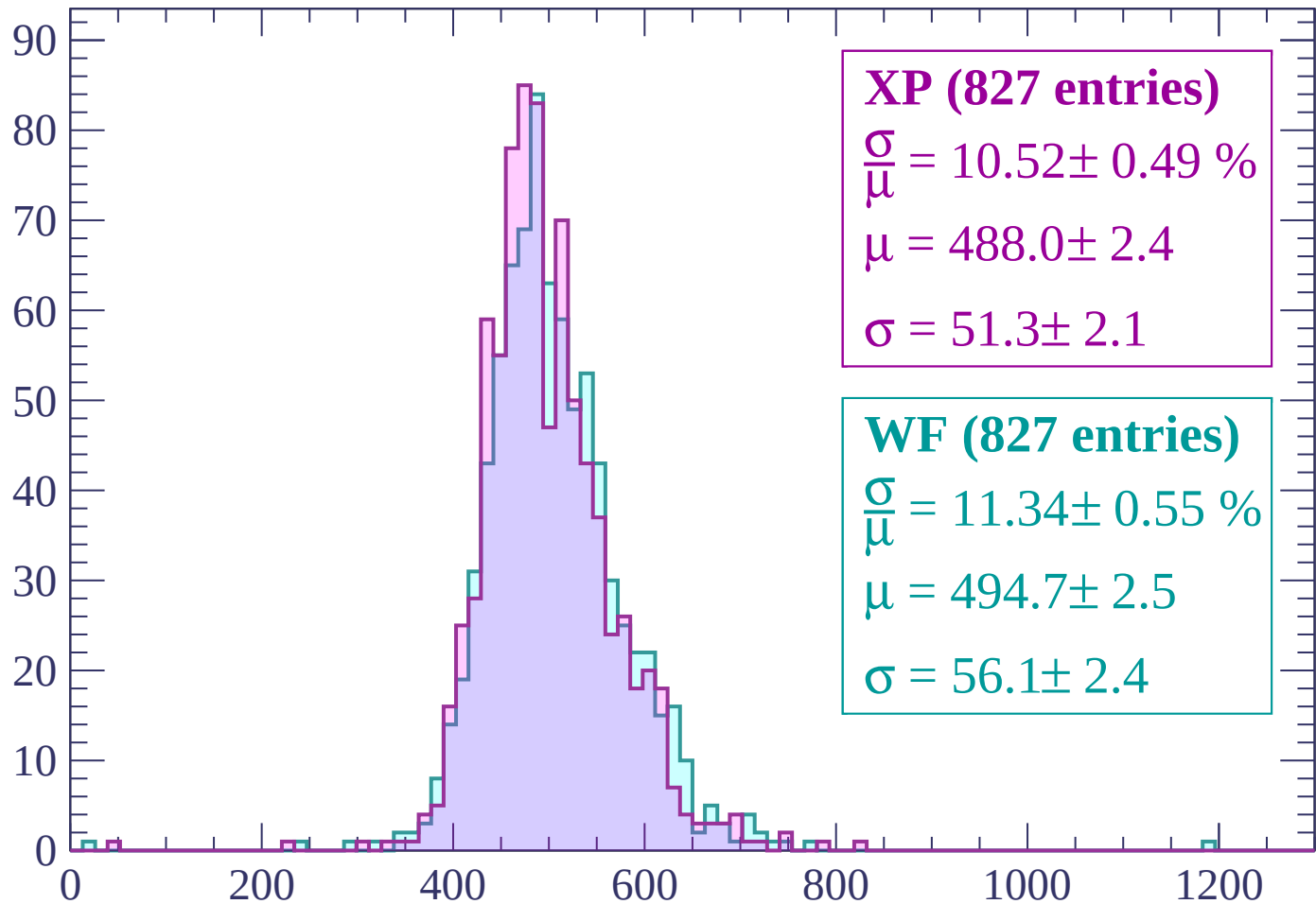
Energy loss | -2040 < p < -1980

Count



Energy loss | $-1980 < p < -1920$

Count



XP (827 entries)

$$\frac{\sigma}{\mu} = 10.52 \pm 0.49 \%$$

$$\mu = 488.0 \pm 2.4$$

$$\sigma = 51.3 \pm 2.1$$

WF (827 entries)

$$\frac{\sigma}{\mu} = 11.34 \pm 0.55 \%$$

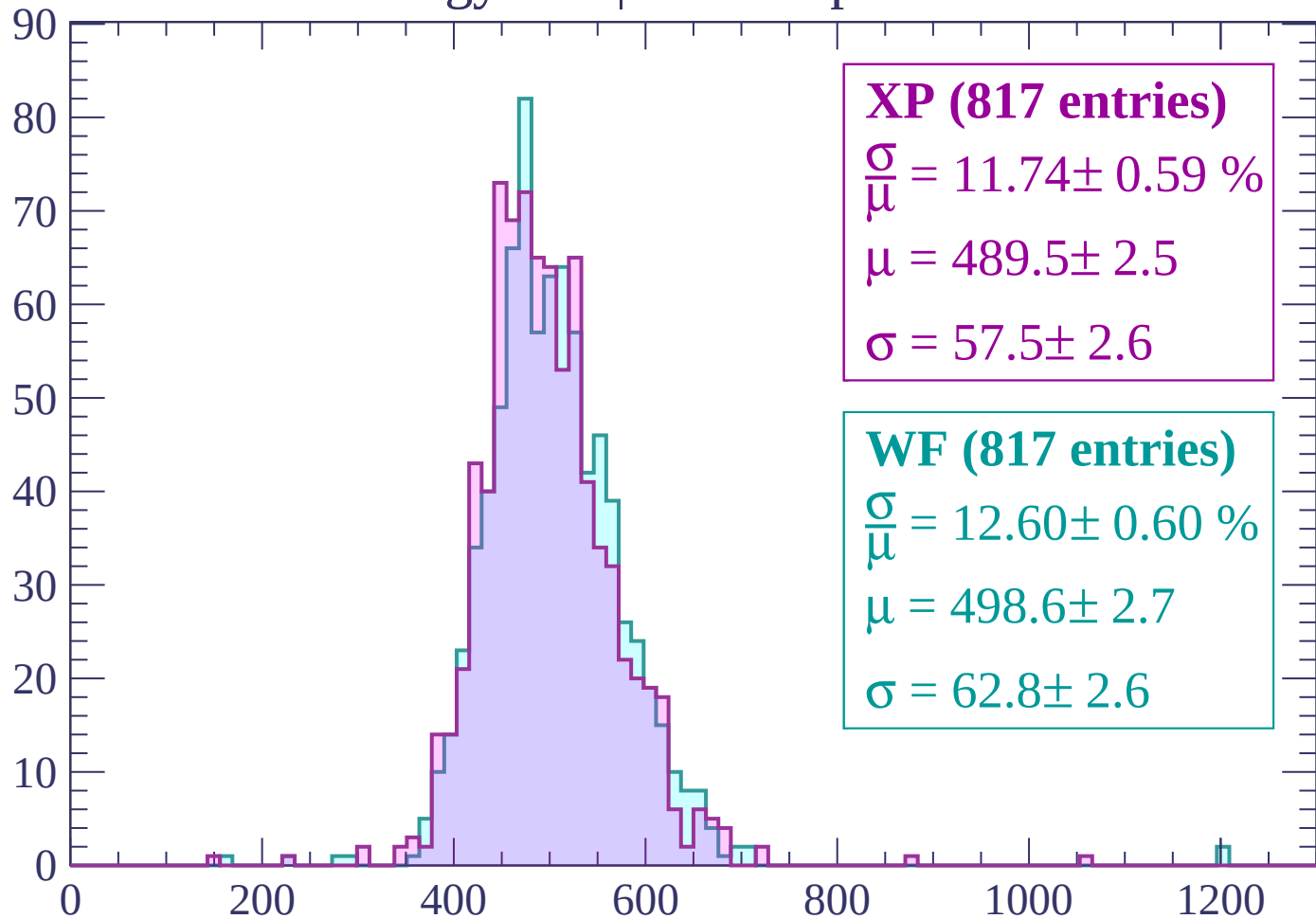
$$\mu = 494.7 \pm 2.5$$

$$\sigma = 56.1 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1860

Count



XP (817 entries)

$$\frac{\sigma}{\mu} = 11.74 \pm 0.59 \%$$

$$\mu = 489.5 \pm 2.5$$

$$\sigma = 57.5 \pm 2.6$$

WF (817 entries)

$$\frac{\sigma}{\mu} = 12.60 \pm 0.60 \%$$

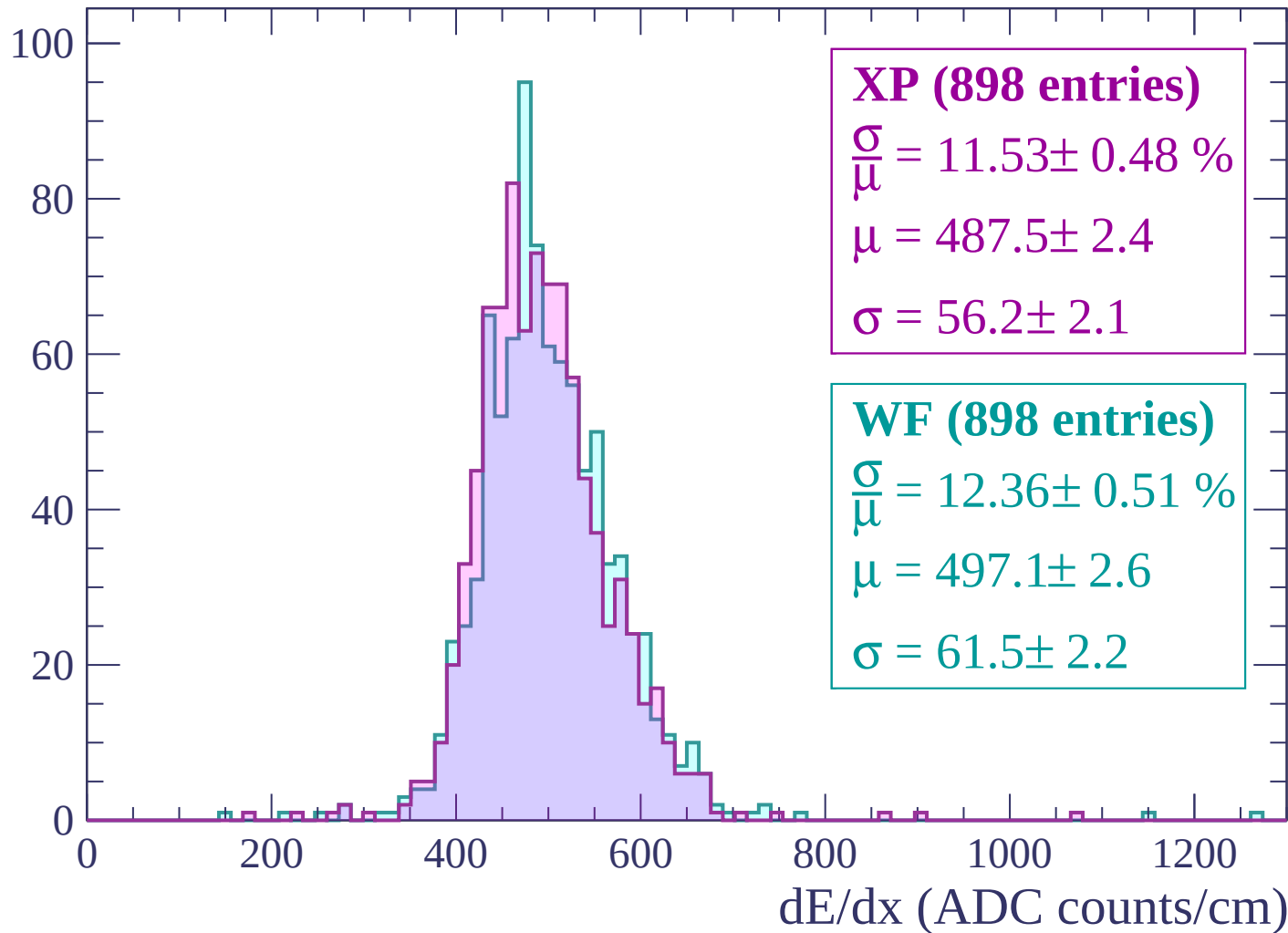
$$\mu = 498.6 \pm 2.7$$

$$\sigma = 62.8 \pm 2.6$$

dE/dx (ADC counts/cm)

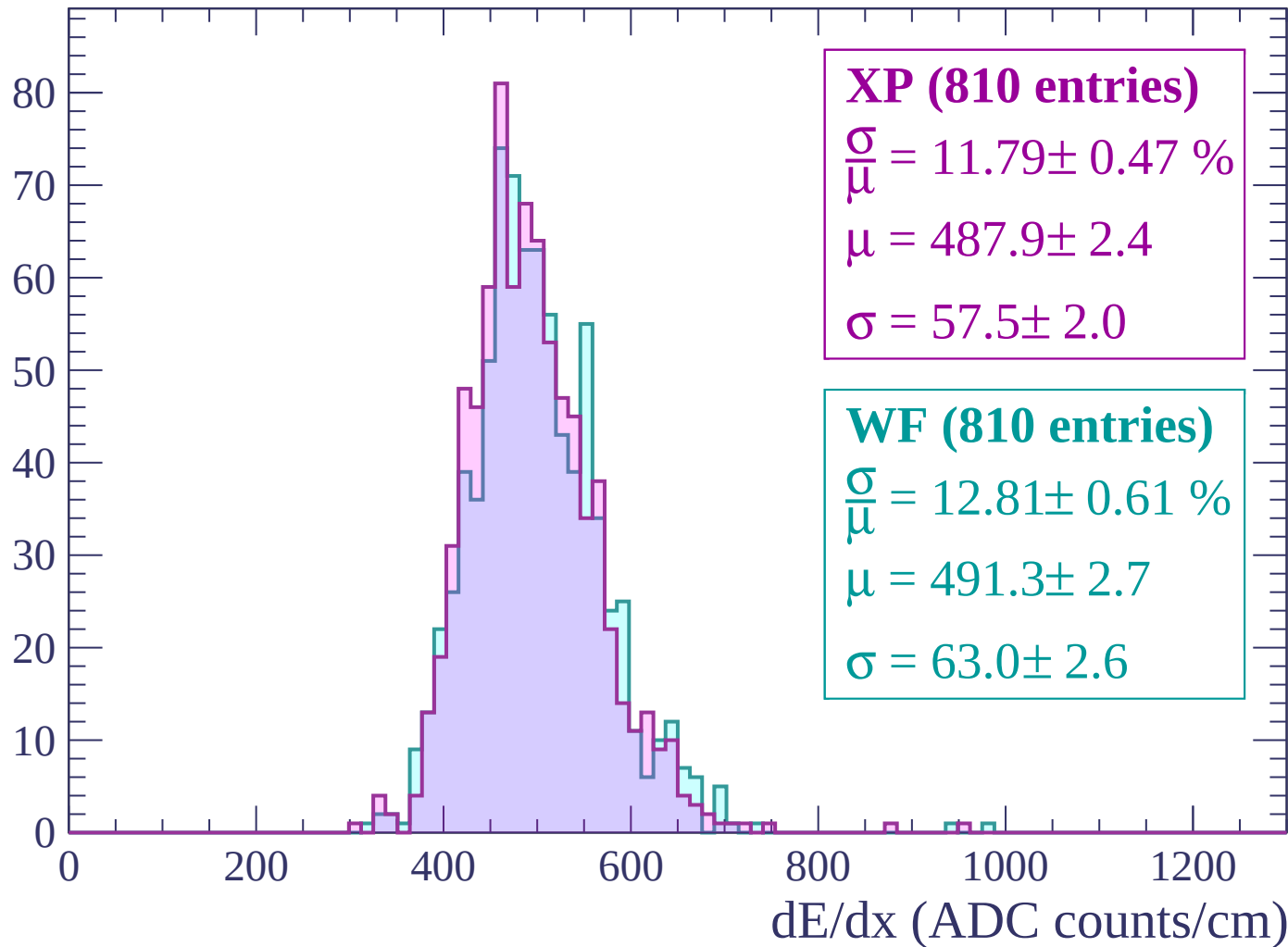
Energy loss | -1860 < p < -1800

Count



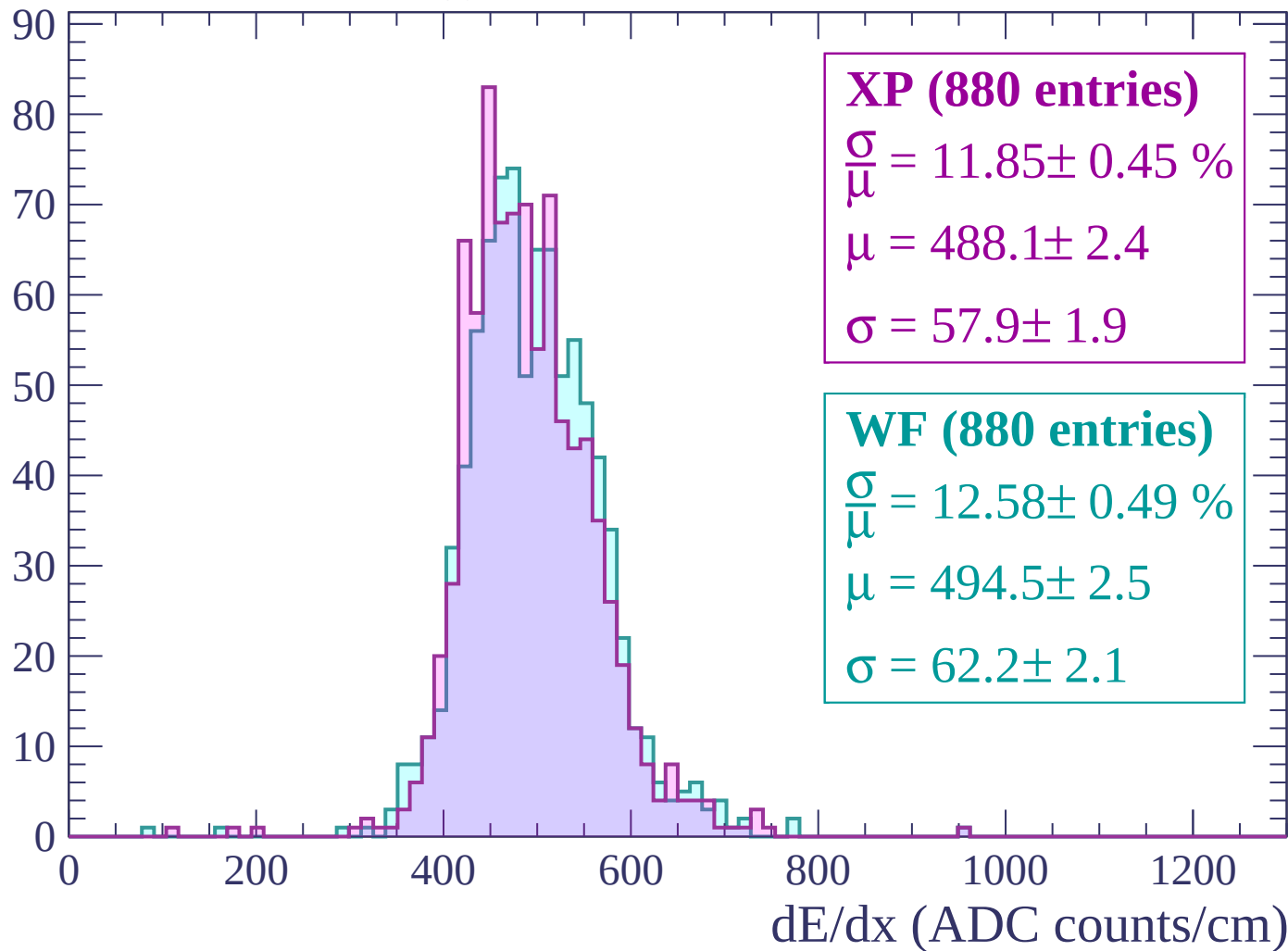
Energy loss | $-1800 < p < -1740$

Count



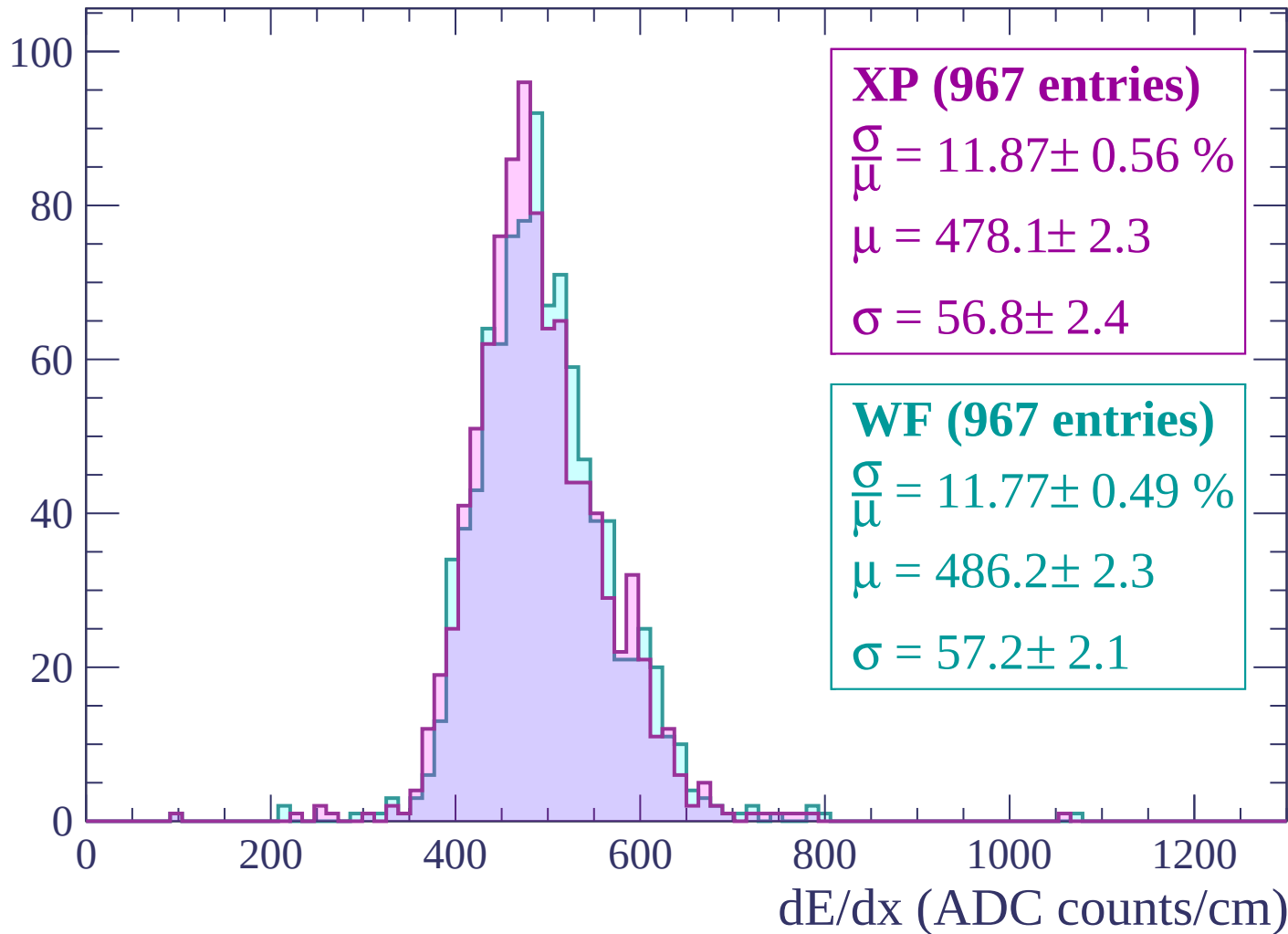
Energy loss | $-1740 < p < -1680$

Count



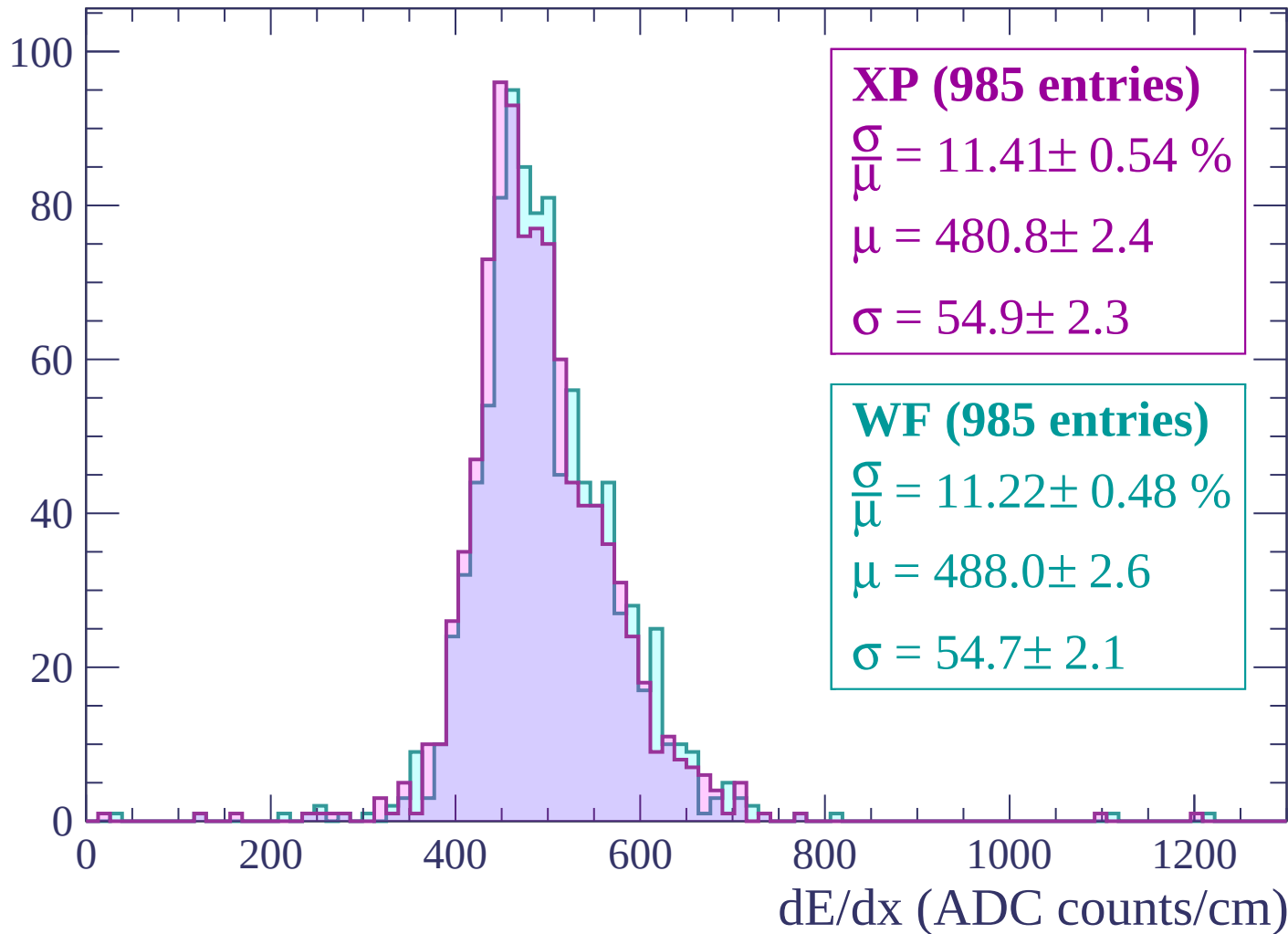
Energy loss | -1680 < p < -1620

Count



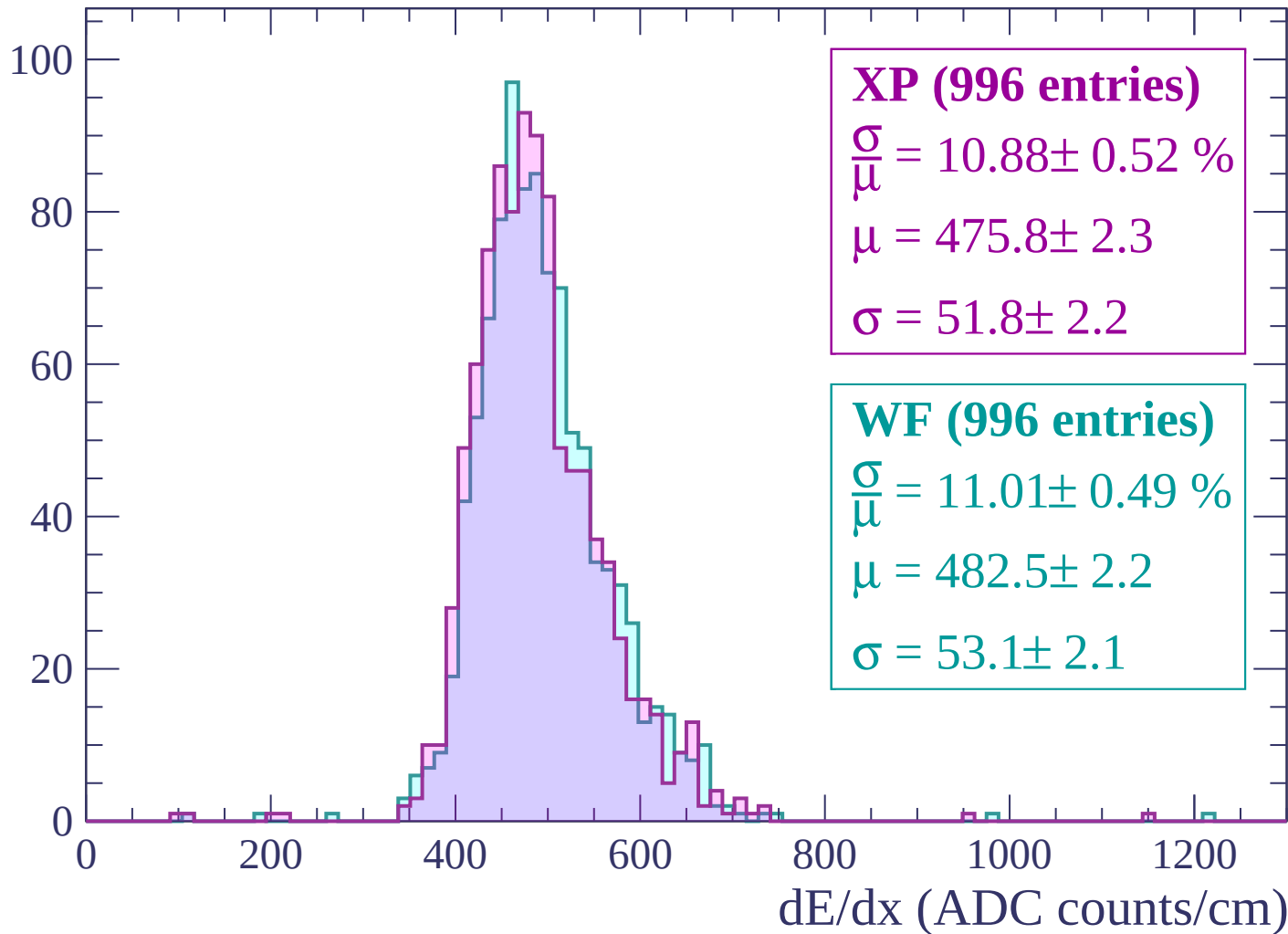
Energy loss | -1620 < p < -1560

Count



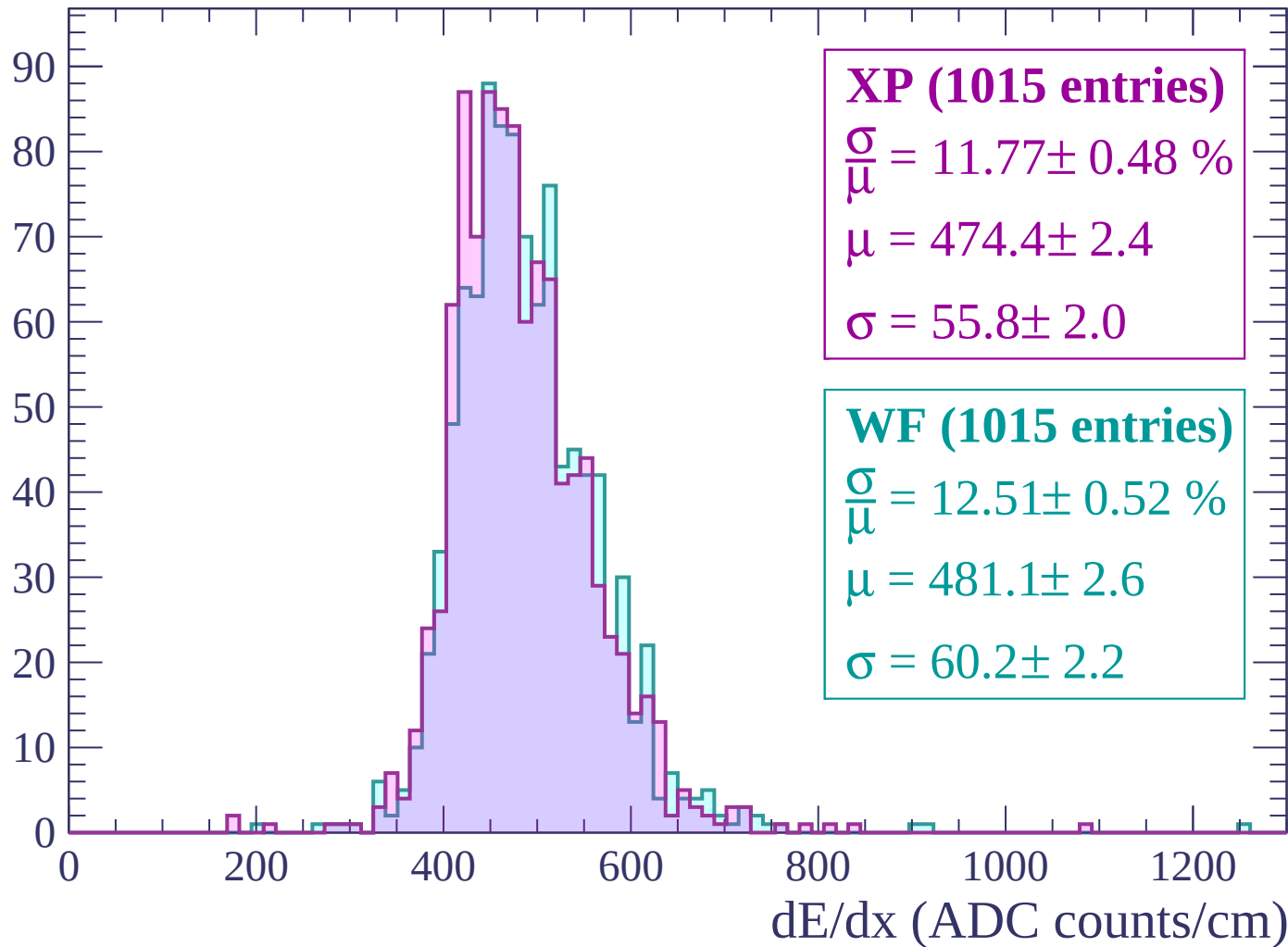
Energy loss | -1560 < p < -1500

Count



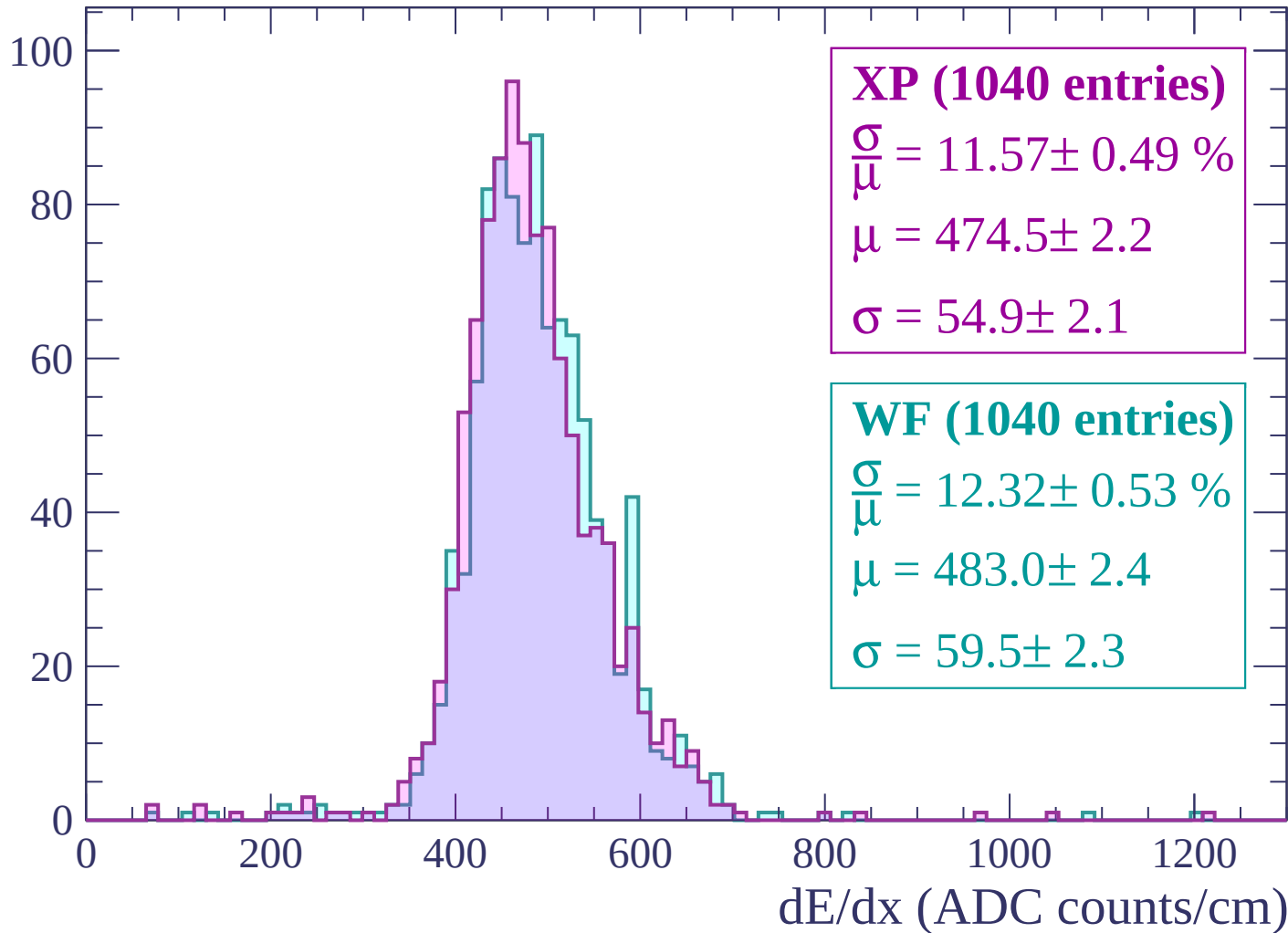
Energy loss | $-1500 < p < -1440$

Count



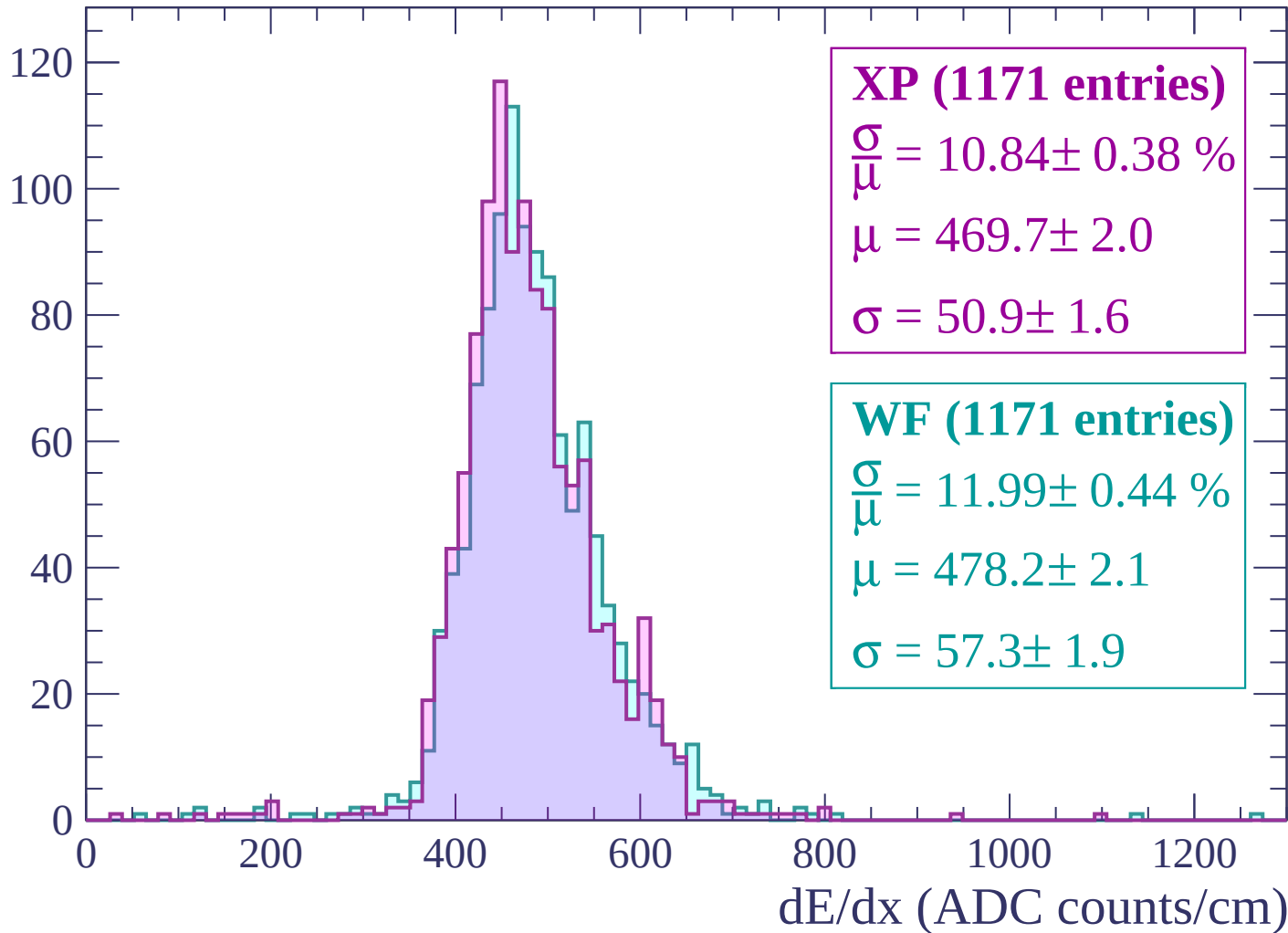
Energy loss | $-1440 < p < -1380$

Count



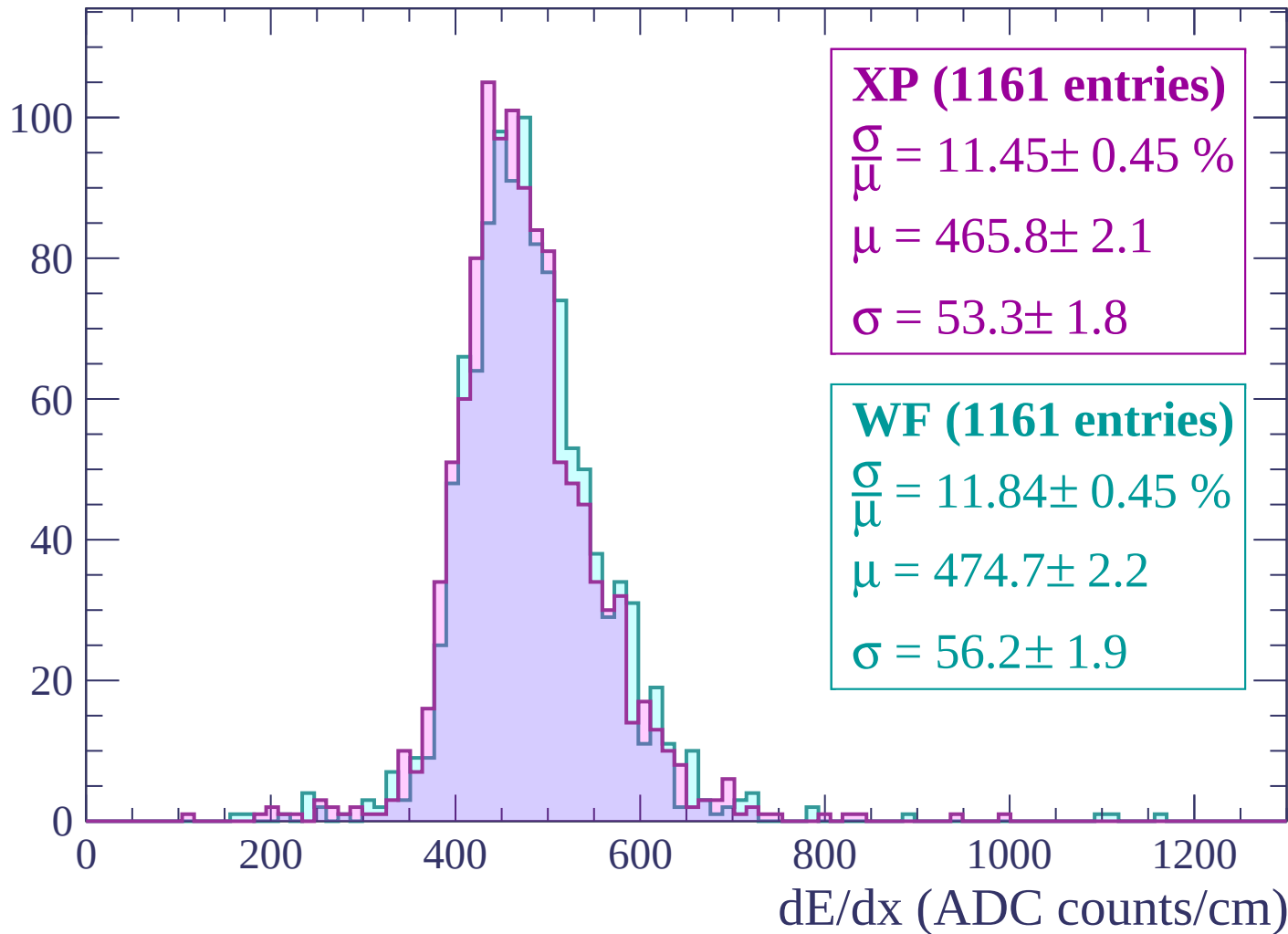
Energy loss | -1380 < p < -1320

Count



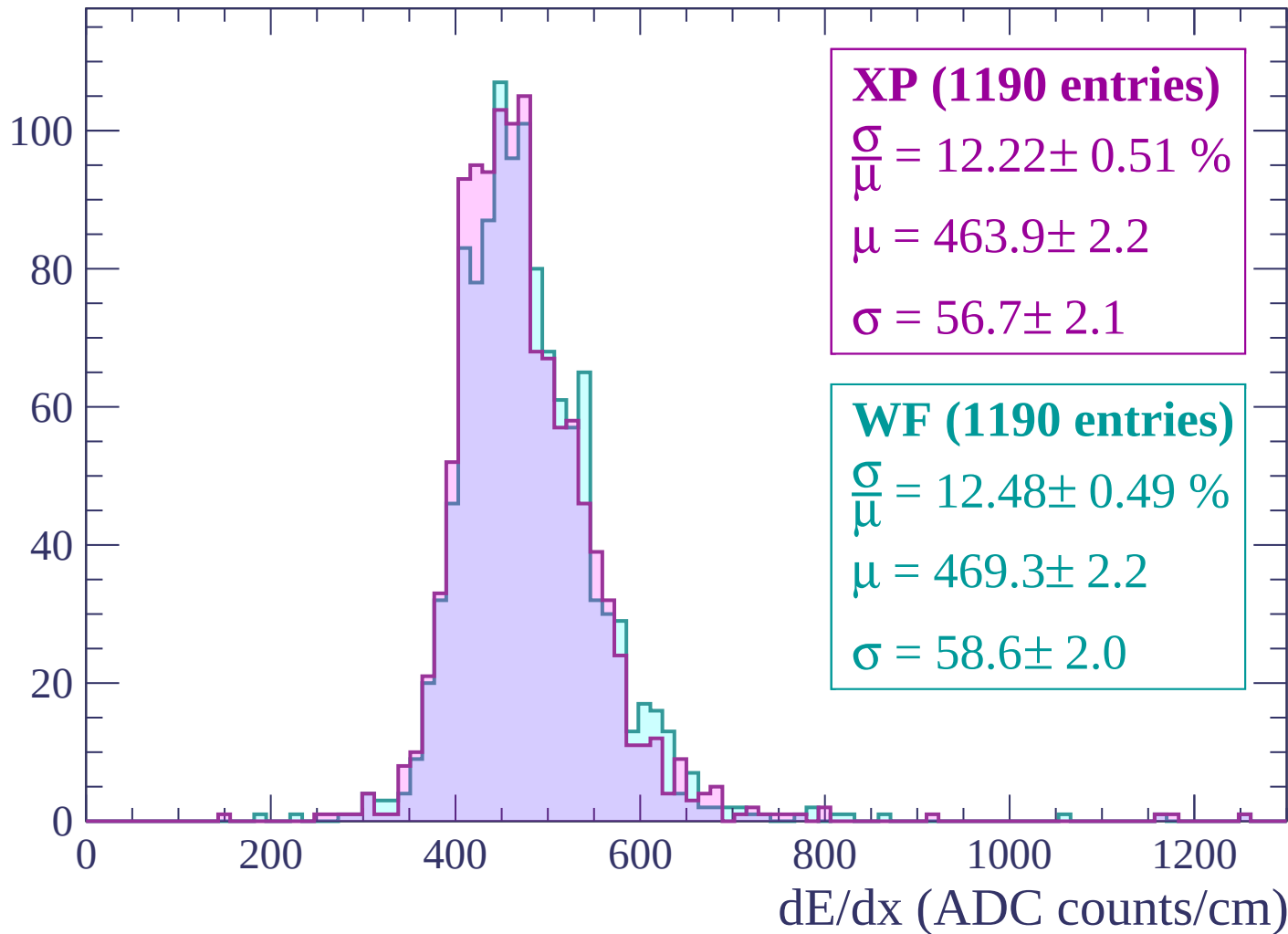
Energy loss | $-1320 < p < -1260$

Count



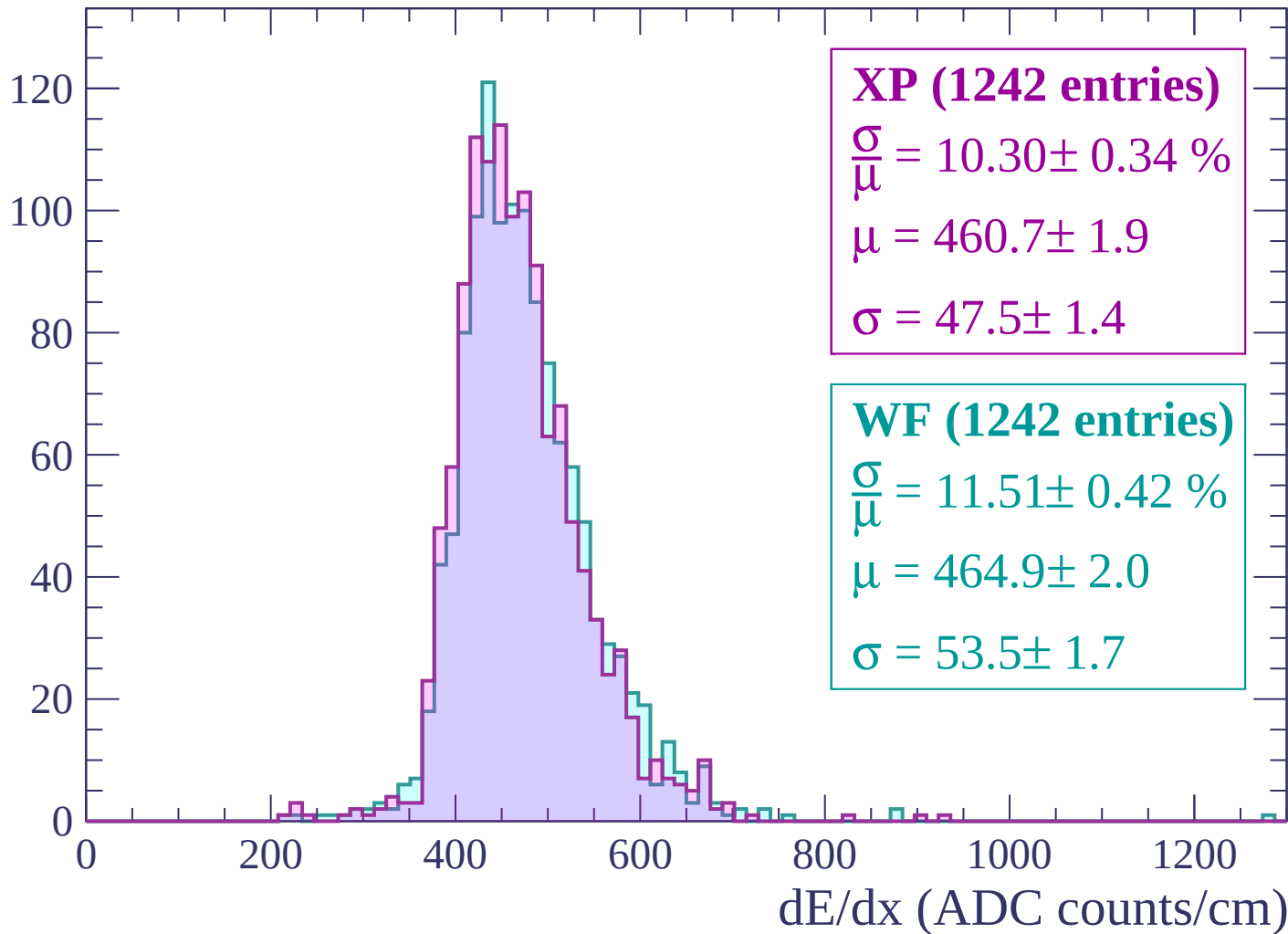
Energy loss | $-1260 < p < -1200$

Count



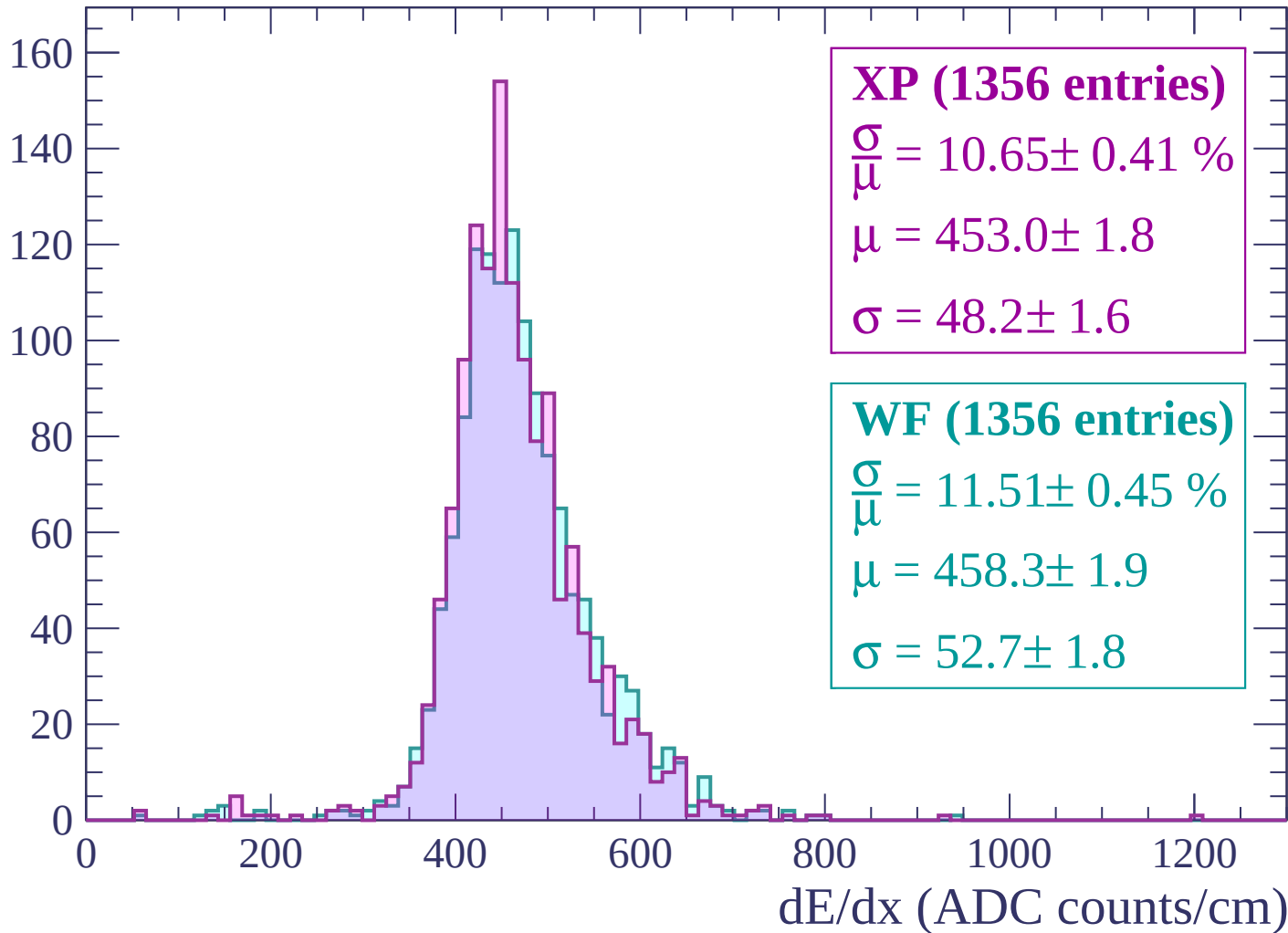
Energy loss | $-1200 < p < -1140$

Count



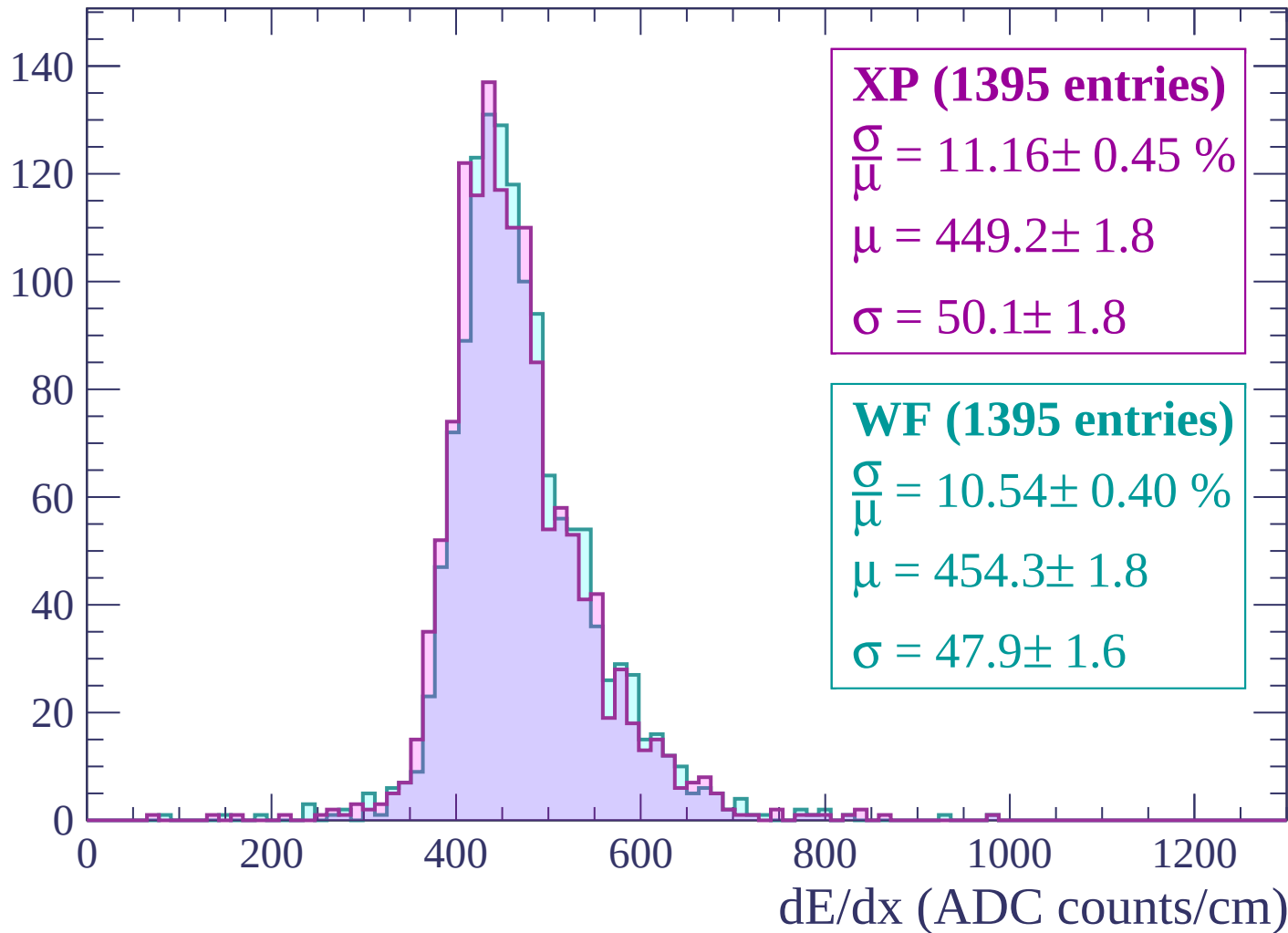
Energy loss | -1140 < p < -1080

Count



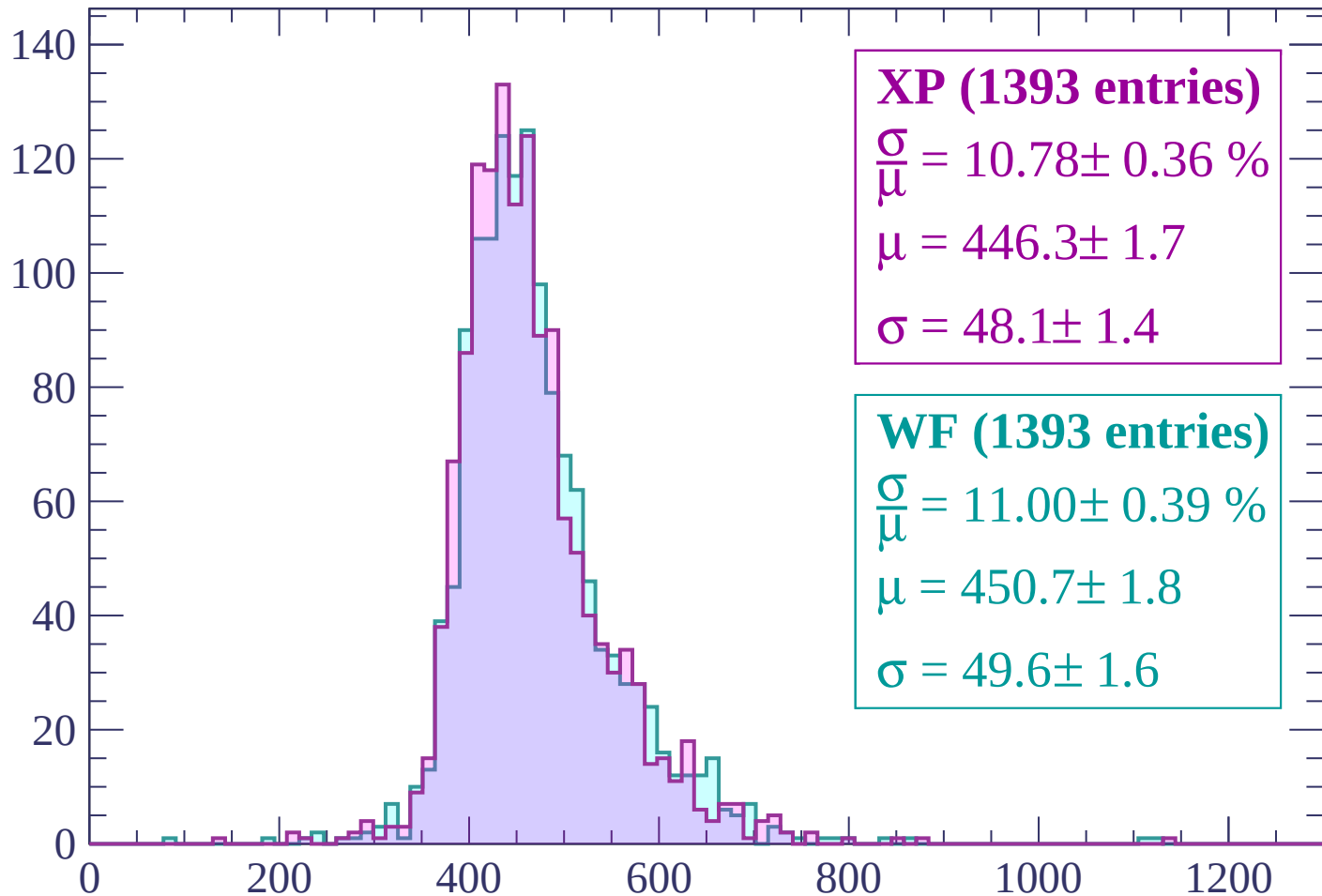
Energy loss | $-1080 < p < -1020$

Count



Energy loss | $-1020 < p < -960$

Count



XP (1393 entries)

$$\frac{\sigma}{\mu} = 10.78 \pm 0.36 \%$$

$$\mu = 446.3 \pm 1.7$$

$$\sigma = 48.1 \pm 1.4$$

WF (1393 entries)

$$\frac{\sigma}{\mu} = 11.00 \pm 0.39 \%$$

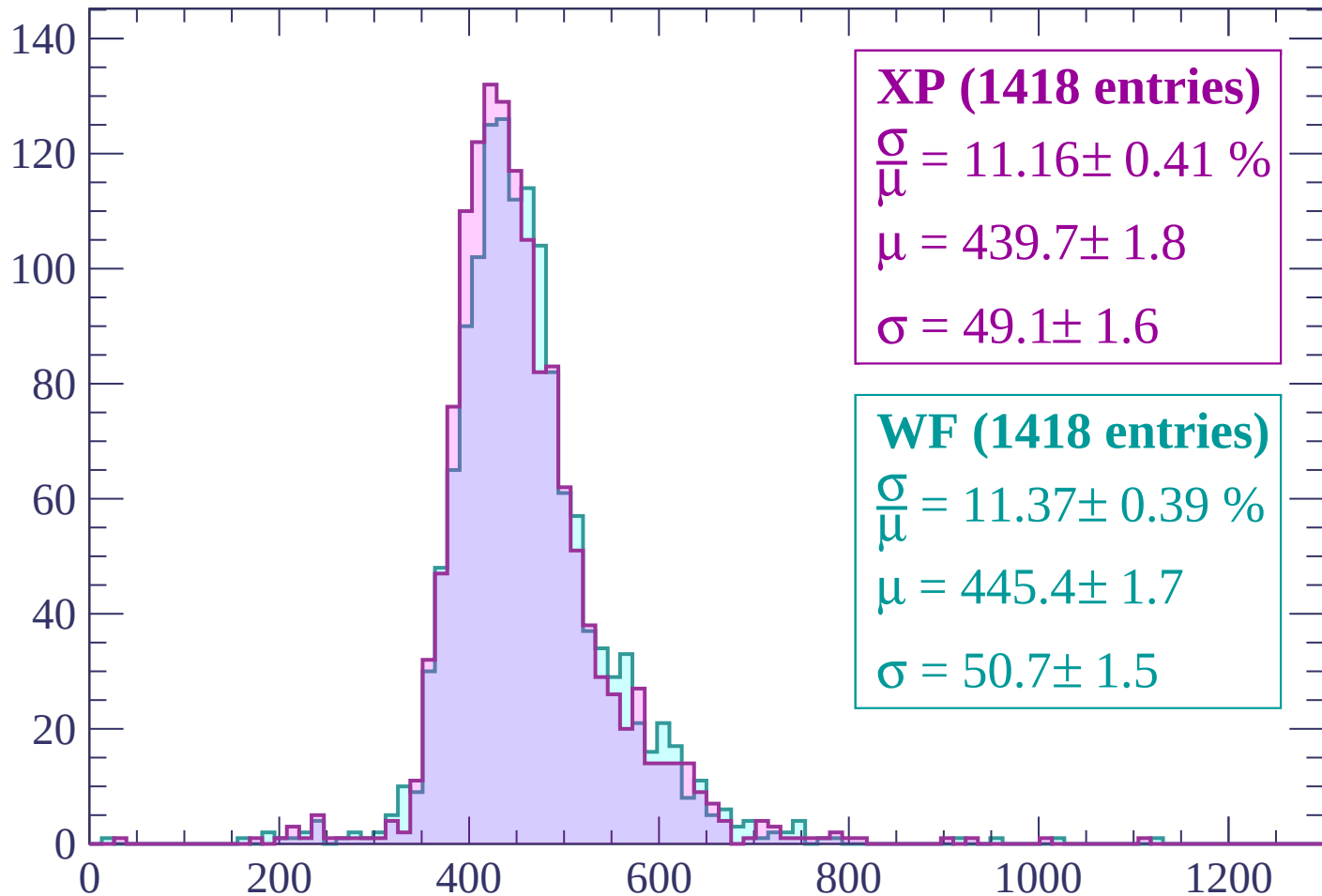
$$\mu = 450.7 \pm 1.8$$

$$\sigma = 49.6 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (1418 entries)

$$\frac{\sigma}{\mu} = 11.16 \pm 0.41 \%$$

$$\mu = 439.7 \pm 1.8$$

$$\sigma = 49.1 \pm 1.6$$

WF (1418 entries)

$$\frac{\sigma}{\mu} = 11.37 \pm 0.39 \%$$

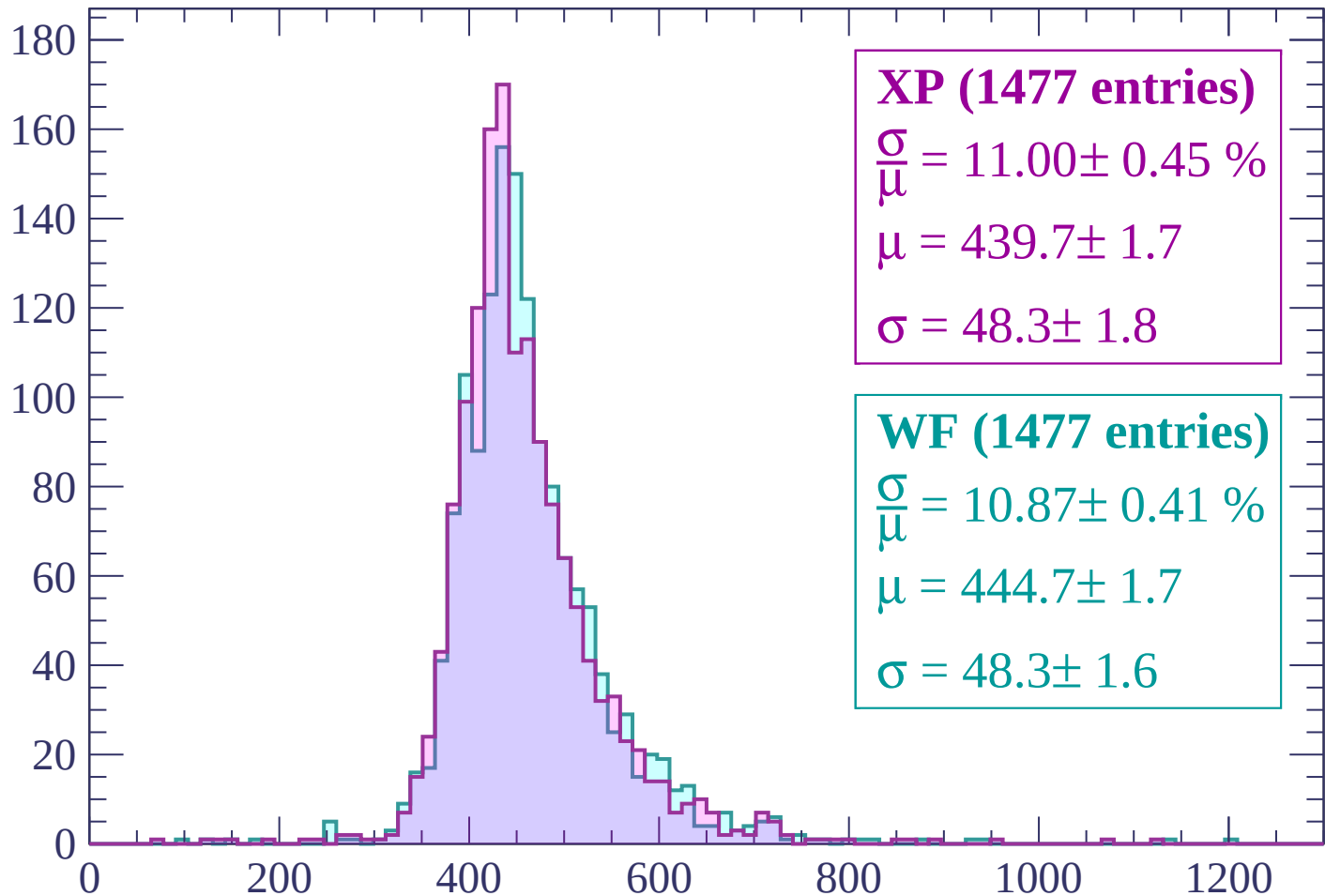
$$\mu = 445.4 \pm 1.7$$

$$\sigma = 50.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (1477 entries)

$$\frac{\sigma}{\mu} = 11.00 \pm 0.45 \%$$

$$\mu = 439.7 \pm 1.7$$

$$\sigma = 48.3 \pm 1.8$$

WF (1477 entries)

$$\frac{\sigma}{\mu} = 10.87 \pm 0.41 \%$$

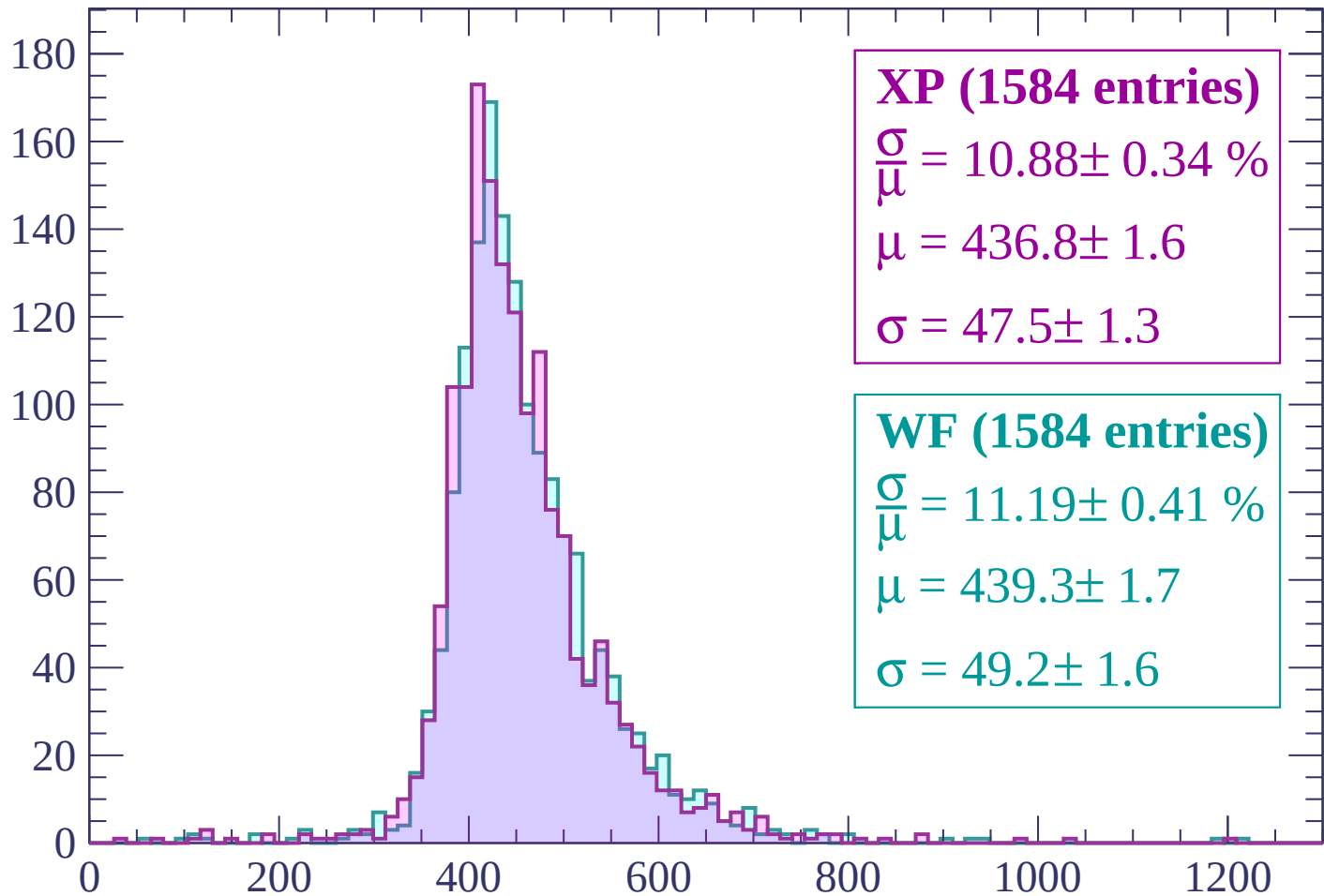
$$\mu = 444.7 \pm 1.7$$

$$\sigma = 48.3 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (1584 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.34 \%$$

$$\mu = 436.8 \pm 1.6$$

$$\sigma = 47.5 \pm 1.3$$

WF (1584 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.41 \%$$

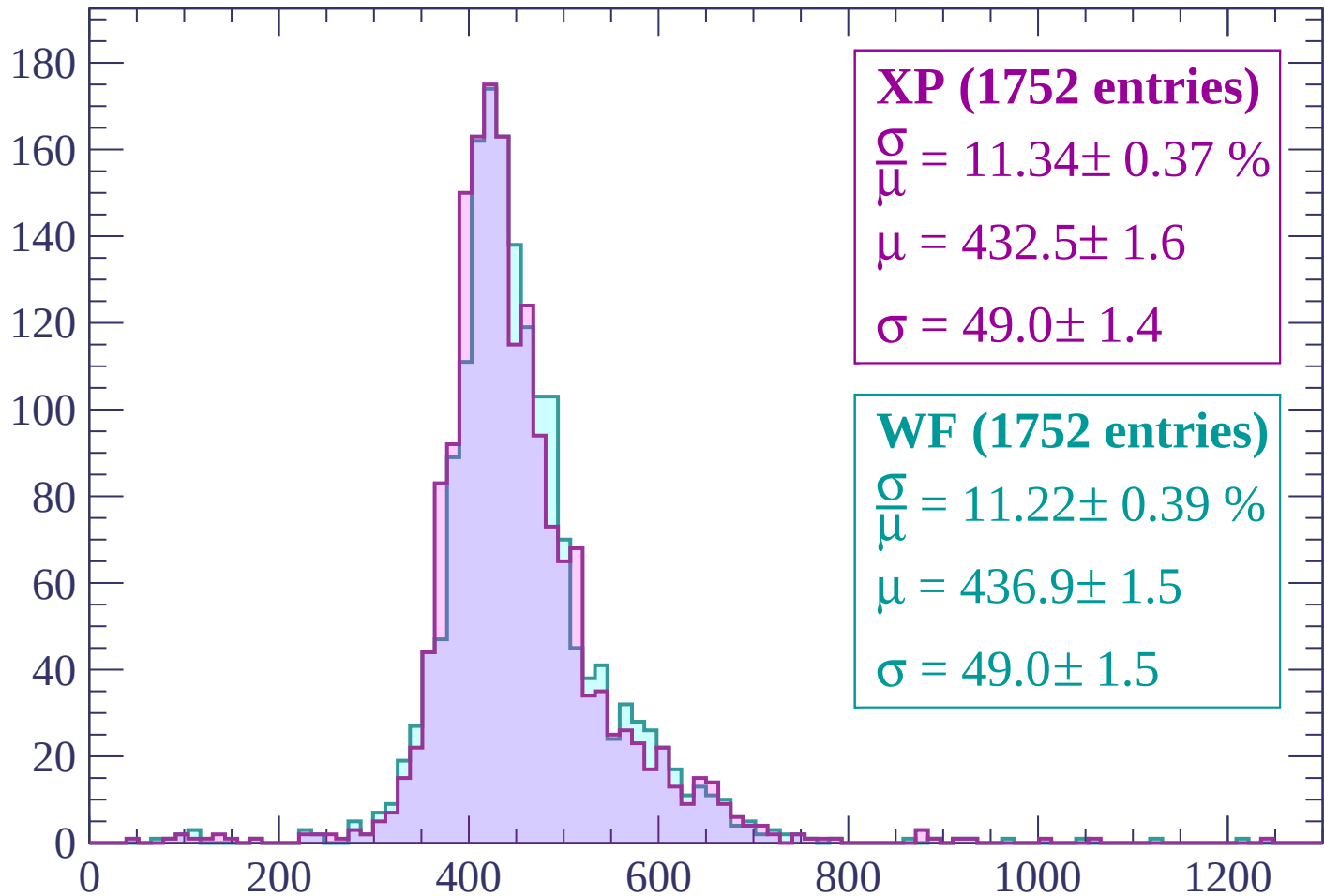
$$\mu = 439.3 \pm 1.7$$

$$\sigma = 49.2 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (1752 entries)

$$\frac{\sigma}{\mu} = 11.34 \pm 0.37 \%$$

$$\mu = 432.5 \pm 1.6$$

$$\sigma = 49.0 \pm 1.4$$

WF (1752 entries)

$$\frac{\sigma}{\mu} = 11.22 \pm 0.39 \%$$

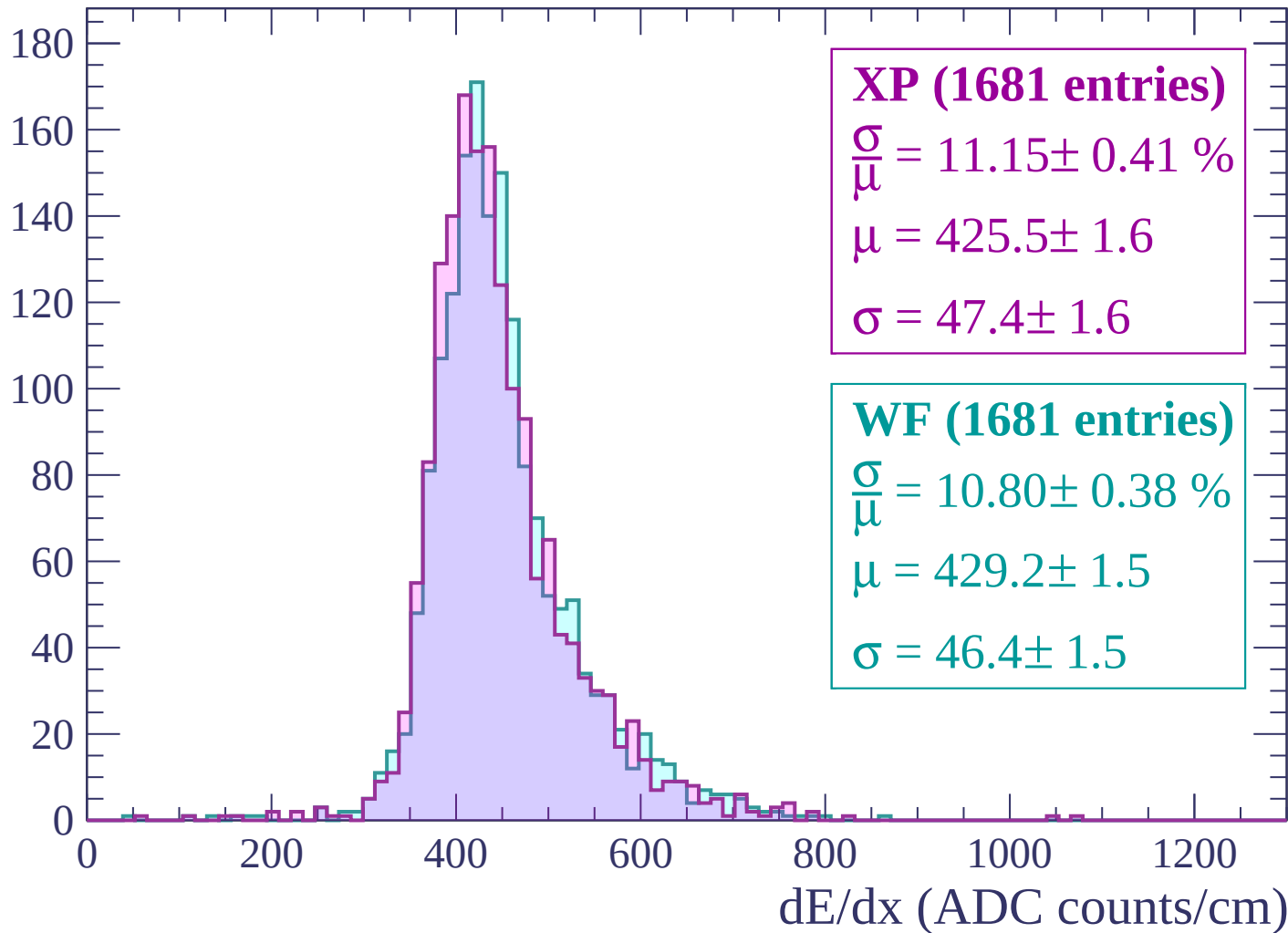
$$\mu = 436.9 \pm 1.5$$

$$\sigma = 49.0 \pm 1.5$$

dE/dx (ADC counts/cm)

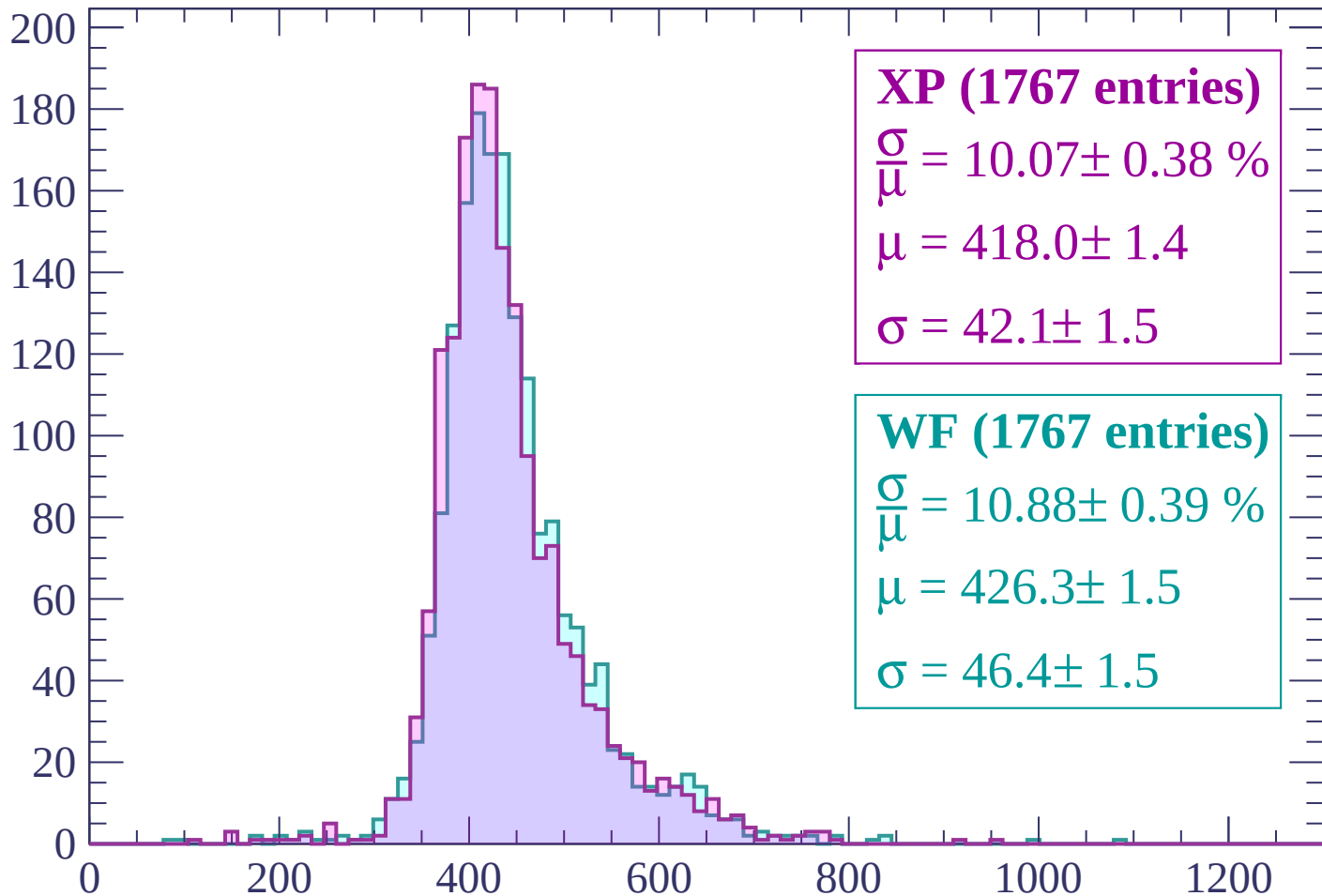
Energy loss | $-720 < p < -660$

Count



Energy loss | $-660 < p < -600$

Count



XP (1767 entries)

$$\frac{\sigma}{\mu} = 10.07 \pm 0.38 \%$$

$$\mu = 418.0 \pm 1.4$$

$$\sigma = 42.1 \pm 1.5$$

WF (1767 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.39 \%$$

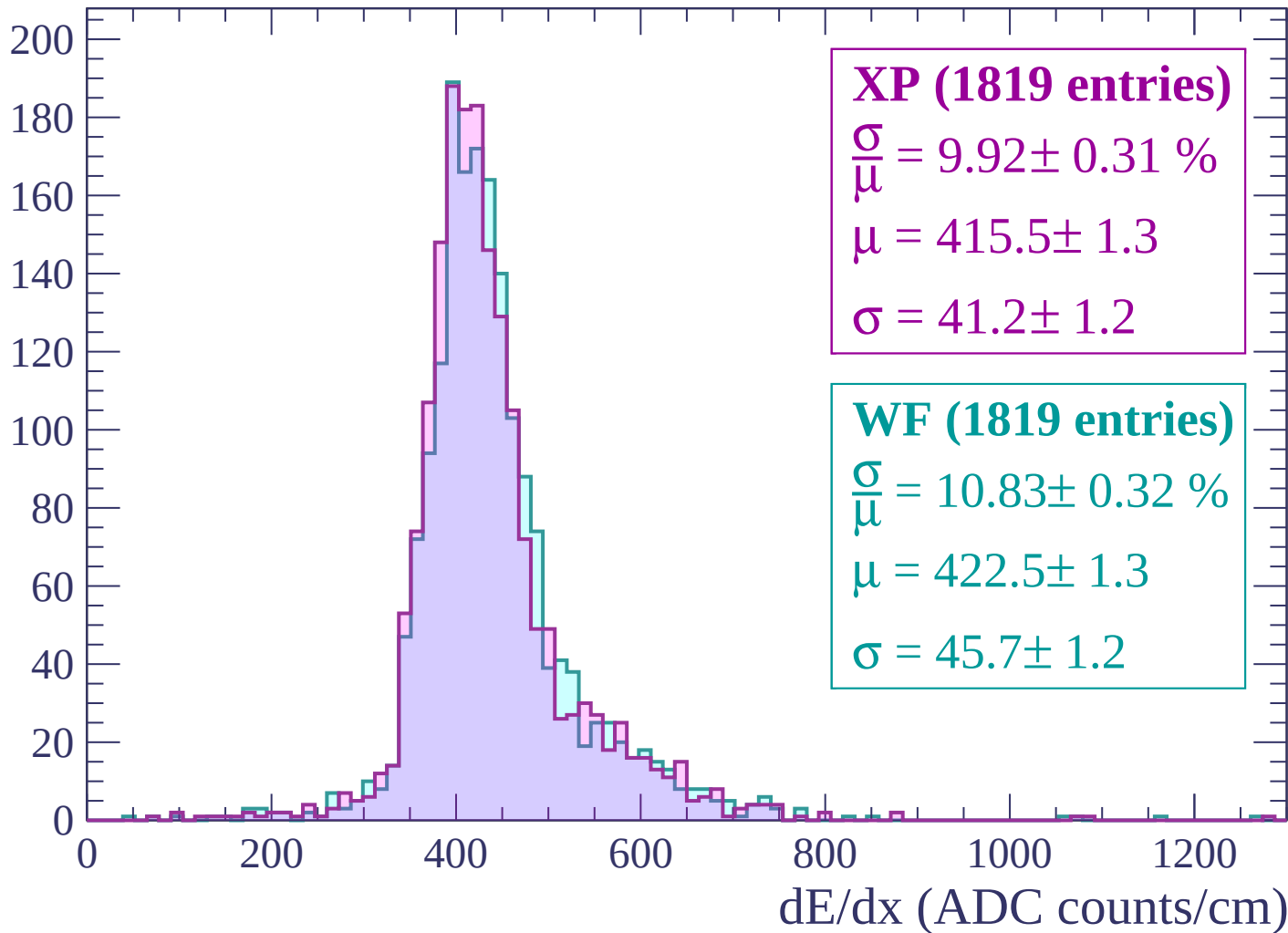
$$\mu = 426.3 \pm 1.5$$

$$\sigma = 46.4 \pm 1.5$$

dE/dx (ADC counts/cm)

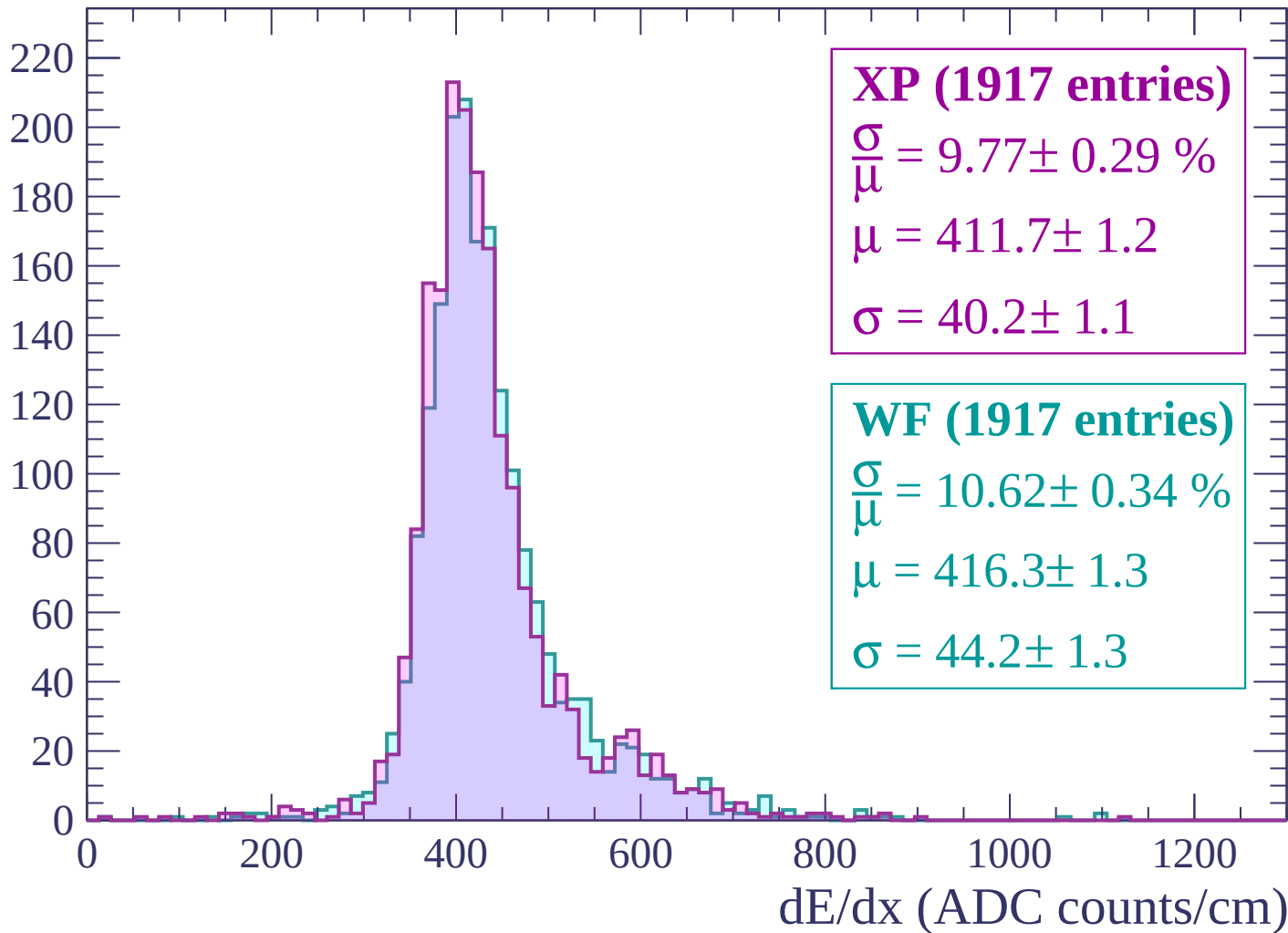
Energy loss | $-600 < p < -540$

Count



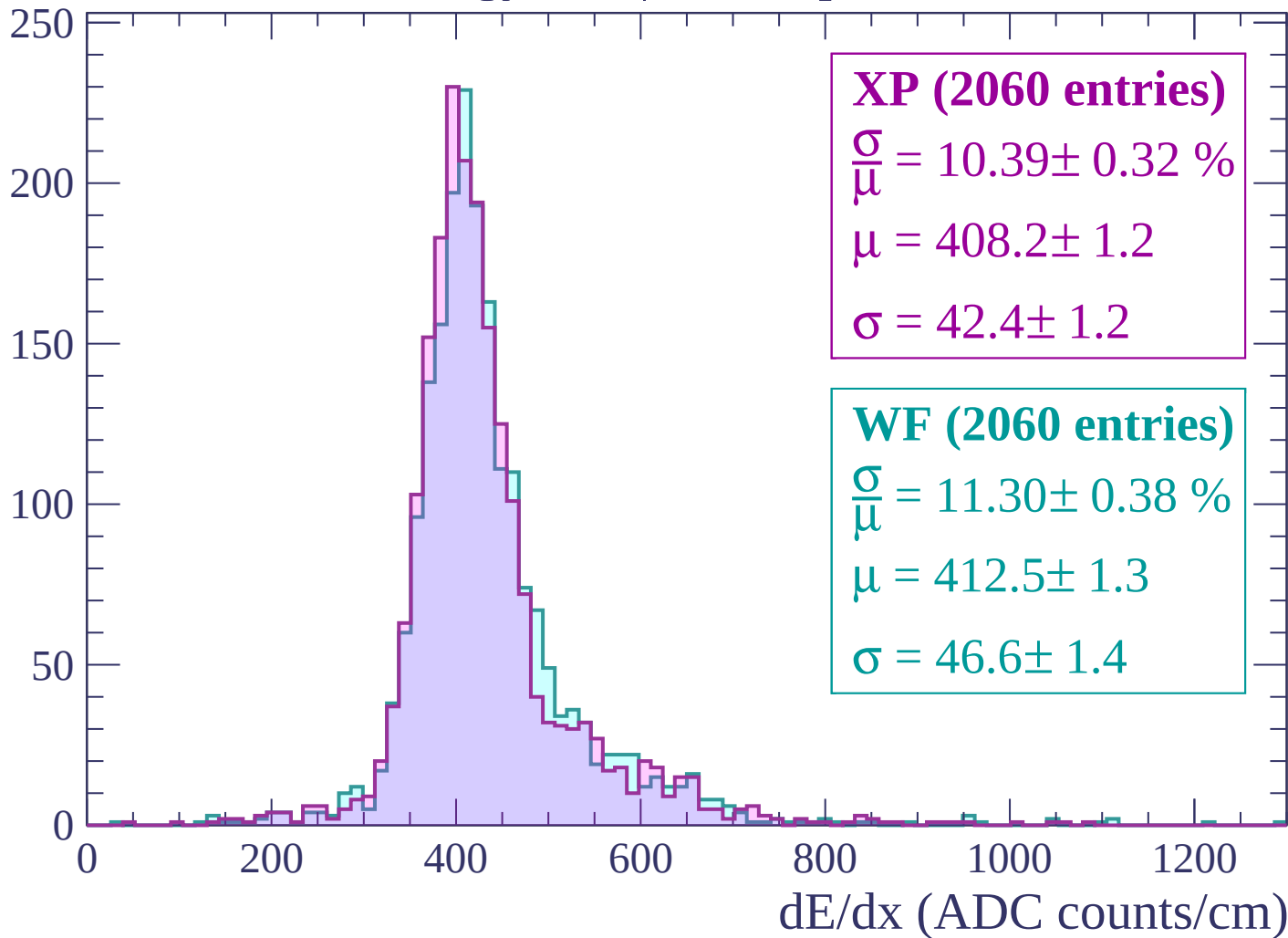
Energy loss | $-540 < p < -480$

Count



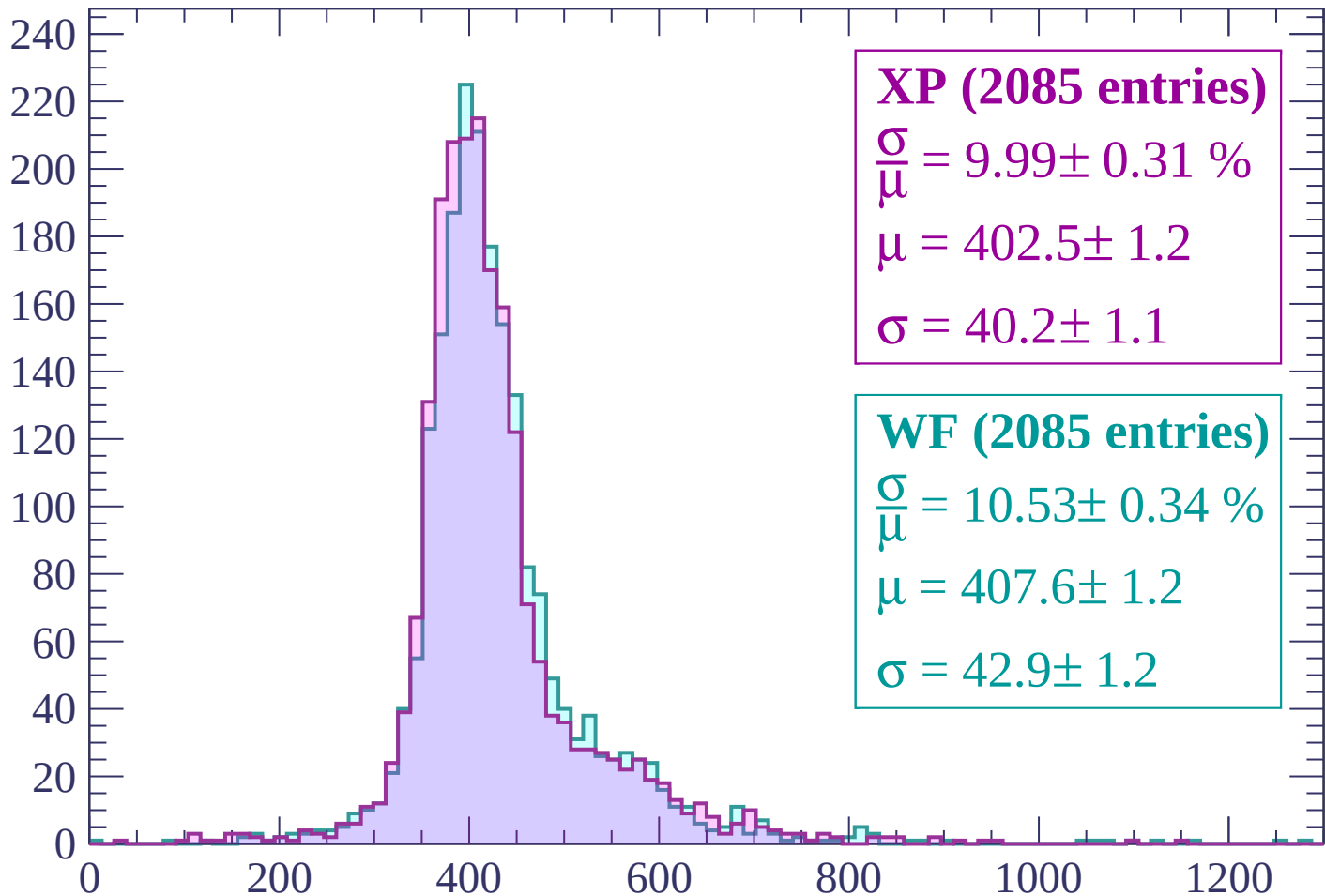
Energy loss | $-480 < p < -420$

Count



Energy loss | $-420 < p < -360$

Count



XP (2085 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.31 \%$$

$$\mu = 402.5 \pm 1.2$$

$$\sigma = 40.2 \pm 1.1$$

WF (2085 entries)

$$\frac{\sigma}{\mu} = 10.53 \pm 0.34 \%$$

$$\mu = 407.6 \pm 1.2$$

$$\sigma = 42.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (2184 entries)

$\frac{\sigma}{\mu} = 10.22 \pm 0.34 \%$

$\mu = 399.0 \pm 1.2$

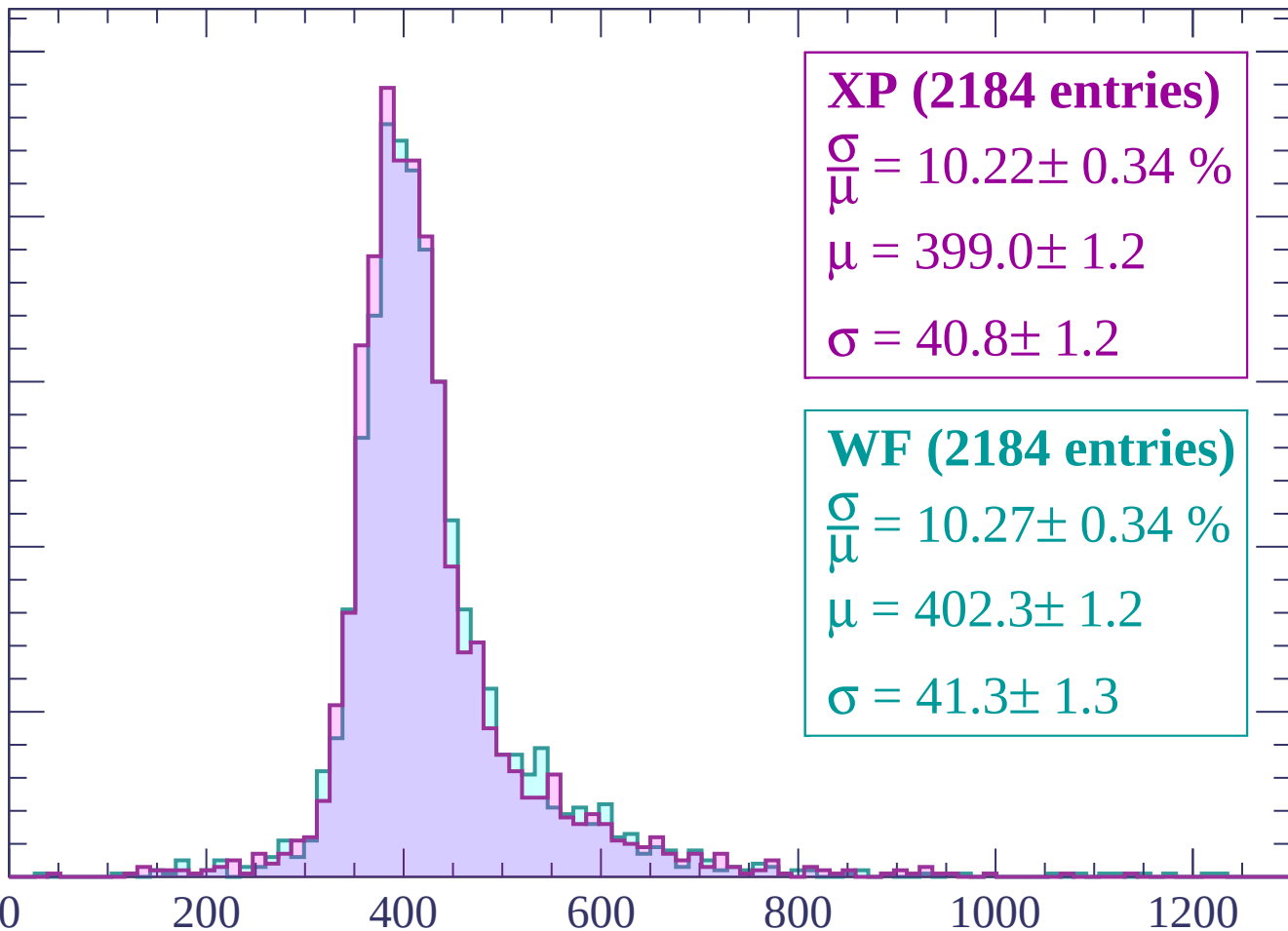
$\sigma = 40.8 \pm 1.2$

WF (2184 entries)

$\frac{\sigma}{\mu} = 10.27 \pm 0.34 \%$

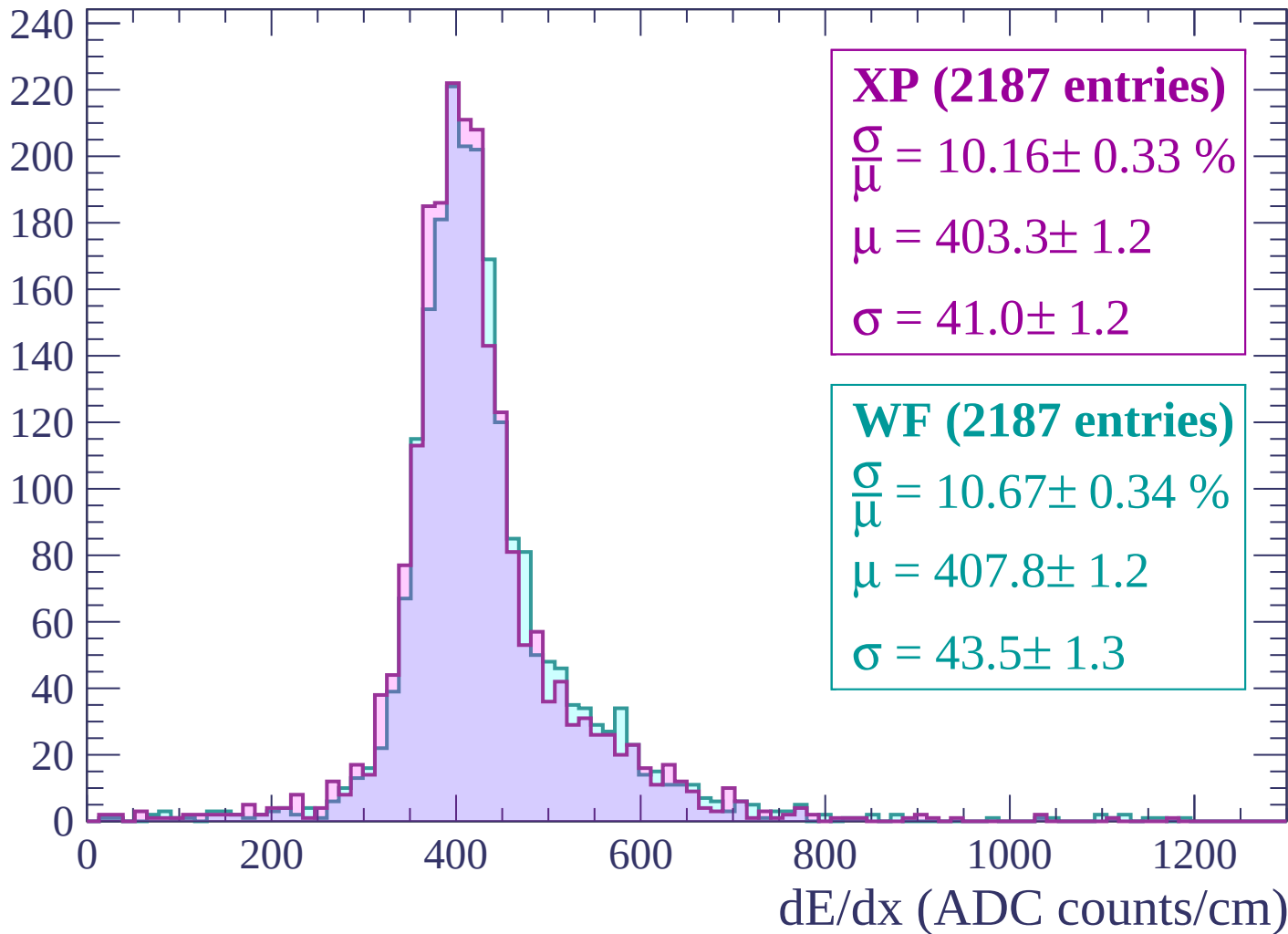
$\mu = 402.3 \pm 1.2$

$\sigma = 41.3 \pm 1.3$



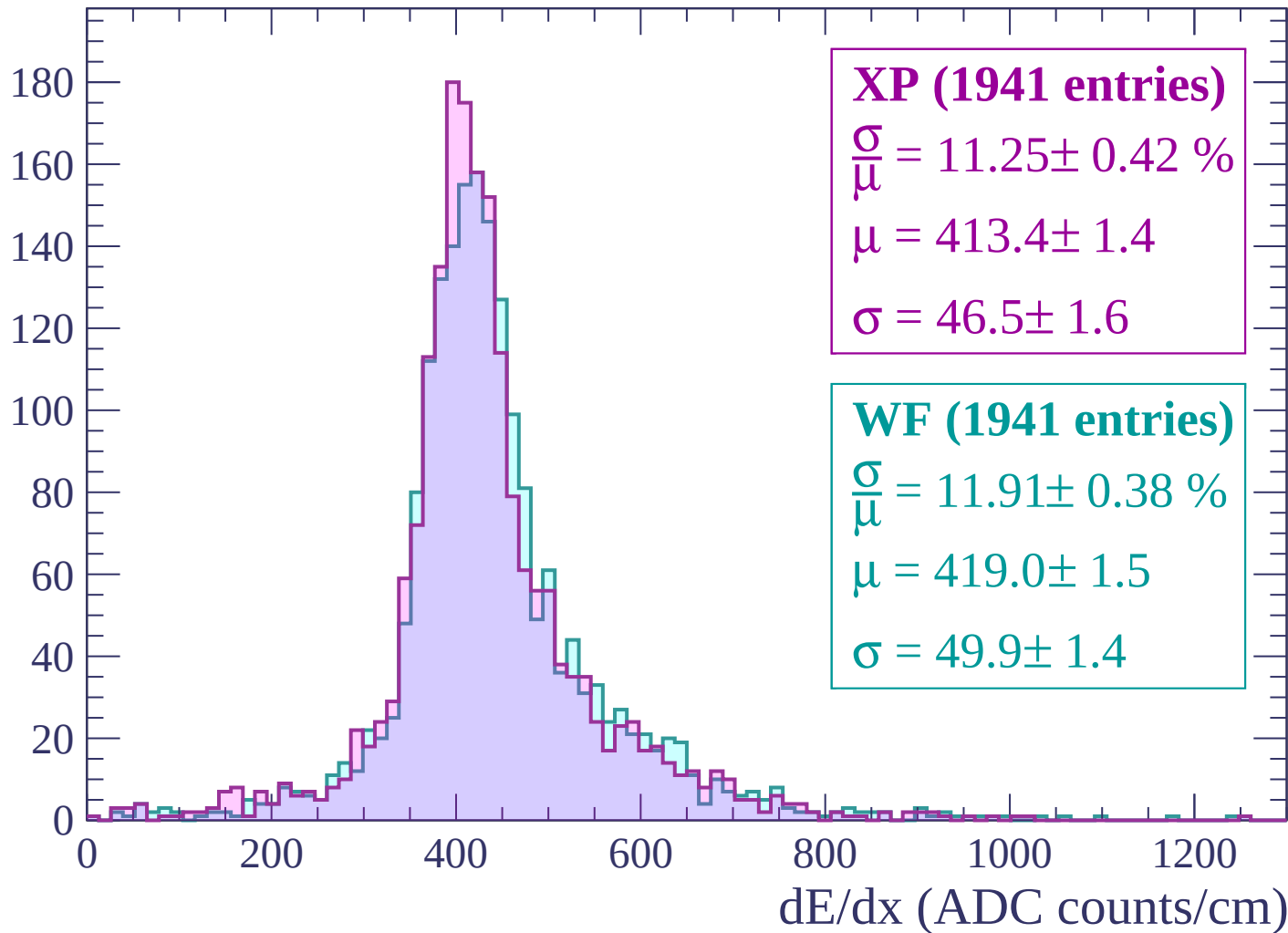
Energy loss | $-300 < p < -240$

Count



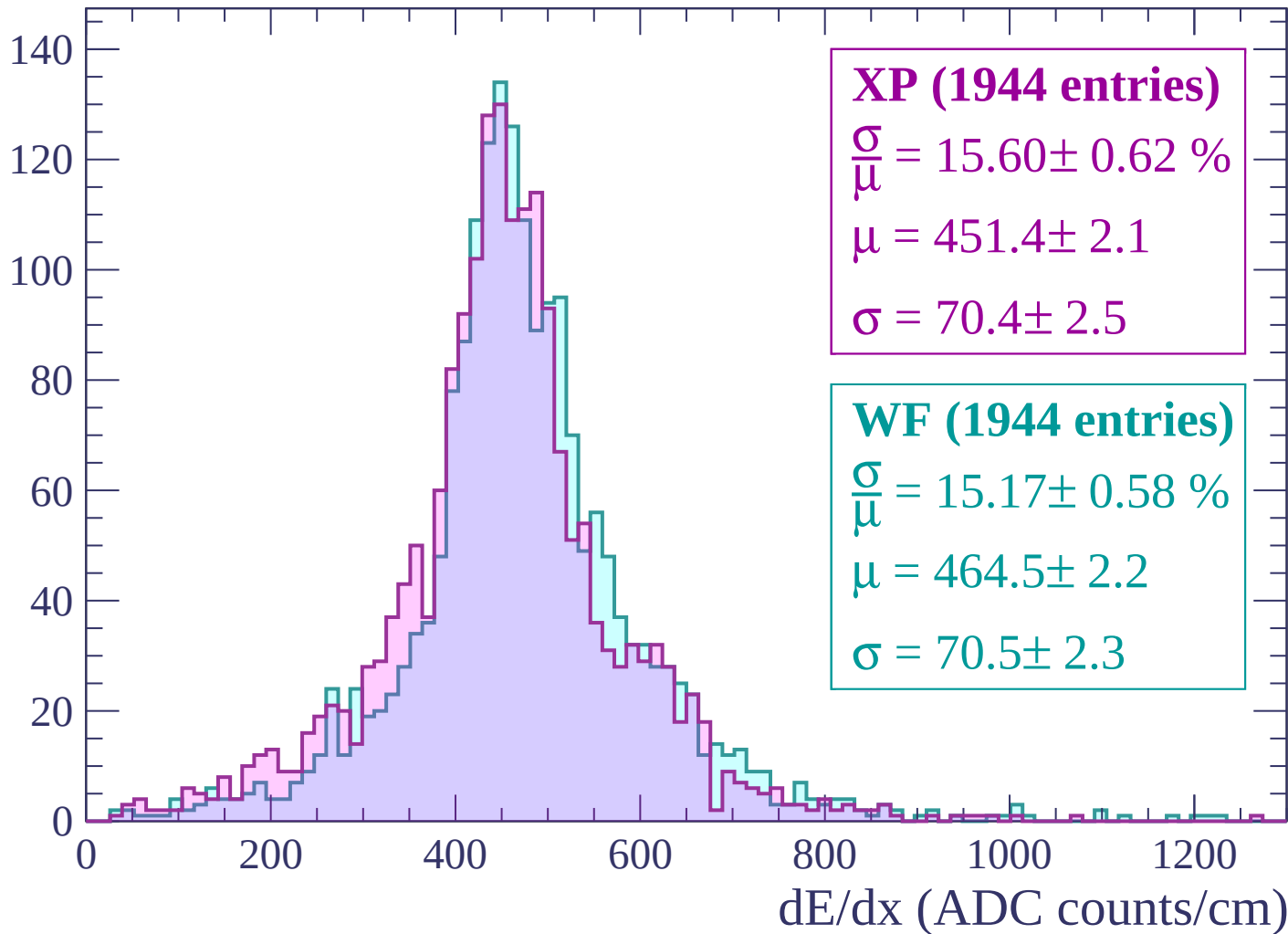
Energy loss | $-240 < p < -180$

Count



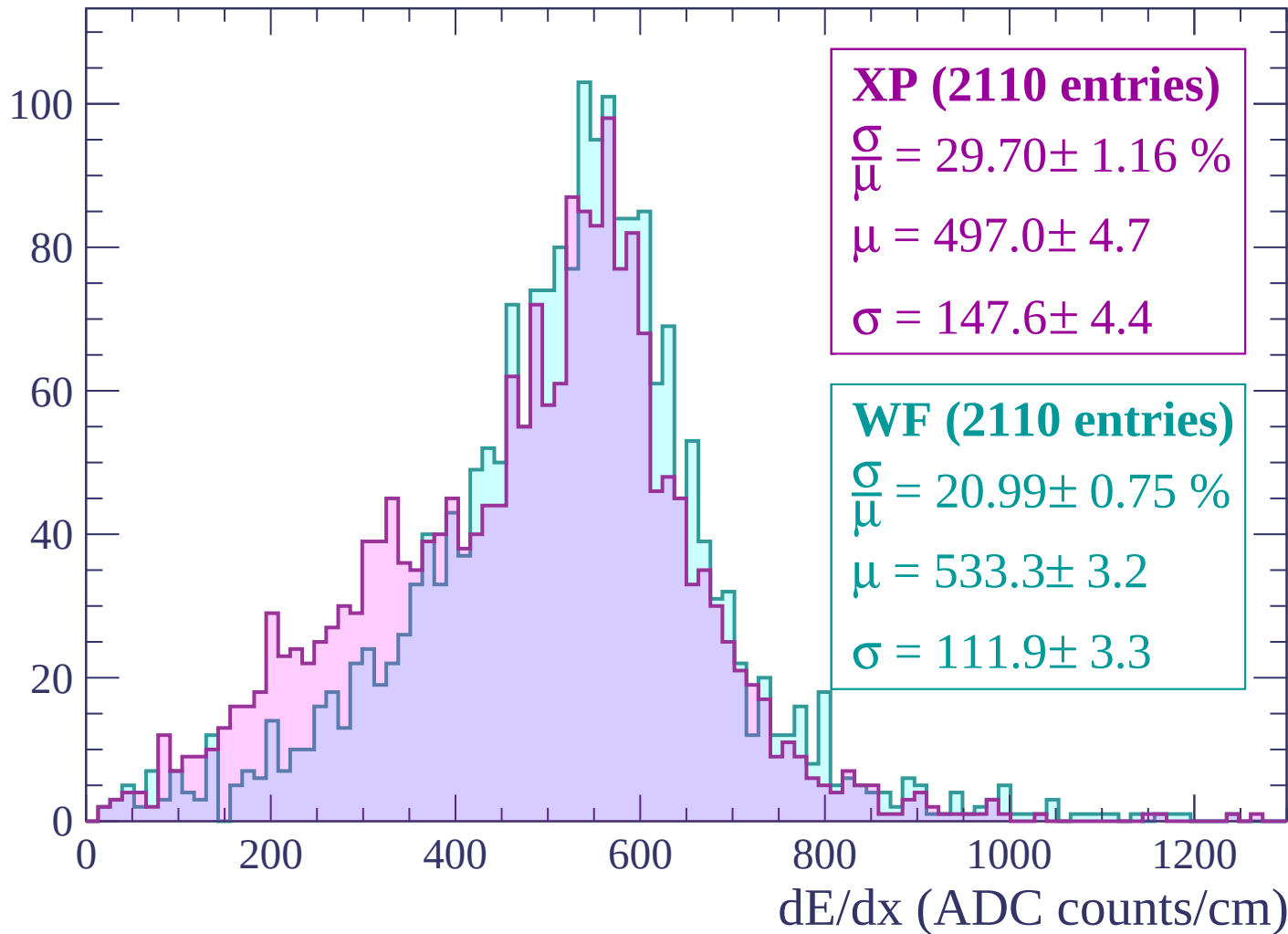
Energy loss | $-180 < p < -120$

Count



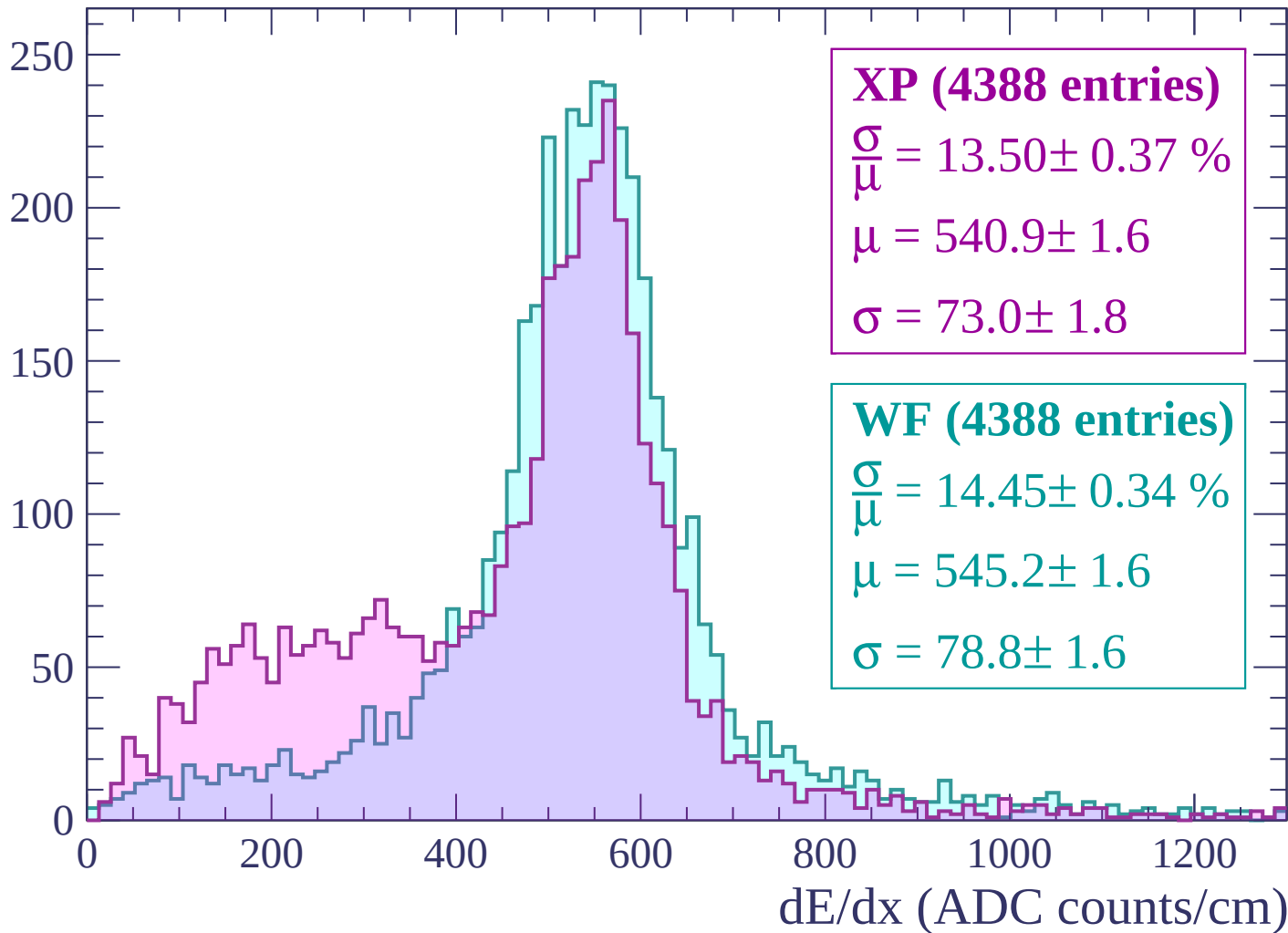
Energy loss | $-120 < p < -60$

Count



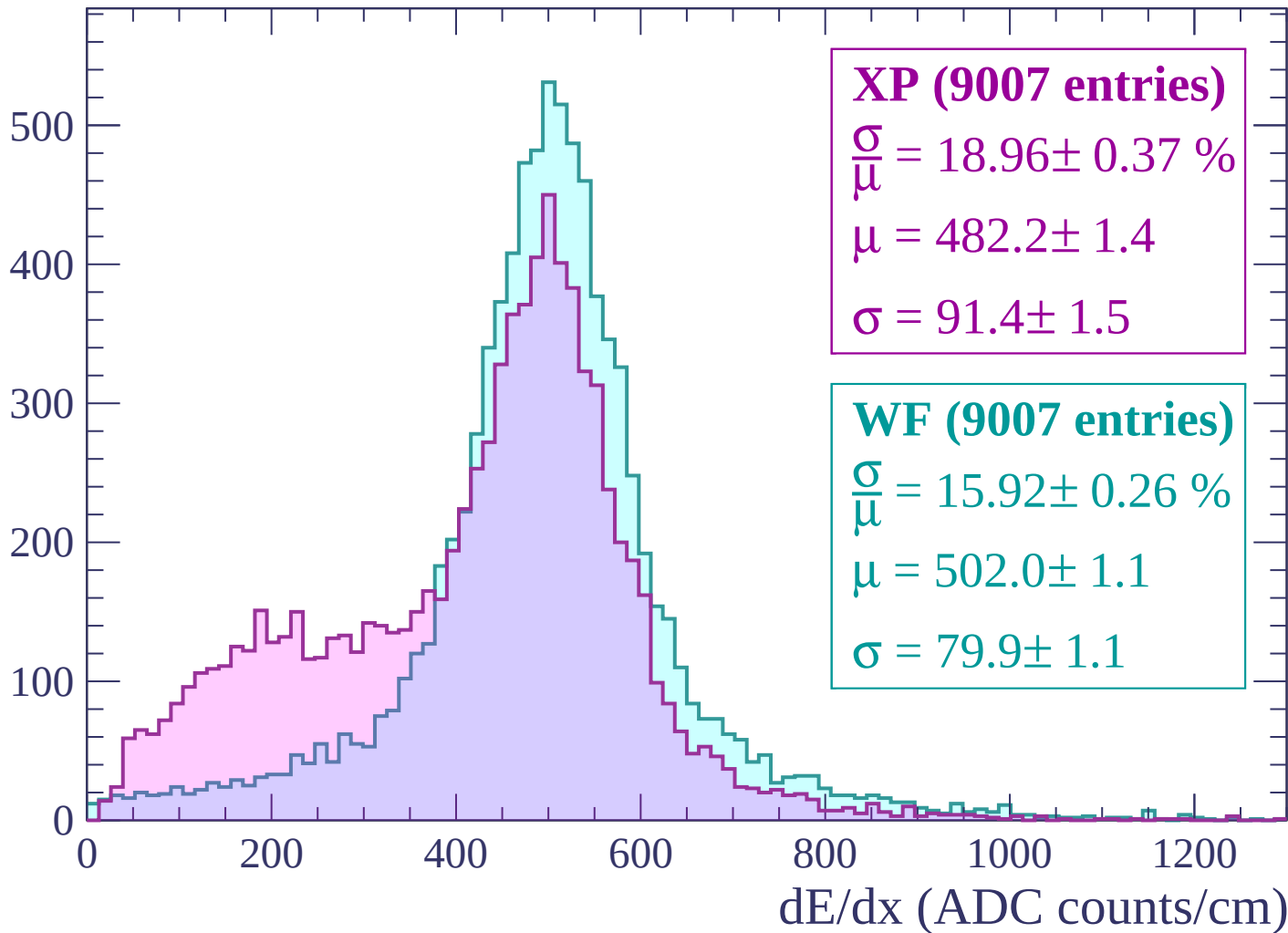
Energy loss | $-60 < p < 0$

Count



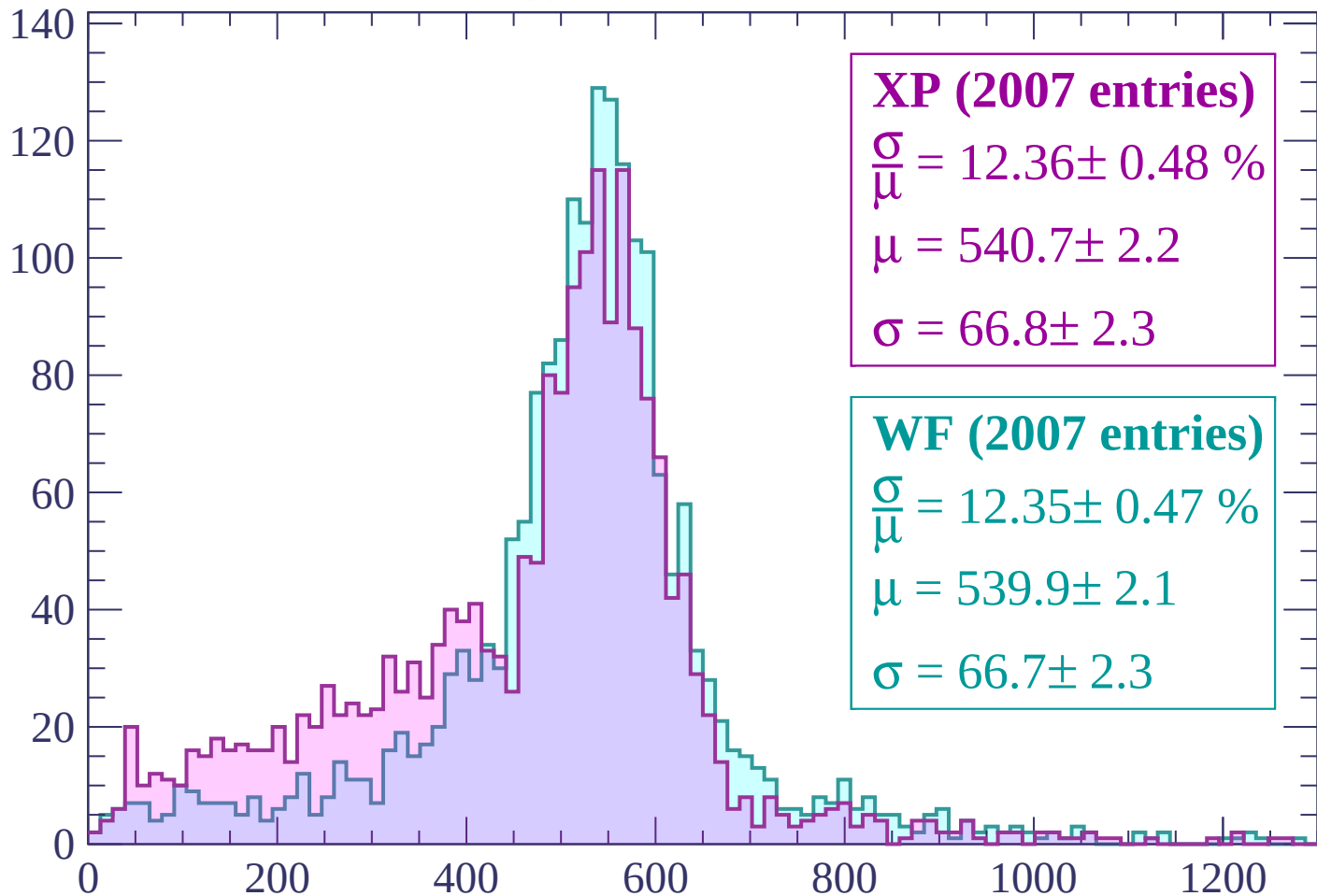
Energy loss | $0 < p < 60$

Count



Energy loss | $60 < p < 120$

Count



XP (2007 entries)

$$\frac{\sigma}{\mu} = 12.36 \pm 0.48 \%$$

$$\mu = 540.7 \pm 2.2$$

$$\sigma = 66.8 \pm 2.3$$

WF (2007 entries)

$$\frac{\sigma}{\mu} = 12.35 \pm 0.47 \%$$

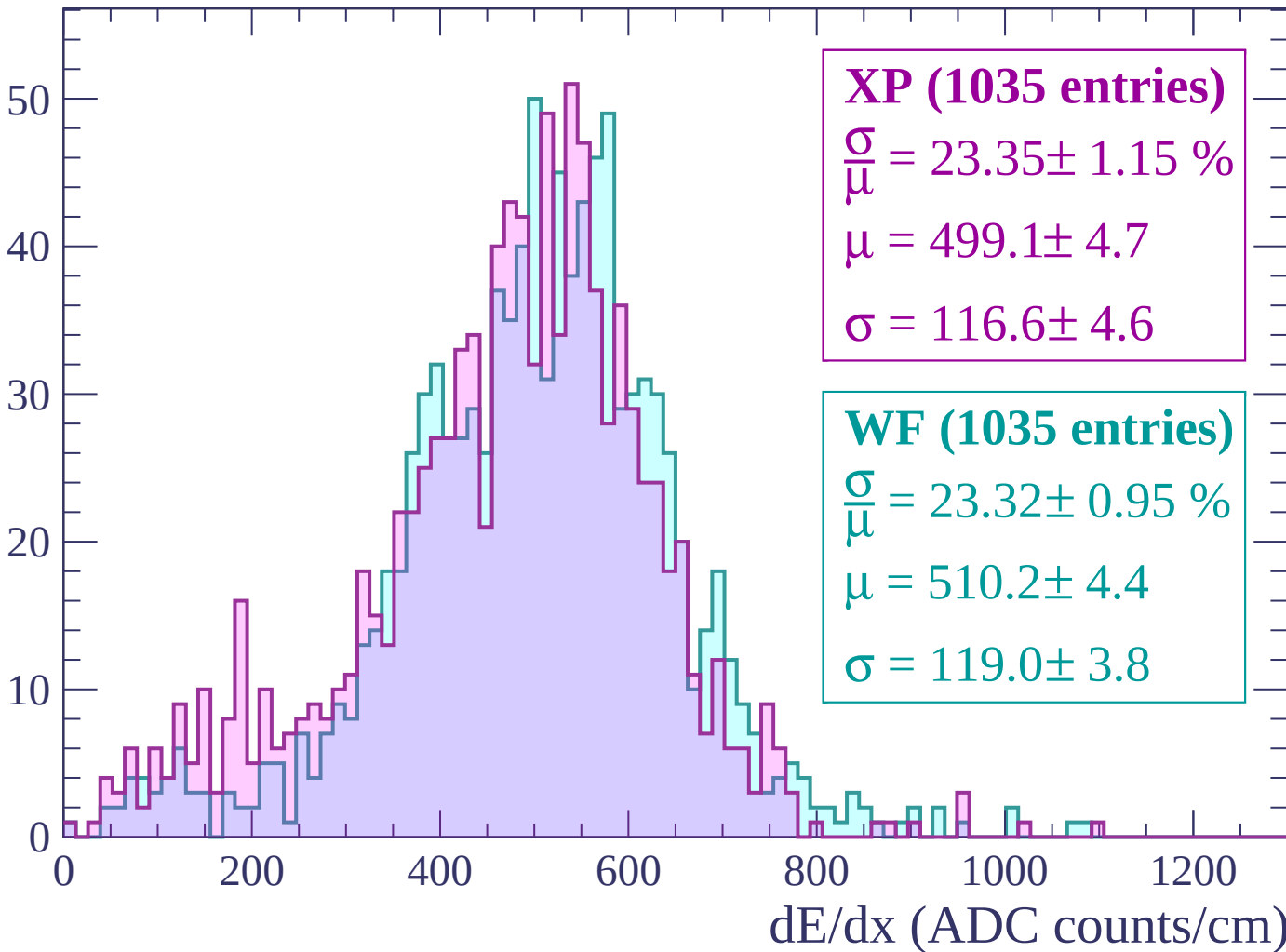
$$\mu = 539.9 \pm 2.1$$

$$\sigma = 66.7 \pm 2.3$$

dE/dx (ADC counts/cm)

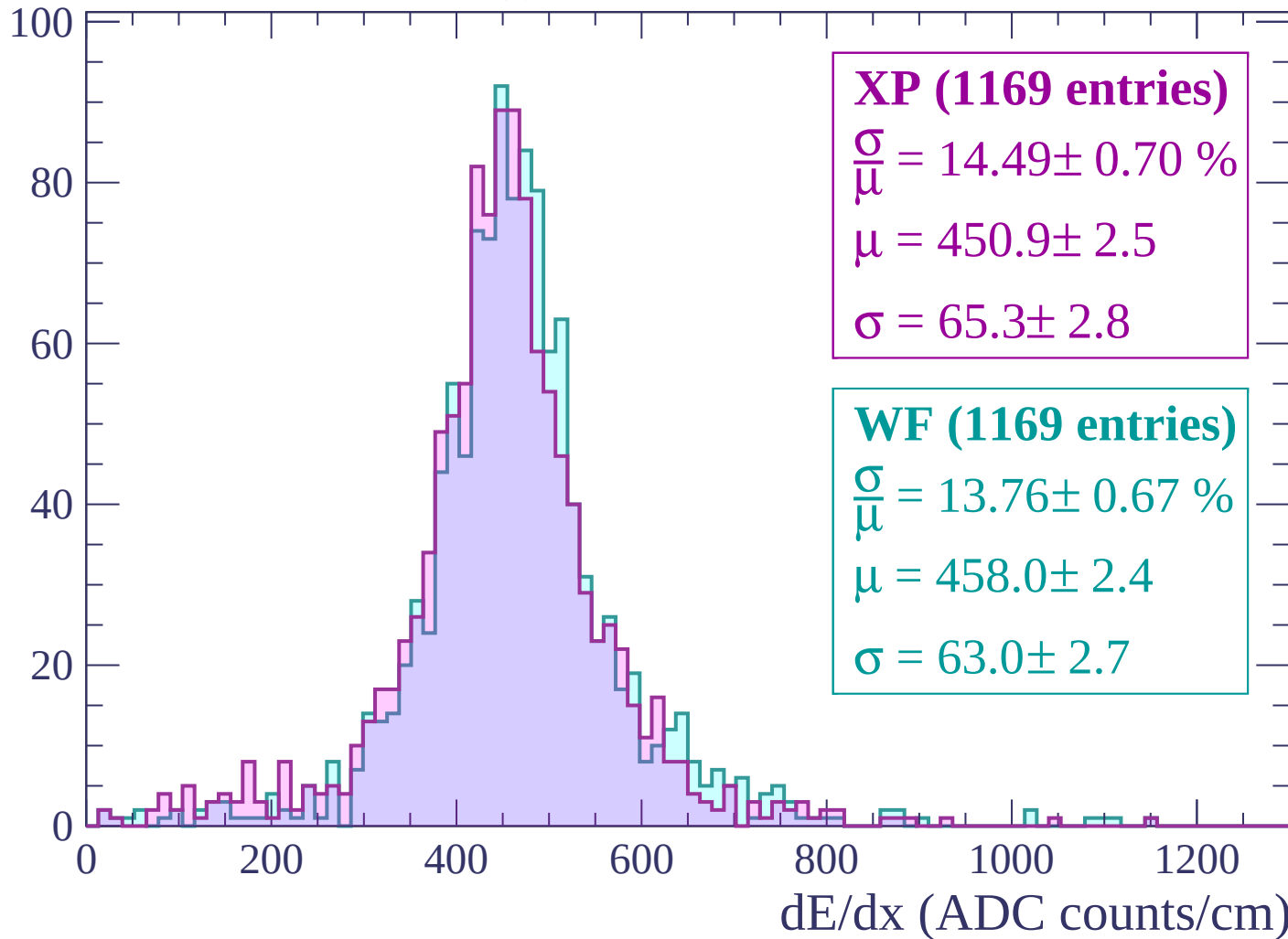
Energy loss | $120 < p < 180$

Count



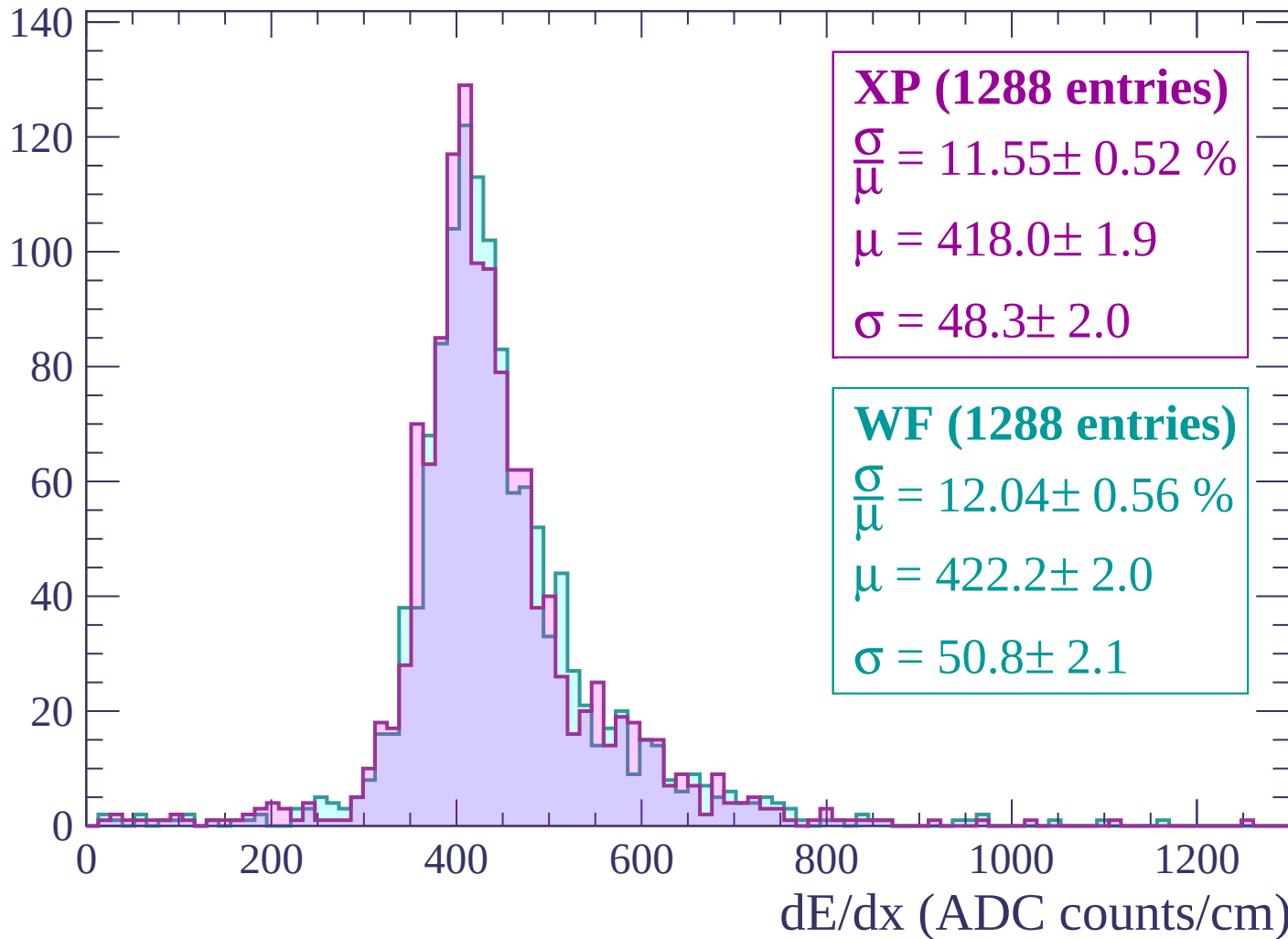
Energy loss | $180 < p < 240$

Count



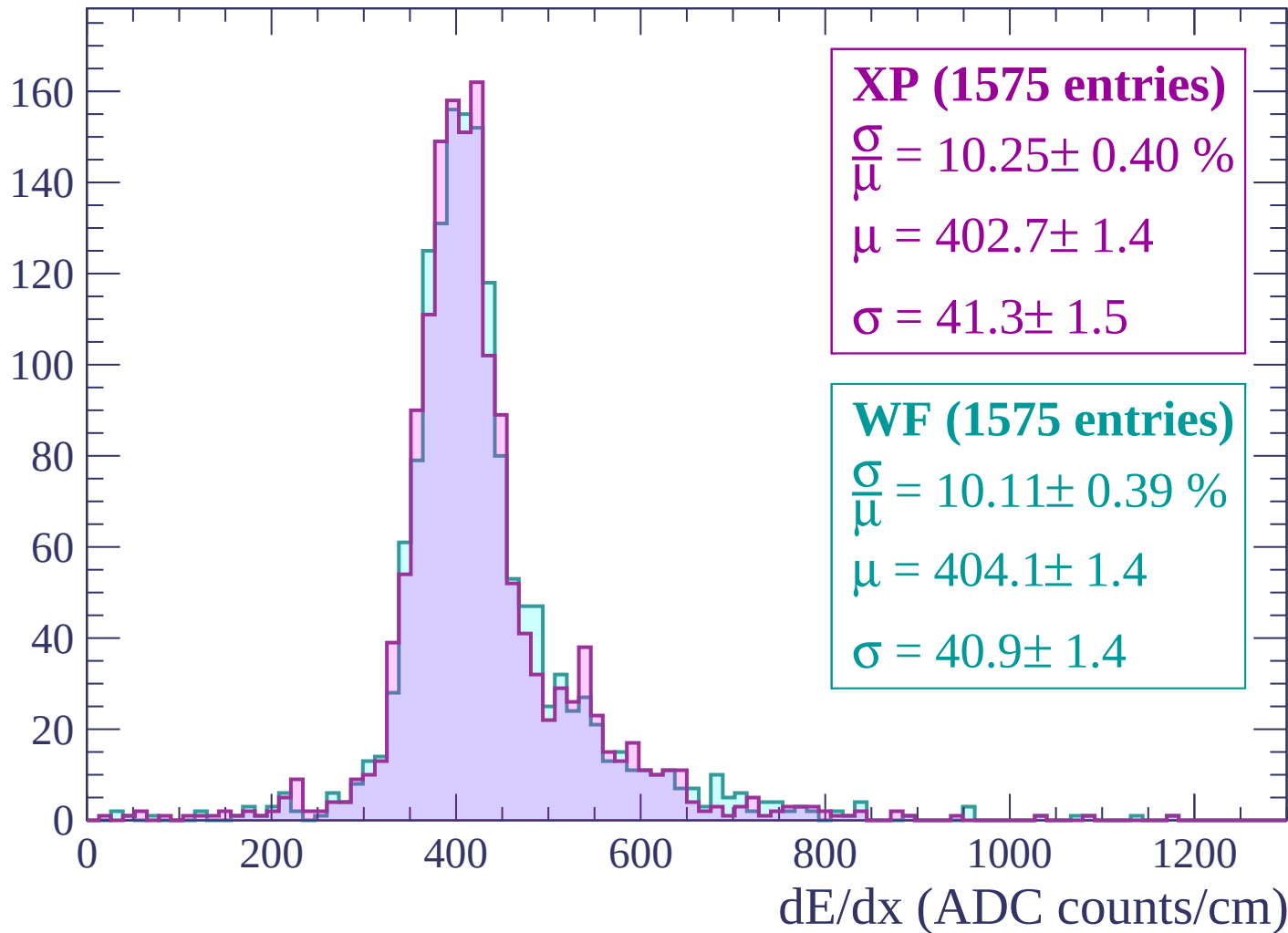
Energy loss | $240 < p < 300$

Count



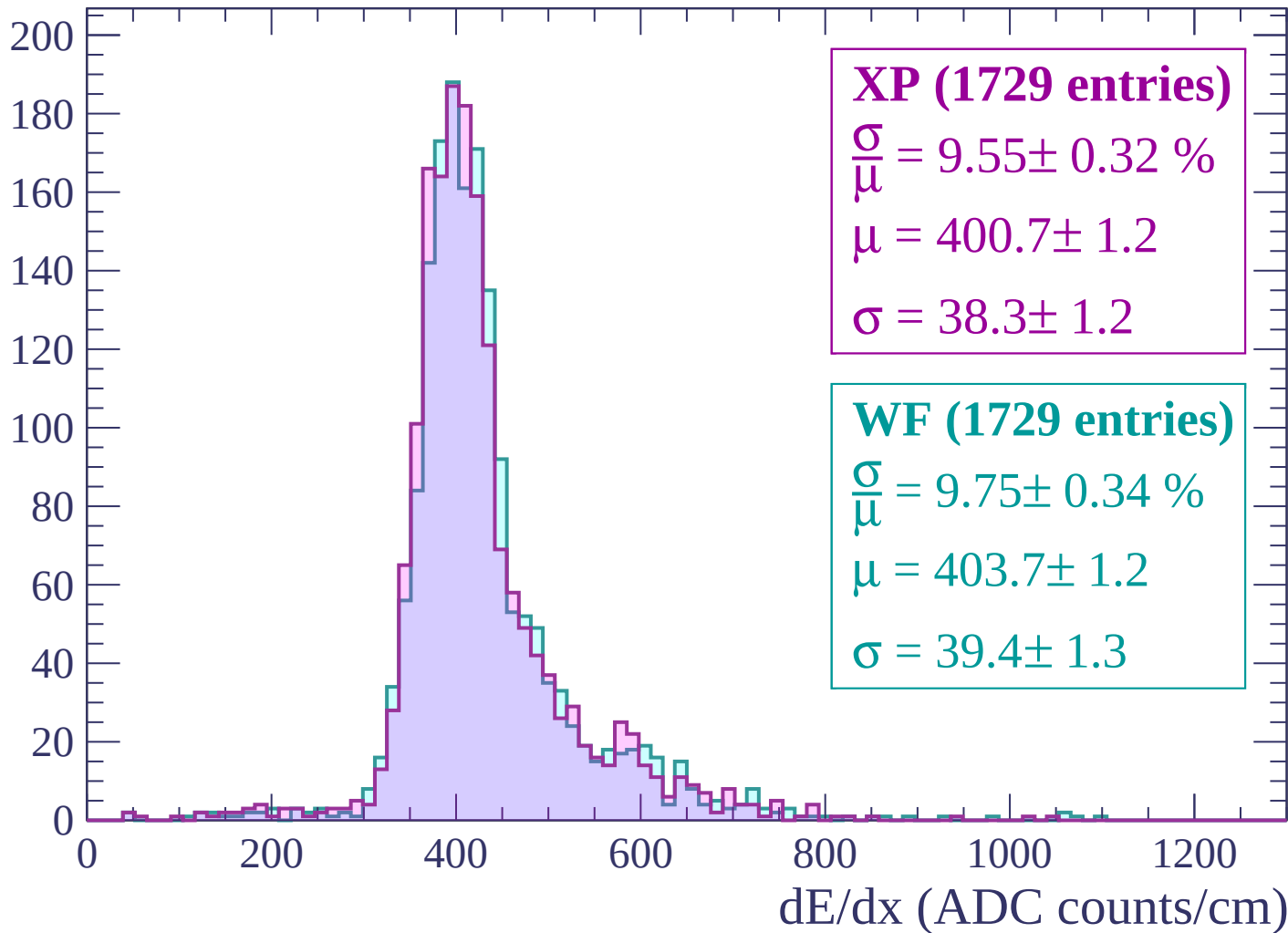
Energy loss | $300 < p < 360$

Count



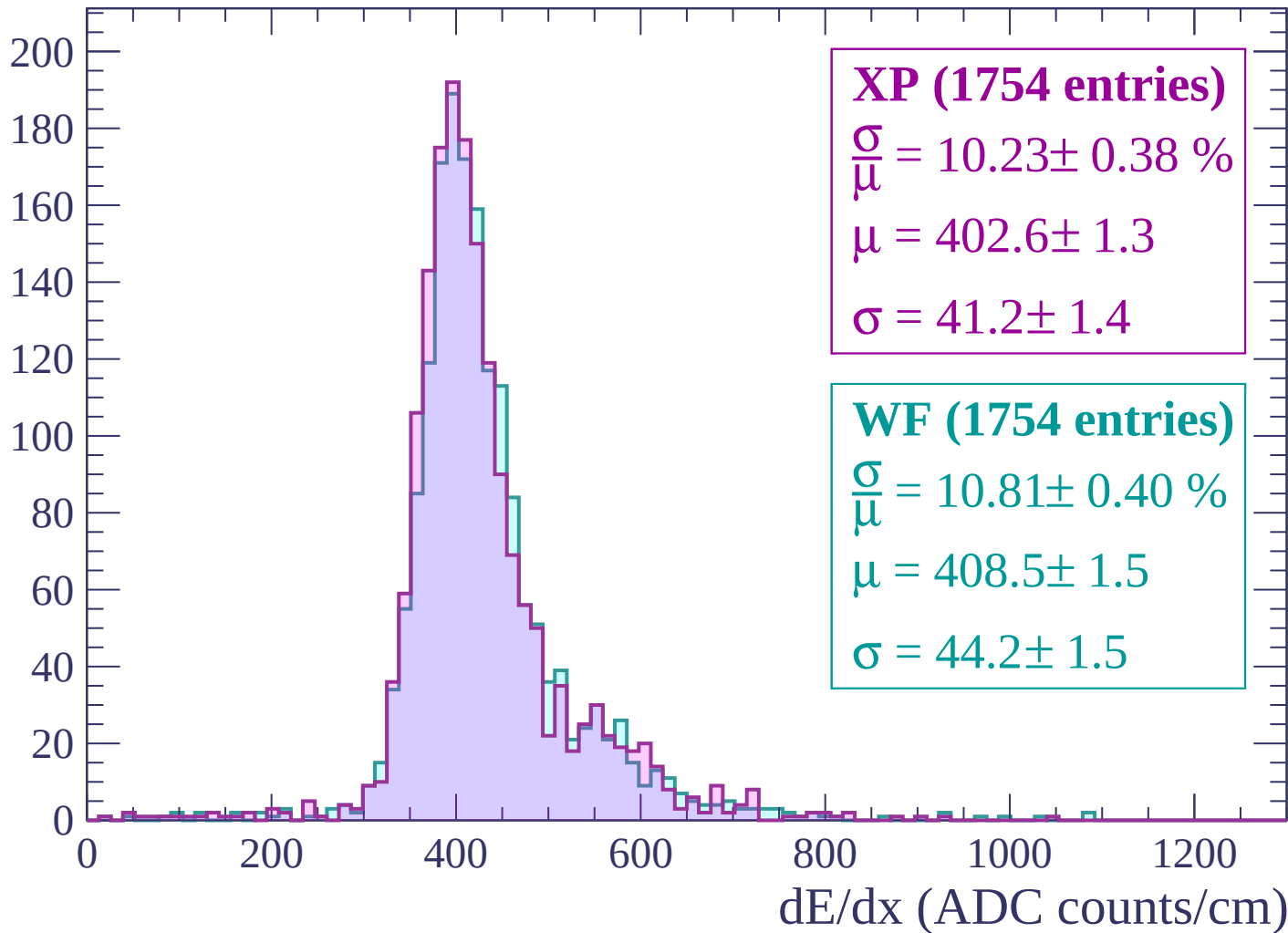
Energy loss | $360 < p < 420$

Count



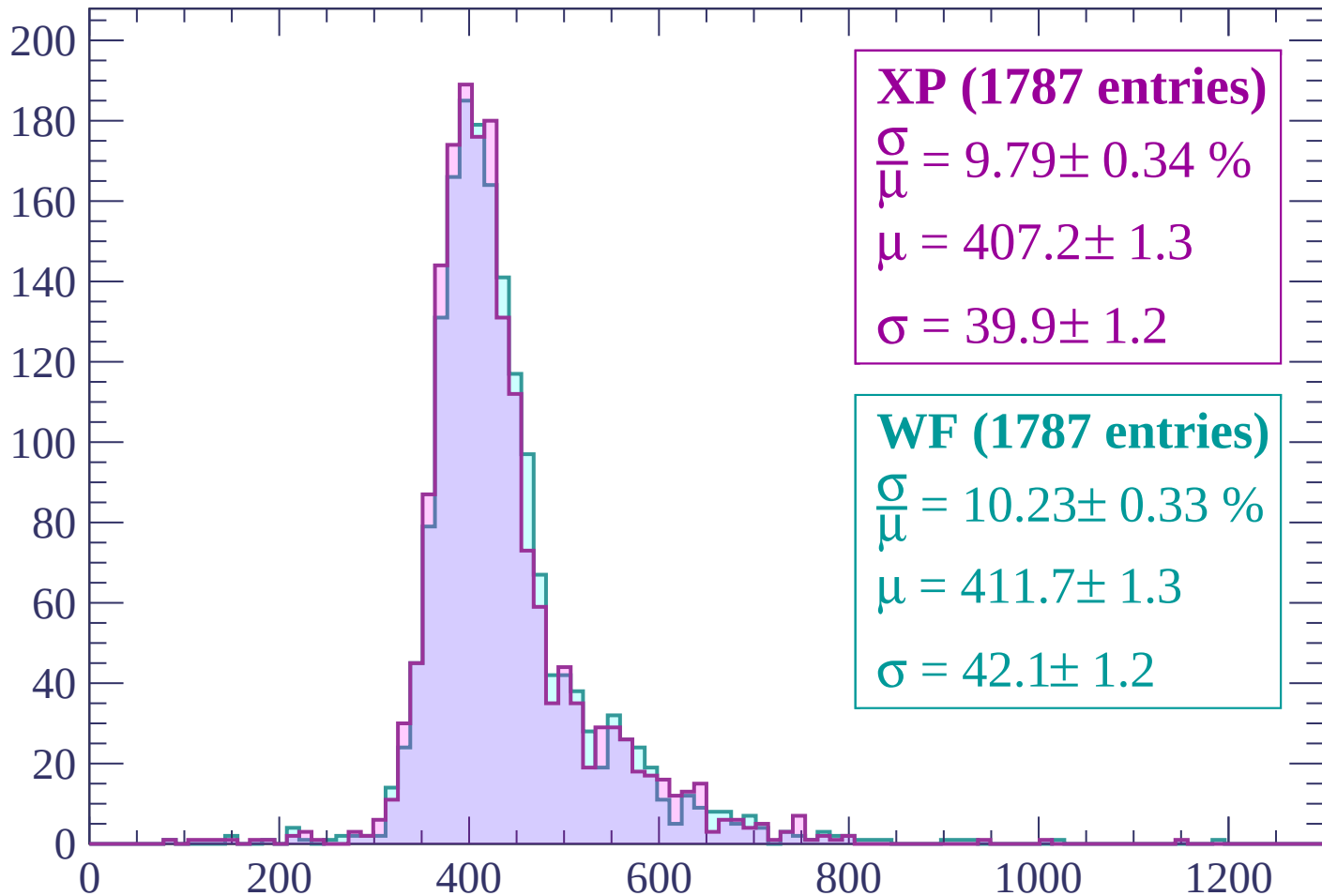
Energy loss | $420 < p < 480$

Count



Energy loss | $480 < p < 540$

Count



XP (1787 entries)

$$\frac{\sigma}{\mu} = 9.79 \pm 0.34 \%$$

$$\mu = 407.2 \pm 1.3$$

$$\sigma = 39.9 \pm 1.2$$

WF (1787 entries)

$$\frac{\sigma}{\mu} = 10.23 \pm 0.33 \%$$

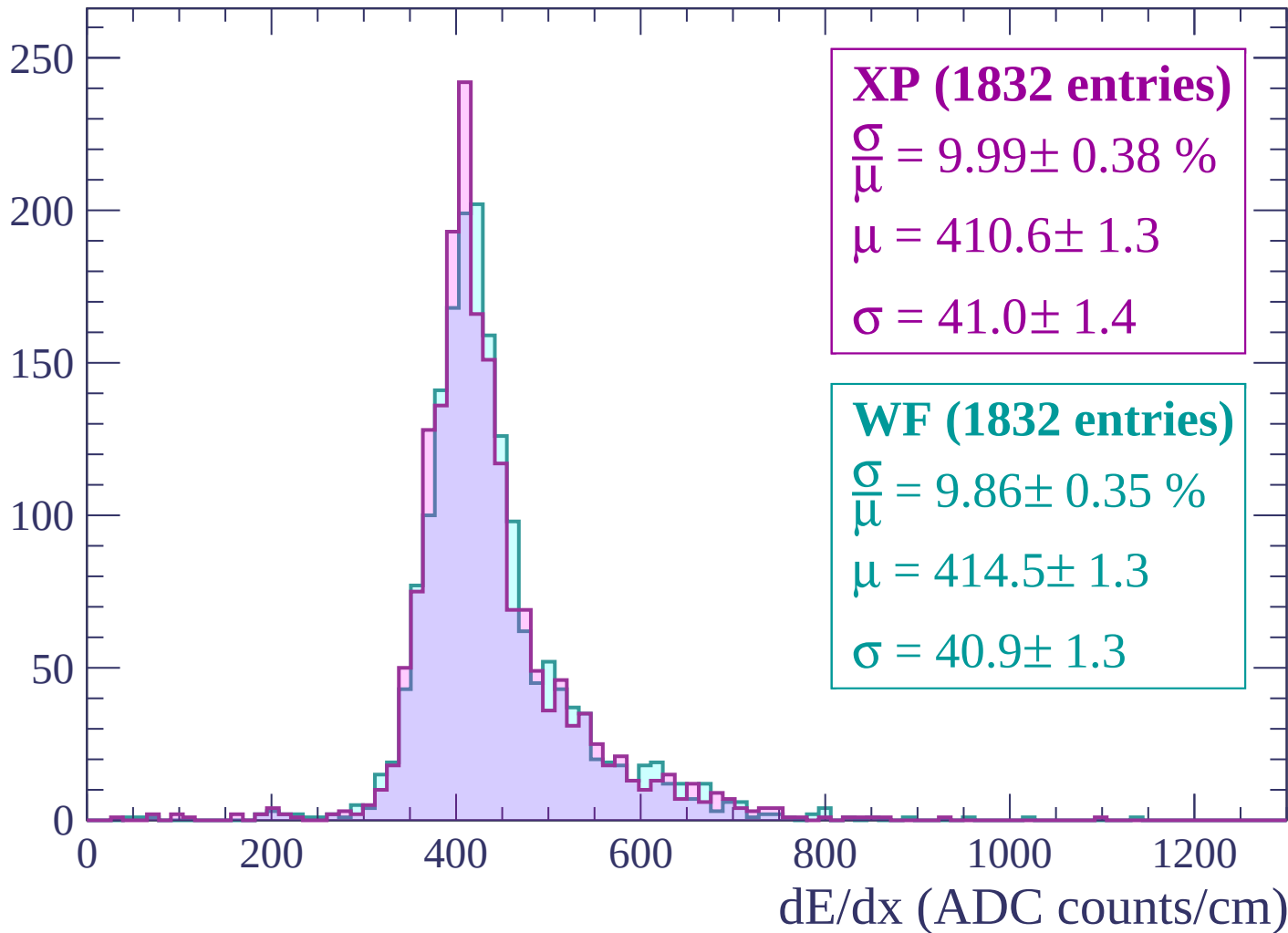
$$\mu = 411.7 \pm 1.3$$

$$\sigma = 42.1 \pm 1.2$$

dE/dx (ADC counts/cm)

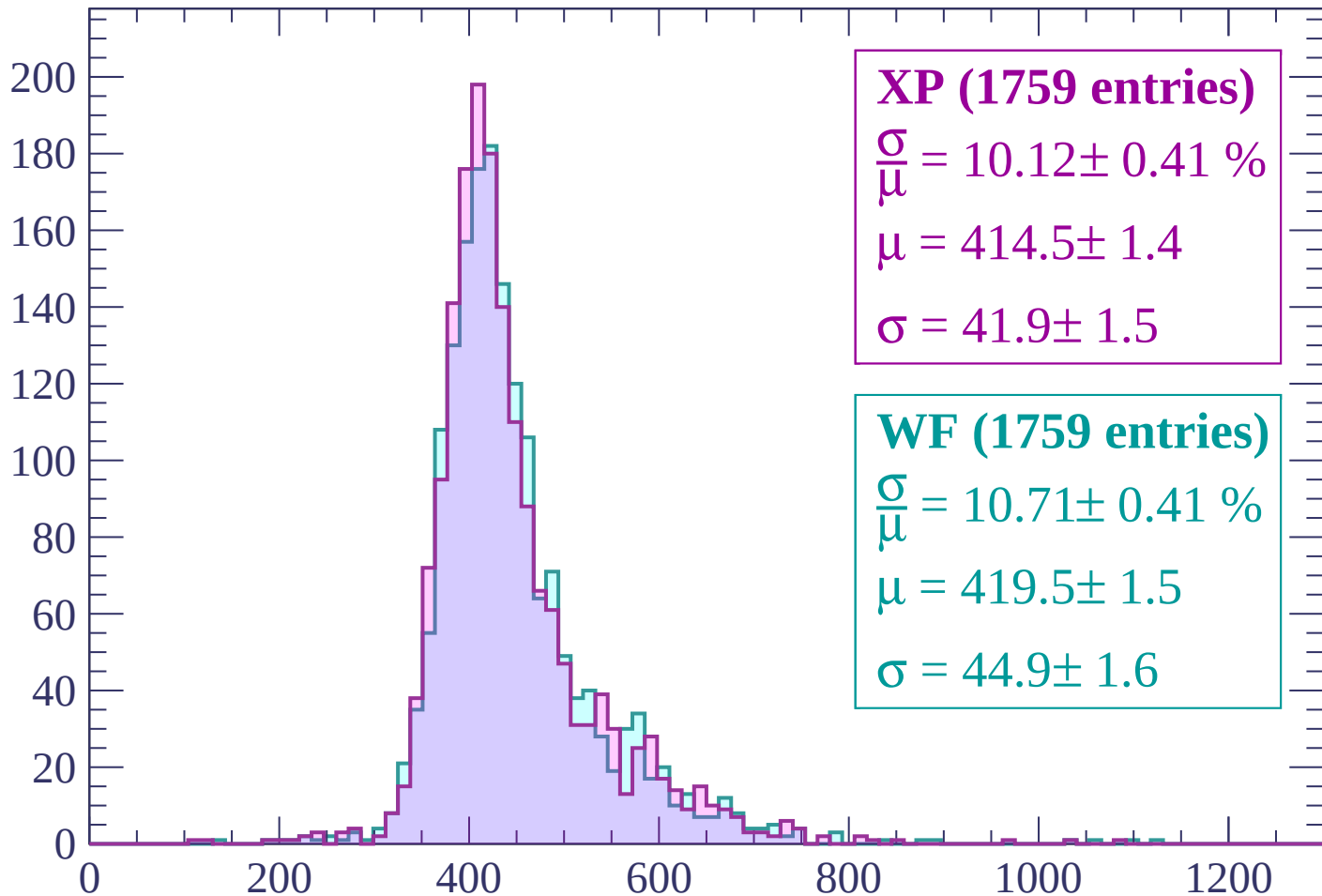
Energy loss | $540 < p < 600$

Count



Energy loss | $600 < p < 660$

Count



XP (1759 entries)

$\frac{\sigma}{\mu} = 10.12 \pm 0.41 \%$

$\mu = 414.5 \pm 1.4$

$\sigma = 41.9 \pm 1.5$

WF (1759 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.41 \%$

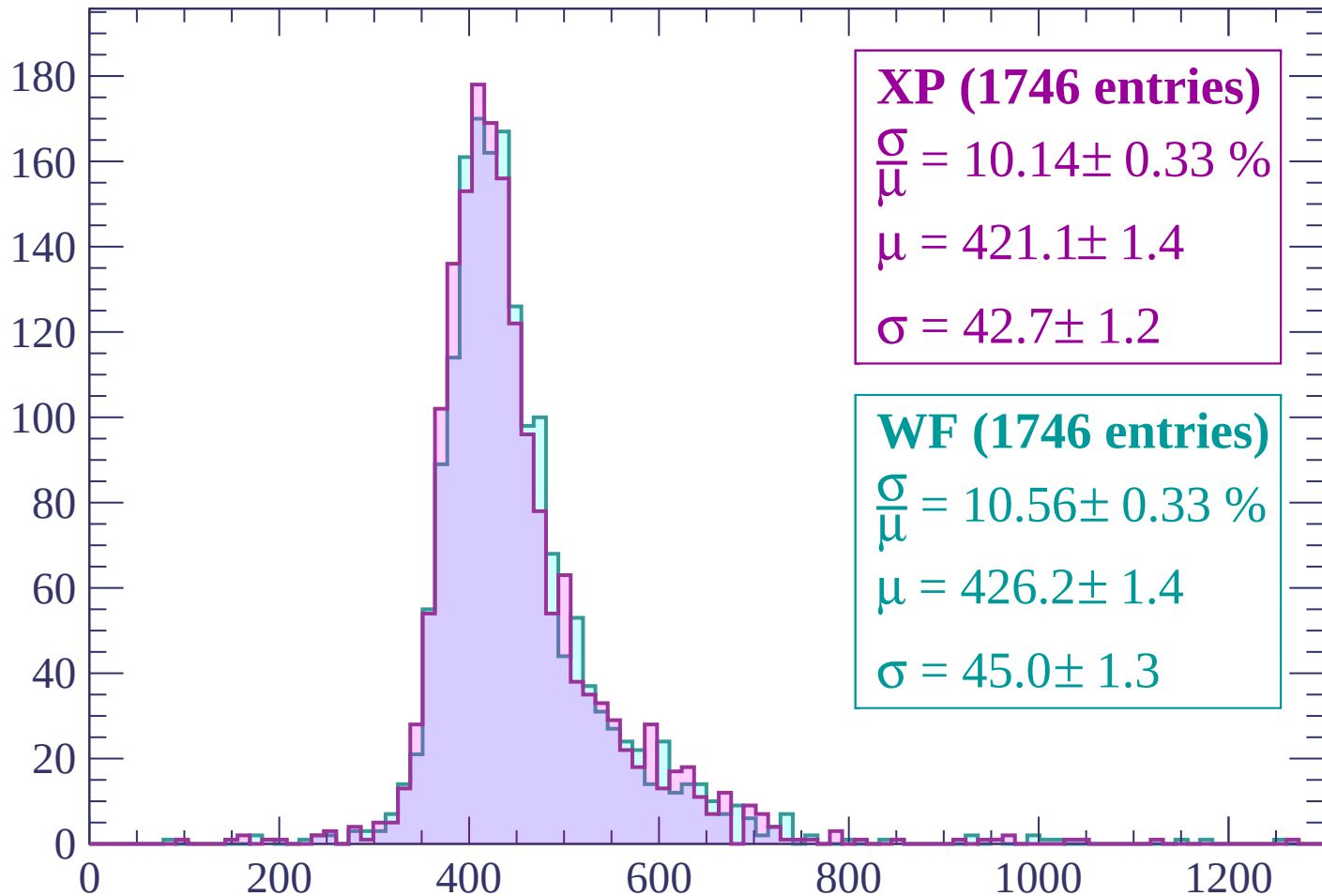
$\mu = 419.5 \pm 1.5$

$\sigma = 44.9 \pm 1.6$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (1746 entries)

$\frac{\sigma}{\mu} = 10.14 \pm 0.33 \%$

$\mu = 421.1 \pm 1.4$

$\sigma = 42.7 \pm 1.2$

WF (1746 entries)

$\frac{\sigma}{\mu} = 10.56 \pm 0.33 \%$

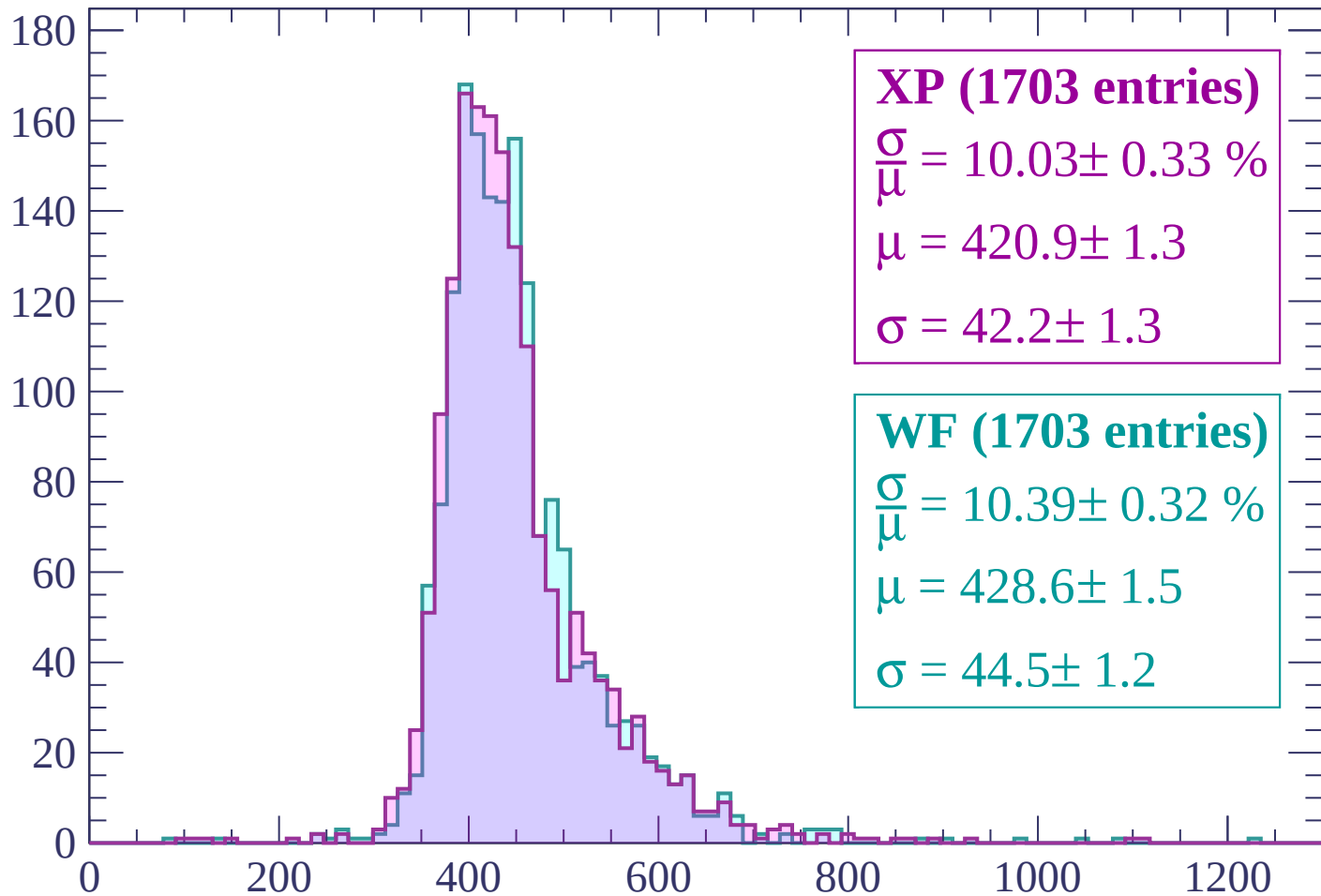
$\mu = 426.2 \pm 1.4$

$\sigma = 45.0 \pm 1.3$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (1703 entries)

$$\frac{\sigma}{\mu} = 10.03 \pm 0.33 \%$$

$$\mu = 420.9 \pm 1.3$$

$$\sigma = 42.2 \pm 1.3$$

WF (1703 entries)

$$\frac{\sigma}{\mu} = 10.39 \pm 0.32 \%$$

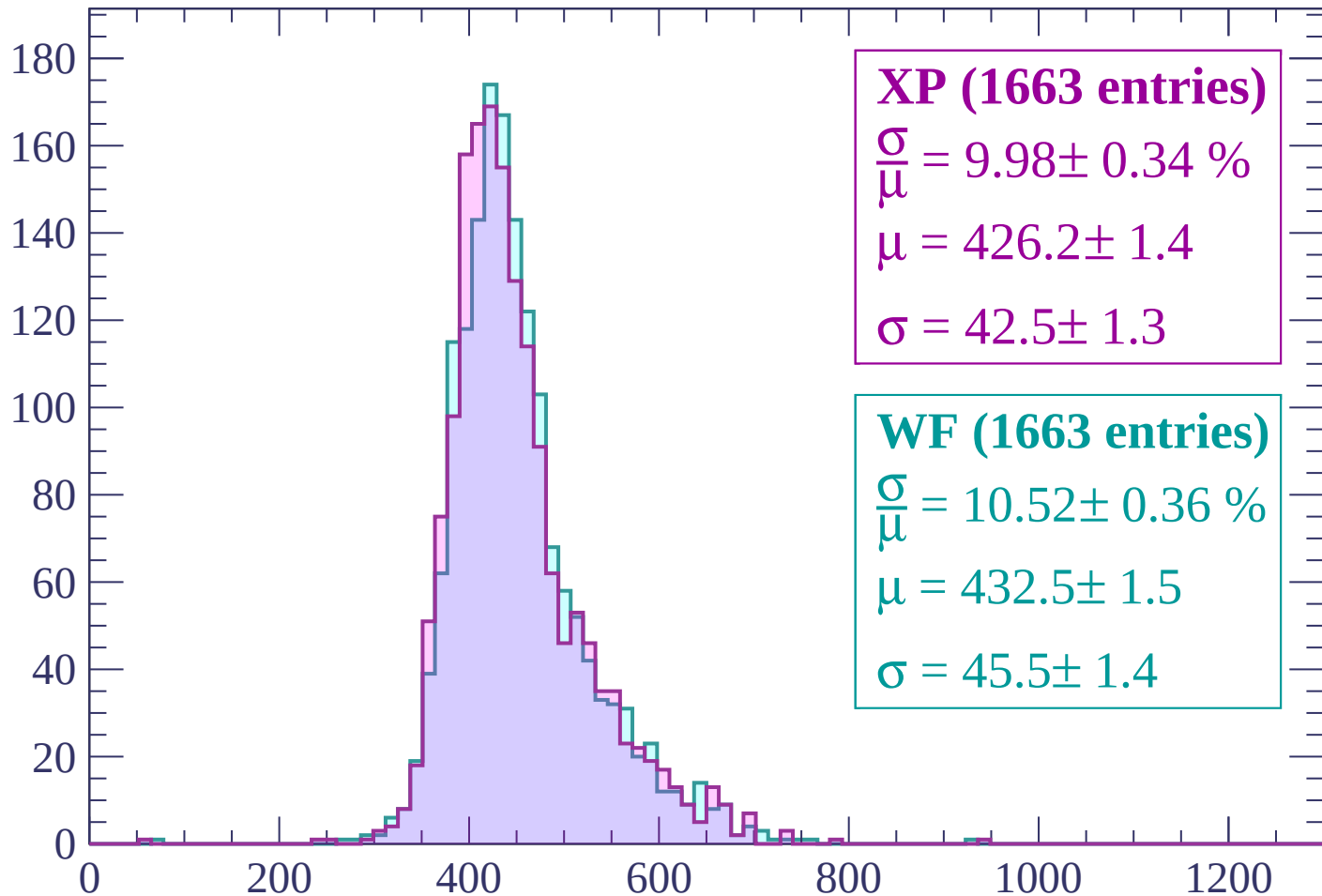
$$\mu = 428.6 \pm 1.5$$

$$\sigma = 44.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (1663 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.34 \%$$

$$\mu = 426.2 \pm 1.4$$

$$\sigma = 42.5 \pm 1.3$$

WF (1663 entries)

$$\frac{\sigma}{\mu} = 10.52 \pm 0.36 \%$$

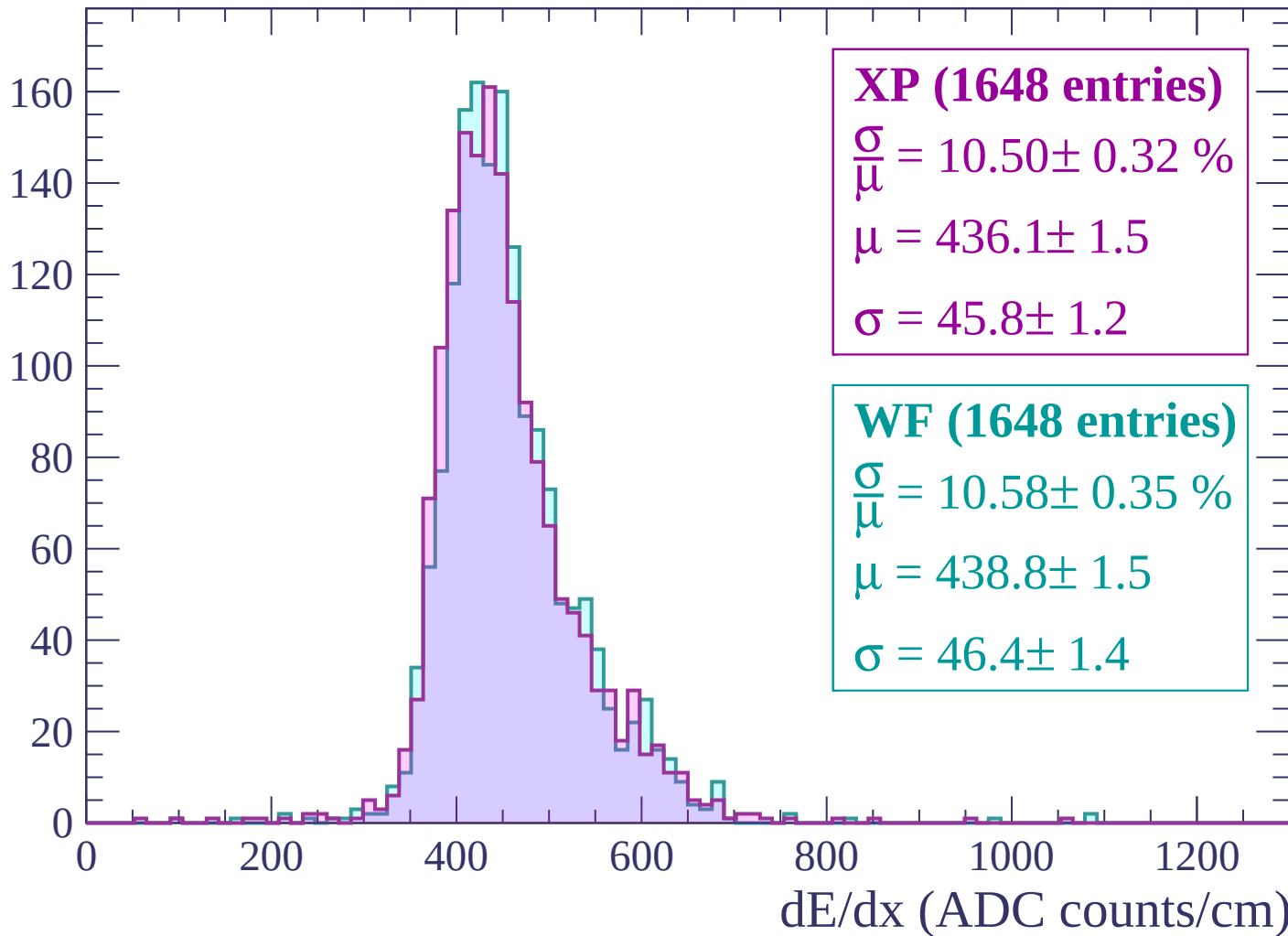
$$\mu = 432.5 \pm 1.5$$

$$\sigma = 45.5 \pm 1.4$$

dE/dx (ADC counts/cm)

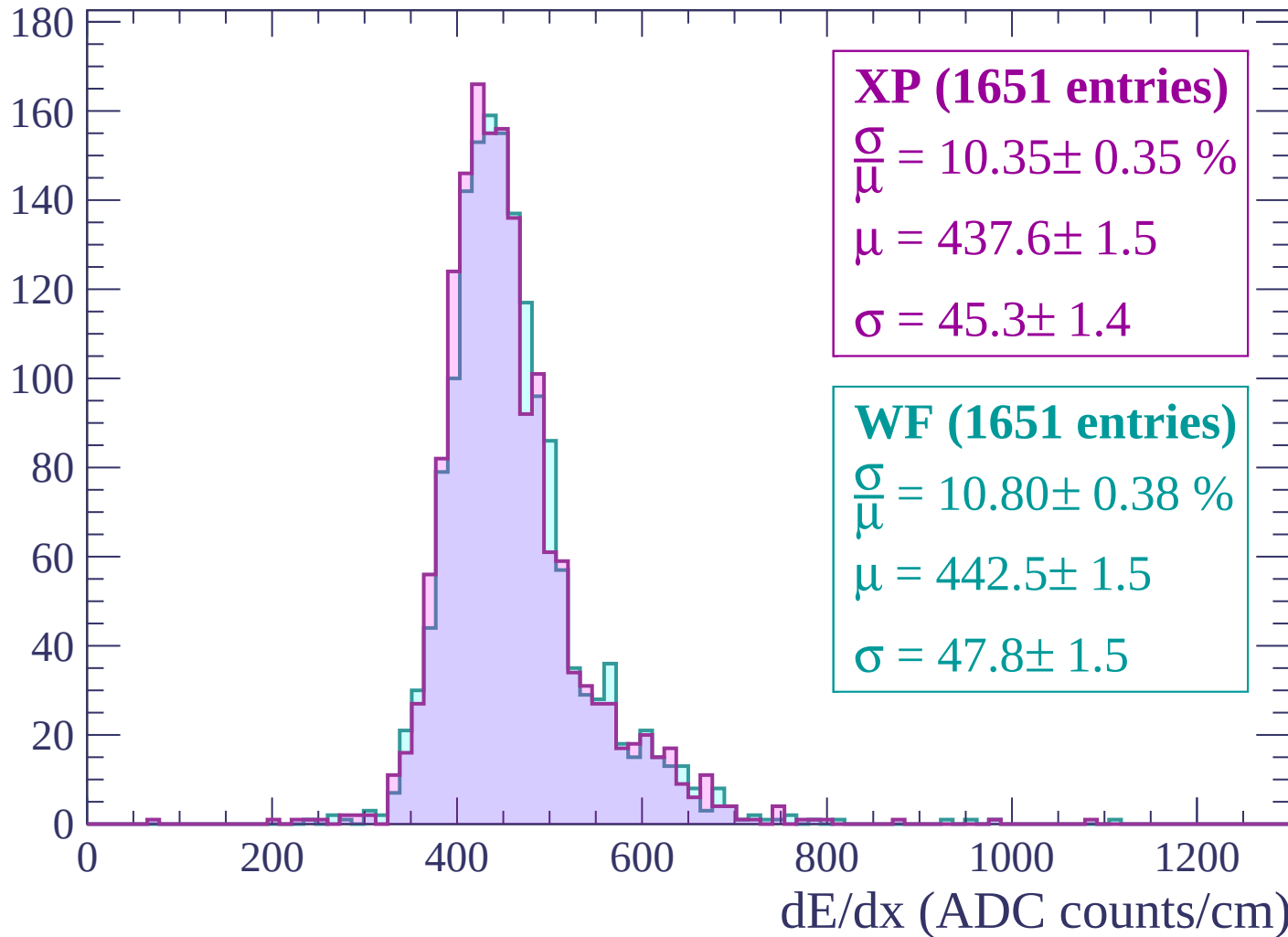
Energy loss | $840 < p < 900$

Count



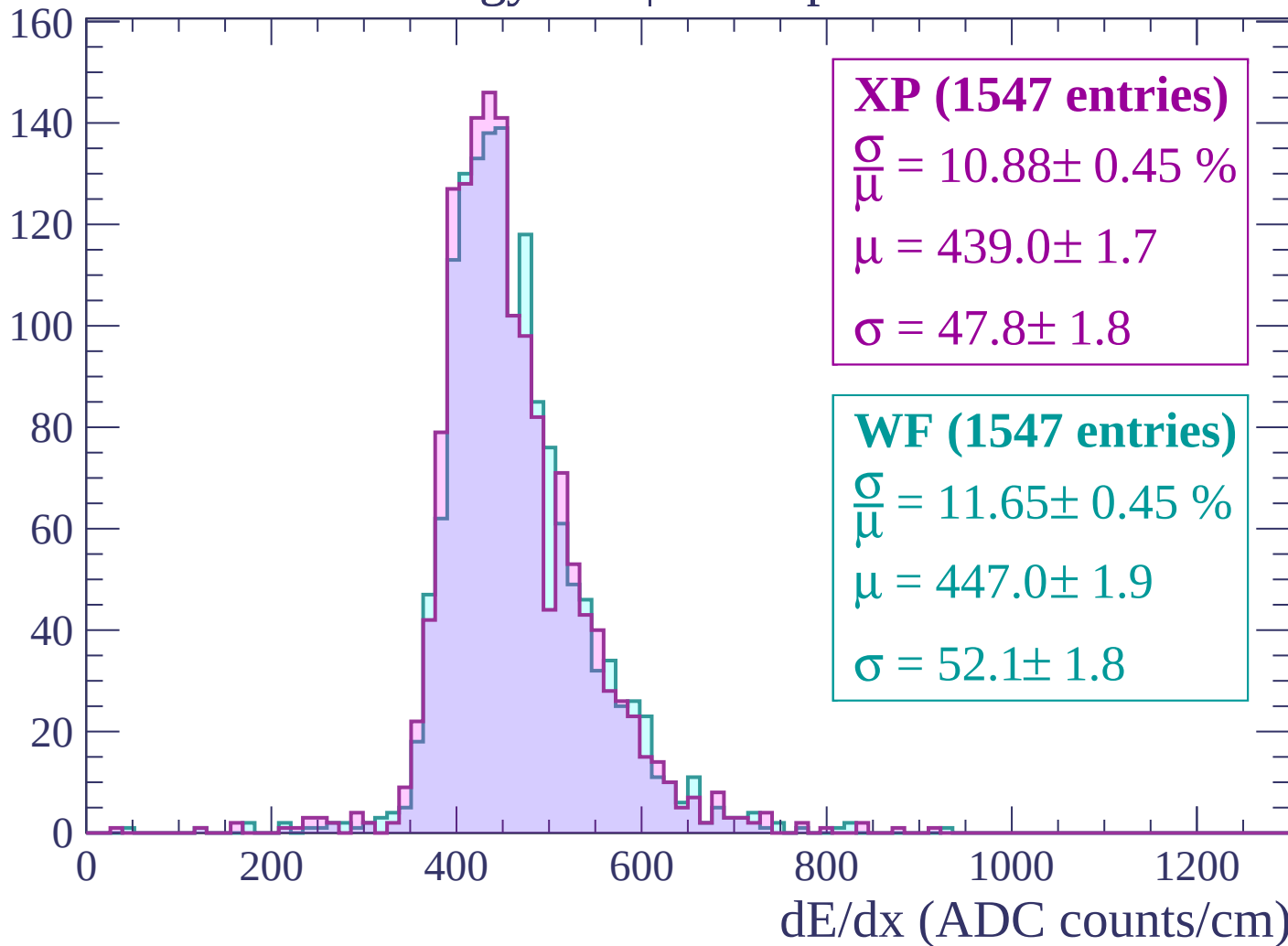
Energy loss | $900 < p < 960$

Count



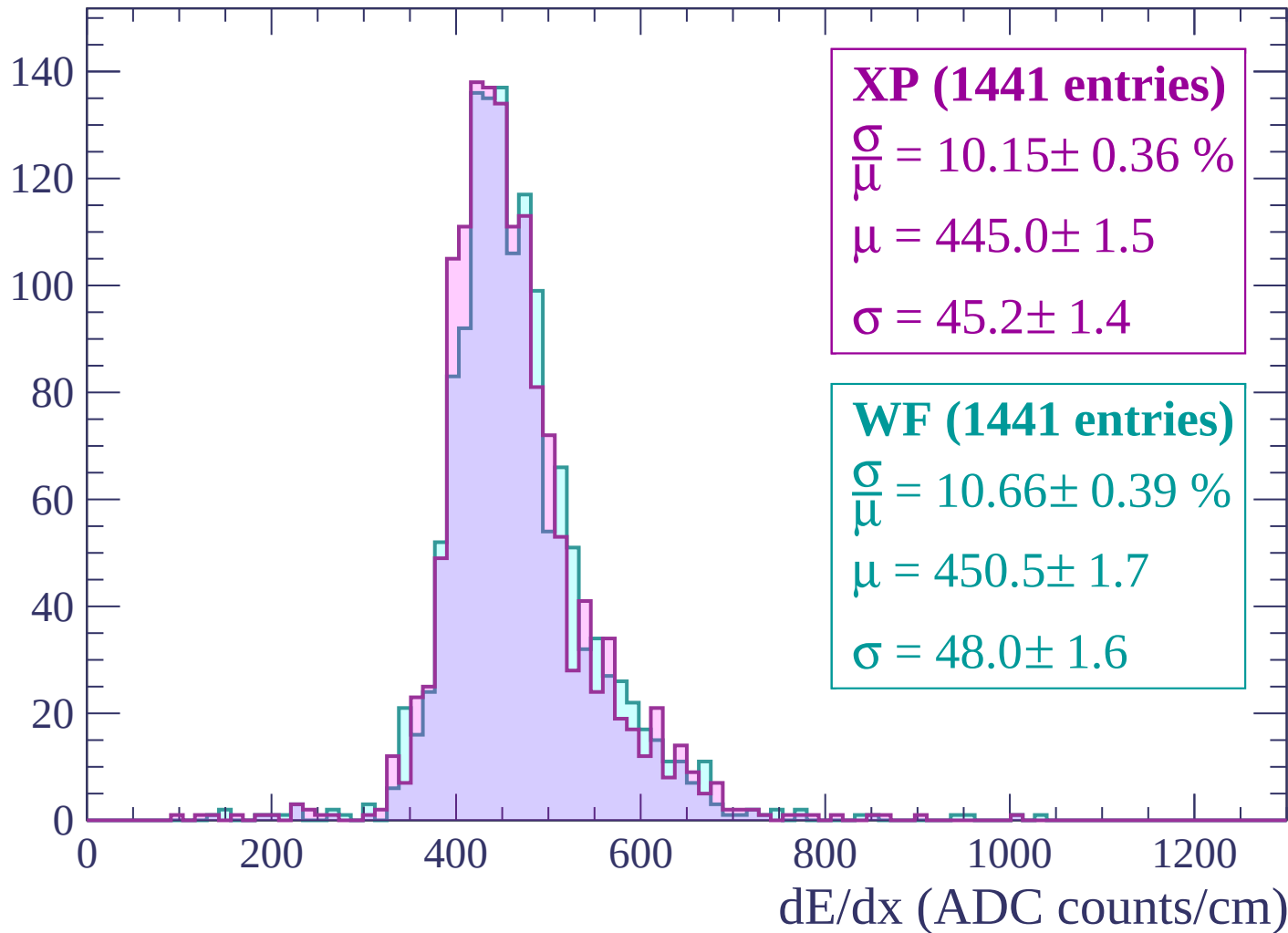
Energy loss | $960 < p < 1020$

Count



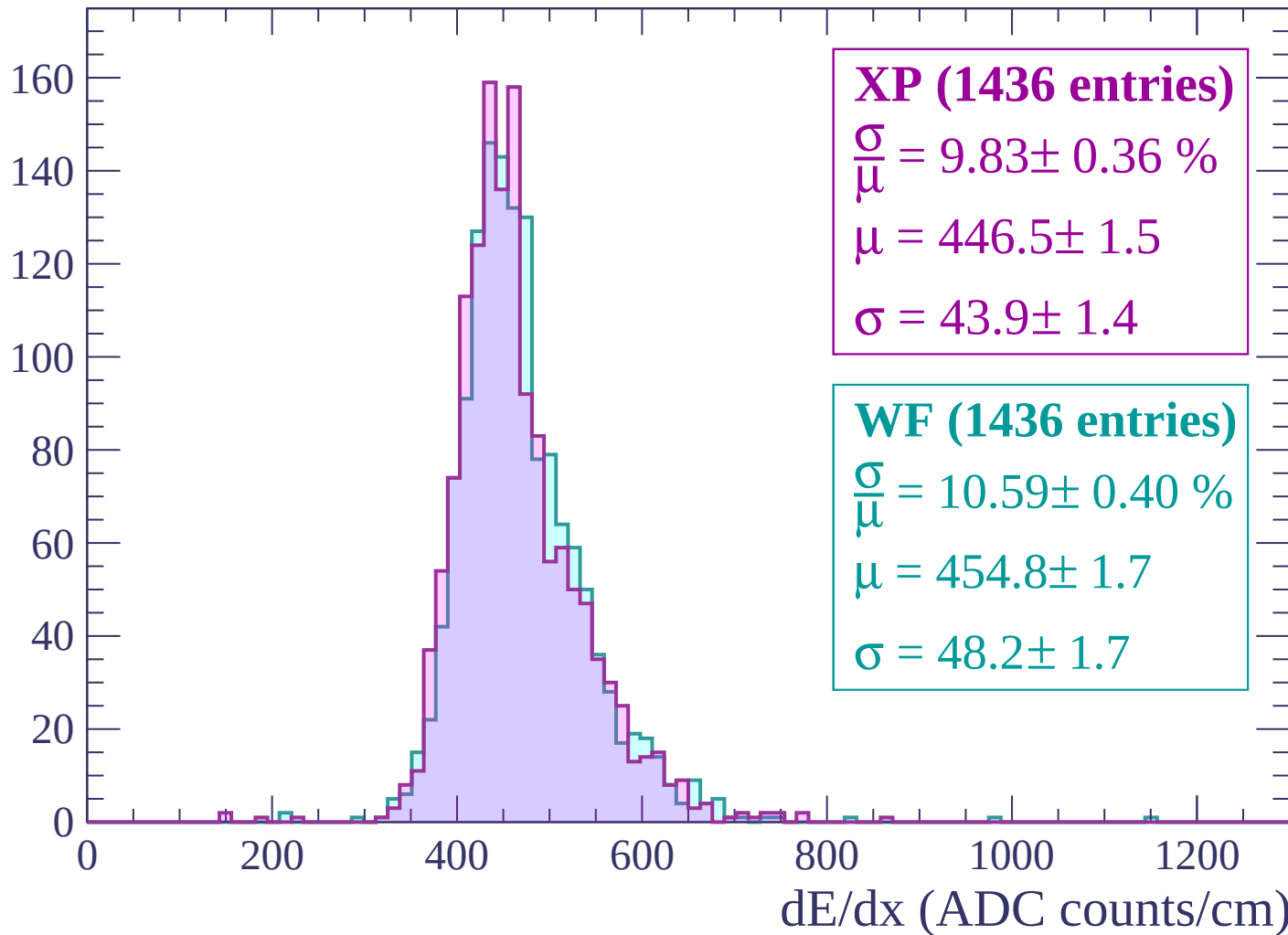
Energy loss | $1020 < p < 1080$

Count



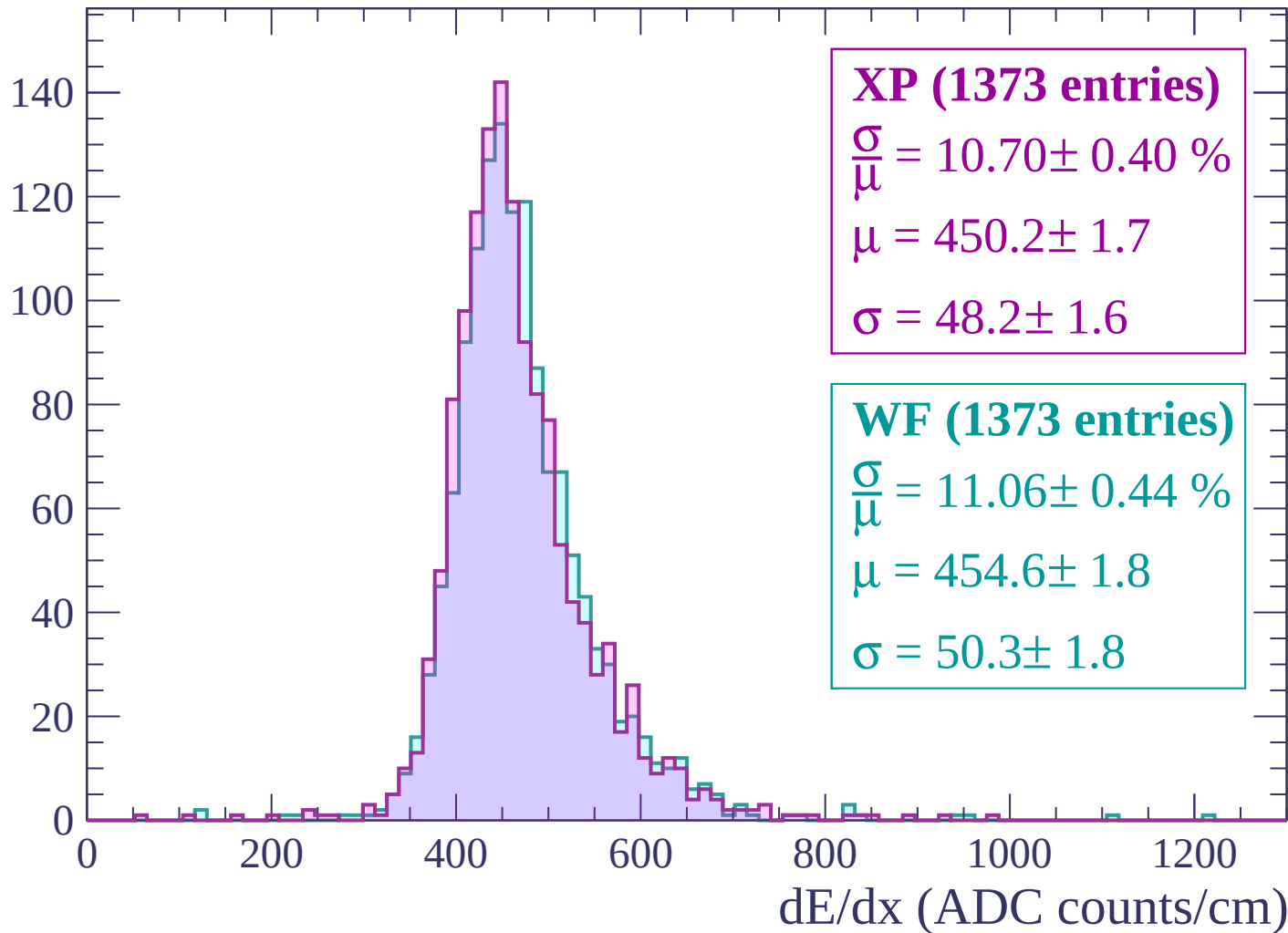
Energy loss | $1080 < p < 1140$

Count



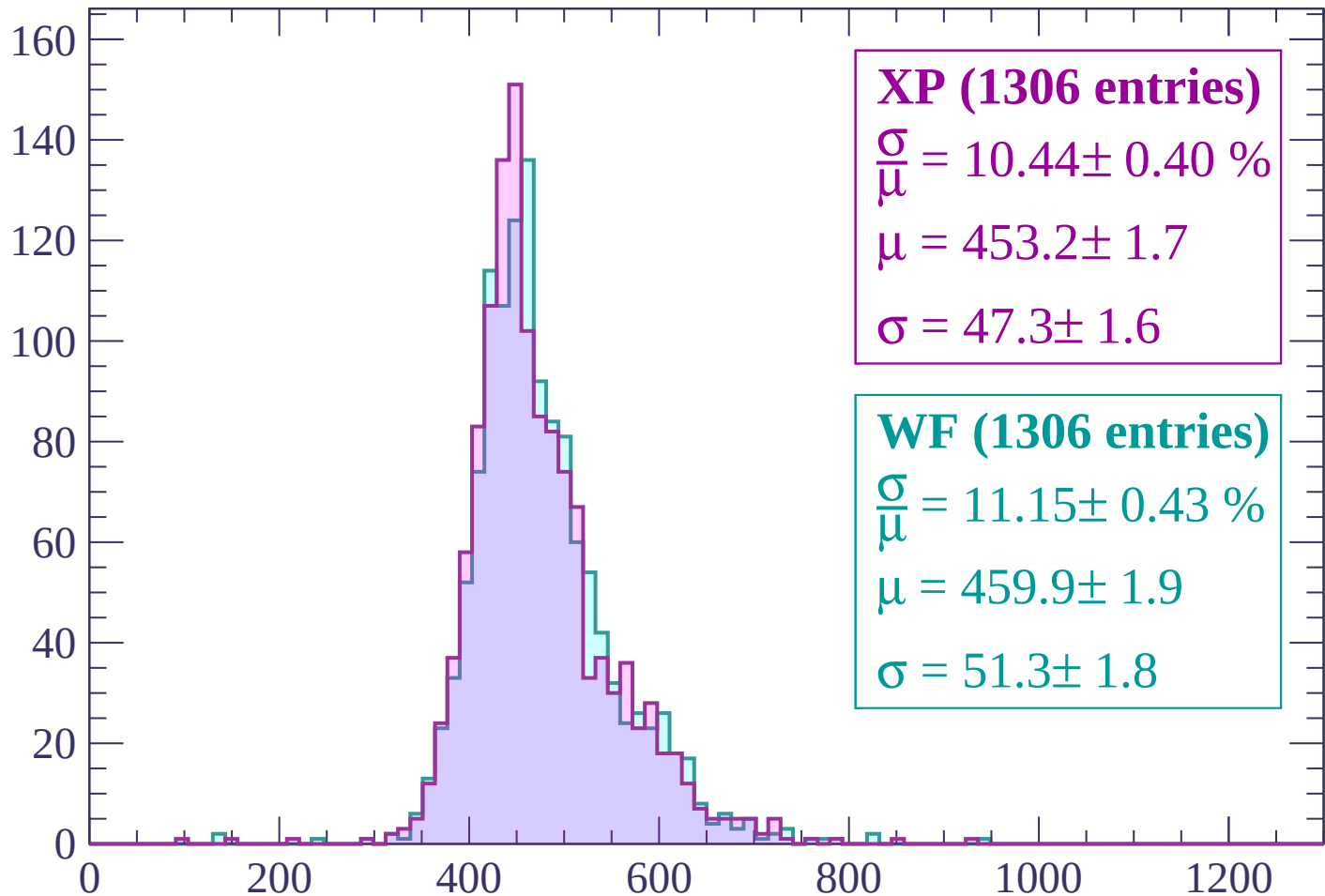
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP (1306 entries)

$$\frac{\sigma}{\bar{\mu}} = 10.44 \pm 0.40 \%$$

$$\mu = 453.2 \pm 1.7$$

$$\sigma = 47.3 \pm 1.6$$

WF (1306 entries)

$$\frac{\sigma}{\bar{\mu}} = 11.15 \pm 0.43 \%$$

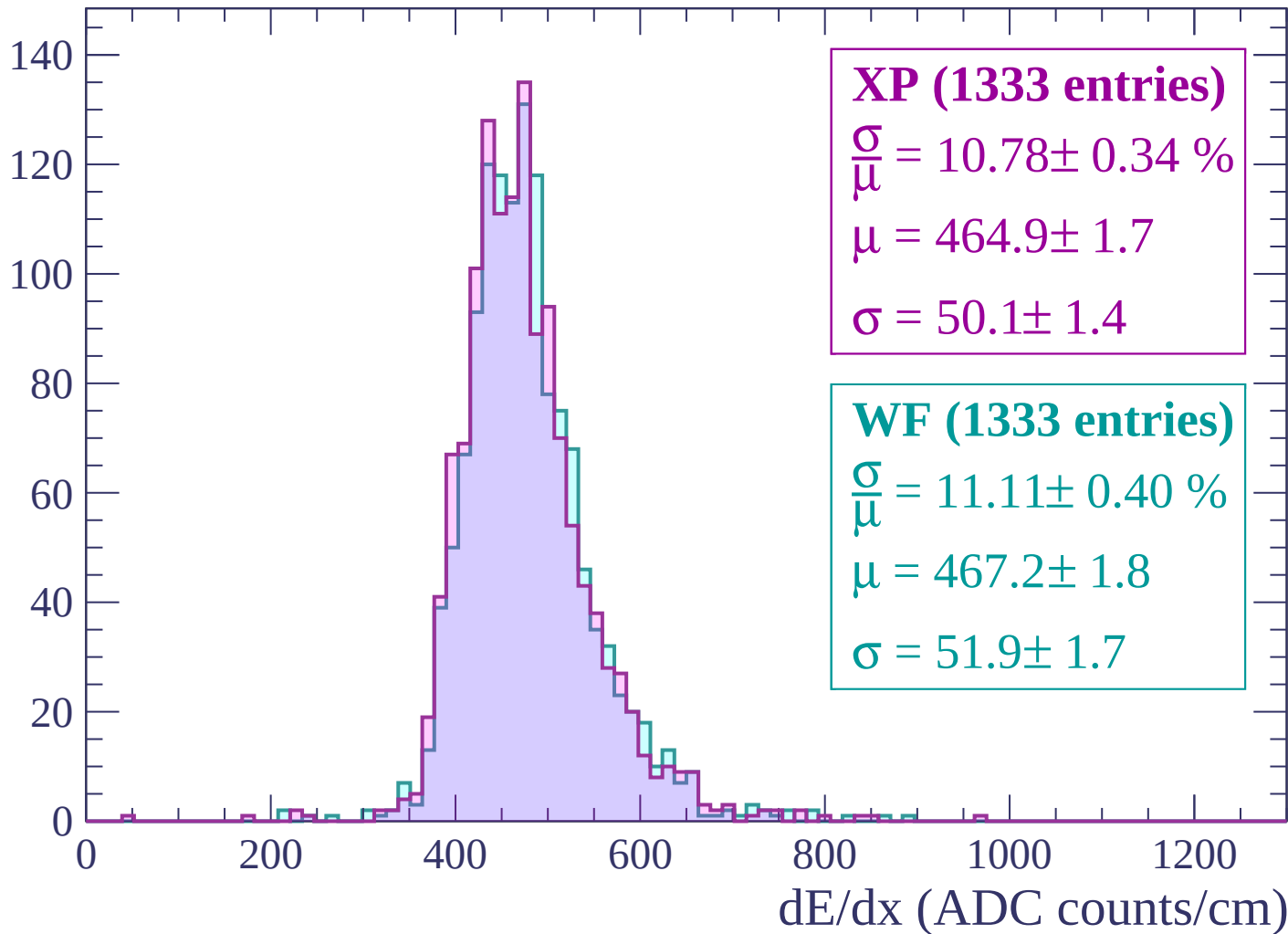
$$\mu = 459.9 \pm 1.9$$

$$\sigma = 51.3 \pm 1.8$$

dE/dx (ADC counts/cm)

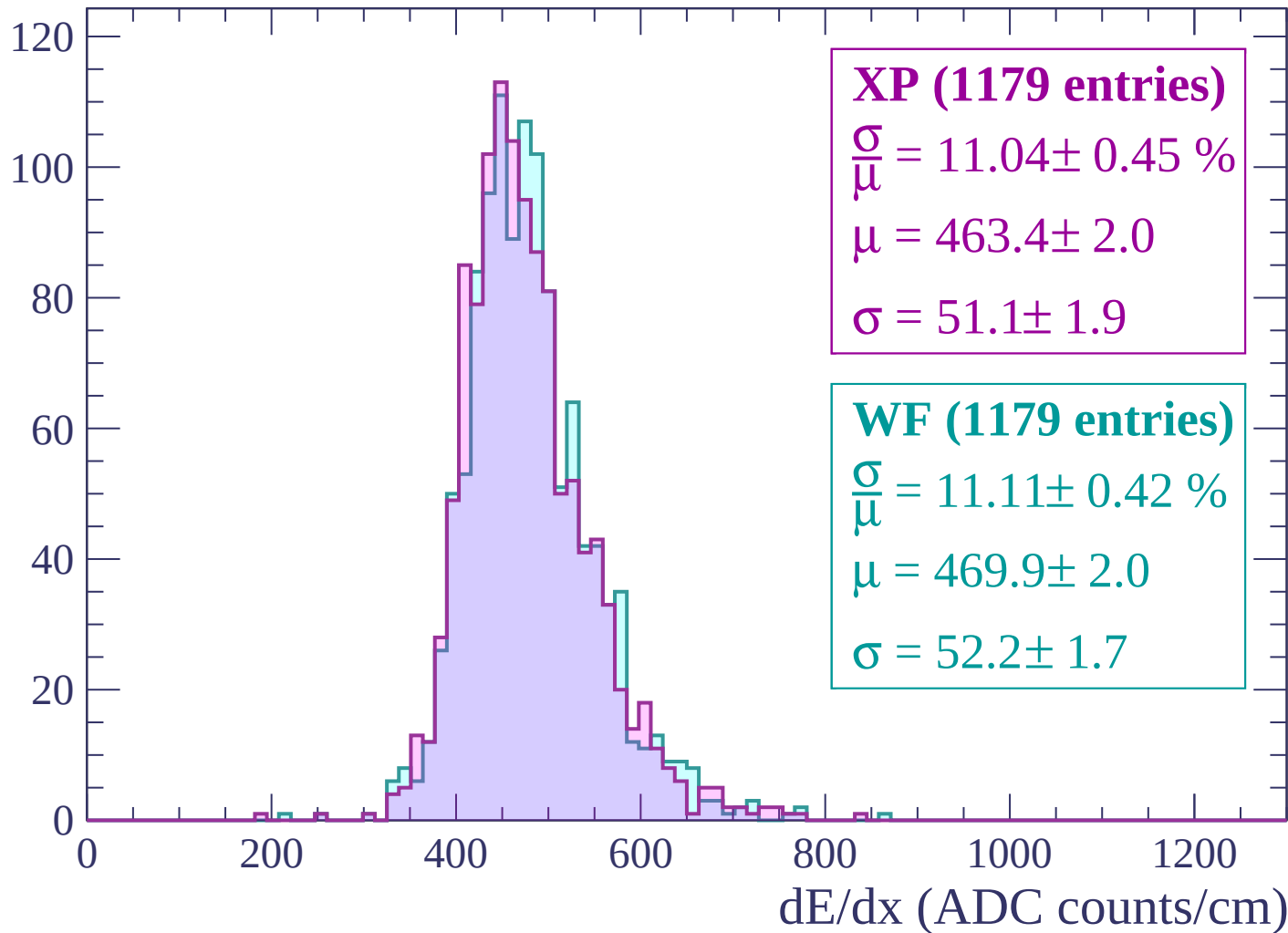
Energy loss | $1260 < p < 1320$

Count



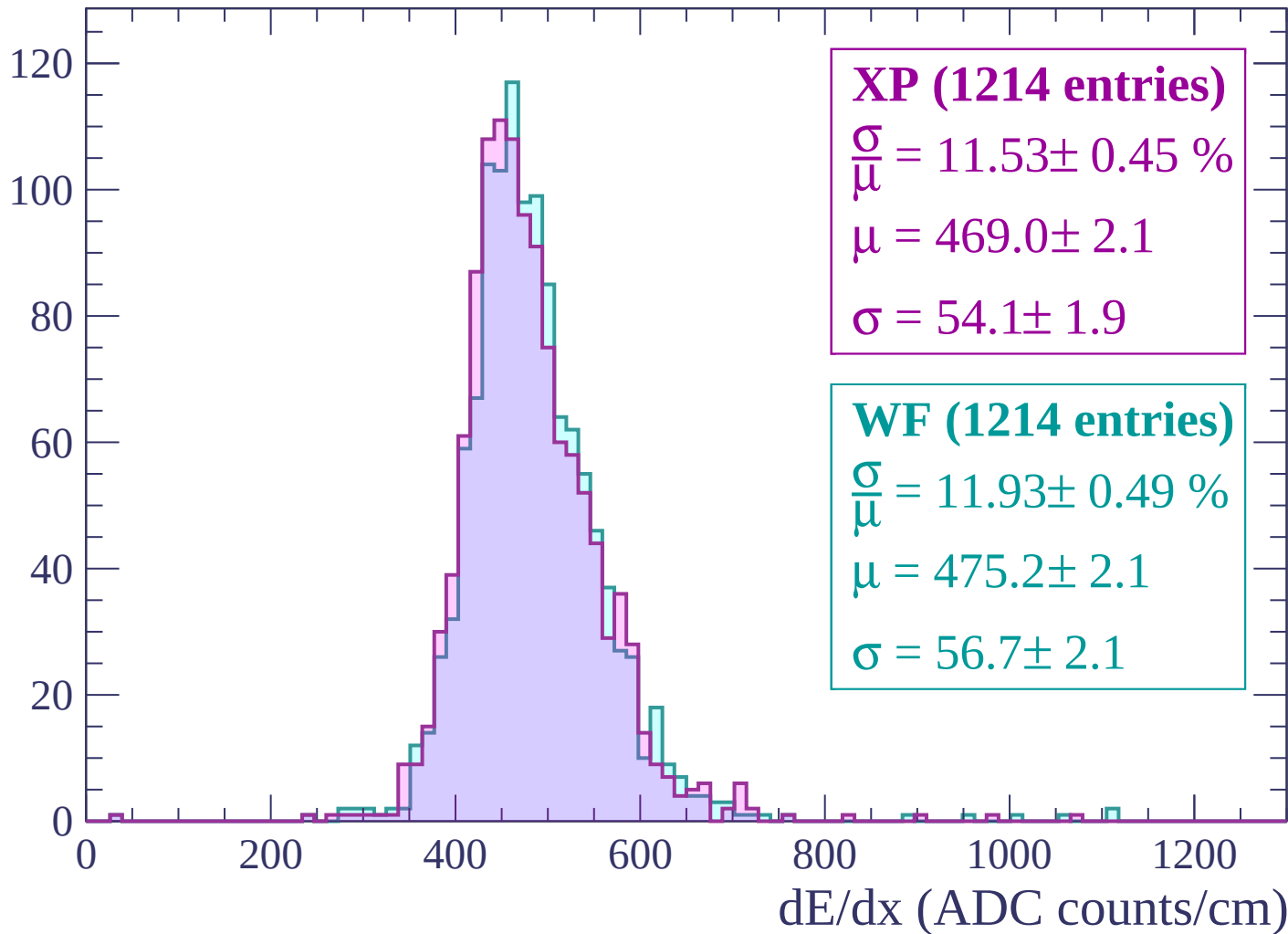
Energy loss | $1320 < p < 1380$

Count



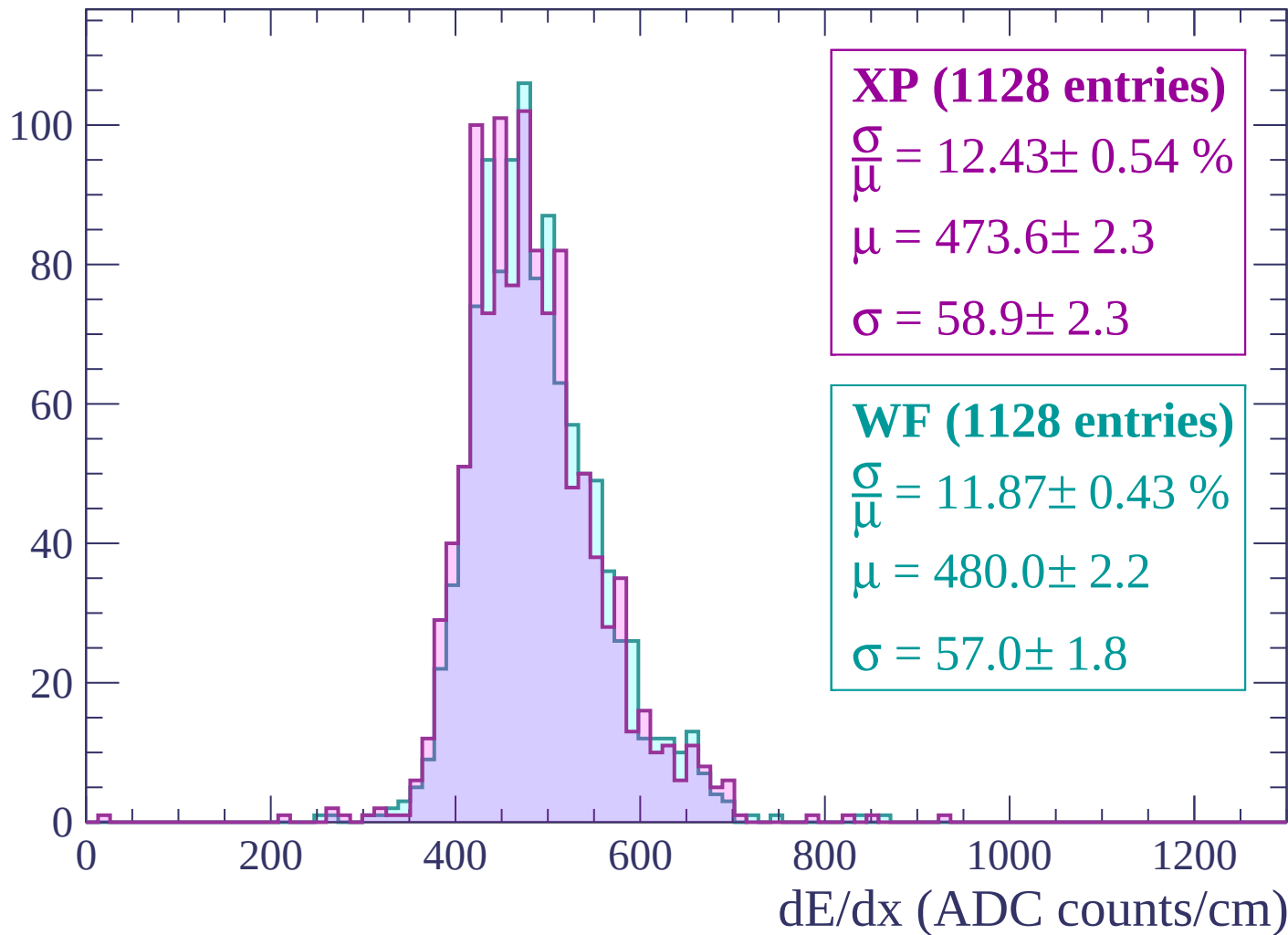
Energy loss | $1380 < p < 1440$

Count



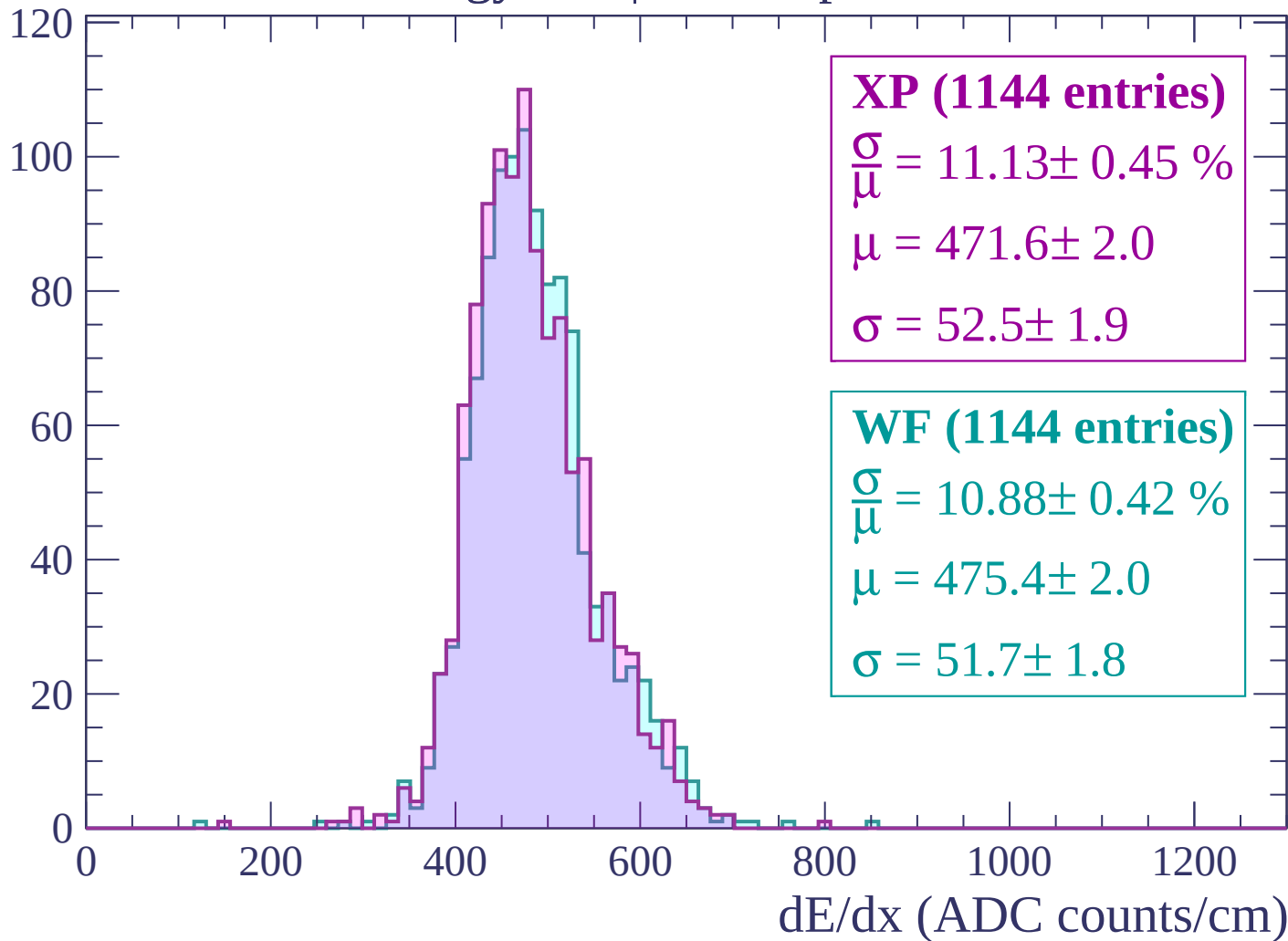
Energy loss | $1440 < p < 1500$

Count



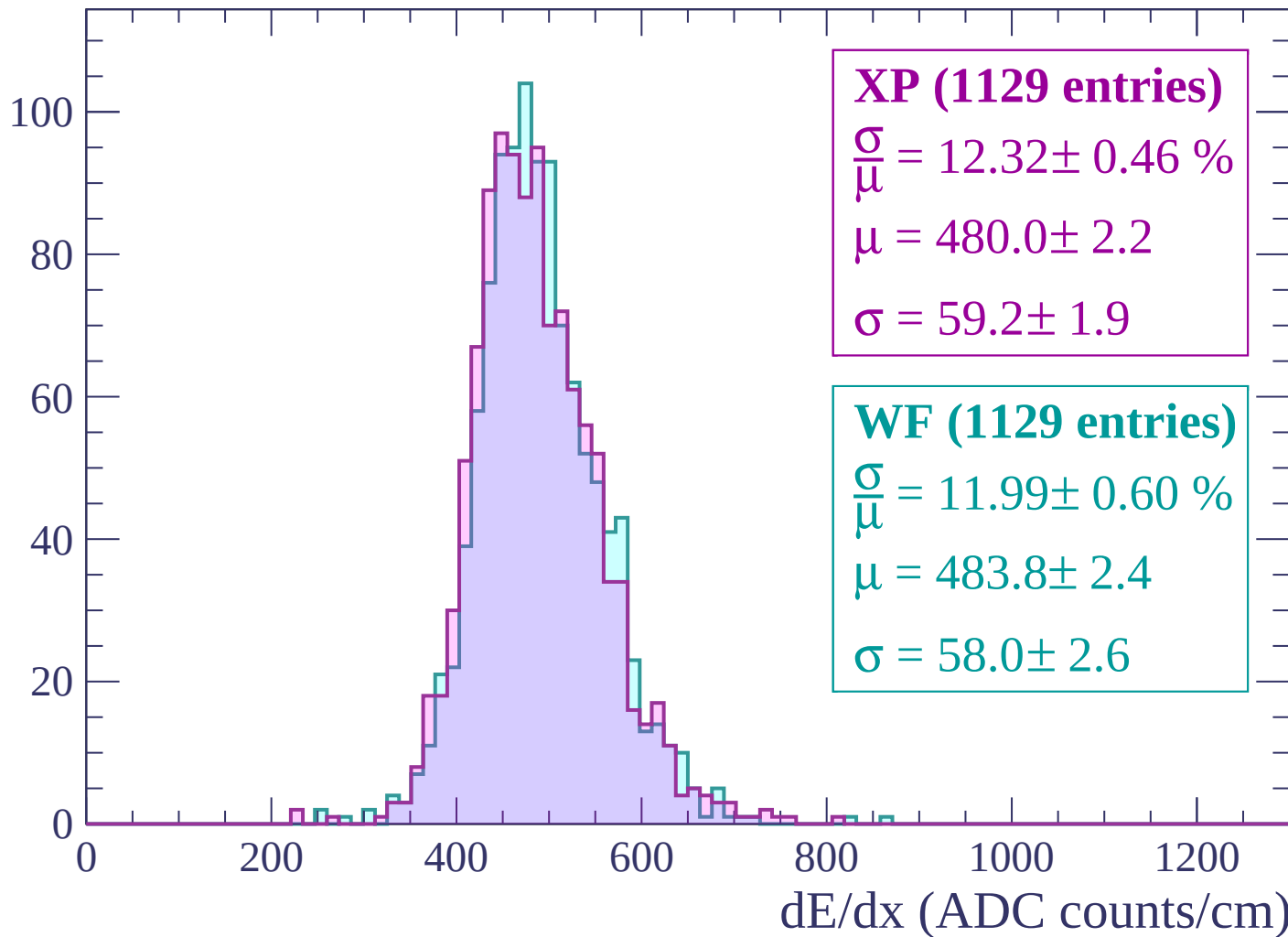
Energy loss | $1500 < p < 1560$

Count



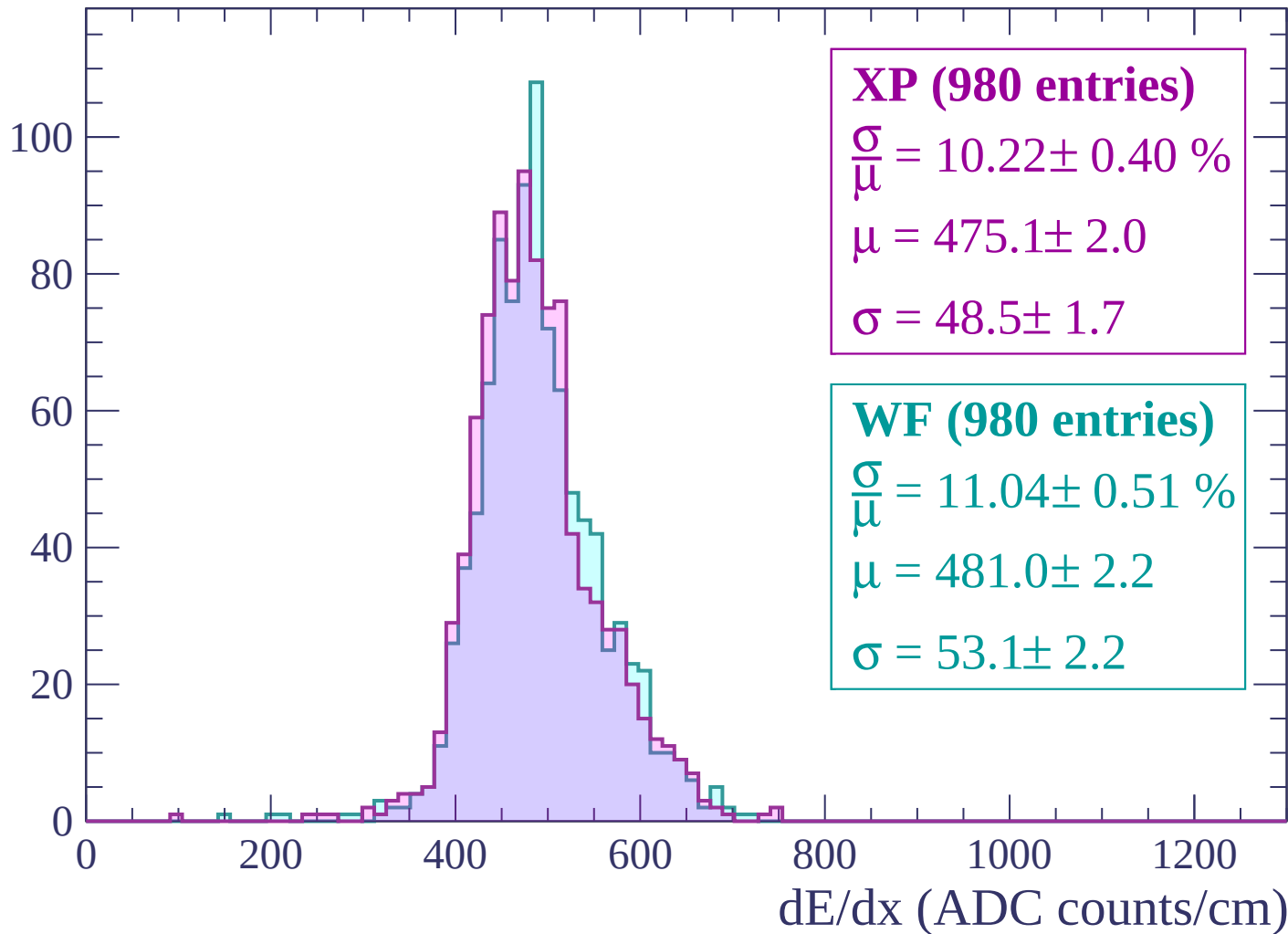
Energy loss | $1560 < p < 1620$

Count



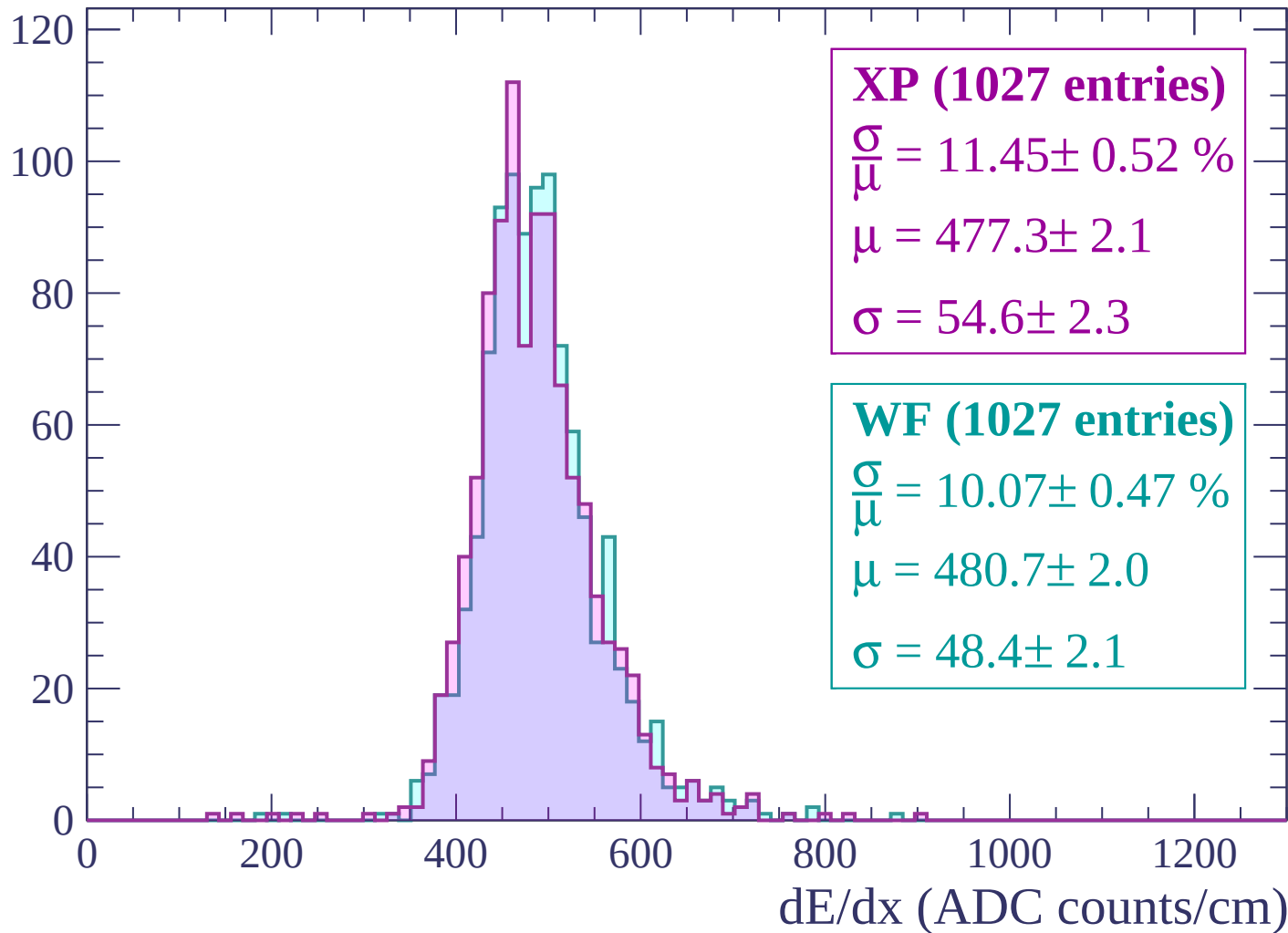
Energy loss | $1620 < p < 1680$

Count



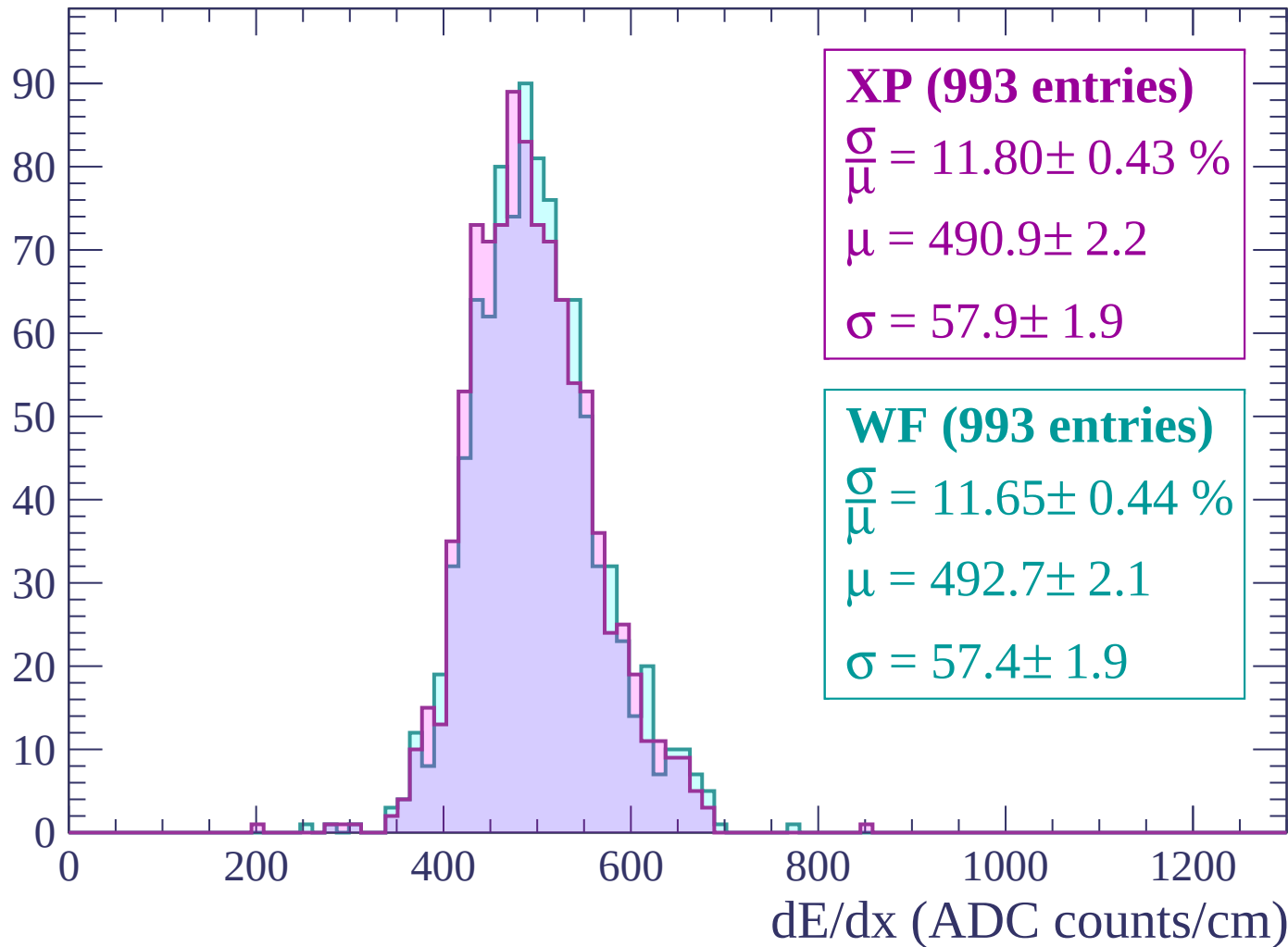
Energy loss | $1680 < p < 1740$

Count



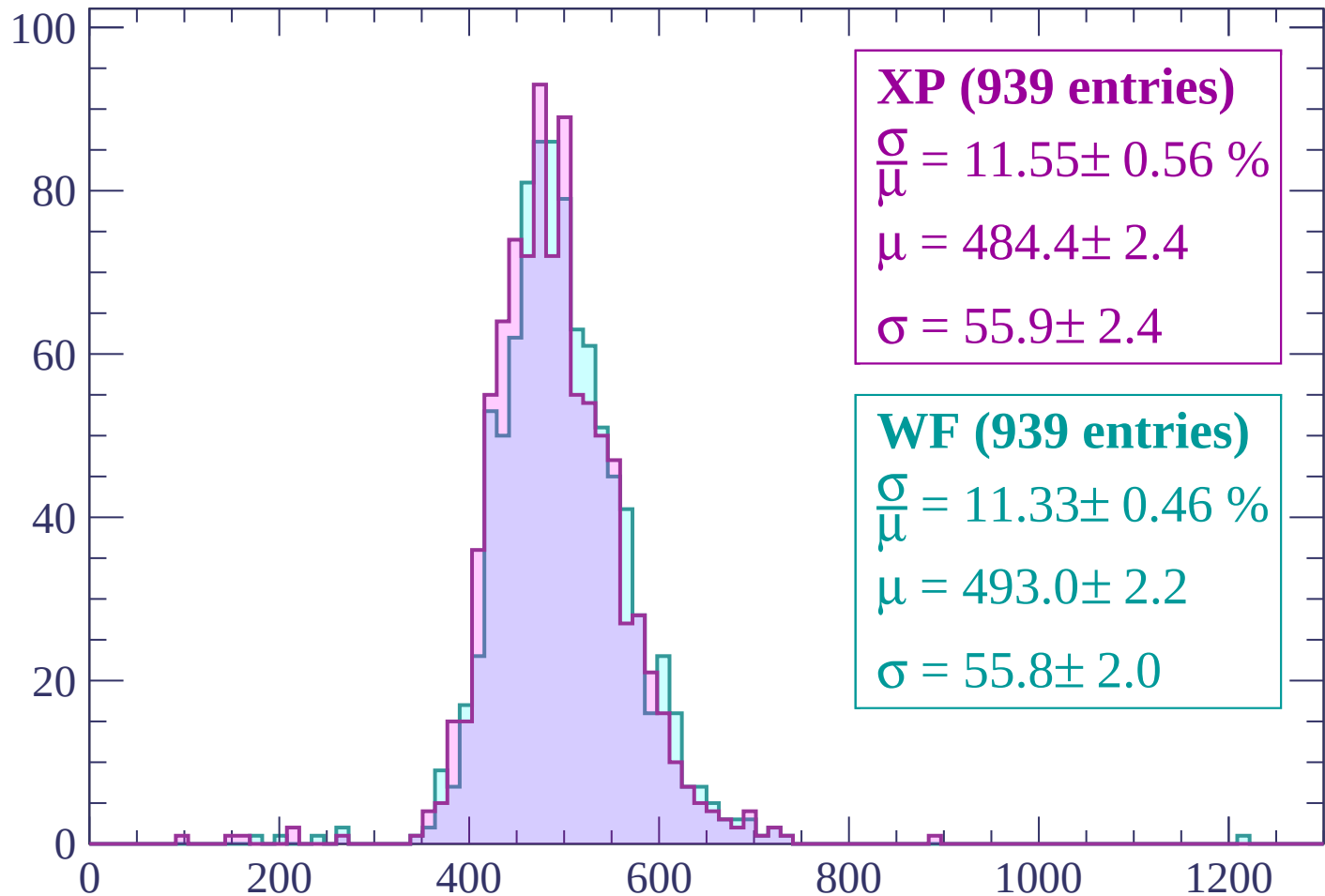
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP (939 entries)

$\frac{\sigma}{\mu} = 11.55 \pm 0.56 \%$

$\mu = 484.4 \pm 2.4$

$\sigma = 55.9 \pm 2.4$

WF (939 entries)

$\frac{\sigma}{\mu} = 11.33 \pm 0.46 \%$

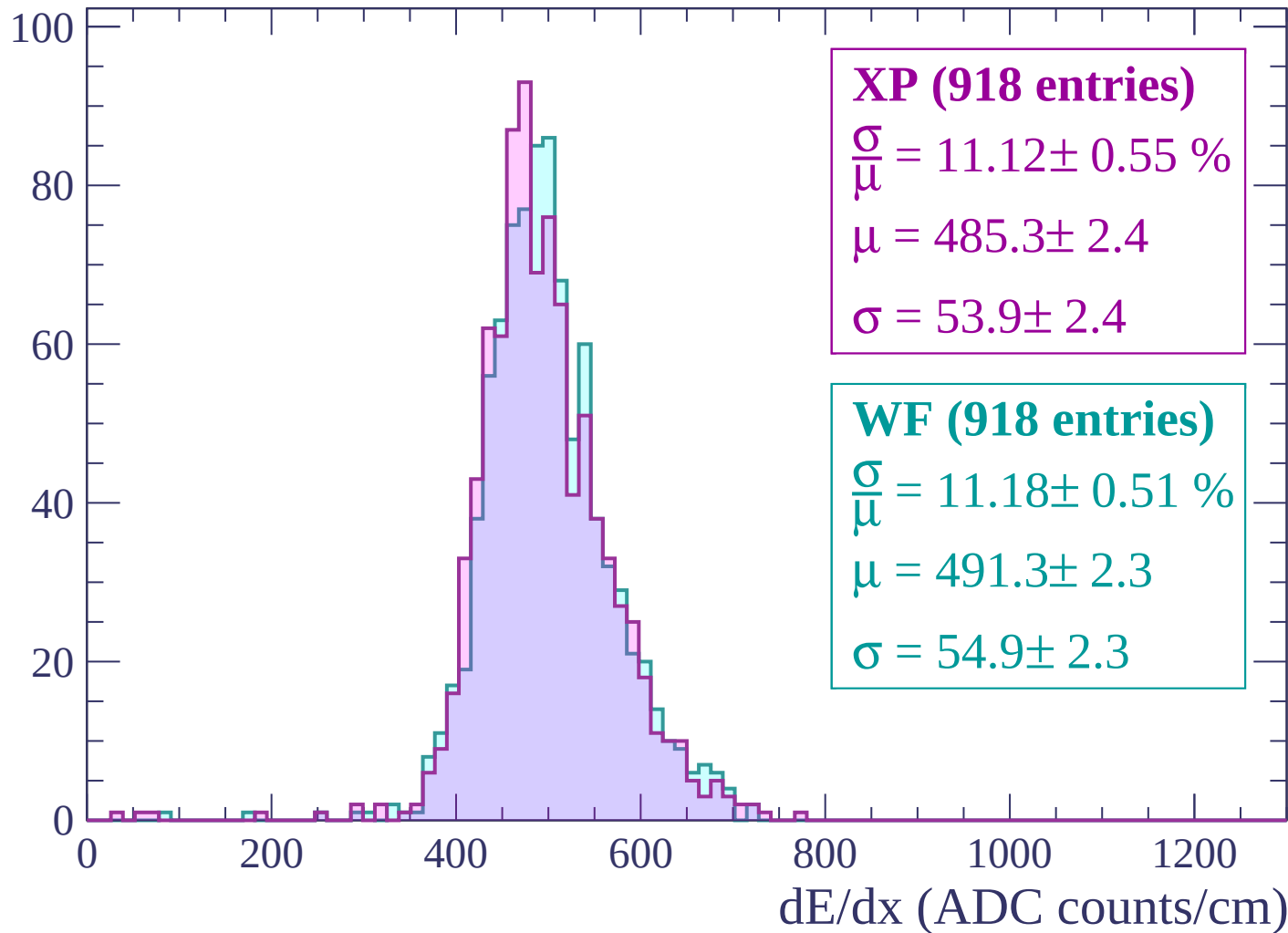
$\mu = 493.0 \pm 2.2$

$\sigma = 55.8 \pm 2.0$

dE/dx (ADC counts/cm)

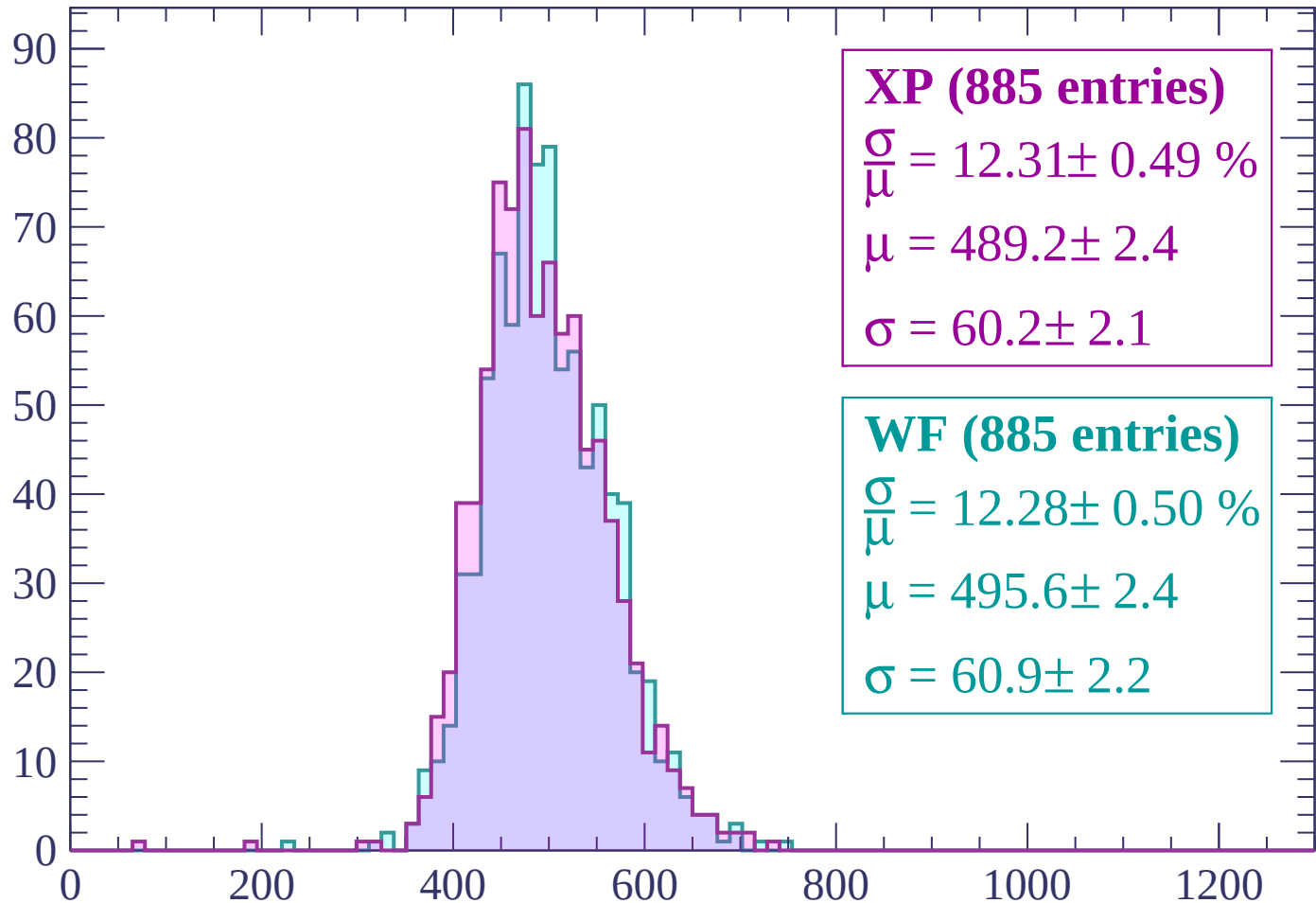
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP (885 entries)

$$\frac{\sigma}{\mu} = 12.31 \pm 0.49 \%$$

$$\mu = 489.2 \pm 2.4$$

$$\sigma = 60.2 \pm 2.1$$

WF (885 entries)

$$\frac{\sigma}{\mu} = 12.28 \pm 0.50 \%$$

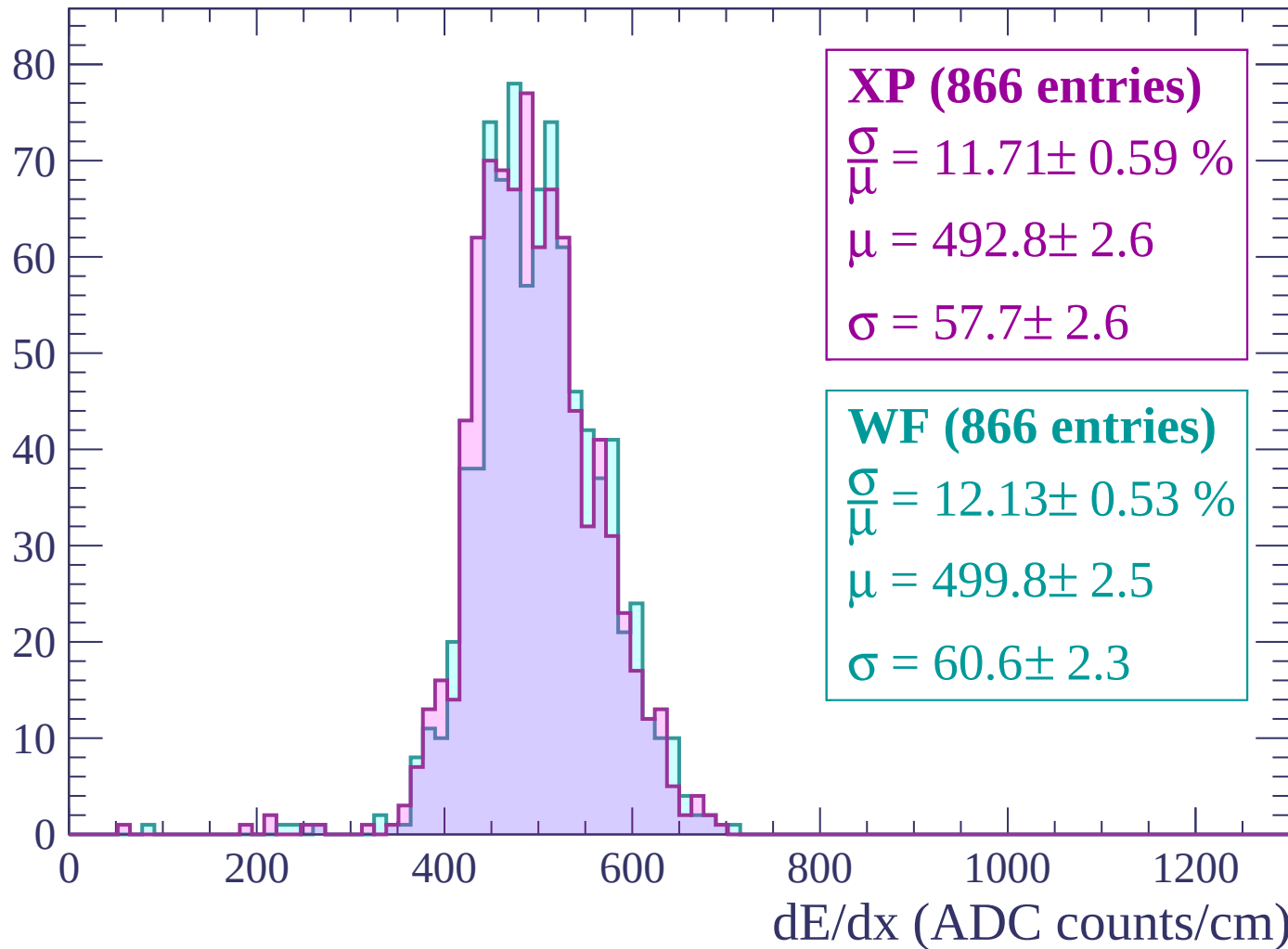
$$\mu = 495.6 \pm 2.4$$

$$\sigma = 60.9 \pm 2.2$$

dE/dx (ADC counts/cm)

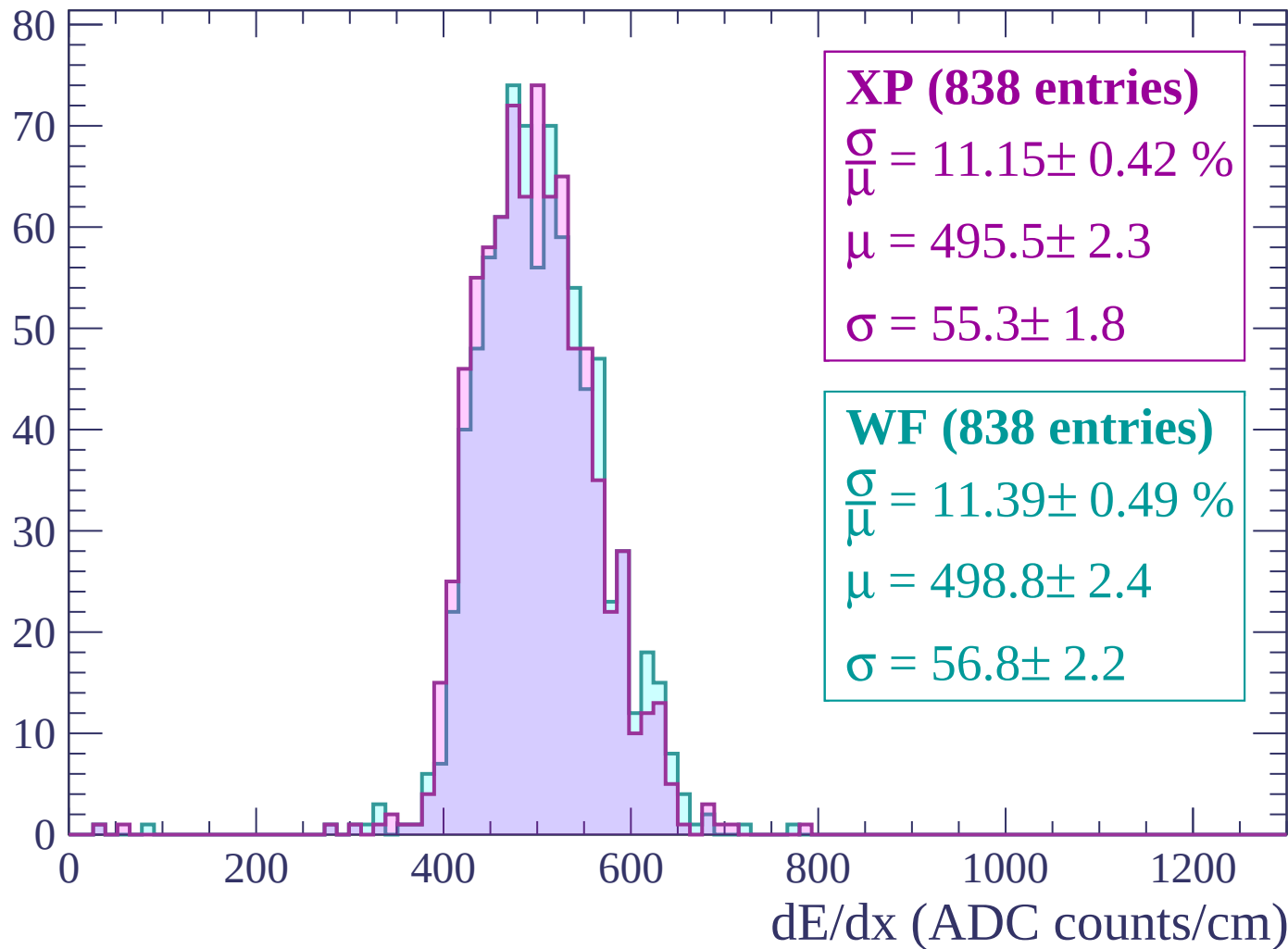
Energy loss | $1980 < p < 2040$

Count



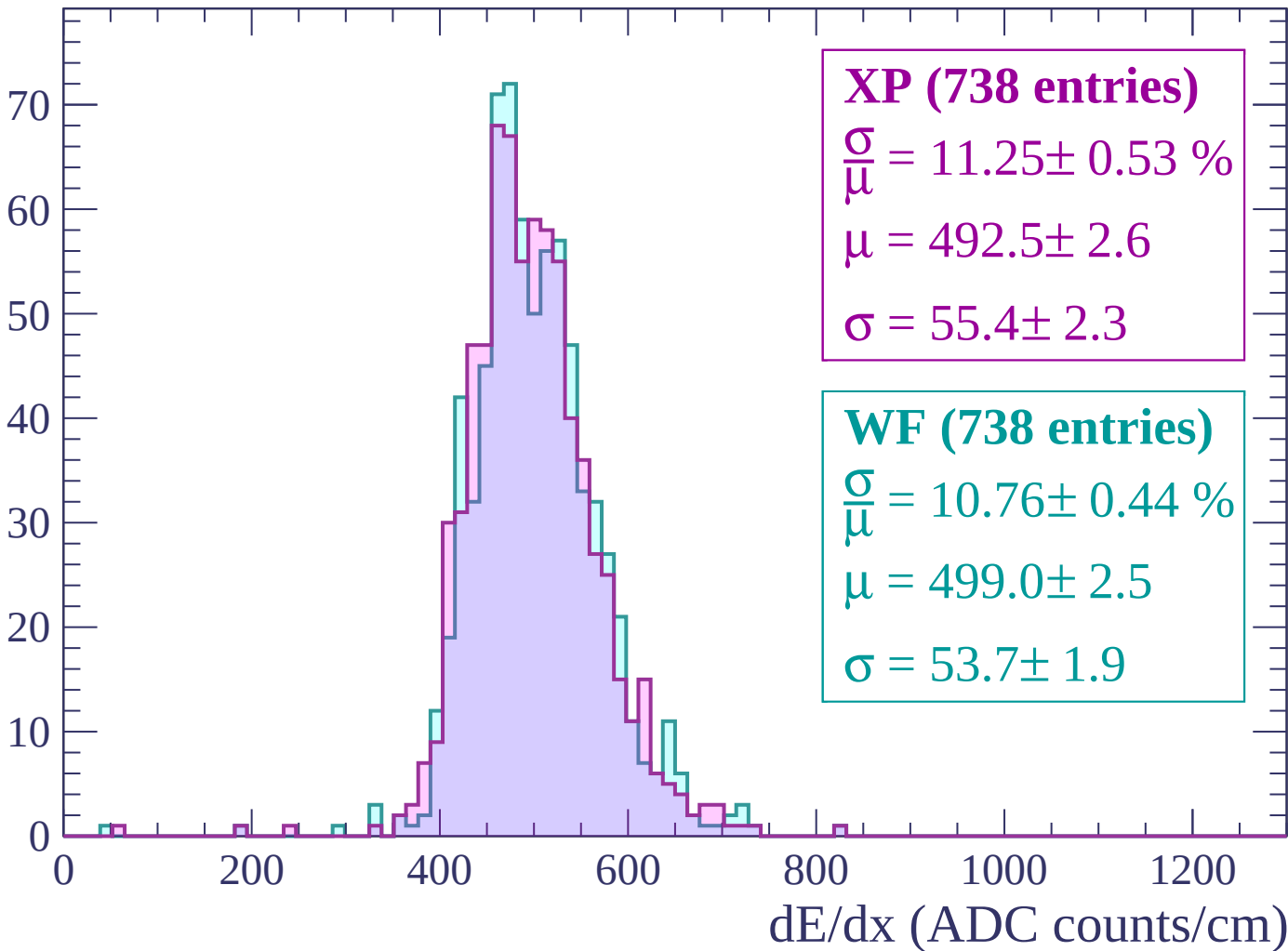
Energy loss | $2040 < p < 2100$

Count



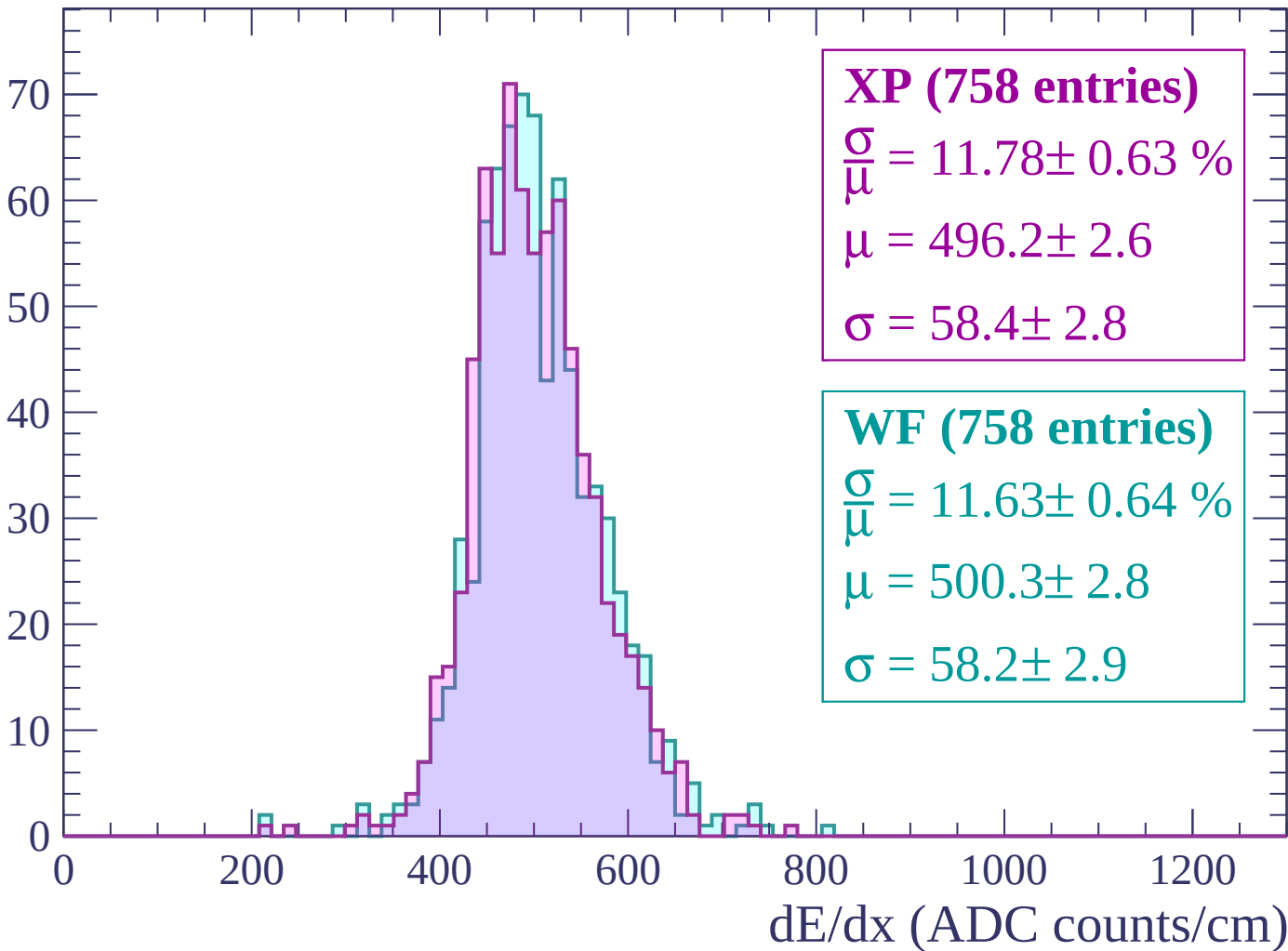
Energy loss | $2100 < p < 2160$

Count



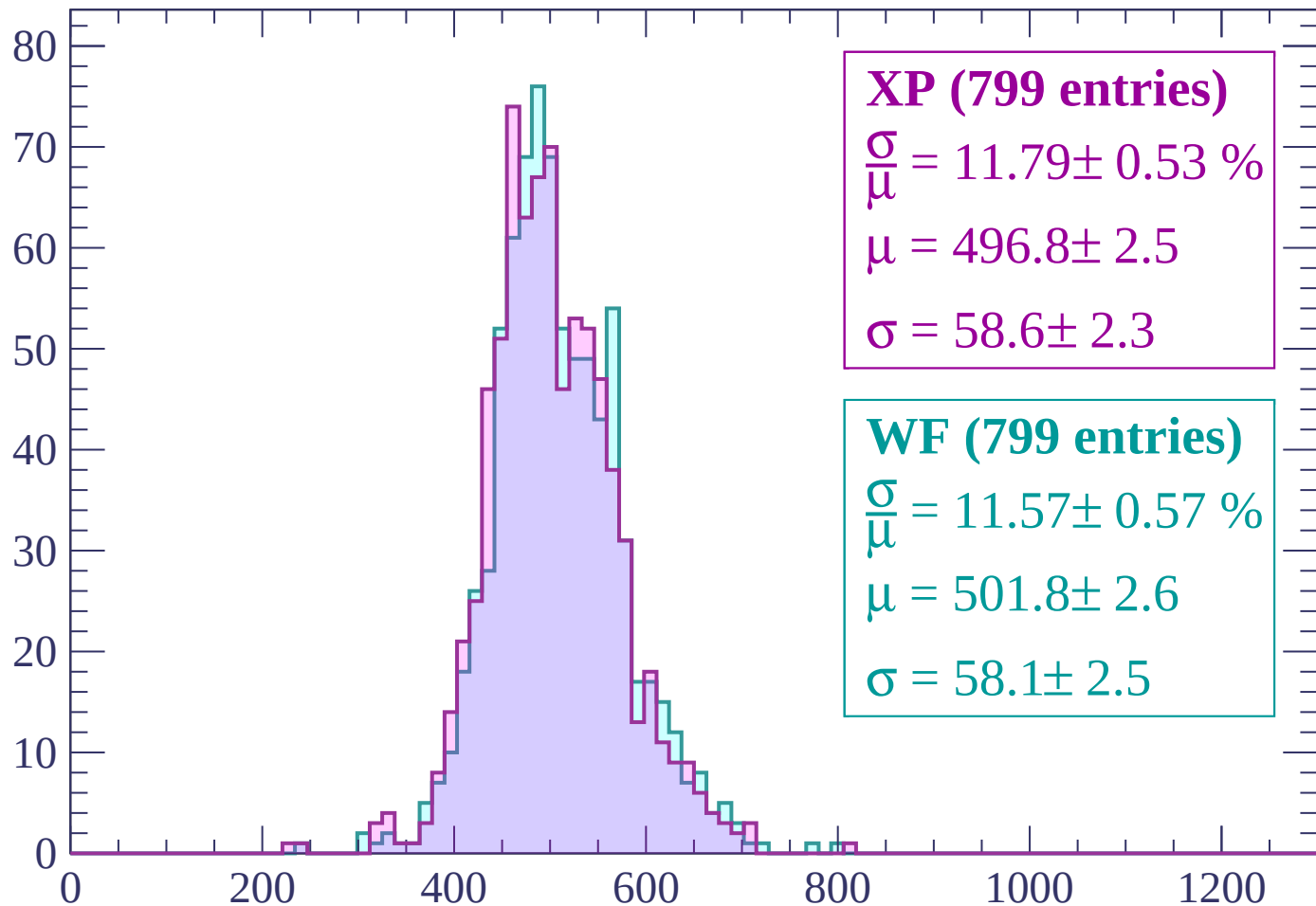
Energy loss | $2160 < p < 2220$

Count



Energy loss | $2220 < p < 2280$

Count



XP (799 entries)

$$\frac{\sigma}{\mu} = 11.79 \pm 0.53 \%$$

$$\mu = 496.8 \pm 2.5$$

$$\sigma = 58.6 \pm 2.3$$

WF (799 entries)

$$\frac{\sigma}{\mu} = 11.57 \pm 0.57 \%$$

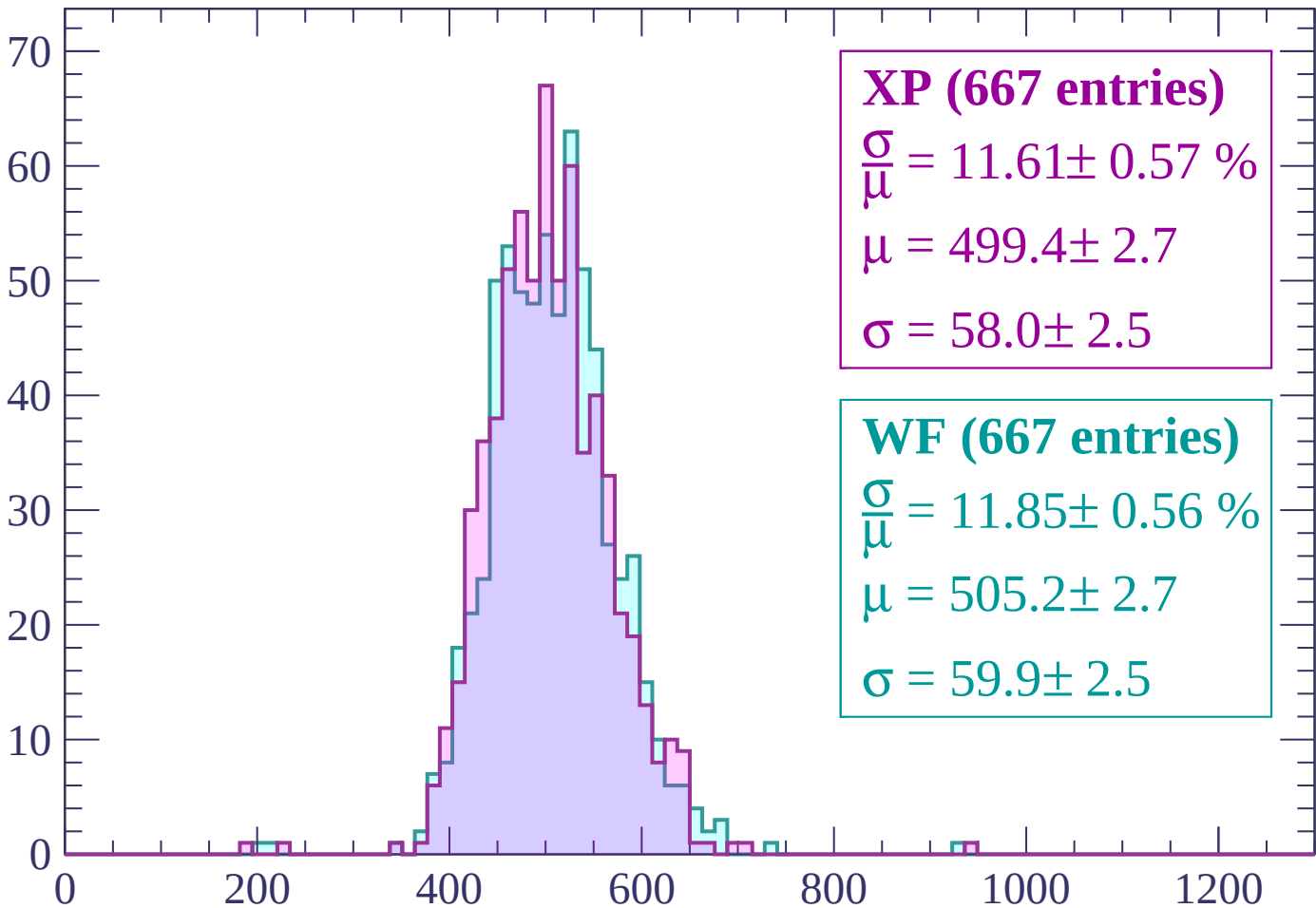
$$\mu = 501.8 \pm 2.6$$

$$\sigma = 58.1 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (667 entries)

$$\frac{\sigma}{\mu} = 11.61 \pm 0.57 \%$$

$$\mu = 499.4 \pm 2.7$$

$$\sigma = 58.0 \pm 2.5$$

WF (667 entries)

$$\frac{\sigma}{\mu} = 11.85 \pm 0.56 \%$$

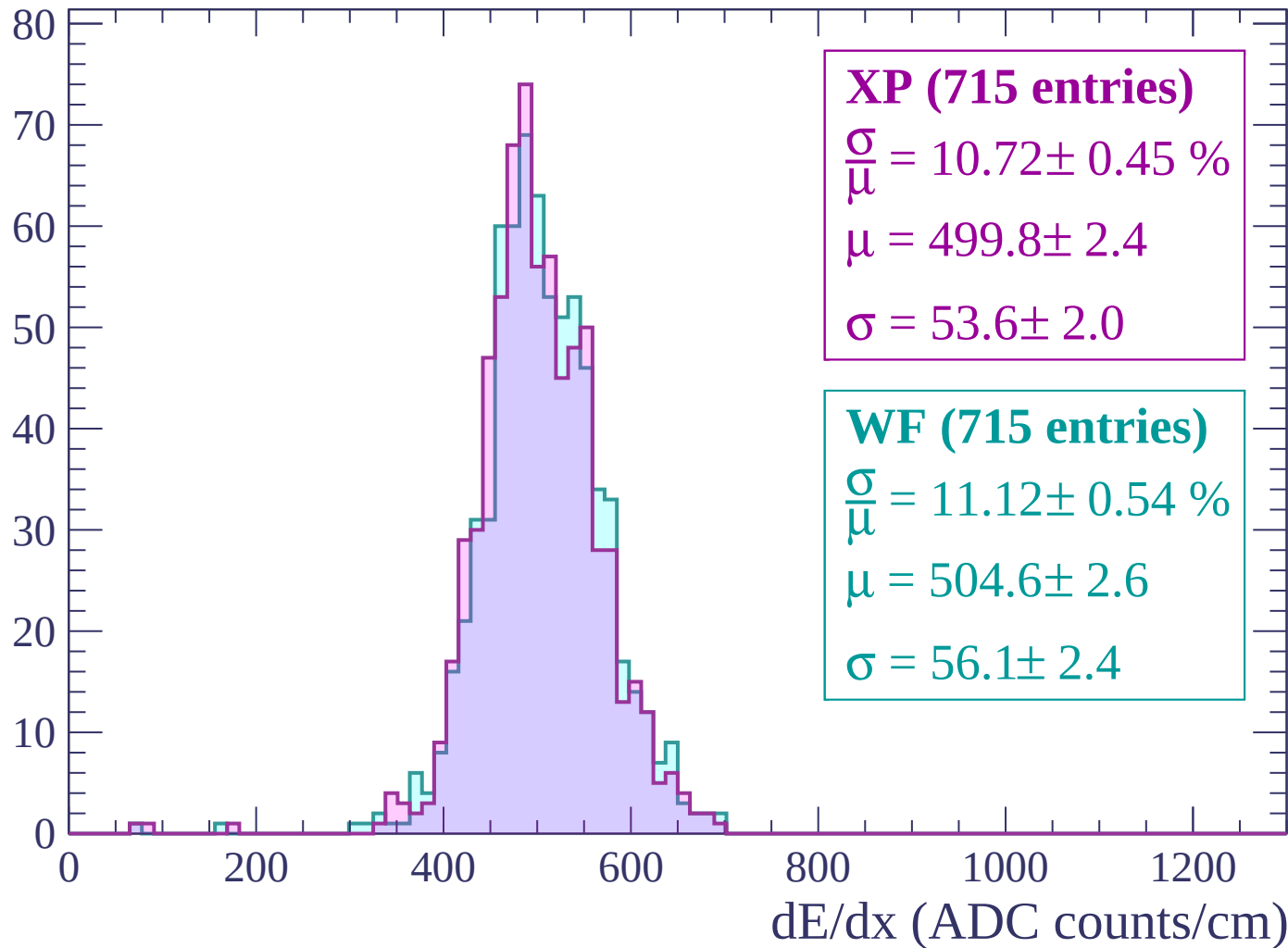
$$\mu = 505.2 \pm 2.7$$

$$\sigma = 59.9 \pm 2.5$$

dE/dx (ADC counts/cm)

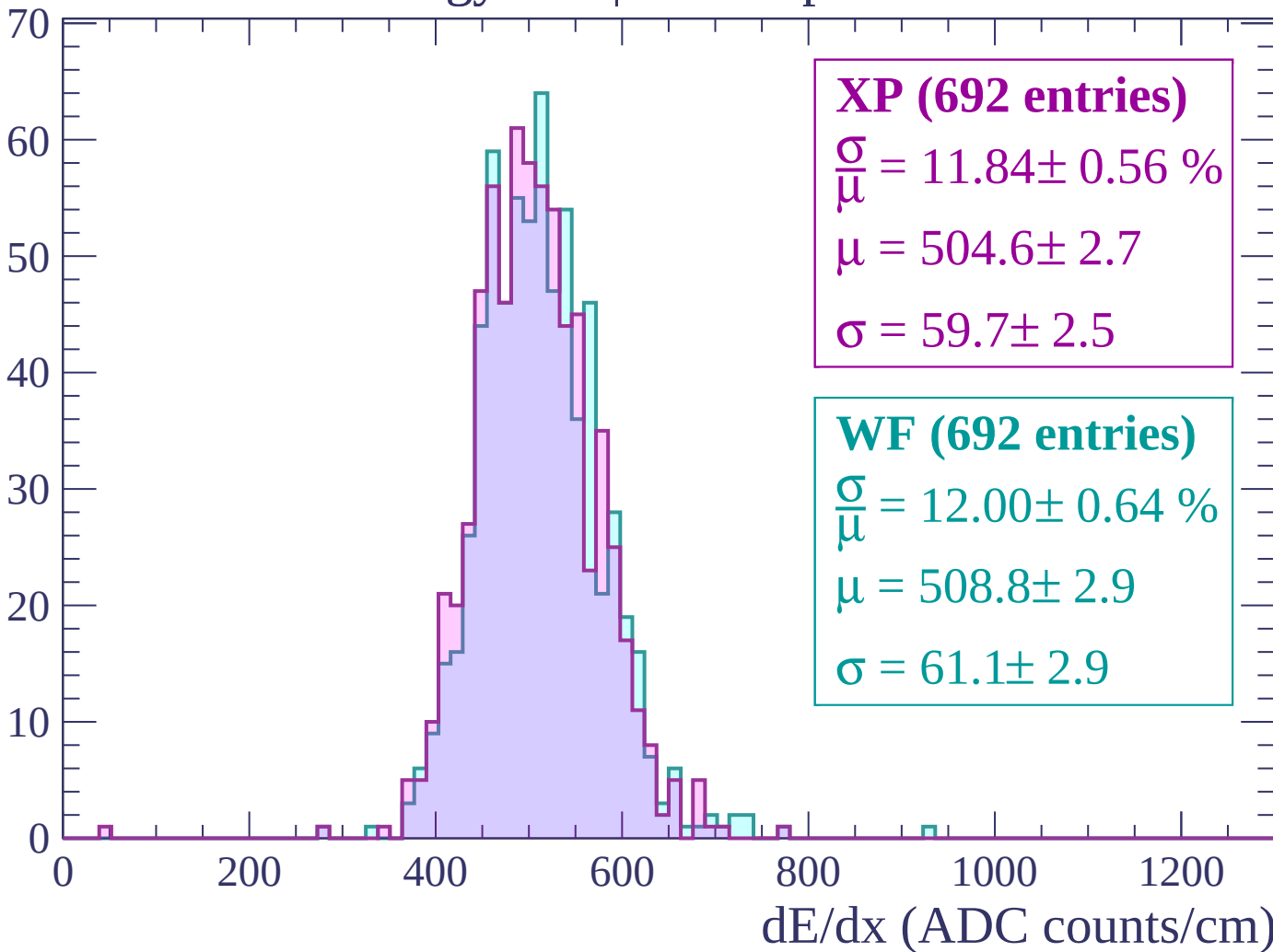
Energy loss | $2340 < p < 2400$

Count



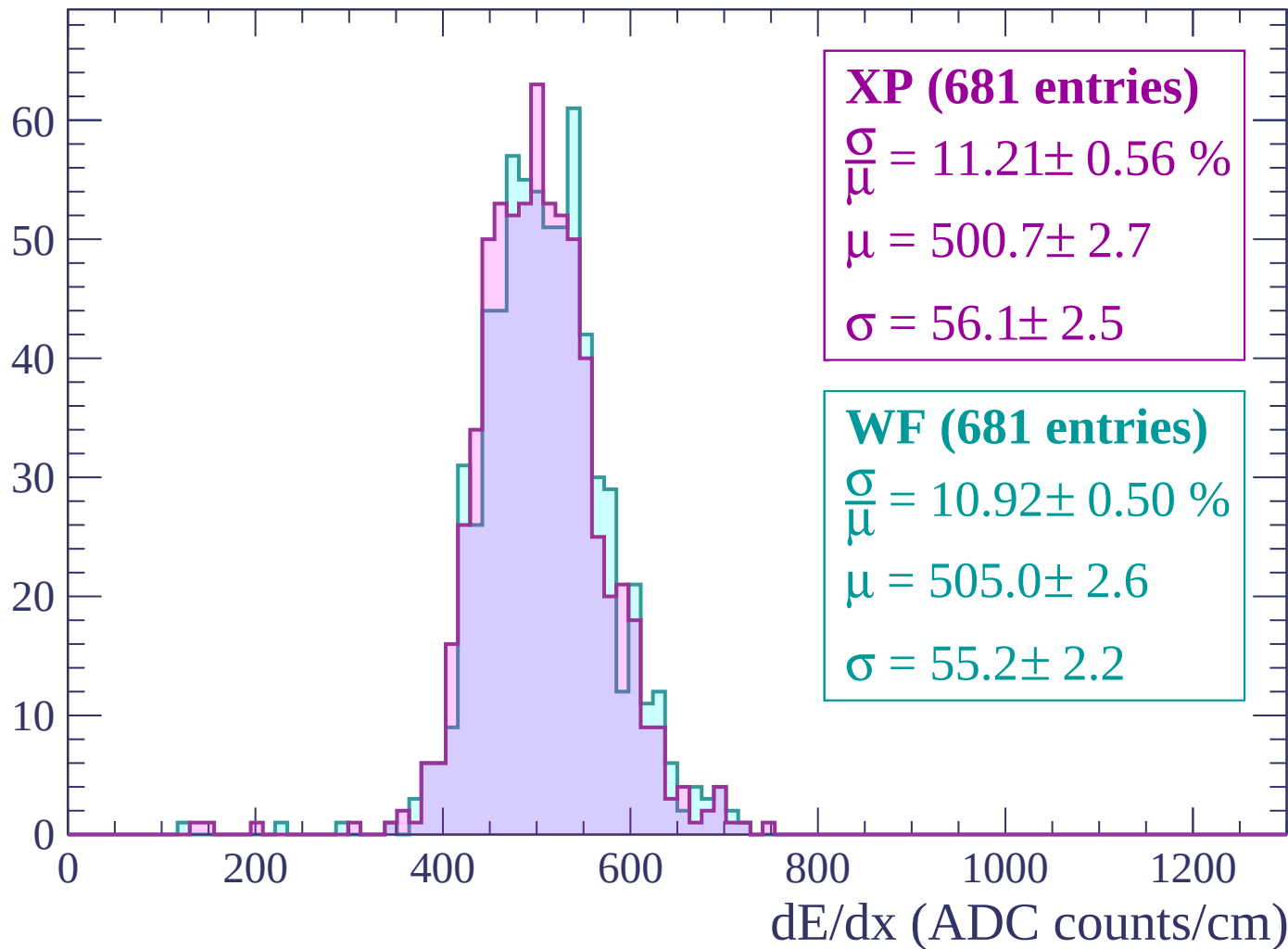
Energy loss | $2400 < p < 2460$

Count



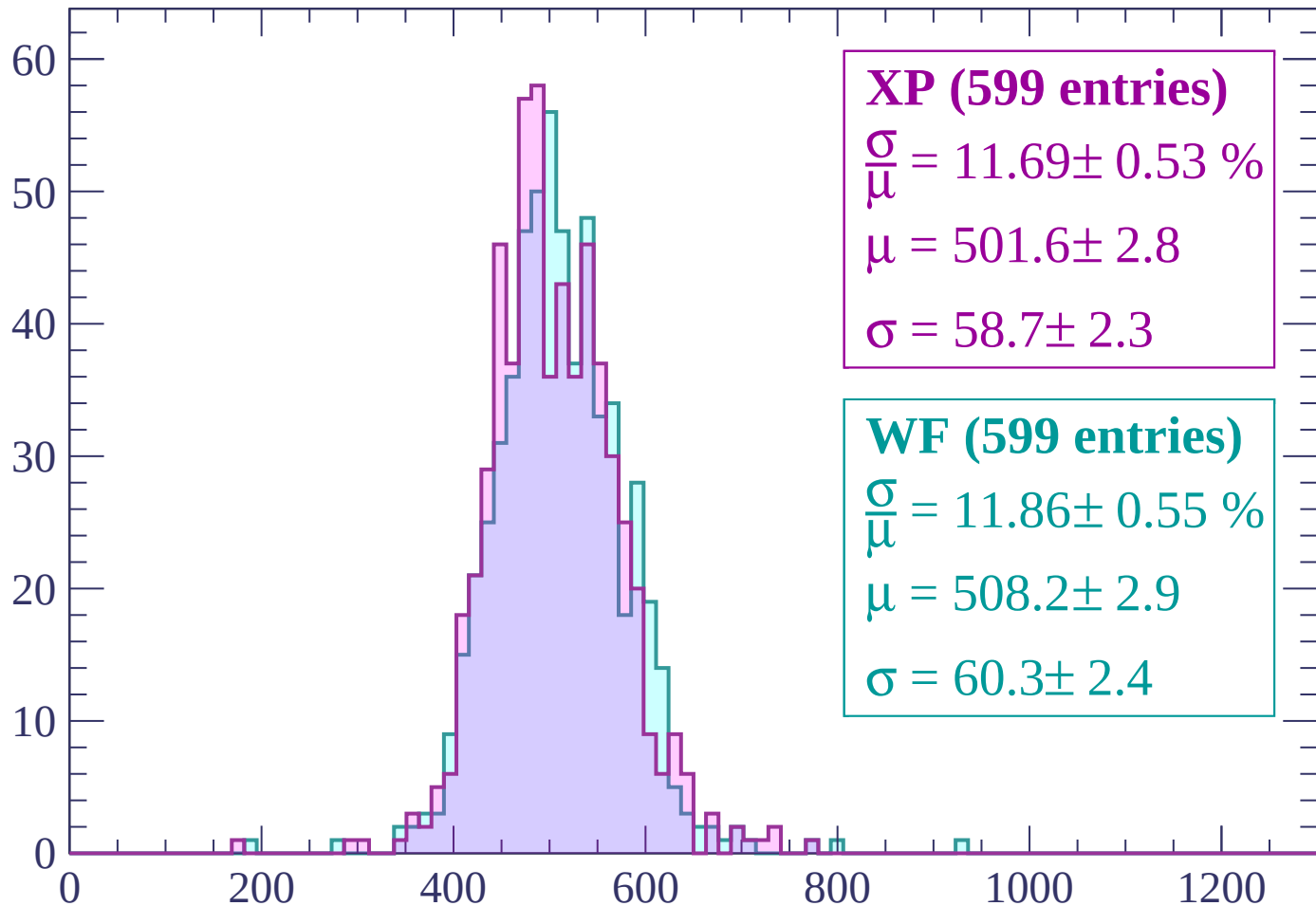
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (599 entries)

$$\frac{\sigma}{\mu} = 11.69 \pm 0.53 \%$$

$$\mu = 501.6 \pm 2.8$$

$$\sigma = 58.7 \pm 2.3$$

WF (599 entries)

$$\frac{\sigma}{\mu} = 11.86 \pm 0.55 \%$$

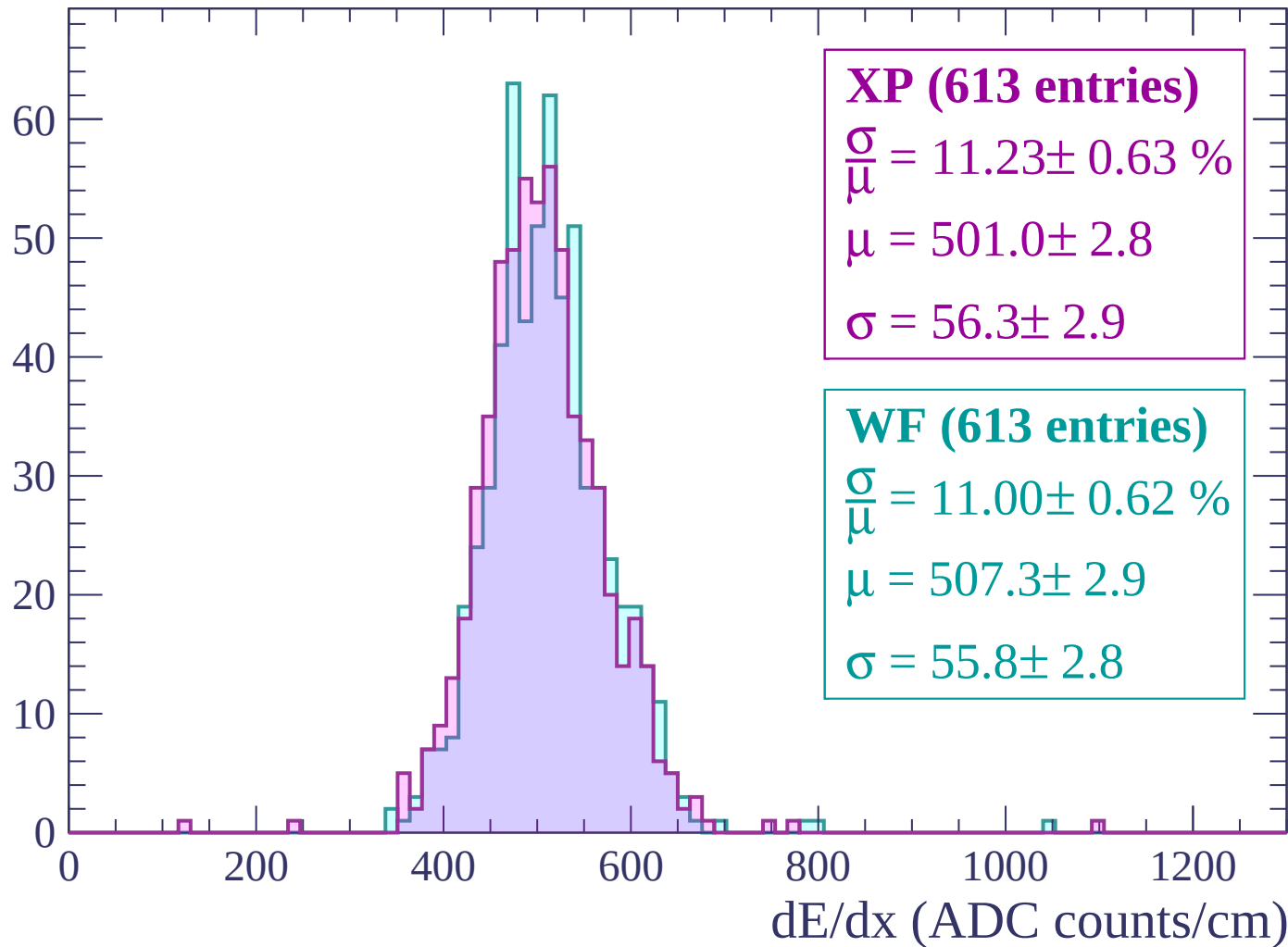
$$\mu = 508.2 \pm 2.9$$

$$\sigma = 60.3 \pm 2.4$$

dE/dx (ADC counts/cm)

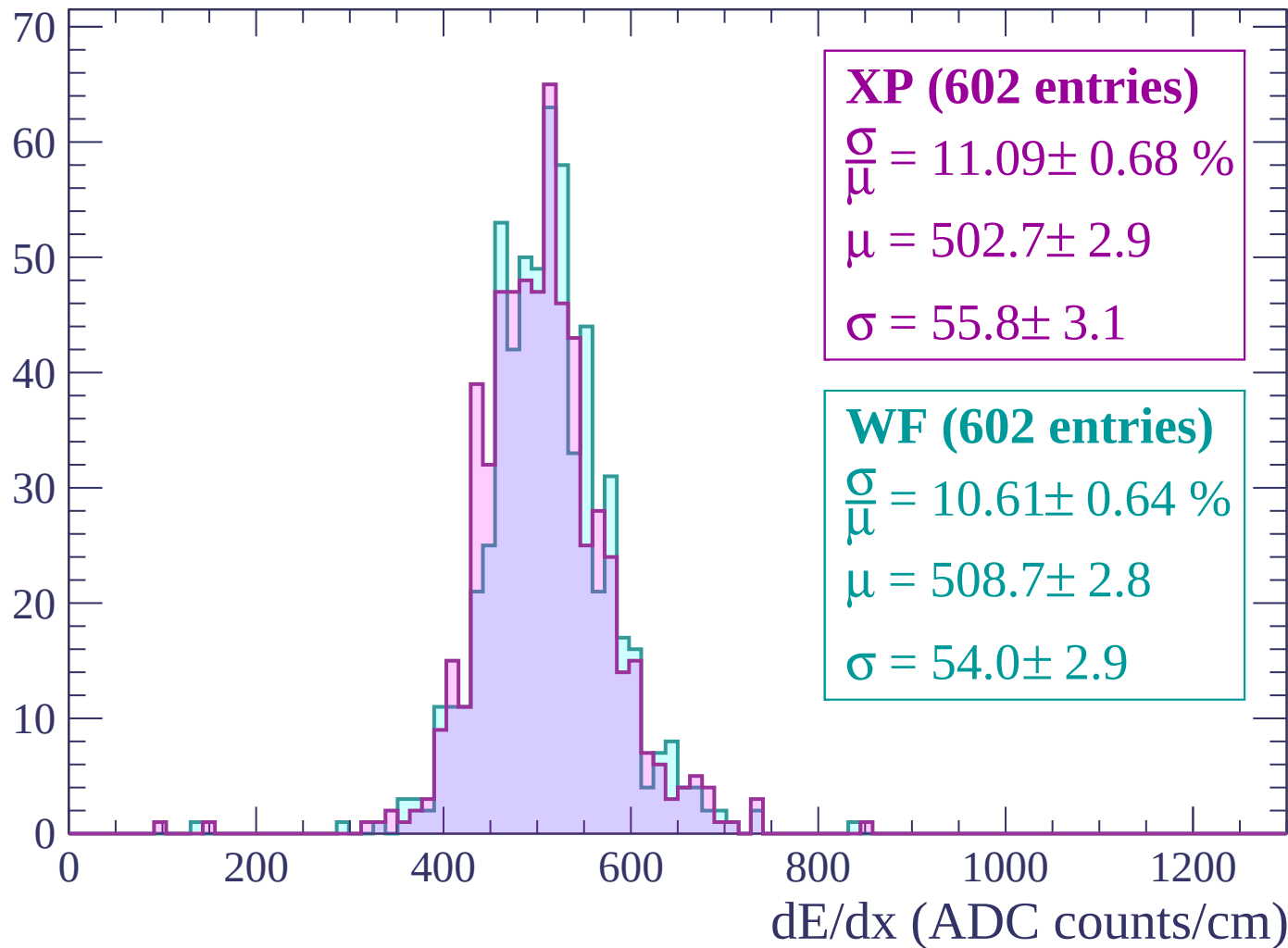
Energy loss | $2580 < p < 2640$

Count



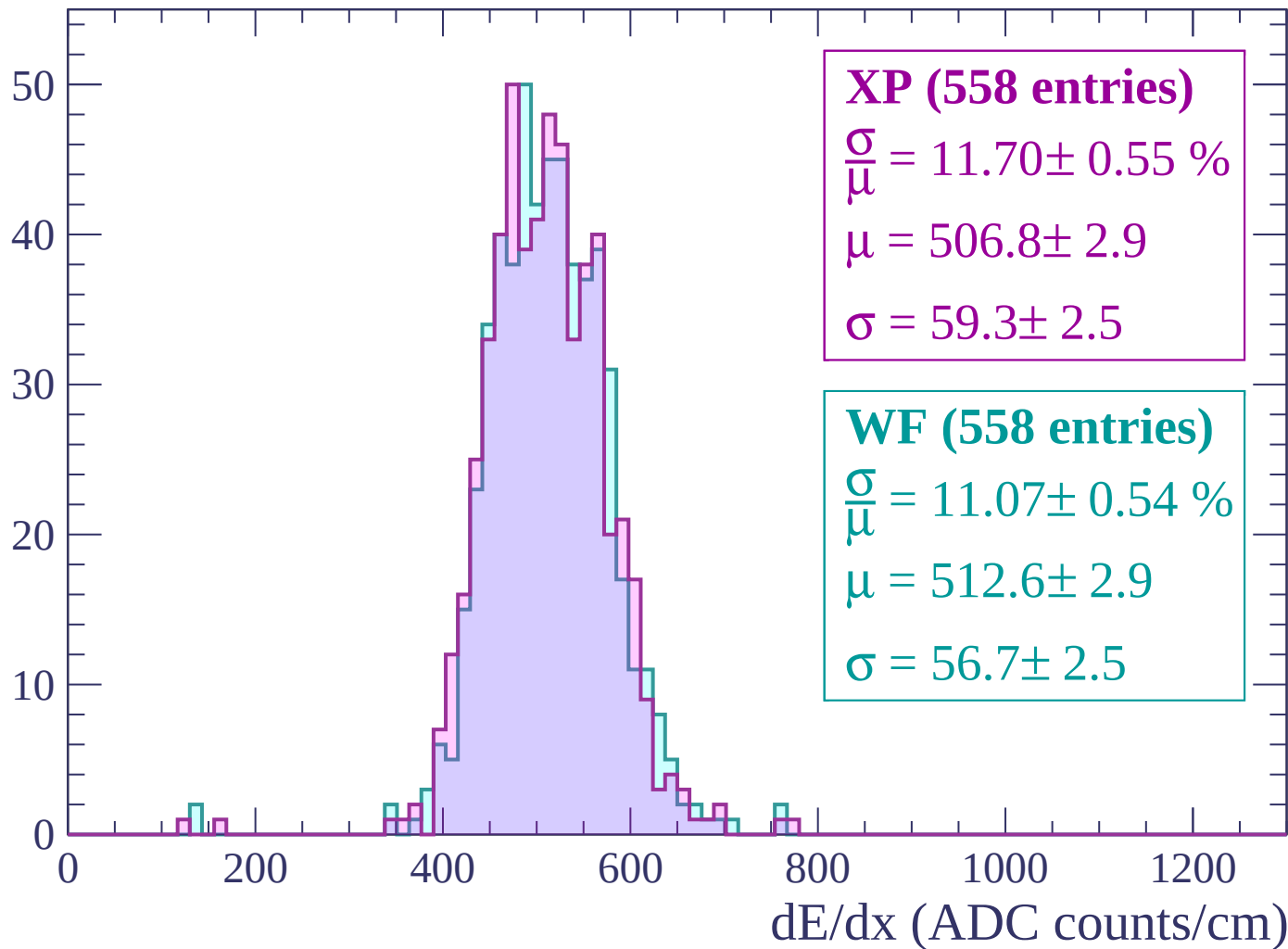
Energy loss | $2640 < p < 2700$

Count



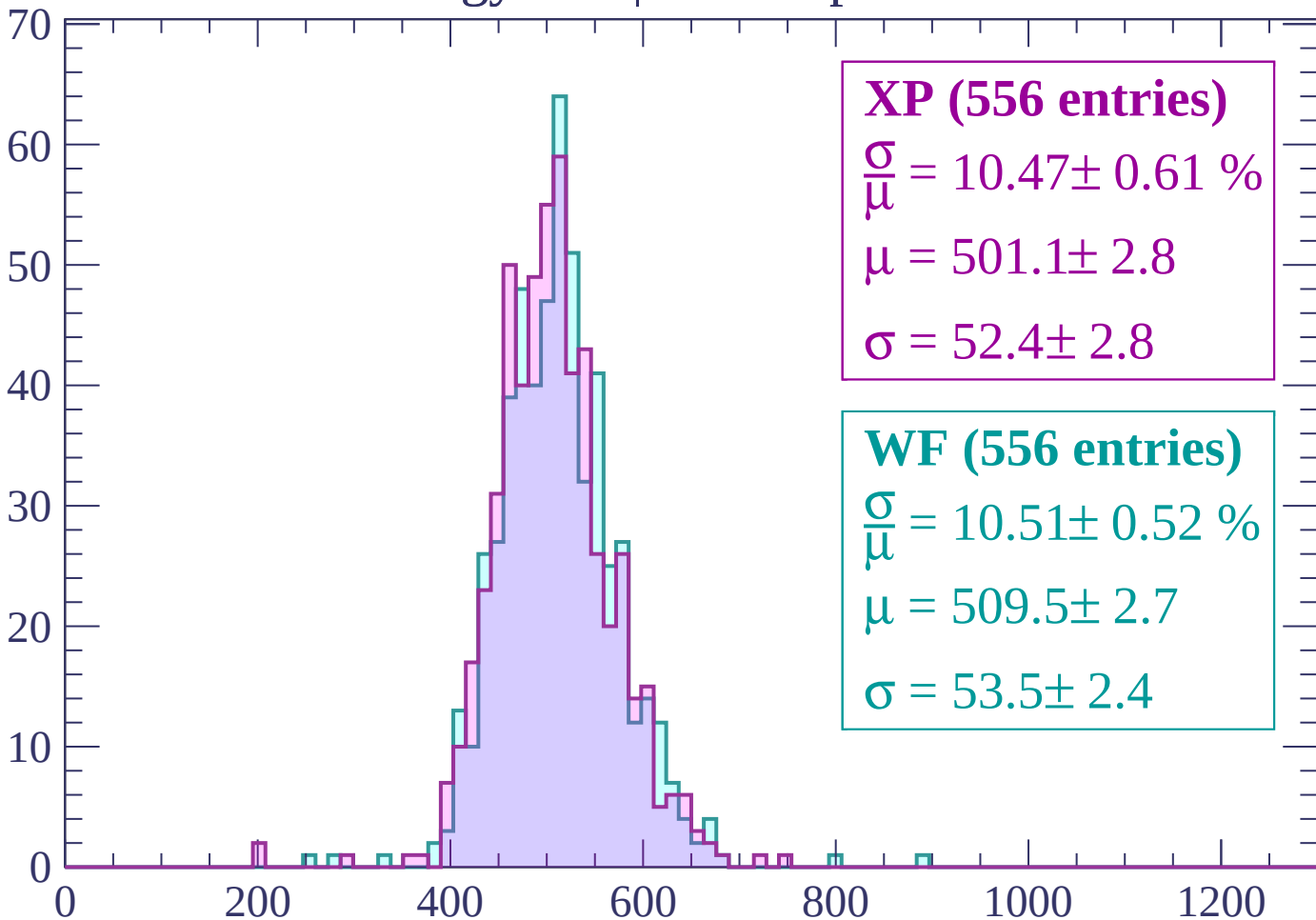
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP (556 entries)

$$\frac{\sigma}{\mu} = 10.47 \pm 0.61 \%$$

$$\mu = 501.1 \pm 2.8$$

$$\sigma = 52.4 \pm 2.8$$

WF (556 entries)

$$\frac{\sigma}{\mu} = 10.51 \pm 0.52 \%$$

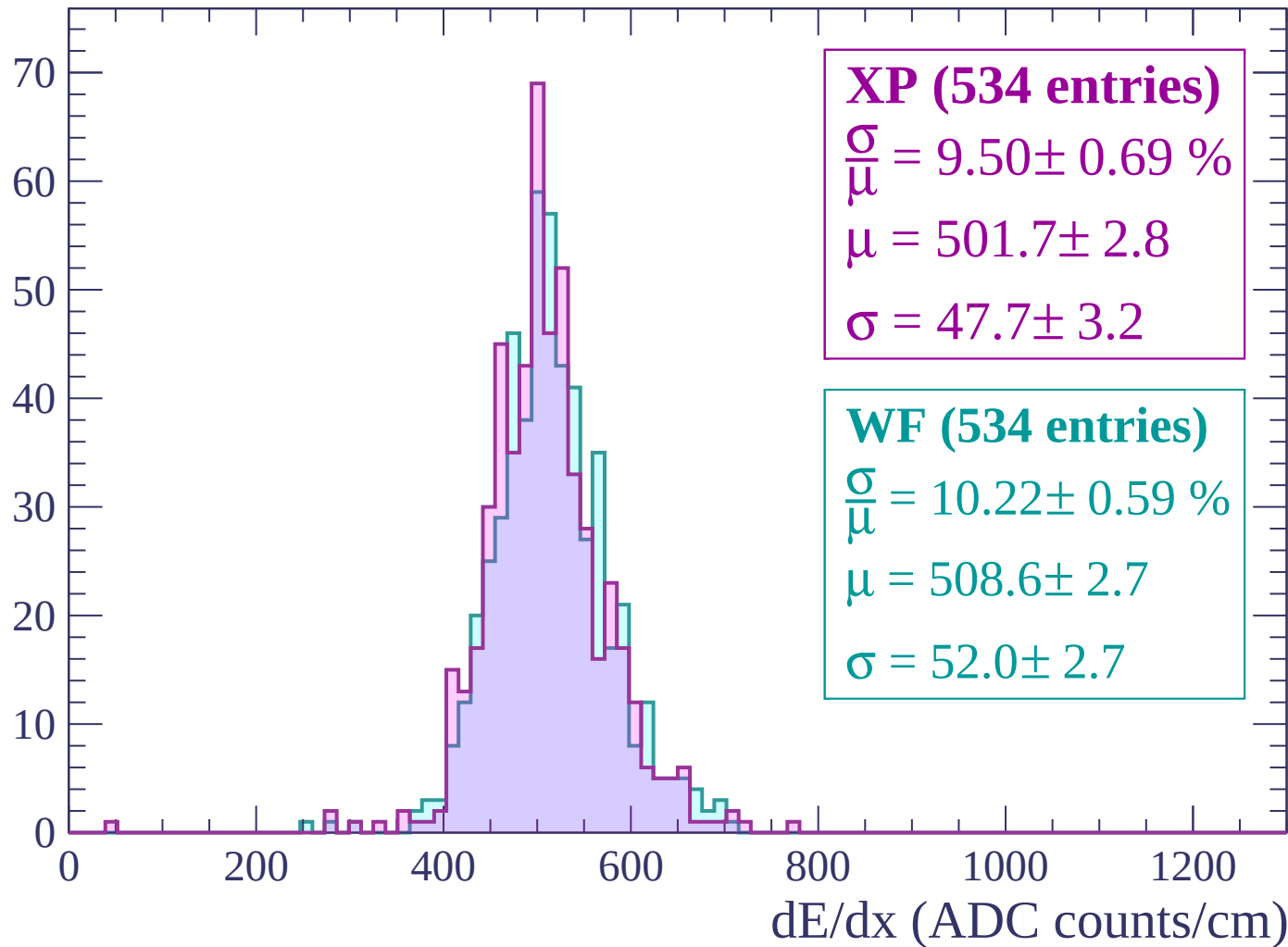
$$\mu = 509.5 \pm 2.7$$

$$\sigma = 53.5 \pm 2.4$$

dE/dx (ADC counts/cm)

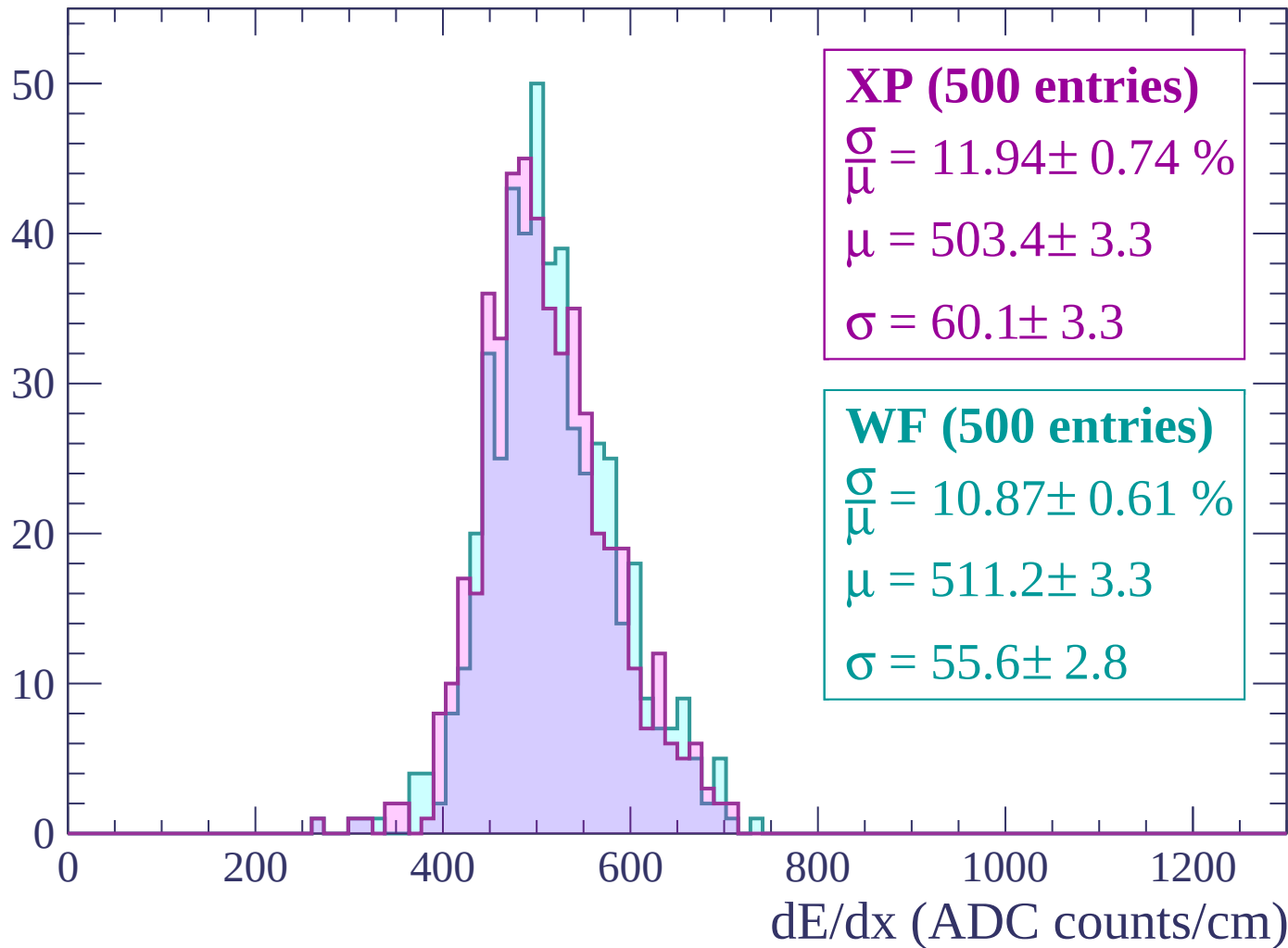
Energy loss | $2820 < p < 2880$

Count



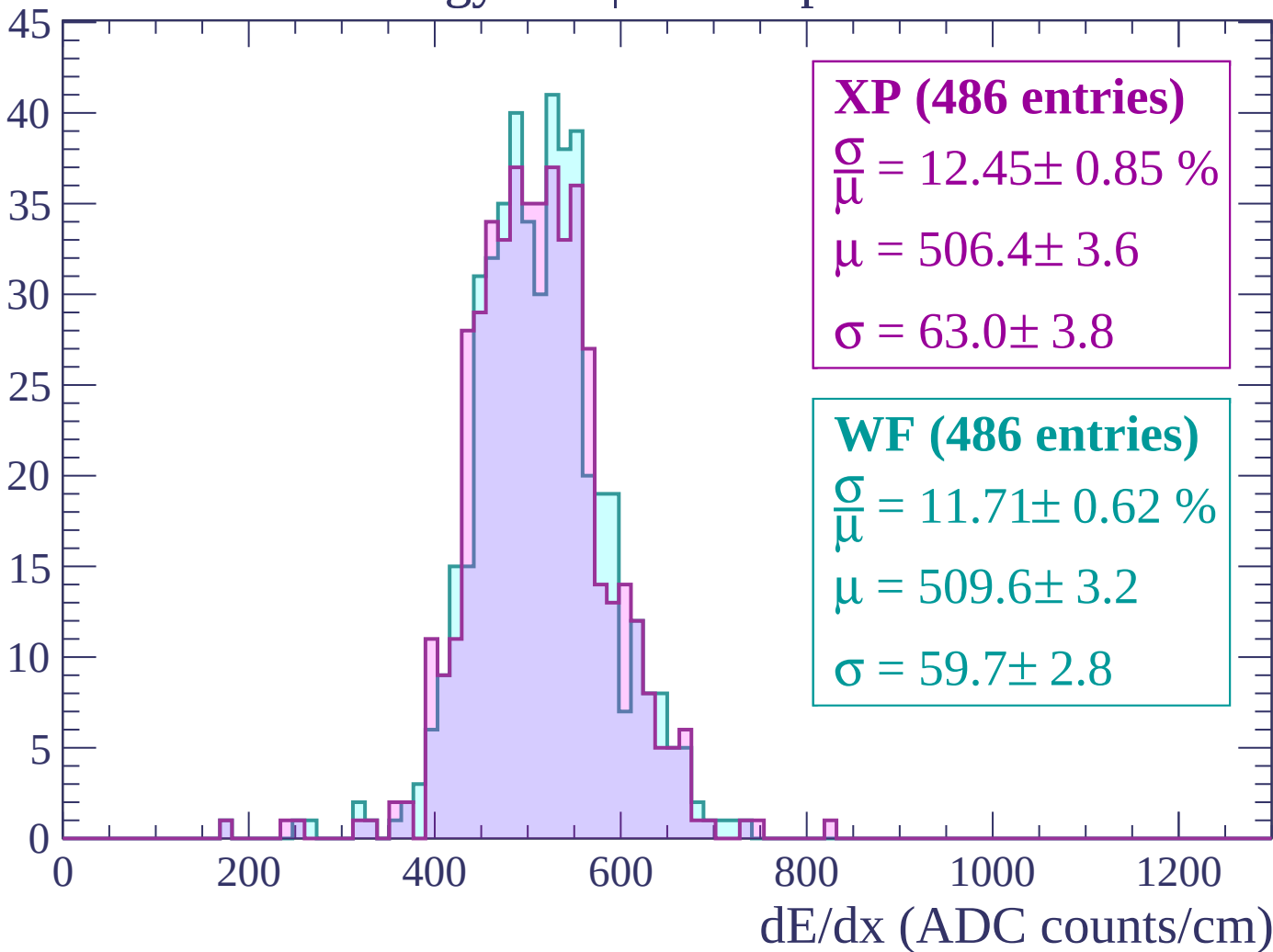
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

